

Dynamic treatment effects in Education

Anzheng Li

September 2026

Abstract

This paper examines the uncertainty involved in extrapolating education effects measured within short observation windows to lifetime welfare. I document the timing information available in existing studies and develop a model of educational attainment with and without a policy. Calibrated logistic curves describe alternative paths that match selected education estimates while allowing permanent changes in eventual attainment and temporary changes in acquisition speed or timing. Two deterministic simulations show how these alternatives change lifetime earnings values and how additional observations can narrow their range. I then apply the model to 38 policy calculations, reconstruct ranges for the marginal value of public funds (MVPF), and propose an ordering rule based on those ranges. Under the specified path and valuation assumptions, the same short-window education evidence can support conflicting welfare judgments even when the estimated effects are treated as known.

1 Introduction

A policy that raises bachelor's degree attainment at age 25 can create a permanent attainment gain, a temporary advantage from earlier graduation, or both. These possibilities have different economic consequences. Earlier completion can generate earnings before the comparison group catches up, while a permanent attainment gain can affect income and tax revenue over a much longer period. Educational costs also depend on when additional schooling occurs. An effect estimated at a particular age therefore answers an important causal question, but does not by itself determine the policy's lifetime value. This distinction matters when education estimates inform long-term earnings projections and comparisons of public spending, as in Hendren and Sprung-Keyser (2020).

This paper asks whether education effects measured within a finite window can rule out conflicting lifetime welfare judgments for the same policy. I take selected causal estimates from existing studies as inputs and construct education paths that match them under explicit outcome conversions and age mappings. Within each reconstruction, I hold these inputs, the external comparison profile, path restrictions, and the valuation rule fixed. I then compare earnings net present values (NPVs) and the marginal value of public funds (MVPF), which relates beneficiaries' willingness to pay to net fiscal cost. The main policy replacement changes projected earnings and fiscal returns while retaining the specified transfers, tuition, and public education-cost components. The question is whether different paths can place beneficiary welfare above or below public cost, or change whether a policy finances itself. A comparison with the original welfare calculation can additionally reflect changes in conversion or valuation; the comparison among paths within a reconstruction holds those choices fixed.

How much the reported education effects reveal about lifetime attainment depends on when education is measured. I document this timing information in a review of 459 studies that combines a search of 12 economics and education journals over 2000–2025 with additional welfare-source papers and an author manuscript. Among the 190 studies reporting bachelor’s-or-higher outcomes, 132 report one distinct estimation time or window for that outcome. More directly relevant to welfare extrapolation, 18 of the 31 studies reporting education-based lifetime or career valuation provide only one window for each valued education series. These counts describe the assembled sample rather than all education research. The original authors also discuss dynamics and extrapolation limits. Cohodes and Goodman (2014) interpret persistence and completion patterns as evidence of dropout rather than pure delay, while Angrist et al. (2022) and Schoonover et al. (2026) qualify their projected returns and discuss education timing. Their evidence informs the reconstruction without determining the unobserved continuation.

To connect the observed education effects to their possible continuation, I represent educational attainment for the same target population by two cumulative age profiles: $H_c^0(a)$ without the policy and $H_c^1(a)$ with it. Their difference is the education effect at each age. Both attainment levels are nondecreasing, but the difference can widen, narrow, or reverse sign. Reported effects constrain this difference at the source’s observation ages, over its age windows, or through changes between observations, using the specified conversions. For the numerical exercises, I use logistic continuations anchored at the last unaffected age and restrict their levels, acquisition rates, and empirical support. Census–ACS profiles supply the comparison curves and calibration restrictions, matched to policy geography and birth cohorts in the applications. These population profiles are external calibrations, not counterfactuals identified by the source study’s research design. The resulting paths match the selected and converted inputs; they need not jointly match every education result reported in the source paper.

Two deterministic simulations then examine how education effects and their measurement ages restrict possible lifetime earnings values. The first compares permanent effects on eventual schooling with temporary effects generated by changes in acquisition speed or timing, while retaining the same age-25 measurement rule. In each of the three illustrated scenarios, the measured effect is compatible with both positive and negative education NPVs under a common valuation rule. Some compatible paths reverse the initial education advantage at later ages, so the alternatives extend beyond different durations of a positive effect. These reversals are possibilities within the specified path family. The second exercise holds the generating path fixed and varies the observation ages. In the illustrated comparisons, adding an effect estimate at one additional age can substantially restrict the compatible paths and narrow the resulting NPV range. The reduction depends on when that additional effect is measured. Even when attainment eventually converges, earnings forgone during an earlier delay need not be recovered.

Applying the model to existing policy evaluations, I reconstruct MVPF ranges for 38 policy calculations and propose a rule for ordering the 37 with an original welfare valuation (L1). The reconstructions build on Hendren and Sprung-Keyser (2020) or the original authors’ valuations, replacing education-related earnings and fiscal returns with values implied by compatible education paths. Among these 37 calculations, 27 initial reconstructions (L2) admit positive-cost MVPFs on both sides of one, and 29 admit both beneficiary losses with public costs and beneficiary gains with self-financing.¹ Single-parameter comparisons show that permanent attainment changes often yield valuations close to L1, whereas matching the same education

¹Eleven tax calculations require an explicit policy-unit conversion, Early Head Start uses monetary components aligned to L1, and algorithmic placement adds a long-run earnings channel; Section 4.1 explains these adjustments. Several calculations share a source study. The remaining application values only the education channel of cannabis legalization and has no whole-policy L1.

input through temporary changes in acquisition can produce substantially different lifetime earnings and MVPFs. Additional comparable observations narrow some ranges by excluding continuations allowed by the initial inputs; twelve calculations with L1 support this richer reconstruction (L3), including converted inputs with explicit timing mappings. The ordering uses L3 where available and L2 otherwise, prioritizing higher lower endpoints and then higher upper endpoints among the paths found. Separately, 1,000 specified input repetitions per calculation examine how imprecision in education or accounting inputs changes the paths and their welfare values. These repetitions distinguish archived source draws from constructed perturbations. They address uncertainty in the inputs without supplying observations of the unmeasured education path.

The main contribution is a systematic analysis of how extrapolating finite-window education effects changes lifetime welfare judgments. Representing attainment with and without a policy as H^1 and H^0 , I construct paths that match the same education inputs and show how permanent and temporary effects differ in their implications for earnings and fiscal returns. This makes explicit which parts of a welfare conclusion follow from the selected education evidence and which depend on continuation assumptions. Research on time to degree and dynamic educational choices establishes the importance of when schooling is acquired (Bound et al., 2012; Bárány et al., 2023; Heckman et al., 2016). Work on long-term extrapolation studies how short-term observations can inform later outcomes under additional restrictions (Athey et al., 2026; Obradović, 2026; Fan et al., 2026). I bring this question to the education-based valuations in Hendren and Sprung-Keyser (2020), using the literature review to document the available information and the simulations and policy reconstructions to assess its welfare consequences. Separating uncertainty about continuation from imprecision in estimated inputs also complements research on inference for calibrated parameters and welfare metrics (Cocci and Plagborg-Møller, 2025; Vohra, 2026).

Two verified education paths that satisfy the same restrictions but imply opposing welfare conclusions establish that those restrictions do not resolve the comparison. The sensitivity exercises assess whether such conflicting conclusions remain under alternative calibration, valuation, and search specifications. They cover all 38 calculations and use additional education observations where available. Across the common specifications, 19 calculations continue to admit beneficiary losses with public costs and beneficiary gains with self-financing. These checks support the existence of conflicting judgments under several stated assumptions, but do not validate every assumption or exhaust the feasible paths. A search that finds only favorable paths cannot establish that every compatible path is favorable. The reported extrema are candidate values, no probabilities are assigned to paths, and input-repetition summaries do not establish confidence coverage.

The comparison of alternative education paths also relates to research on lifetime earnings measurement, long-term prediction, and policy choice with incomplete information. Haider and Solon (2006) show that the relationship between current and lifetime earnings varies with age, while Bhuller et al. (2017) document how additional schooling changes the age profile of earnings. These studies examine earnings over the life cycle; the reconstruction here also varies the education path entering the valuation. To forecast beyond experimental follow-up, García et al. (2020) combine experimental and nonexperimental data with economic models to estimate lifetime program benefits and costs. Chen and Ritzwoller (2023) develop efficient estimators combining short-term experimental and longer-term observational outcomes, and Imbens et al. (2025) use the sequential structure of short-term outcomes to address persistent confounding in such data combinations. These approaches clarify how additional data and identifying restrictions support long-term estimates. The present analysis examines which welfare judgments remain unresolved when selected education observations leave room for different attainment continuations. It also relates to sufficient-statistics welfare analy-

sis, which connects reduced-form estimates to welfare through economic restrictions (Chetty, 2009; Kleven, 2021). Projecting lifetime earnings and fiscal returns from an education effect similarly requires a restriction on its continuation. Finally, Manski (2011) studies policy choice when treatment responses are only partly known and no probability distribution is assigned to that uncertainty. This distinction between what the evidence determines and what a decision rule adds also guides the interpretation of the policy ordering in this paper.

The remaining sections move from the available education evidence to path construction, welfare applications, and their interpretation. Section 2 presents the motivating cases and literature statistics, and Section 3 develops the model and simulations. Sections 4 and 5 present the three main applications and the full policy collection. Section 6 discusses assumptions, robustness, and policy implications, and Section 7 concludes.

2 From Observed Education Effects to Lifetime Valuation

Assessing the lifetime value of education policies requires connecting effects estimated within finite observation windows to education paths over the life cycle. Figure 1 shows that educational attainment continues to rise after age 25. Among the 1985–1987 birth cohorts, bachelor’s-or-higher attainment (BA+) rises from 29.34 percent at 25 to 40.80 percent at 35; approximate schooling rises from 13.30 to 13.97 years. The profiles use the full-population American Community Survey (ACS) extract for 2000–2022, with fixed cohort weights.² The cohort comparison in Figure A1 shows that gains in degree attainment and schooling between ages 25 and 35 are larger for later-born groups.

To document the time information used in higher-education research and its role in longer-run interpretation and valuation, I review the studies described in Appendix A. The sample contains 459 papers: 447 from a search of 12 economics journals over 2000–2025, 11 additional welfare-source papers, and one author manuscript. Among 190 papers with BA+ outcomes, 132 report one distinct BA+ estimation time or window (69.5 percent); one window can span several years.³ Separately, 31 papers report education-based lifetime or career valuation. Among these, 18 report one age, follow-up, or stage-specific window per valued education series, and 12 report multiple outcome times.⁴ The following three scholarship studies illustrate the calculations and the dynamic evidence accompanying them.

Angrist et al. (2022) study Susan Thompson Buffett Foundation (STBF) scholarships in a randomized experiment. Among applicants targeting four-year colleges, awards raise bachelor’s degree completion by 8.1 percentage points six years after high school. Combining educational credentials and time enrolled with an ACS earnings model, they project a lifetime earnings gain of \$21,150 per award recipient at a 3 percent discount rate and a marginal value of public funds (MVPF) of 1.75 under hypothetical public funding. The authors examine graduation dynamics and describe the valuation as provisional: observed education histories inform the calculation, while later attainment and earnings require extrapolation.

Cohodes and Goodman (2014) estimate effects of Massachusetts’ Adams Scholarship on college choice, persistence, and completion. Scholarship eligibility lowers the probability of earning a bachelor’s degree

²Each cohort contributes ages 16–35 with equal weight at every age. Appendix A.5 gives sample, person-weighting, and education-crosswalk details.

³The journal-only share is 130 of 186 BA+ papers (69.9 percent). Coding distinguishes attained age, individual follow-up, and policy event time.

⁴One further paper values institutional degree flows in policy event time. Appendix A.3 records the valued outcomes and their observation clocks.

within six years of high-school graduation by 2.5 percentage points near the eligibility threshold. The authors interpret the persistence and graduation patterns as evidence of dropout rather than pure delay, and use ACS earnings differences to illustrate lifetime losses for students induced to switch colleges. Hendren and Sprung-Keyser (2020) construct a policy-average MVPF from the scholarship’s enrollment and completion effects, which together imply a 0.018-year reduction in schooling. They project the associated earnings and tax changes, then account for transfers and private and public education costs to obtain an MVPF of 0.72.

Schoonover et al. (2026) estimate the effects of Tennessee Promise in overlapping two-year age groups from 18–19 through 23–24. At ages 23–24, the estimated increase in bachelor’s-or-higher attainment is 2.7 percentage points, statistically significant at the 10% level under permutation inference. The authors combine the education effects with Tennessee-specific returns to schooling to project lifetime income and tax revenue. Their valuation yields positive willingness to pay and negative net public cost, classifying the program as self-financing with an infinite MVPF. They acknowledge that outcomes observed in early adulthood cannot fully distinguish lasting attainment gains from changes in the timing of schooling and labor-market entry.

These studies use different education outcomes and populations, and their authors already examine graduation dynamics or qualify the lifetime projections. The question pursued here is how much those projections can vary when the observed education evidence is retained but its unobserved continuation changes. The next section holds observed education effects fixed and varies age paths within stated economic restrictions. Applying the same valuation rule to each path isolates the role of those paths in the calculation.

3 Education Paths and Lifetime Valuation

A finite observation window leaves the evolution of educational attainment partly unobserved, even when the effects within that window are measured without error. To separate this information problem from sampling uncertainty, I hold a comparison path fixed and construct treated paths consistent with stipulated education effects. In each of the three displayed scenarios, the same age-25 effect admits paths with positive and negative education-channel net present values (NPV) under a common valuation rule. Adding the corresponding age-35 effect reduces the width of the numerical candidate range by 93.5–97.8 percent. These comparisons treat the observed effects as exact and hold the baseline and path restrictions fixed. The first experiment varies the education process while observing it at age 25; the second holds each generating process fixed and varies the observation ages. Together, they show which valuation conclusions follow from the observed effects and which depend on the unobserved schooling path.

3.1 Education Paths, the Baseline, and Path Restrictions

An education effect measured at a particular age describes the attainment difference at that age, while leaving its earlier and later evolution unresolved. For a fixed population in cohort c , let $H_c^0(a)$ and $H_c^1(a)$ denote the mean of a specified educational attainment outcome at age a without and with the policy, respectively. The outcome may record cumulative college entry, degree attainment, or completed years of schooling.⁵ The numerical illustration below uses years of schooling. For any of these cumulative outcomes, I distinguish the

⁵For binary attainment, $H_c^j(a)$ is a population share and $100\Delta H_c(a)$ is a percentage-point effect. Schooling levels and effects are measured in years. Current enrollment is not cumulative and need not rise with age.

policy effect on the attainment stock from its age- a increment:

$$\Delta H_c(a) \equiv H_c^1(a) - H_c^0(a), \quad \Delta h_c(a) \equiv \Delta H_c(a) - \Delta H_c(a-1). \quad (1)$$

Cumulative attainment satisfies $H_c^j(a) - H_c^j(a-1) \geq 0$ for $j \in \{0, 1\}$ in a fixed population. This restriction does not imply monotonicity of the treatment effect $\Delta H_c(a)$, because the two potential attainment paths can accumulate at different rates. The increment $\Delta h_c(a)$ may therefore be positive, negative, or zero, according to whether attainment accumulates faster, more slowly, or at the same rate under treatment. In particular, $\Delta h_c(a) < 0$ indicates a decline in the treatment–comparison attainment gap; neither level path need decline. Appendix B.1 gives the detailed definitions, and Figure 2 illustrates the two levels, their age-specific gaps, and the change in the gap between adjacent ages.

I represent the treated path with a three-parameter logistic continuation anchored to the comparison level. For a given starting age a_0 , write $\tau_c = a_0 - 1$ and $S_c = H_c^0(\tau_c)$. With $\theta_c = (\Delta K_c, \Delta b_c, \Delta d_c)$, define $L_c(a; \theta_c) = \Lambda[(b_c + \Delta b_c)(a - d_c - \Delta d_c)]$, where $\Lambda(x) = (1 + e^{-x})^{-1}$. The treated path is

$$H_c^1(a; \theta_c, a_0) = \begin{cases} H_c^0(a), & a \leq \tau_c, \\ S_c + (K_c + \Delta K_c - S_c) \frac{L_c(a; \theta_c) - L_c(\tau_c; \theta_c)}{1 - L_c(\tau_c; \theta_c)}, & a > \tau_c \end{cases}. \quad (2)$$

The normalization makes the continuation pass through S_c at τ_c and approach $K_c + \Delta K_c$ at older ages. Thus, the first affected observation reflects acquisition since a common starting point, rather than a level difference imported when switching curves. The parameter K is measured in attainment units, b in inverse age-years, and d in age-years. They govern the plateau and the speed and timing of accumulation conditional on the anchor. I call an education effect permanent when it changes eventual attainment, so that $\lim_{a \rightarrow \infty} \Delta H_c(a) = \Delta K_c \neq 0$. An effect is temporary when the attainment gap eventually vanishes. In the single-parameter exercises, a nonzero K change is permanent, whereas b or d changes with K fixed are temporary.⁶ Holding K , b , and the common anchor fixed, a higher d delays subsequent accumulation: attainment is lower at each finite age after the anchor, and any given level between the starting value and plateau is reached later. The delay varies across attainment levels, so an increase in d does not imply an equally sized delay at every point on the path. These are population-path parameters rather than individual graduation dates. Figure B1 illustrates their roles under the normalization; Appendix B.2 gives the derivatives.

The choice of a_0 reflects when the policy can first affect the specified attainment outcome for the fixed target population. I assume no policy effect or anticipation through τ_c , which justifies the common starting level in Equation (2). This compares the same population under two policy states; it does not require the observed treatment and comparison samples to have identical pre-policy means. In the simulation, the hypothetical policy first affects the population at age 18, making age 17 the anchor. The anchor is therefore determined by exposure timing, not by the age at which the researcher happens to observe an effect.⁷

For the years-of-schooling simulation, $H_c^0(a)$ is the equally weighted average of fitted ACS profiles for the 1985–1987 birth cohorts, using the full-population data at ages 16–35 introduced in Figure 1. Here c indexes the synthetic three-cohort group. Each cohort is fitted separately, and the fixed one-third weights

⁶Temporary describes eventual convergence, not disappearance within the valuation horizon. Such an effect may persist beyond age 65 and can have a nonzero lifetime NPV. A permanent effect can also involve changes in acquisition speed or timing.

⁷Observation at 25 does not justify anchoring at 24. The anchor must precede possible effects and anticipation. If exposure ages differ, use a common unaffected age or cohort-specific anchors before aggregating with fixed weights.

preserve its contribution at every age; values above age 35 are extrapolated.⁸ I vary H_c^1 while holding this comparison curve fixed. Appendix C, Table C1, and Figure C1 document the construction and fit.

The admissible path class combines the logistic specification with restrictions on attainment levels and annual increments. Both experiments set the first possible effect age to $a_0 = 18$ and evaluate paths through age 65. I require completed schooling under treatment to be nondecreasing, to lie between zero and 25 years, and to increase by no more than 1.5 years between adjacent ages, including the transition from age 17 to age 18. The total-parameter domain is $K \in [13.3370, 15.1395]$, $b \in [0.12, 0.65]$, and $d \in [8, 30]$.⁹ Let C collect these maintained conditions. They define which paths enter the comparison; satisfying them does not establish that a particular path is the response to an actual policy.

3.2 What a Single Observed Effect Leaves Unresolved

An education estimate restricts a feature of $\Delta H_c(\cdot)$ rather than revealing the entire path. Let the observation operator $\mathcal{M}$ map a candidate path into the effects represented by a study:

$$\mathbf{m} = \mathcal{M}\{\Delta H_c(\cdot)\}. \quad (3)$$

For attainment measured at one age, the relevant entry of $\mathbf{m}$ is $\Delta H_c(a)$. An average effect over an age window $\mathcal{A}$ instead enters as $\sum_{a \in \mathcal{A}} w_a \Delta H_c(a)$, using the source's weights.¹⁰ An effect on attainment acquired between ages a_1 and a_2 restricts $\Delta H_c(a_2) - \Delta H_c(a_1)$. These restrictions preserve the source study's outcome definition and time scale.¹¹ The source-specific mappings for credits, enrollment, and credentials enter the applications in Section 4. For the fixed comparison path and target population, the compatible path set is

$$\mathcal{P}(\mathbf{m}; a_0, C) = \{H_c^1(\cdot; \theta_c, a_0) \in C : \mathcal{M}\{H_c^1(\cdot; \theta_c, a_0) - H_c^0(\cdot)\} = \mathbf{m}\}. \quad (4)$$

An additional observed effect changes the moment restrictions; changing the parameter domain or the path conditions changes C . With one or two exact age-specific effects, the matching equations generally leave two or one free parameter dimensions. Appendix B.3 gives the minimum-distance formulation and the algebra used to solve these equations directly. I impose no probability distribution over compatible paths or their parameters. The resulting NPV and MVPF candidate ranges describe conditional possibilities, without assigning probabilities to a positive NPV or an MVPF above one.¹²

The first experiment fixes the observation age at 25 and varies the education process generating that observation. Holding two generating parameters at their reference values, I vary the third over its entire conditional feasible interval. Each scenario supplies its own age-25 effect. I then set aside knowledge of the generating parameters and search the common three-parameter class for paths matching that effect. Only

⁸The reference triple averages parameters, whereas H_c^0 averages fitted curves. Their anchored continuations differ by at most 0.000226 schooling years over ages 18–65; zero parameter increments need not imply an exactly zero effect.

⁹The plateau limits are weighted first and 99th percentiles of a historical state-cohort fit bank. Transporting this support to the full-population baseline is a calibration assumption (Appendix C.3). Speed, timing, and annual-increment limits are design restrictions. Binary outcomes require share-consistent bounds instead.

¹⁰An exact 0.1-year effect at 25 imposes $\Delta H_c(25) = 0.1$. An equally weighted two-percentage-point BA effect at ages 23–24 imposes $[\Delta H_c(23) + \Delta H_c(24)]/2 = 0.02$, not a restriction at the midpoint.

¹¹Six years after college entry need not mean age 24. Converting follow-up into age requires entry-age information or source-specific averaging. Policy event time also differs from individual follow-up.

¹²Grid spacing does not assign path probabilities. Here the stipulated effects have no sampling error, so candidate ranges are not confidence intervals. Finite search also does not certify global bounds; Appendix B.7 distinguishes these issues.

one generating parameter varies at a time, but all three candidate parameters may vary: the observer is not assumed to know which parameter the policy changed. Appendix B.6 specifies the finite grids used to span the intervals.

An education effect observed at age 25 leaves open whether the effect is permanent or temporary. Panel A of Figure 3 compares a permanent increase in completed schooling, a temporary gain from faster accumulation, and a temporary loss from delayed accumulation followed by catch-up, from left to right.¹³ The orange curves describe these stipulated responses. In the middle column, the early advantage largely disappears as the comparison population catches up; in the right column, the treated population initially falls behind and subsequently catches up. The shaded regions summarize the alternative paths found when only the age-25 effect is observed. Within each column, every candidate reproduces that effect and satisfies the same maintained restrictions, yet the subsequent education outcomes can differ. For both positive early effects, the candidates include persistent gains and later reversals of the attainment advantage. For the negative early effect, they include persistent shortfalls, catch-up, and a later attainment advantage. These are possible evolutions of population means; they do not identify particular individuals' graduation or return-to-school decisions. Observing the gap at one age therefore leaves uncertainty about both eventual schooling and how long an advantage or disadvantage lasts. Under the valuation rule introduced next, these differences change the duration of earnings gains or losses and the timing of schooling costs.

3.3 From Education Paths to Net Present Value

Education paths that reproduce the same observed effects can imply different economic values under a common earnings and cost schedule. For college policies without observed earnings effects, Hendren and Sprung-Keyser (2020) project income changes from education outcomes using returns estimated by Zimmerman (2014). The superscripts point and path distinguish the benchmark and its path-based extension. I write both values as functions of the schooling-effect path ΔH_c , holding the observation age a^* fixed. The benchmark uses only the observed gap $\Delta H_c(a^*)$ in completed schooling years per member of the target population.¹⁴ Let $Y_0(a)$ be annual comparison earnings in a common dollar year and r_t^{HSK} the proportional earnings effect per additional schooling year in period t after policy onset. The dollar premium is $\pi_{\text{YS}}^{\text{HSK}}(a) = Y_0(a)r_{a-a_0+1}^{\text{HSK}}$. With first possible effect age a_0 , terminal valuation age A , and discount rate ρ , write the common discount factor as $D_\rho(a; a_0) = (1 + \rho)^{-(a-a_0)}$. The single-effect benchmark is

$$\text{NPV}_c^{\text{point}}(\Delta H_c; a_0, A, \rho) = \sum_{a=a_0}^A D_\rho(a; a_0) \pi_{\text{YS}}^{\text{HSK}}(a) \Delta H_c(a^*). \quad (5)$$

The observation age a^* stays fixed as the valuation age a runs from a_0 to A . The benchmark therefore assigns the same NPV to all paths with the same observed gap. The retained schedule assigns negative returns in the first seven periods and positive returns thereafter.¹⁵ Thus, a single observed education difference generates

¹³Only the named parameter moves to its conditional-interval midpoint. Relative to (13.9222, 0.2575, 12.5649), the increments are (0.3160, 0, 0), (0, 0.1113, 0), and (0, 0, 6.4351). The rule is independent of valuations and does not imply equal-sized policy shocks. Table B1 reports all scenario parameters.

¹⁴Appendix B.4 gives schooling conversions and the treatment of age-window averages.

¹⁵The schedule uses approximately -0.0667 in periods 1–7 and 0.1126 thereafter, following the Hendren and Sprung-Keyser replication's conversion of Zimmerman's estimates. It is an illustrative benchmark; their policy accounts also use other education inputs and observed earnings.

an income profile that varies with age through earnings, returns, and discounting. Its value depends on using that same education input in periods for which the education path is unobserved.

I retain the earnings-profile and discounting logic but allow completed schooling to differ at every age. The stock $\Delta H_c(a)$ determines the education premium available at that age; the flow $\Delta h_c(a)$ is used to approximate the difference in schooling-related forgone work. For example, if treatment postpones attainment, the temporary schooling deficit already lowers the earnings-premium component until the comparison and treated paths converge. The opportunity-cost component concerns the earnings displaced while education is being acquired, which is a separate object from the return to a completed credential. To distinguish these channels, I use the positive dollar premium $\pi_{YS}^+(a) = Y_0(a)r^+$, where r^+ is the retained positive return per completed schooling year, and a cost c_{HSK} per additional schooling-year flow. The path-based value is

$$\text{NPV}_c^{\text{path}}(\Delta H_c; a_0, A, \rho) = \sum_{a=a_0}^A D_\rho(a; a_0) [\pi_{YS}^+(a)\Delta H_c(a) - c_{HSK}\Delta h_c(a)]. \quad (6)$$

Comparisons across candidate paths hold Equation (6) fixed. Relative to the single-effect benchmark, the path account changes both the education input and the treatment of forgone earnings. The positive premium $\pi_{YS}^+(a)$ replaces the signed schedule before the explicit flow cost is subtracted, avoiding a second charge for the same early earnings adjustment. Thus comparisons with the benchmark do not isolate continuation alone. Both terms inside the brackets are in dollars per member of the target population. A negative $\Delta h_c(a)$ produces a relative opportunity-cost saving because schooling acquisition is lower under treatment at that age; it does not imply a negative absolute cost of education. The first experiment uses the observation age $a^* = 25$. For all candidate paths, I use the same earnings profile, terminal age $A = 65$, annual discount rate $\rho = 0.03$, and cost $c_{HSK} = \$4,172.26$ per additional schooling-year flow, in 2000 dollars.¹⁶ For other paths, Equation (6) is an additional valuation assumption: completed schooling does not directly measure enrollment or forgone work, and applying the positive premium at early ages also requires a calibration. Appendix B.4 gives the calibration details and age-by-age decomposition. These education-channel values precede the transfers, taxes, tuition, and public expenditures required for a policy's full MVPF.

Applying the common path valuation to the candidates in the first experiment produces both positive and negative values in all three displayed scenarios. Table 1 separates each scenario's single-effect benchmark, its known generating-path value, and the values of other paths matching the observed effect. The age-25 candidate NPV ranges, in thousands of 2000 dollars, are $[-6.28, 46.44]$ for the plateau change, $[-3.19, 46.80]$ for the speed change, and $[-24.11, 43.16]$ for the timing change. Finding paths with opposite signs establishes that the stipulated education evidence and maintained restrictions do not determine the NPV sign. The generating-path value exceeds the single-effect benchmark in the plateau and timing examples and is below it in the speed example, so the direction of that comparison is not uniform. Table B2 summarizes the full scans; the complete scenario results remain in the replication files.

The annual accounts in Figure 4 show how schooling accumulated earlier produces earnings gains over more years, while additional acquisition incurs a contemporaneous cost. The illustration combines a higher plateau with changes in the speed and timing of accumulation; Figure B2 shows its corresponding level

¹⁶I match the signed-premium and stock-flow accounts on a reference path acquiring one schooling year evenly over ages 18–24. This does not equate the fixed-input benchmark with every path NPV. The coefficient is a calibration, not a universal policy cost estimated by Hendren and Sprung-Keyser.

paths.¹⁷ Its schooling gap is 0.364 years at age 25 and remains 0.263 years at age 65. The single-effect and full-path NPVs are approximately \$15,390 and \$12,397, respectively. Panel A's orange bars are the early losses embedded in the signed earnings schedule; Panel B's are costs assigned to schooling-acquisition flows. Later negative flow costs represent relative savings as the comparison group partly catches up. The initial charge concerns additional acquisition after the common age-17 anchor. These components illustrate the maintained accounting assumptions rather than observed earnings losses.

The distinction between permanent and temporary effects can also be examined with the education inputs used in the policy applications. Figure B3 in the appendix compares each reference earnings NPV with the value obtained by matching its education input through a plateau, speed, or timing change alone. Table B3 reports the corresponding normalized NPVs and parameter changes, including calculations without a comparable pair. The other two parameter families remain at their calibrated baseline values. The K specification produces a permanent effect; the b and d specifications produce temporary effects as the paths eventually converge. For Adams, the converted schooling input implies an earnings loss of about \$1,109 under a plateau change, compared with \$188 and \$150 under speed and timing changes, against a reference loss of \$1,204 in 2004 dollars. Both axes use each account's monetary units and the same reference-expenditure normalization. These comparisons retain the applications' own valuation schedules, rather than imposing the simulation's common schedule on every policy. Appendix B.5 explains the construction, unit bridges, and differing matched samples. Differences from the reference projection can reflect both continuation and the treatment of earnings and opportunity costs; they are not estimates of which mechanism actually occurred.

3.4 What Additional Observation Ages Reveal

Additional observations can distinguish paths that fit the same early effect, but the information they provide depends on when schooling is measured. The second experiment holds each generating path fixed and changes the observation ages, retaining the same baseline, path restrictions, and valuation rule. I use five positions spanning each parameter's conditional feasible interval and examine every single age and every pair of distinct ages from 18 through 35. The path remains fixed as observations change; its effects at different ages need not be equal. Each added observation restricts the same underlying path set. Appendix B.6 gives the full design and the numerical procedure for preserving this nesting.

Adding an age-35 effect sharply narrows the NPV ranges for the same three midpoint scenarios. Panel B of Figure 3 retains the generating parameters, baseline, restrictions, valuation rule, and axis scales of Panel A. The candidate NPV ranges in Table 1 become [11.25, 13.44], [2.13, 3.23], and [-5.65, -1.30] thousand dollars, reductions in width of 95.8, 97.8, and 93.5 percent. The second observation matters because a single schooling gap does not determine how quickly it emerged or what follows it. Timing also matters for value: a year acquired earlier can produce earnings gains over more years but also moves forgone earnings closer to the discount base. In the timing example, the age-35 gap is only -0.043 schooling years, yet the candidates found have negative NPVs: near-convergence in later schooling need not erase the value lost during earlier delays. The candidates found for the plateau and speed examples are positive. These findings do not prove sign identification: the search is finite, and three of the five timing scenarios still admit both signs after adding age 35.

¹⁷The joint increments are $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$. This illustration satisfies the common path restrictions but lies outside both experiment grids.

The location of the additional observation matters alongside its availability. Table 2 compares selected designs within each scenario before summarizing across scenarios.¹⁸ Across the three parameter groups, adding age 20 to age 25 reduces the NPV width by a median 28.6–86.3 percent, whereas adding age 24 reduces it by 36.0–90.6 percent. Adding age 30 yields median reductions of 76.3–96.3 percent. The age-24 pair is more informative than the age-20 pair in these summaries despite the shorter distance between observations. It restricts the curve near the first observation, while later ages constrain how much of the schooling gap persists. Neither comparison supplies a general rule about nearby and distant observations: the ranking depends on the logistic family, anchor, and support. Figure B4 summarizes all observation combinations, showing both interior-scenario medians and maximum widths across all five scenarios; the individual-scenario results remain in the replication files.

Finite-window evidence can therefore leave economically consequential differences among admissible education paths, while observations at other ages can narrow the differences found. The exercise conditions on the fitted baseline, logistic family, parameter domain, and valuation calibration. It does not evaluate source-estimate sampling error or sensitivity to alternative schooling functions and opportunity-cost mappings. The policy applications in Section 4 add source-specific outcome definitions, timing bridges, observed mechanisms, and full welfare accounting.

4 From Education Paths to Policy Welfare

Calculating the earnings NPV of an education path is only one step toward evaluating a policy: the welfare comparison also depends on transfers, private education payments, and the net cost to the government. Building on the marginal value of public funds (MVPF) framework of Hendren and Sprung-Keyser (2020), I incorporate alternative education paths into beneficiaries’ willingness to pay and the government’s fiscal account. The broader exercise reconstructs education-related welfare calculations for 38 policy cases, whose results are brought together in Table 3 and Section 5. This section examines three cases introduced in Section 2: the STBF scholarship, the Adams Scholarship, and Tennessee Promise.¹⁹ These cases illustrate how the reconstruction procedure operates and how alternative education paths can alter welfare conclusions. Under the reference specification, the MVPF is finite and greater than one for STBF, finite and less than one for the Adams Scholarship, and classified as self-financing for Tennessee Promise. The discussion relates the candidate paths to each study’s dynamic evidence and mechanisms, distinguishing the observed education effects from the assumptions needed to project lifetime benefits and fiscal costs.

4.1 Welfare Accounting and the Education-Path Replacement

The marginal value of public funds (MVPF) measures beneficiaries’ willingness to pay for a policy change per dollar of its net cost to the government. Following Hendren and Sprung-Keyser (2020), let WTP_i denote willingness to pay and N_i net fiscal cost for policy contrast i . Both are present values for the same population

¹⁸Reductions are calculated within scenarios relative to age 25 alone, then summarized by the median at the 25, 50, and 75 percent interval positions. Largest widths and sign counts include both boundaries.

¹⁹The collection table normalizes both monetary components by fixed reference program costs; this section uses source dollars. MVPF is unchanged.

and policy unit. For a positive net cost,

$$\text{MVPF}_i = \frac{\text{WTP}_i}{N_i}, \quad N_i > 0. \quad (7)$$

Willingness to pay is the monetary value of the policy to its beneficiaries under the maintained welfare assumptions. Net fiscal cost includes the initial expenditure and subsequent fiscal effects of behavioral responses; it is not simply the program's budget. For example, higher earnings can increase private disposable income while also reducing net public cost through higher taxes and lower means-tested transfers. A policy with positive willingness to pay and nonpositive net fiscal cost is classified as self-financing and assigned an infinite MVPF.²⁰

The education-policy accounts can be organized into earnings-related components and the remaining benefits and costs. Use the superscript ref for L1 and write

$$\text{WTP}_i^{\text{ref}} = B_i^{\text{ref}} + \text{NPV}_i^{\text{after,ref}} - C_i^{\text{priv,ref}}, \quad (8)$$

where B_i^{ref} contains the value of program transfers and other private benefits included in the source account, $\text{NPV}_i^{\text{after,ref}}$ is the disposable-income component generated by the education-related earnings projection, and $C_i^{\text{priv,ref}}$ is the associated private education payment. The corresponding public account is

$$N_i^{\text{ref}} = G_i^{\text{ref}} + C_i^{\text{pub,ref}} - \text{NPV}_i^{\text{tax,ref}}. \quad (9)$$

Here G_i^{ref} contains direct program expenditure and other net fiscal costs, $C_i^{\text{pub,ref}}$ is additional public education expenditure not already included in G_i^{ref} , and $\text{NPV}_i^{\text{tax,ref}}$ is the earnings-related fiscal return. The superscript tax includes changes in transfers when the source uses a combined tax-and-transfer rate. All components are signed policy-induced changes: a reduction in tuition payments raises willingness to pay, whereas lost tax revenue raises net cost.²¹

I assemble the applications from education-related entries in the Policy Impacts Library and their underlying studies, focusing on calculations that project lifetime earnings and fiscal returns from education effects measured within finite windows. Of the 38 policy calculations, 37 have an original welfare calculation, denoted L1. Most use the reconstruction by Hendren and Sprung-Keyser (2020), as in Adams; STBF and Tennessee Promise use their original authors' welfare calculations (Angrist et al., 2022; Schoonover et al., 2026). The remaining application values the education channel of cannabis legalization.²²

The reconstructions use a common finite, anchored logistic path family, with case-specific education inputs and valuation schedules. Twenty-four calculations use the monetary components and policy units underlying L1 without the additional adjustments described below. Eleven HOPE/AOTC calculations use that same finite family after converting their marginal attendance coefficients into an explicit finite policy experiment. Early Head Start uses an account aligned to its reference values, and algorithmic placement adds a long-run earnings channel. All 37 calculations with a reference MVPF enter the ordering; the separate

²⁰Self-financing is a welfare convention, not the arithmetic ratio at negative cost. If both components are nonpositive, I report them without an assigned MVPF. Hendren and Sprung-Keyser also apply the marginal framework to finite policy contrasts.

²¹Components vary by policy, and transfers must not be counted twice. Scholarship benefits need not equal expenditure dollar for dollar: Hendren and Sprung-Keyser distinguish induced students from those whose behavior is unchanged. Appendix D.1 gives accounting conventions.

²²The 20 September 2026 library snapshot and Appendix E supply 23 source-study families. Several tax contrasts can come from one study. Inclusion requires reconstructable education and accounting inputs; the collection is not a census of education policies.

groups make these differences visible. Appendix E.4 documents the units and accounts.²³ Where a source observes earnings or uses several education outcomes, its components are retained or replaced separately; the single-schooling-input projection in Equation (5) is not imposed on every account.

I first translate the reported education effects into inputs for the candidate paths. For Adams, I retain the conversion of enrollment and completion effects into schooling years used by Hendren and Sprung-Keyser (2020). For STBF, I treat additional time enrolled as an equivalent increase in completed schooling. This conversion is an additional assumption. Tennessee Promise instead retains separate paths for some college or higher, associate's or higher, and bachelor's or higher. These paths are valued jointly. Each input constrains the corresponding path at the ages or windows specified by the case's age mapping. The candidate paths then determine the projected education-related earnings and fiscal returns.²⁴

I replace the projected education-related income components with values generated by the candidate path $\Delta H_i(a) = H_i^1(a) - H_i^0(a)$, using i to index the policy's target population. For schooling-year paths, Equation (6) provides one such earnings account; its policy-specific application requires a stated earnings schedule and schooling-to-forgone-work calibration. Applying the retained tax-and-transfer mapping gives $NPV_i^{\text{after,path}}(\Delta H_i)$ and $NPV_i^{\text{tax,path}}(\Delta H_i)$. Holding the other accounting components at their reference values, the replacements are

$$WTP_i^{\text{path}}(\Delta H_i) = WTP_i^{\text{ref}} + \left[NPV_i^{\text{after,path}}(\Delta H_i) - NPV_i^{\text{after,ref}} \right] \quad (10)$$

and

$$N_i^{\text{path}}(\Delta H_i) = N_i^{\text{ref}} - \left[NPV_i^{\text{tax,path}}(\Delta H_i) - NPV_i^{\text{tax,ref}} \right]. \quad (11)$$

Thus, a path with larger projected disposable-income gains increases willingness to pay, and one with larger fiscal returns reduces net public cost. Figure D1 in the appendix plots these paired components for the three applications. At positive net cost, points above the diagonal have MVPF greater than one; positive benefits with nonpositive net cost indicate self-financing. This view distinguishes changes in dollar benefits from a large ratio caused by net cost approaching zero. The original projected component is removed before its replacement is added. Directly adding the path NPV to the original willingness to pay would count the original education return twice. Appendix D.2 derives the earnings and fiscal mappings, including the treatment of forgone earnings.

These replacements isolate the income and fiscal-return channels conditional on the retained account. They hold fixed the other transfer, tuition, and public education components, even when those components were originally calculated from education estimates. This is an additional restriction on the comparison. A path that changes time enrolled or the institution attended may also change private payments, public resource use, or grant expenditure; evaluating those changes requires further source-specific mappings, given in Appendix D.3. The Section 3 schooling-flow proxy does not supply them automatically. Likewise, comparisons among paths under one earnings rule isolate continuation differences, whereas a comparison with the source benchmark may also change the valuation rule. I keep these two comparisons distinct, particularly when the path account replaces embedded early earnings losses with explicit opportunity costs.

²³The tax experiment fixes a schedule displacement calibrated to one point-estimate credit dollar, assumes local-linear transport, and retains the two-year schooling conversion. Actual source ages determine its cohort calibration, while the second acquisition year is explicitly assumed. Early Head Start's price year remains unresolved. Cannabis values only the education channel and has no whole-policy L1.

²⁴A BA-completion effect need not equal four schooling years, especially when enrollment already enters the conversion. Appendices D.6–D.8 give case-specific mappings.

The comparison curves H_i^0 and empirical restrictions come from the Census–ACS calibration library, matched to each application’s geographic target and birth cohorts. STBF uses Nebraska and Adams uses Massachusetts, retaining all races, sexes, and birthplaces in the external population data. Tennessee requires an additional adjustment because its education effects are conditional on high school graduation: I match that denominator, the source’s prior-year state definition, and its latest-window control means. These adjustments improve correspondence to the reported outcomes without recovering the source design’s untreated potential path. Where a comparable control level is unavailable, a statewide population mean remains an external calibration rather than a measured mean for selected scholarship applicants.²⁵

L1 comprises the original WTP, net fiscal cost, and MVPF calculation. L2 reconstructs the education inputs retained for L1; L3 preserves L2 and adds separately reported source windows or dynamics. For accounts based on Hendren and Sprung-Keyser (2020), both reconstructions retain their outcome-conversion rules; author-reported accounts use the case-specific conversions above. Additional information need not be a directly estimated schooling-years effect: an earlier enrollment estimate can enter through the retained duration conversion, provided its placement on the schooling path is stated explicitly. The same coefficient may already enter the reference total; L3 then retains its separate timing rather than introducing a previously unused estimate. Both reconstructions hold the comparison curve, empirical support, welfare account, and L2 constraints fixed.²⁶

For the three applications below, the replacement takes the simple form $WTP = w_0 + (1 - t)E$ and $N = n_0 - tE$, where E is the education-related earnings NPV and w_0, n_0 retain all other private and fiscal components. At positive net cost, $MVPF > 1$ if and only if $E > n_0 - w_0$. This threshold makes the economic comparison transparent: the earnings gain must cover the difference between the fixed public cost and the fixed private benefit. It does not apply by dividing across a negative denominator. Self-financing additionally requires positive WTP and nonpositive N .²⁷ Welfare comparisons across beneficiary groups also require social welfare weights (Hendren and Sprung-Keyser, 2020). Appendix D.4 distinguishes verified path pairs from certified bounds and statistical confidence regions.

4.2 STBF Scholarship

Angrist et al. (2022) study how Susan Thompson Buffett Foundation scholarships affect college enrollment, time in school, and degree completion. Awards are randomly assigned among Nebraska applicants in the 2012–2016 high-school graduating cohorts. The reconstruction uses the 2012–2014 four-year-college strata, which enter the authors’ welfare calculation and have six-year follow-up. At that horizon, the archived replication estimates imply that an award offer raises time enrolled by 0.3648 years, bachelor’s completion by 8.13 percentage points, and completion of any degree by 5.18 percentage points. The authors find an initial graduation delay followed by higher completion. Table D1 records the education inputs and their windows.

The authors value these effects under hypothetical public funding, following Hendren and Sprung-Keyser (2020). Their earnings model combines credentials and time enrolled with Nebraska-born ACS earnings

²⁵The library covers 50 states, DC, and the nation. Multi-state targets use fixed weights; identification control states need not belong to the target population. Table E1 and Appendix D.5 document matching and support.

²⁶Appendix E.7 records the observation windows, conversions, and any temporal or population mapping. Institutional policy lags and conditional college-year outcomes require such mappings; they are not relabeled as individual follow-up. Splitting a single entry observation into assumed study years alone does not create L3, as in Tennessee Pell and Georgia HOPE.

²⁷The simplification requires the same earnings NPV and complementary private and fiscal shares; it is not imposed on all 38 accounts. Thresholds use each account’s own currency year and policy unit and cannot rank policies by their dollar size.

profiles. With 3 percent discounting, projected earnings rise by \$21,150 per treated applicant. The \$32,250 scholarship transfer enters private benefits and public expenditure, while a 20 percent tax rate divides the earnings gain between recipients and the government. The resulting willingness to pay is \$49,170 and net public cost is \$28,020, giving the L1 MVPF of 1.75. Table D2 records this account.

I retain the award-offer contrast, scholarship transfer, and tax rate, and replace the earnings projection with the NPV generated by each candidate schooling path. Writing this NPV as E in 2019 dollars gives $WTP = 32,250 + 0.8E$ and $N = 32,250 - 0.2E$. Here the threshold for an MVPF above one is $E > 0$, because the retained transfer enters both accounts equally; higher lifetime earnings raise recipients' disposable income and offset public expenditure through taxes. To construct the schooling input, I treat the 0.3648-year increase in time enrolled as an equal increase in completed-schooling equivalents. This additional assumption permits the stock–flow valuation in Section 3. Comparisons with L1 therefore change the earnings mapping as well as the continuation, while comparisons among candidate paths hold that mapping fixed.

L2 matches the six-year schooling input using anchored logistic paths around cohort-matched Nebraska ACS profiles. Fixed entry-year and high-school-age weights map the observation into an average over ages 23–25. Each path is anchored before college exposure and satisfies the state-calibrated empirical restrictions (Figure 5). The statewide profiles remain external approximations for scholarship applicants. A separate L3 would require another comparable schooling-time moment or an explicit link from credentials to schooling years; the retained degree endpoints alone do not provide that link. Appendix D.6 gives the cohort weights, age mapping, and auxiliary degree check.

The same six-year input admits earnings NPVs from \$8.46 thousand to \$60.26 thousand. Net public cost remains positive, and the candidate MVPFs range from 1.28 to 3.98, compared with 1.75 in L1 (Table 3). These paths illustrate uncertainty about the magnitude of welfare gains even when all retained candidates lie above one. Further gains or comparison-group catch-up remain conditional possibilities beyond the observed graduation dynamics. A finite candidate set does not establish that every admissible continuation has an MVPF above one.²⁸

4.3 Adams Scholarship

Cohodes and Goodman (2014) use test-score eligibility thresholds to estimate the effects of Massachusetts' Adams Scholarship on college choice, persistence, and completion. The estimates used here concern the 2005–2006 high-school graduating classes. Eligibility shifts students toward participating public colleges. Within two years of high school, two-year enrollment rises by 0.7 percentage points and four-year enrollment changes little. At six years, bachelor's completion falls by 2.5 percentage points, while the two-year-completion effect is approximately zero. The authors interpret similar declines in persistence and graduation as evidence favoring dropout over pure delay within follow-up.

Hendren and Sprung-Keyser (2020) convert these enrollment and completion effects into -0.018 schooling-equivalent years per eligible student, using assumed durations for each enrollment and completion margin. Their earnings projection gives an NPV of $-\$1,204.04$. Combining this loss with scholarship benefits, private education payments, public costs, and tax-and-transfer effects yields willingness to pay of \$1,364.23 and net cost of \$1,896.35, in 2004 dollars. The L1 MVPF is 0.72. This calculation concerns scholarship eligibility rather than the separate effect among students induced to switch colleges.

²⁸Table D3 pairs welfare components from complete paths; an envelope edge need not be feasible. Appendix E.9 reports calibration and valuation sensitivities.

I retain the education conversion and the account's transfer and education-cost components, and replace its earnings projection. Let E be the candidate earnings NPV in 2004 dollars and $E^{\text{ref}} = -1,204.04$ the reference value just reported. With the retained effective earnings tax-and-transfer share $t = 0.1979$, the replacement gives $\text{WTP} = 1,364.23 + (1 - t)(E - E^{\text{ref}})$ and $N = 1,896.35 - t(E - E^{\text{ref}})$. Thus, the change in projected earnings raises private benefits by its after-tax share and lowers net public cost by its fiscal share. Appendix D.7 derives the equivalent constants of \$2,329.97 and \$1,658.06 by removing the original income components from these totals. As in STBF, the path valuation changes the earnings-timing rule relative to L1; that rule remains fixed across L2 and L3 candidate paths.

The paths use the Massachusetts ACS profile for the 1987 birth cohort, anchored at age 18, under state-calibrated empirical restrictions. L2 places the converted six-year total at age 24. L3 also retains the source's earlier entry window, assigning its 0.007-year converted contribution to age 20 (Figure 6). This uses the same enrollment and credential multipliers as L2, but preserves two timing restrictions rather than only their final aggregate. Entry within two years does not directly measure completed schooling at age 20: that placement is an explicit temporal allocation of the conversion. Appendix D.7 gives both restrictions; Appendix E.9 reports the calibration and valuation checks.

The L2 candidates leave the welfare comparison unresolved. Adding the early converted input in L3 narrows the searched MVPFs to -0.62–0.30, with positive net cost throughout. This result depends on the source's early entry estimate and its stated placement on the schooling path. In this account, an MVPF above one requires $E > -\$671.91$; the L3 earnings losses are larger than that threshold permits. Positive willingness to pay can therefore coexist with an MVPF below one because the retained private benefit need not cover public cost. The original authors' dropout interpretation concerns the observed persistence and completion history. A candidate that later catches up describes a possibility beyond that history, not evidence against their mechanism.

4.4 Tennessee Promise

Schoonover et al. (2026) study the expansion of tuition-free community college from a Tennessee county program in 2009 to a statewide program in 2015. Their attainment analysis uses 2000–2024 ACS data for high-school graduates aged 18–24, with exposure defined by eligibility at age 17. They estimate effects for overlapping age pairs from 18–19 through 23–24, controlling for state and survey year. At ages 23–24, some college or higher rises by 3.5 percentage points, associate's or higher by 4.0 points, and bachelor's or higher by 2.7 points. The authors examine earlier windows and acknowledge that some participants may complete degrees after age 24.

The authors use Tennessee-specific education returns to project lifetime income within the framework of Hendren and Sprung-Keyser (2020). Private benefits include tuition relief for students who would have enrolled anyway and after-tax earnings for induced students. Scholarship and public education expenditure are offset by projected tax receipts. With a 26 percent tax rate and 3 percent discounting, they report willingness to pay of \$72,837 and net public cost of -\$10,053, giving a self-financing L1 classification. The reported tax gain of \$24,845 implies a gross earnings NPV of $E^{\text{ref}} = 24,845/0.26 = 95,557.69$ under this fiscal mapping.

I replace that income projection with the NPV E of the three credential paths, valued using Tennessee earnings differences between adjacent credential categories. Keeping tuition relief and education expenditure fixed gives $\text{WTP} = 72,837 + 0.74(E - E^{\text{ref}})$ and $N = -10,053 - 0.26(E - E^{\text{ref}})$. These expressions change

the reported totals only by the private and fiscal shares of the revised income projection. The schedules are multiplied by 4.883134 so that constant endpoint gaps reproduce E^{ref} ; this preserves the reference earnings total without reproducing its full forecasting model. Appendix D.8 derives the equivalent private remainder of \$2,124.31 and pretax fiscal cost of \$14,792. Under this account, positive willingness to pay and earnings NPV above approximately \$56,892 yield self-financing.

L2 matches all three latest-window effects using anchored logistic paths for the 1992–1998 birth cohorts, with fixed equal cohort weights. The Tennessee ACS reference matches the high-school-graduate denominator, prior-year residence definition, and reported control levels. Both policy states share the age-17 anchor, and credential levels remain nested at every age (Figure 7). State-calibrated support and a restriction based on remaining completion in earlier cohorts limit the continuations; the main bachelor’s plateau cannot exceed 45.4 percent. These are maintained restrictions on an external reference, not recovered untreated predictions from the source regressions.²⁹

The latest-window effects alone admit earnings NPVs from -74.20 to 262.53 thousand dollars, including beneficiary losses with positive net cost and gains with self-financing. L3 adds fifteen earlier age-window estimates using the source’s marginal permutation intervals, while retaining the latest effects exactly. Figure 7 focuses on ages 17–25, where these restrictions differ; the appendix atlas shows the full life-cycle ranges. With the one-pixel inward extraction rule, positive-cost MVPFs start at approximately 0.63. The extraction sensitivity in Appendix E.9 places the lower endpoint on different sides of one. Earlier observations remove some adverse candidates, but the threshold comparison depends on graphical precision.³⁰

Under this account, an MVPF above one requires $E > \$12,667.69$ while net cost is positive. Removing the earnings multiplier yields no self-financing candidate in the sensitivity analysis (Appendix E.9). Consequently, the fiscal classification depends on the earnings calibration as well as the admissible education history. The source’s cautions about marginal returns and later completion apply directly to this interpretation.

Input uncertainty is examined separately from continuation at fixed education effects. Table 4 summarizes 1,000 prespecified input repetitions per application. Adams propagates linked source inputs, whereas STBF and Tennessee use researcher-constructed perturbations under stated dependence and scale choices. Tennessee’s normal perturbations are not draws from the source permutation distribution. Appendix F documents these differences and unsuccessful searches; the endpoint summaries do not establish confidence coverage for the continuous path class.

The cases show how continuation and its restrictions enter distinct welfare questions: the size of the gains in STBF, the assumptions needed to locate Adams’ converted schooling loss in time, and the fiscal classification of Tennessee Promise. Within each path comparison, projected earnings and taxes change while the other transfer and education-cost components remain fixed. Section 5 asks how frequently comparable fixed-input paths leave these judgments unresolved across the policy collection.

5 Welfare Judgments across Policies and Populations

The welfare implications of an education effect depend on its persistence, the ages at which attainment changes, and uncertainty in the education inputs. This section asks whether the observed information de-

²⁹Table D4 and Appendix D.8 document donor cohorts, plateau extrapolation, level scaling, and age weights.

³⁰The permutation intervals are marginal, not a joint 95 percent region. Inward adjustment narrows a digitized interval and is a sensitivity to its boundary, not a uniformly conservative confidence procedure. Appendix D.8 gives the extraction rules.

termines which side of MVPF one a policy lies on, whether it can finance itself, and how these conclusions change when additional comparable observations restrict the path. The collection contains 38 finite-path calculations, including 37 with L1. The tax-input conversions and account adjustments described in Section 4.1 remain identified within the full comparison. Single-parameter comparisons distinguish permanent effects on eventual attainment from temporary effects on acquisition speed or timing. Input repetitions examine a separate source of variation under each case's specified uncertainty scheme. Appendix G compares men and women within seven policies, holding terminal education effects and valuation prices fixed.

5.1 Continuation Values and Policy Comparisons

Figure 8 places the L2 and available L3 ranges beside the reference MVPFs, ordered by descending L1. All 38 calculations use finite attainment paths. Among the 37 with L1, 27 admit positive-cost paths on both sides of MVPF one at fixed L2 inputs. Of these, 15 belong to the 24-account group without the additional tax-unit and account adjustments. Taking the education coefficients as known therefore need not determine whether beneficiary welfare exceeds one dollar per dollar of net public cost.³¹

The uncertainty can concern the signs of the monetary components, rather than only the size of their ratio. In 29 of the 37 calculations with L1, including 17 in the 24-account group, one compatible path gives negative willingness to pay and positive public cost, whereas another gives positive willingness to pay and nonpositive public cost. These paired outcomes follow from different projected earnings and tax receipts under the same education restrictions. The transfer, tuition, and public education-cost components remain fixed. A gain that persists into high-earning ages can benefit recipients and repay the public expenditure; an initial gap that disappears or reverses can produce a different conclusion. These are possible aggregate continuations, not estimates of individual dropout or return-to-school behavior.

A large MVPF need not represent a large dollar benefit. Numerical checks find positive-cost approaches to zero net cost with a positive limiting WTP in 28 of the 37 calculations, and with a negative limiting WTP in 12. The resulting ratio diverges while both monetary components remain finite. Figure 8 therefore separates positive-cost ratio spans from self-financing states. The span does not assert that every intermediate ratio is attained, and a finite searched endpoint does not certify the boundary of the continuous feasible set.³²

Figure 9 applies the same lower-endpoint ordering to all 37 calculations with a reference MVPF. It first compares lower endpoints and then upper endpoints, both descending, using L3 where available and L2 otherwise. Finite endpoints are rounded to two decimals, and equal endpoint pairs share a rank. Eight of the eleven tax calculations retain an unbounded positive-cost ratio span, although their education levels, WTP, and net cost are bounded. The divergence comes from finite paths approaching zero net cost, not from an unrestricted policy derivative. These calculations share an endpoint pair with four other policies.³³ Both axes increase from small to large ranks, so a point below the diagonal has higher priority under this rule than under

³¹Several tax contrasts come from one study, so the calculation count is not a count of independent studies. The 24-account group spans 20 source-study families, with threshold pairs in 13. MVPF one weights beneficiary dollars and public resources equally; welfare priorities can also depend on distributional weights and financing alternatives (Hendren and Sprung-Keyser, 2020).

³²Appendix E.3 documents complete paired paths, constraint checks, and one-sided ratio sequences. A large finite ratio or a rounded zero cost is insufficient evidence of an infinite limit.

³³The tax calculations assume that the source's local attendance and accounting responses apply over a fixed finite schedule change calibrated to one point-estimate credit dollar. Appendix E.4.1 states the conversion and age allocation. The full ranking includes these assumptions; it does not infer policy-response magnitudes from ACS levels alone.

L1. The ordering expresses a preference over conditional endpoint summaries; it is not a social-welfare or certified worst-case ranking.

Twelve calculations with L1 admit L3 under the common rule for additional source windows or dynamics. Replacing one calculation's L2 by L3 improves its displayed priority in ten cases and leaves it unchanged in two, holding the others fixed. This is a consequence of the endpoint-ordering rule, not evidence that information raises true welfare. Additional restrictions can exclude favorable as well as adverse continuations. Five of these L3 reconstructions separate converted source components or dynamics under explicit time or population mappings, including Adams; their narrowing therefore depends on these mappings as well as on the additional source information. Appendix E.2 records the inputs and conversions; Figures E1–E2 display H^0 and H^1 for all 38 calculations. No probability distribution is assigned to the paths.

Table 3 reports willingness to pay and net cost alongside MVPF. The policy-type columns in the table and Figure 8 distinguish 14 direct student-aid calculations, 17 education tax-benefit calculations, four tuition or public-college-funding calculations, one college-assessment and course-placement intervention, one early-childhood intervention, and one education channel of another policy. These types describe the intervention, whereas the panels distinguish accounting specifications. Several calculations share a source study, so the category counts are not independent-study counts. Panel A contains the 24 comparisons without additional tax-unit or account adjustments. Panel B reports the eleven finite tax reconstructions, whose source marginal coefficients require the stated finite-policy conversion. Their WTP and net-cost extrema are finite, even when their ratios are not. Panel C distinguishes the aligned Early Head Start account, the algorithmic-placement extension, and the cannabis education channel. Monetary components in Panels A and B use a fixed reference program-cost divisor within each calculation; Panel C uses native thousands. Every MVPF combines the two components of the same path. Very large normalized amounts indicate that the allowed future response can dominate a small reference expenditure; they do not establish that such a response is likely. The main finding is the existence of conflicting welfare judgments within a fixed reconstruction, conditional on both its path restrictions and its input conversion.

5.2 Permanent and Temporary Education Effects

The same finite-window education input can support a permanent or a temporary effect, with different implications for lifetime welfare. I examine this distinction by holding two parameter families at their calibrated comparison values and solving for a change in the third that matches the retained L2 education input. A nonzero change in K produces a permanent effect by altering eventual attainment. Changing b or d with K fixed produces a temporary effect: the population acquires education faster or more slowly, but the attainment gap eventually vanishes. The observation operator, empirical support, and welfare account remain unchanged across these three exercises. This makes the continuation assumption explicit without treating the observed input as evidence that one particular parameter changed. Appendix B.5 gives the matching procedure and its relation to the main path construction.³⁴

Interpreting the observed education input as a permanent effect tends to preserve the reference welfare comparison. In Panel A of Figure 10, the finite MVPF pairs follow an upward pattern broadly aligned with the 45-degree equality line. Changing K leaves a permanent difference in eventual attainment, so the

³⁴The exercise preserves the individual components of each fitted H^0 , so a zero change reproduces that baseline exactly. K and d change additively; b changes by a common proportion. Tennessee Promise permits a separate change in the active family for each of its three credential paths, subject to joint restrictions. This is a diagnostic rather than a decomposition of the main candidate ranges.

education gain or loss continues to affect projected earnings beyond the observed window. This resembles a reference calculation that uses the measured education effect to project sustained earnings gains or losses. The alignment is consistent with interpreting the reference valuations in this collection as assigning a permanent effect to the short-window education input. Shared education inputs and related return calibrations contribute to this resemblance, so it does not independently establish that the education effect is permanent.³⁵

Temporary effects produce a different welfare comparison. In Panels B and C, many finite reconstructed MVPFs cluster near one across a wide range of reference values, rather than following the near-diagonal pattern in Panel A. With eventual attainment fixed, changes in b or d match the education input through faster or slower acquisition, and the attainment gap eventually fades. Figure B3 shows that these temporary effects still generate dispersed earnings NPVs, even though many corresponding MVPFs lie close to one. This pattern reflects how earnings enter the retained welfare accounts. When earnings NPV is small relative to the retained transfers and other non-earnings components, those components largely determine willingness to pay relative to net public cost. Several accounts consequently remain near MVPF one even as their earnings NPVs differ.³⁶ Temporary effects therefore retain economic value; their influence on MVPF depends on their duration and on the scale and composition of the welfare account under the specified valuation rule. The contrast with Panel A separates two explanations of the same education input: a permanent change in eventual attainment and a temporary change in when education is acquired. A lifetime welfare calculation must distinguish them through additional evidence or an explicit continuation assumption. Greater precision within the observed window does not by itself resolve that distinction.

Table 5 gives each reference MVPF, its single-parameter reconstruction, and the required parameter change. It also records calculations without a matching solution. The matched samples differ, which limits comparisons of the plotted patterns. Holding two parameters fixed can leave no solution within the empirical support; four, eleven, and nineteen calculations have no matched plateau, speed, or timing solution, respectively. Their absence is reported rather than interpreted as a zero response. Red rings identify four plateau solutions that also pass the additional L3 restrictions; none of the matched speed or timing solutions does so. More complete source evidence can therefore distinguish some otherwise admissible single-family explanations. The results do not establish that the original lifetime projections are incorrect or that their authors ignored dynamics. They show how the welfare interpretation of a retained short-window input depends on the continuation assigned to it, while the source studies' observed dynamics and mechanism evidence remain relevant to choosing among those continuations.

5.3 From Uncertain Inputs to Paths and Welfare

Uncertainty in an estimated education effect carries through to the education paths that match it and to their welfare values. This section examines that propagation, together with sensitivity to selected accounting inputs, using the specified input repetitions. Figure 11 displays point-input values in dark colors and input-repetition endpoint summaries in lighter colors. Its Type column repeats the policy categories, separately from the bracketed input-method labels. Its L1 panel reports available source 95 percent MVPF intervals.

³⁵Panel A contains 18 finite pairs, eight calculations that move from a finite reference MVPF to self-financing, and seven that are self-financing in both accounts. The diagonal denotes equality on the common asinh scale; it is not an estimated regression line.

³⁶Among finite pairs, 13 of 20 temporary speed results and 9 of 14 temporary timing results lie between 0.75 and 1.25. For $W = w_0 + (1 - t)E$ and $N = n_0 - tE$, MVPF approaches w_0/n_0 as earnings NPV E approaches zero, provided $n_0 > 0$. That ratio need not equal one. This accounting relationship does not imply general insensitivity to temporary effects: the ratio can change sharply near zero net cost.

The L2 and L3 panels instead summarize the existing 1,000 prespecified repetitions per calculation, keeping the ACS calibration and specified conversion and valuation rules fixed. The eleven tax calculations use the same input IDs in the finite path model, keeping the point policy displacement fixed across repetitions. Early Head Start keeps its reference-earnings multiplier fixed. Each summary runs from the 2.5th percentile of candidate minima to the 97.5th percentile of candidate maxima. The methods are intentionally named separately: propagation of archived joint source/accounting draws, constructed normal or correlated perturbations, and exercises varying economic inputs while holding education fixed.³⁷

Among the 37 calculations with L1, 36 L2 summaries cross MVPF one under their specified repetition schemes. WTP summaries cross zero in 34 calculations and net-cost summaries cross zero in 34; 34 have an infinite upper MVPF endpoint under the self-financing convention or an audited positive-cost limit. Within the 24-account group, the corresponding counts are 23, 21, 22, and 22. These counts describe how the reconstructed endpoints respond to the inputs being varied. They do not measure the chance that a policy is harmful or self-financing. The monetary components remain informative when the ratio becomes unstable near zero net cost.³⁸

Table 4 identifies each repetition method, endpoint summary, and usable count. Across the full collection, all 38,000 scheduled reduced-information repetitions were evaluated; 37,952 yielded a finite-path set, and 37,951 yielded a defined MVPF. Failed searches and unassigned ratios remain in the record.³⁹ Five admitted L3 specifications lack joint repetitions for their complete input sets: Florida, Wisconsin, Kalamazoo, CUNY Pell, and Social Security student benefits. Their point-input paths remain available. Tennessee’s repeated normal shocks use fixed permutation-interval widths; two displayed zero standard errors are assigned a 0.0005 scale based on reporting precision. This is a sensitivity convention, not a recovered standard error or a draw from the source permutation distribution.

Path uncertainty remains even if the original education effect is known exactly: different paths can match that effect and imply different welfare values. Estimation uncertainty arises because the original effect is itself estimated imprecisely. Allowing the input to vary changes the compatible paths and, in turn, the welfare range. The two sources can operate together, since each possible input may still admit several continuations. More precise estimates within the same window address estimation uncertainty; comparable observations at additional ages can help resolve path uncertainty. The repetitions reported here trace this propagation under the specified source-draw or input-sensitivity scheme.

6 Discussion

Education effects observed within a finite window can support different lifetime welfare judgments, depending on assumptions about persistence, attainment timing, and valuation. I take the source studies’ causal identification as given; the additional assumptions link their estimates to education paths for the target population. Conversions must preserve policy units and observation windows, while assumed schooling durations cannot constitute new observations. The external comparison profile must also be relevant to the affected

³⁷Appendix F.1 defines the order-statistic summaries and the inputs varied. These procedures do not share a sampling law or establish confidence coverage for all compatible paths. No probability distribution is assigned to paths within a repetition.

³⁸The source intervals and reconstructed summaries can vary different inputs and use different dependence assumptions. Their widths are not a controlled comparison of inferential methods.

³⁹Social Security student benefits account for 48 failed searches. Texas CC tuition has one repetition with monetary components but no assigned ratio. Cannabis’ six-year-only repetitions are diagnostic; its primary display uses the three-window L3.

population. Matching Census–ACS geography and birth cohorts improves comparability, but population averages may not describe scholarship applicants or marginal students. Historical cohorts’ unobserved early attainment is reconstructed using other cohorts’ shapes. The common anchor imposes no effect or anticipation before the specified exposure age. Fit to ACS profiles cannot establish these population and exposure assumptions.⁴⁰

Path and valuation assumptions further determine the comparisons. Logistic curves may omit interrupted schooling or reentry, and historical variation need not bound a policy response. The single-parameter exercises distinguish permanent from temporary explanations without identifying which applies. Earnings returns may also depend on completion age (Bárány et al., 2023). The policy reconstructions retain specified transfers and education costs as earnings and fiscal receipts change. The simulation’s NPV comparison additionally changes the opportunity-cost treatment, so it combines continuation and valuation. To assess sensitivity to these choices, Appendix E.9 varies calibration, valuation, and search at fixed source inputs. Using L3 where available, positive-cost MVPFs cross one in 21 of 38 calculations, compared with 18 under narrower support. Every common specification admits paths with negative WTP and positive net cost alongside self-financing paths in 19 calculations. These conditional contrasts neither validate all assumptions nor exclude unsearched paths; wider support preserves existing paths by construction.

Interpreting the welfare ranges also requires distinguishing input uncertainty from the rule used to rank policies. Different repetition schemes and incomplete covariance information prevent a common sampling interpretation, although all repetitions hold calibration and the path family fixed. Endpoint percentiles do not establish confidence coverage (Imbens and Manski, 2004), and more repetitions can reduce simulation error without resolving these limits or assigning probabilities to paths. The lower-endpoint ordering gives weight to adverse continuations among the paths found. Its ranks depend on search coverage, calibration, and information available for L3; they are neither expected-welfare nor certified worst-case rankings. Joint reporting of WTP and net cost distinguishes large benefits from ratios driven by near-zero costs, while self-financing and paths without an assigned ratio require separate interpretation. Spending recommendations further require distributional weights, a financing comparison, and assumptions about policy expansion. To assess the underlying extrapolations, later education outcomes withheld from path construction could be used to compare logistic predictions with constant effects and flexible monotone alternatives. Observed earnings could test valuation, while comparable age windows and joint coefficient covariances would improve reconstruction and inference. Such evidence can distinguish permanent from temporary effects with different welfare implications.

7 Conclusion

An education effect measured within a finite window leaves two economically distinct questions open: how much of the effect persists and when the remaining attainment occurs. This paper makes those questions explicit by constructing education paths that match the converted observation restrictions and passing each path through a specified valuation rule. The simulations show how additional observations can distinguish continuations that agree at an earlier age. Single-parameter comparisons separate permanent effects on eventual attainment from temporary effects on acquisition speed or timing. Temporary effects still carry lifetime

⁴⁰Appendix D.5 records age support and donor shapes. The selected literature and policy collections also limit the generalizability of the reported frequencies.

earnings consequences, even when the attainment gap eventually vanishes. In the policy applications, verified paths often support different welfare judgments even when the retained education estimates are treated as known. At fixed inputs, 27 of the 37 calculations with L1 admit positive-cost MVPFs on both sides of one. More precise estimation within the same window would not remove these particular continuation differences.

The contribution is a way to assess which welfare judgments remain dependent on unobserved education paths. Compatible paths with different classifications establish that dependence under the stated calibration and valuation rules; they do not establish either path's probability. Input repetitions answer a separate question about imprecision in the retained estimates. The distinction helps identify whether further progress requires more precise inputs, additional observations of attainment, or stronger evidence on the valuation relationship.

The results favor reporting the observed education window and the assumed continuation separately. When additional outcomes are available, their contribution depends on whether they constrain attainment at a new age for a comparable population. Reporting willingness to pay and net fiscal cost alongside MVPF also distinguishes changes in benefits from ratios driven by costs near zero. Longer follow-up, richer education histories, and observed earnings can each improve the assessment, but they answer different questions. The useful standard is whether the available evidence distinguishes the paths that matter for the welfare comparison.

References

- Abraham, K. G. and Clark, M. A. (2006). Financial aid and students' college decisions: Evidence from the District of Columbia Tuition Assistance Grant Program. *Journal of Human Resources*, 41(3):578–610.
- Acton, R. (2021). Effects of reduced community college tuition on college choices and degree completion. *Education Finance and Policy*, 16(3):388–417.
- Angrist, J. D., Autor, D. H., and Pallais, A. (2022). Marginal effects of merit aid for low-income students. *The Quarterly Journal of Economics*, 137(2):1039–1090.
- Athey, S., Chetty, R., Imbens, G. W., and Kang, H. (2026). The surrogate index: Combining short-term proxies to estimate long-term treatment effects more rapidly and precisely. *Review of Economic Studies*, 93(4):2284–2312.
- Bárány, Z. L., Buchinsky, M., and Corblet, P. (2023). Late bloomers: The aggregate implications of getting education later in life. NBER Working Paper 31874, National Bureau of Economic Research.
- Bartik, T. J., Hershbein, B., and Lachowska, M. (2021). The effects of the Kalamazoo Promise Scholarship on college enrollment and completion. *Journal of Human Resources*, 56(1):269–310.
- Bergman, P., Kopko, E., and Rodríguez, J. (2026). A seven-college experiment using algorithms to track students: Impacts and implications for equity and fairness. Author manuscript, June 2026 version.
- Bettinger, E. (2004). How financial aid affects persistence. In Hoxby, C. M., editor, *College Choices: The Economics of Where to Go, When to Go, and How to Pay for It*, pages 207–238. University of Chicago Press, Chicago.
- Bhuller, M., Mogstad, M., and Salvanes, K. G. (2017). Life-cycle earnings, education premiums, and internal rates of return. *Journal of Labor Economics*, 35(4):993–1030.
- Bound, J., Lovenheim, M. F., and Turner, S. (2012). Increasing time to baccalaureate degree in the United States. *Education Finance and Policy*, 7(4):375–424.
- Bruce, D. J. and Carruthers, C. K. (2014). Jackpot? The impact of lottery scholarships on enrollment in Tennessee. *Journal of Urban Economics*, 81:30–44.
- Bulman, G. B. and Hoxby, C. M. (2015). The returns to the federal tax credits for higher education. *Tax Policy and the Economy*, 29(1):13–88.
- Carruthers, C. K. and Welch, J. G. (2019). Not whether, but where? Pell grants and college choices. *Journal of Public Economics*, 172:1–19.
- Castleman, B. L. and Long, B. T. (2016). Looking beyond enrollment: The causal effect of need-based grants on college access, persistence, and graduation. *Journal of Labor Economics*, 34(4):1023–1073.
- Chen, J. and Ritzwoller, D. M. (2023). Semiparametric estimation of long-term treatment effects. *Journal of Econometrics*, 237(2):105545.

- Chetty, R. (2009). Sufficient statistics for welfare analysis: A bridge between structural and reduced-form methods. *Annual Review of Economics*, 1(1):451–488.
- Cocci, M. D. and Plagborg-Møller, M. (2025). Standard errors for calibrated parameters. *Review of Economic Studies*, 92(5):2952–2978.
- Cohodes, S. R. and Goodman, J. S. (2014). Merit aid, college quality, and college completion: Massachusetts’ Adams Scholarship as an in-kind subsidy. *American Economic Journal: Applied Economics*, 6(4):251–285.
- Cornwell, C., Mustard, D. B., and Sridhar, D. J. (2006). The enrollment effects of merit-based financial aid: Evidence from Georgia’s HOPE program. *Journal of Labor Economics*, 24(4):761–786.
- Deming, D. J. and Walters, C. R. (2017). The impact of price caps and spending cuts on U.S. postsecondary attainment. NBER Working Paper 23736, National Bureau of Economic Research.
- Denning, J. T. (2017). College on the cheap: Consequences of community college tuition reductions. *American Economic Journal: Economic Policy*, 9(2):155–188.
- Dynarski, S. M. (2003). Does aid matter? Measuring the effect of student aid on college attendance and completion. *American Economic Review*, 93(1):279–288.
- Fan, Y., Manzanares, C. A., Park, H., and Qi, Y. (2026). A sensitivity analysis of the surrogate index approach for estimating long-term treatment effects. Working paper, arXiv:2603.00580v1.
- García, J. L., Heckman, J. J., Leaf, D. E., and Prados, M. J. (2020). Quantifying the life-cycle benefits of an influential early-childhood program. *Journal of Political Economy*, 128(7):2502–2541.
- Goldrick-Rab, S., Kelchen, R., Harris, D. N., and Benson, J. (2016). Reducing income inequality in educational attainment: Experimental evidence on the impact of financial aid on college completion. *American Journal of Sociology*, 121(6):1762–1817.
- Haider, S. and Solon, G. (2006). Life-cycle variation in the association between current and lifetime earnings. *American Economic Review*, 96(4):1308–1320.
- Heckman, J. J., Humphries, J. E., and Veramendi, G. (2016). Dynamic treatment effects. *Journal of Econometrics*, 191(2):276–292.
- Hendren, N. and Sprung-Keyser, B. (2020). A unified welfare analysis of government policies. *The Quarterly Journal of Economics*, 135(3):1209–1318.
- Hess, A. J. (2025). *Think of the Children: The Role of Early Childhood Interventions in Long-Term Human Capital Formation*. PhD thesis, University of California, Los Angeles.
- Hoxby, C. M. and Bulman, G. B. (2016). The effects of the tax deduction for postsecondary tuition: Implications for structuring tax-based aid. *Economics of Education Review*, 51:23–60.
- Imbens, G., Kallus, N., Mao, X., and Wang, Y. (2025). Long-term causal inference under persistent confounding via data combination. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):362–388.

- Imbens, G. W. and Manski, C. F. (2004). Confidence intervals for partially identified parameters. *Econometrica*, 72(6):1845–1857.
- Kleven, H. J. (2021). Sufficient statistics revisited. *Annual Review of Economics*, 13(1):515–538.
- Li, A. (2026). Higher education or just higher? The unintended effects of recreational marijuana legalization. Author manuscript, version supplied September 2026.
- Long, B. T. (2004). The impact of federal tax credits for higher education expenses. In Hoxby, C. M., editor, *College Choices: The Economics of Where to Go, When to Go, and How to Pay for It*, pages 101–168. University of Chicago Press, Chicago.
- Manski, C. F. (2011). Choosing treatment policies under ambiguity. *Annual Review of Economics*, 3(1):25–49.
- Marx, B. M. and Turner, L. J. (2018). Borrowing trouble? Human capital investment with opt-in costs and implications for the effectiveness of grant aid. *American Economic Journal: Applied Economics*, 10(2):163–201.
- Obrovčić, F. (2026). Identification of long-term treatment effects via temporal links, observational, and experimental data. Working paper, revised March 2026, arXiv:2411.04380v2.
- Schoonover, P., Attridge, J., Carruthers, C. K., and Welch, J. G. (2026). A promise worth keeping? Impacts of free community college on degrees and earnings. NBER Working Paper 35226, National Bureau of Economic Research.
- Seftor, N. S. and Turner, S. E. (2002). Back to school: Federal student aid policy and adult college enrollment. *Journal of Human Resources*, 37(2):336–352.
- Turner, N. (2011). The effect of tax-based federal student aid on college enrollment. *National Tax Journal*, 64(3):839–861.
- Vohra, V. (2026). Inference for welfare metrics. *Journal of Public Economics*, 256:105603.
- Zimmerman, S. D. (2014). The returns to college admission for academically marginal students. *Journal of Labor Economics*, 32(4):711–754.

Table 1. Education valuation under one and two observed ages

Generating process	$\Delta H_c(25)$ (years)	Single-effect NPV	Generating-path NPV	Candidate NPV range	
				Age 25	Ages 25 and 35
Permanent (K)	0.265	11.20	13.22	[-6.28, 46.44]	[11.25, 13.44]
Temporary (b)	0.336	14.19	3.17	[-3.19, 46.80]	[2.13, 3.23]
Temporary (d)	-0.402	-16.96	-3.67	[-24.11, 43.16]	[-5.65, -1.30]

Notes: NPVs are thousands of 2000 dollars per person. Each generating scenario changes only the named parameter; the other two increments are zero. The increments are (0.3160, 0, 0), (0, 0.1113, 0), and (0, 0, 6.4351), respectively. $\Delta H_c(25)$ is the implied schooling effect. The benchmark uses that effect; the generating-path value uses the full path. Permanent and temporary describe the generating path. All three parameters may vary among candidates matching the stated ages. Ranges summarize verified candidates, not confidence intervals or certified global bounds.

Table 2. How additional observation ages change the NPV range

Observed ages	Median width (\$1,000)	Largest width (\$1,000)	Reduction (percent)	Both signs found (of 5 scenarios)
<i>Panel A. Plateau (ΔK)</i>				
25	52.73	67.40	0.0	3/5
20, 25	21.69	24.34	58.9	1/5
24, 25	16.20	18.18	69.3	1/5
25, 30	6.39	7.17	87.9	0/5
25, 35	2.20	2.46	95.8	0/5
<i>Panel B. Speed (Δb)</i>				
25	49.99	73.07	0.0	3/5
20, 25	6.85	22.24	86.3	2/5
24, 25	4.68	16.83	90.6	2/5
25, 30	1.87	12.30	96.3	1/5
25, 35	1.10	9.22	97.8	0/5
<i>Panel C. Timing (Δd)</i>				
25	67.27	89.68	0.0	5/5
20, 25	53.15	53.59	28.6	4/5
24, 25	43.07	51.33	36.0	4/5
25, 30	15.96	43.68	76.3	4/5
25, 35	4.35	17.47	93.5	3/5

Notes: Only the named generating parameter varies; all candidate parameters may vary. Medians use the three interior scenarios. Proportional reductions are calculated within each scenario relative to age 25 alone, then summarized by their median. Largest widths and sign counts use all five scenarios. A “both signs” count requires NPVs above \$0.01 and below $-\$0.01$; a zero count does not establish sign identification. Values are 2000 dollars per person; ranges summarize the numerical search.

Table 3. L1 welfare values and point-input continuation ranges

Policy	Type	MVFP			WTP			Net fiscal cost		
		L1	L2	L3	L1	L2	L3	L1	L2	L3
Panel A. Primary comparisons: WTP and net cost per reference program dollar										
Tuition deduction: single, lower threshold	Tax	$+\infty^s$	$[-\infty, -4.43]^{s,\dagger}$	–	5.38	$[-150.82, 305.69]$	–	–5.10	$[-80.19, 34.02]$	–
Tennessee Promise	Aid	$+\infty^s$	$[-1.55, +\infty]^s$	$[0.63, +\infty]^s$	4.92	$[-3.57, 13.28]$	$[0.54, 12.10]$	–0.68	$[-3.61, 2.30]$	$[-3.20, 0.86]$
Tuition deduction: single, upper threshold	Tax	$+\infty^s$	$[-\infty, -4.02]^{s,\dagger}$	–	1.00	$[-215.11, 362.04]$	–	–1.13	$[-92.07, 53.56]$	–
Texas CC tuition	Funding	349.51	$[1.70, +\infty]^s$	$[515.98, +\infty]^s$	9.31	$[2.89, 29.01]$	$[9.34, 13.92]$	0.03	$[-5.12, 1.70]$	$[-1.18, 0.02]$
Michigan CC tuition	Funding	29.46	$[-3.01, +\infty]^s$	$[5.00, +\infty]^s$	7.80	$[-34.34, 85.19]$	$[5.00, 11.18]$	0.26	$[-20.17, 11.39]$	$[-0.63, 1.00]$
DC tuition assistance	Aid	22.98	$[3.20, +\infty]^s$	–	7.62	$[2.67, 44.14]$	–	0.33	$[-3.98, 0.83]$	–
HOPE tax credit	Tax	12.58	$[-2.83, +\infty]^s$	–	5.23	$[-16.93, 55.04]$	$[18.93, 22.87]$	0.42	$[-12.05, 5.97]$	$[-4.00, -3.01]$
Florida access grant	Aid	7.42	$[1.42, 99.62]$	$[4.67, 99.62]$	7.40	$[1.54, 32.92]$	$[4.64, 32.92]$	1.00	$[0.33, 1.09]$	$[0.33, 0.99]$
Social Security student benefits	Aid	4.86	$[3.54, 9.56]$	$[4.01, 4.10]$	3.04	$[2.61, 3.87]$	$[2.78, 2.81]$	0.62	$[0.40, 0.74]$	$[0.68, 0.69]$
College appropriations: spending	Funding	4.00	$[-2.54, +\infty]^s$	$[7.91, +\infty]^s$	3.03	$[-10.29, 27.10]$	$[4.04, 10.58]$	0.76	$[-5.18, 4.05]$	$[-1.10, 0.51]$
Georgia HOPE	Aid	4.00	$[-1.66, +\infty]^s$	–	3.12	$[-2.10, 17.10]$	–	0.78	$[-0.19, 1.27]$	–
Ohio Pell grants	Aid	2.49	$[-0.32, +\infty]^s$	–	2.88	$[-0.68, 32.70]$	–	1.16	$[-6.74, 2.10]$	–
Adult Pell introduction	Aid	2.18	$[2.95, +\infty]^s$	–	2.71	$[3.23, 20.37]$	–	1.25	$[-4.02, 1.09]$	–
Kalamazoo Promise	Aid	1.93	$[0.72, 6.49]$	$[0.72, 6.49]$	1.76	$[0.83, 3.34]$	$[0.83, 3.34]$	0.91	$[0.52, 1.14]$	$[0.52, 1.14]$
Tennessee HOPE	Aid	1.86	$[-3.00, +\infty]^s$	–	1.89	$[-22.05, 38.25]$	–	1.02	$[-8.61, 7.36]$	–
STBF scholarship	Aid	1.75	$[1.28, 3.98]$	–	1.52	$[1.21, 2.49]$	–	0.87	$[0.63, 0.95]$	–
Wisconsin Scholars Grant	Aid	1.43	$[-1.05, +\infty]^s$	$[1.91, +\infty]^s$	1.42	$[-1.91, 6.74]$	$[1.74, 6.74]$	0.99	$[-0.34, 1.82]$	$[-0.34, 0.91]$
CUNY Pell grants	Aid	1.39	$[-3.11, +\infty]^s$	$[-1.77, -0.19]$	1.42	$[-24.67, 38.47]$	$[-4.62, -0.28]$	1.02	$[-8.80, 7.93]$	$[1.47, 2.62]$
College appropriations: tuition	Funding	1.02	$[-2.92, +\infty]^s$	$[-1.53, +\infty]^s$	1.02	$[-13.00, 24.98]$	$[-3.08, 6.35]$	1.00	$[-4.91, 4.46]$	$[-0.32, 2.01]$
Tennessee Pell eligibility	Aid	0.85	$[-3.06, +\infty]^s$	–	0.79	$[-16.04, 25.93]$	–	0.93	$[-5.47, 5.25]$	–
Tuition deduction: joint, upper threshold	Tax	0.77	$[-4.07, +\infty]^s$	–	1.00	$[-121.13, 204.84]$	–	1.29	$[-46.21, 29.75]$	–
Adams Scholarship	Aid	0.72	$[-2.36, +\infty]^s$	$[-0.62, 0.30]$	1.21	$[-11.22, 22.62]$	$[-1.44, 0.55]$	1.68	$[-3.60, 4.75]$	$[1.84, 2.33]$
Tuition deduction: joint, lower threshold	Tax	–0.02	$[-4.12, +\infty]^s$	–	–0.03	$[-144.70, 235.01]$	–	1.38	$[-53.39, 35.10]$	–
HOPE and Lifetime Learning credits	Tax	–8.81	$[-\infty, -4.36]^{s,\dagger}$	–	–42.82	$[-294.55, 129.04]$	–	4.86	$[-38.38, 67.57]$	–

Continued on next page

Table 3 (continued)

Policy	Type	MVPF			WTP			Net fiscal cost		
		L1	L2	L3	L1	L2	L3	L1	L2	L3
Panel B. Finite tax reconstructions: per reference program dollar										
HOPE: joint, lower kink ^f	Tax	+∞ ^s	[−∞, +∞] ^{s,†}	−	18.41	[−26,240.82, 50,551.57]	−	−3.15	[−11,781.65, 6,099.22]	−
AOTC: dependent single, lower kink ^f	Tax	+∞ ^s	[−∞, +∞] ^{s,†}	−	23.39	[−25,930.65, 52,839.68]	−	−1.61	[−12,306.75, 6,017.33]	−
AOTC: joint, lower kink ^f	Tax	+∞ ^s	[−∞, +∞] ^{s,†}	−	9.96	[−32,961.38, 66,976.52]	−	−1.37	[−15,301.00, 7,514.79]	−
HOPE: independent single, lower kink ^f	Tax	18.86	[−4.07, +∞] ^s	−	5.91	[−49,505.38, 100,845.90]	−	0.31	[−24,720.50, 12,159.52]	−
HOPE: dependent single, upper kink ^f	Tax	11.83	[−∞, +∞] ^s	−	4.59	[−21,779.63, 41,833.66]	−	0.39	[−10,458.42, 5,431.12]	−
AOTC: simulated instrument ^f	Tax	10.05	[−∞, +∞] ^s	−	5.36	[−97,565.74, 177,781.69]	−	0.53	[−41,614.36, 22,882.67]	−
AOTC: independent single, lower kink ^f	Tax	6.75	[−4.10, +∞] ^s	−	2.45	[−81,288.51, 176,116.56]	−	0.36	[−42,932.93, 19,817.56]	−
HOPE: joint, upper kink ^f	Tax	2.37	[−4.30, +∞] ^s	−	8.21	[−28,147.39, 54,108.62]	−	3.47	[−12,605.33, 6,555.89]	−
AOTC: dependent single, upper kink ^f	Tax	−0.02	[−∞, +∞] ^{s,†}	−	−0.02	[−27,114.47, 54,980.56]	−	1.12	[−12,807.75, 6,289.69]	−
HOPE: dependent single, lower kink ^f	Tax	−1.64	[−∞, +∞] ^{s,†}	−	−1.91	[−20,564.64, 39,427.55]	−	1.16	[−9,857.51, 5,127.53]	−
AOTC: joint, upper kink ^f	Tax	−1.77	[−∞, +∞] ^{s,†}	−	−7.63	[−33,866.98, 68,622.36]	−	4.31	[−15,675.18, 7,723.09]	−
Panel C. Aligned or extended accounts and education channel: native thousands										
Algorithmic college placement ^x	Placement	+∞ ^s	[−∞, −3.94] ^{s,†}	−	0.22	[−10.36, 21.89]	−	−0.09	[−5.68, 2.63]	−
Early Head Start ^r	Childhood	0.35	[−1.53, 1.96]	−	3.91	[−25.56, 17.13]	−	11.22	[8.74, 16.73]	−
Cannabis legalization: education channel	Other	−	−	[−∞, −20.31] ^{s,†}	−	−	[−4.02, 0.51]	−	−	[−0.05, 0.20]

Policy types: Aid, direct student financial aid; Tax, education tax benefits; Funding, tuition and public-college funding; Placement, college assessment and course placement; Childhood, early childhood services; Other, the education channel of a noneducation policy. Types describe interventions, whereas panels distinguish accounting specifications. *Notes:* L1 is the source valuation; brackets show fixed-input candidate spans, not confidence intervals or certified bounds. Component extrema need not come from the same path and must not be combined. Infinite endpoints include audited positive-cost ratio limits; *s* denotes separate self-financing paths ($W > 0$, $N \leq 0$), and $\dagger$ undefined-ratio paths. Policy types are student aid, education tax benefits, college tuition/funding, assessment/placement, early childhood services, and other policies. Panel A normalizes monetary values by reference program expenditure. Panel B also uses this normalization but requires finite-credit, local-linear, and duration mappings, including the simulated-instrument unit correction. Panel C reports thousands of source-account dollars: Early Head Start uses aligned costs with an unresolved price year; college placement adds earnings omitted from L1; cannabis values only the education channel. Superscripts *f*, *r*, and *x* mark tax, alignment, and extension cases. Dashes indicate unavailable or inapplicable entries; missing L3 means no further source-window restriction is admitted under the retained mapping. Appendix E.4 details units and account differences.

Table 4. Reported intervals and conditional input-repetition endpoint summaries

Policy	L1: reported 95%	L2: endpoints	n_2	L3: endpoints	n_3	Scope
HOPE: joint, lower kink	$[-3.51, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
Tuition deduction: single, lower threshold	$[-\infty, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
AOTC: dependent single, lower kink	$[-0.84, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
AOTC: joint, lower kink	$[-5.96, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
Tennessee Promise	$[+\infty, +\infty]$	$[-1.71, +\infty]$	1000	$[-0.58, +\infty]$	1000	P
Algorithmic college placement	–	$[-\infty, +\infty]$	1000	–	–	P
Tuition deduction: single, upper threshold	$[-\infty, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
Texas CC tuition	$[0.93, +\infty]$	$[-1.71, +\infty]$	999*	$[1.56, +\infty]$	999*	J
Michigan CC tuition	$[-2.33, +\infty]$	$[-4.02, +\infty]$	1000	$[0.75, +\infty]$	1000	J
DC tuition assistance	$[1.12, +\infty]$	$[0.51, +\infty]$	1000	–	–	X
HOPE: independent single, lower kink	$[-3.03, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
HOPE tax credit	$[-29.15, +\infty]$	$[-3.41, +\infty]$	1000	$[0.98, +\infty]$	1000	J
HOPE: dependent single, upper kink	$[-5.36, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
AOTC: simulated instrument	$[-18.36, +\infty]$	$[-\infty, +\infty]$	1000	–	–	P
Florida access grant	$[0.69, +\infty]$	$[-7.66, +\infty]$	1000	NR	–	J
AOTC: independent single, lower kink	$[-1.61, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
Social Security student benefits	$[0.98, 52.39]$	$[0.35, +\infty]$	952*	NR	–	J
College appropriations: spending	$[1.25, 20.44]$	$[-3.15, +\infty]$	1000	$[1.08, +\infty]$	1000	J
Georgia HOPE	$[0.46, 22.38]$	$[-2.26, +\infty]$	1000	–	–	C
Ohio Pell grants	$[0.82, 5.54]$	$[-1.04, +\infty]$	1000	–	–	J
HOPE: joint, upper kink	$[-2.22, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
Adult Pell introduction	$[0.71, 6.11]$	$[-0.87, +\infty]$	1000	–	–	C
Kalamazoo Promise	$[0.97, 5.61]$	$[-0.14, 65.79]$	1000	NR	–	J
Tennessee HOPE	$[0.92, 5.08]$	$[-3.38, +\infty]$	1000	–	–	J
STBF scholarship	–	$[1.10, 4.34]$	1000	–	–	P
Wisconsin Scholars Grant	$[0.96, 2.18]$	$[-1.82, +\infty]$	1000	NR	–	J
CUNY Pell grants	$[-2.95, 12.88]$	$[-3.99, +\infty]$	1000	NR	–	J
College appropriations: tuition	$[-1.11, 5.13]$	$[-3.52, +\infty]$	1000	$[-2.97, +\infty]$	1000	J
Tennessee Pell eligibility	$[-1.59, 3.57]$	$[-3.69, +\infty]$	1000	–	–	C
Tuition deduction: joint, upper threshold	$[-1.83, 57.84]$	$[-4.16, +\infty]$	1000	–	–	J
Adams Scholarship	$[-0.92, 3.05]$	$[-2.66, +\infty]$	1000	$[-1.82, 6.06]$	1000	J
Early Head Start	–	$[-1.94, 1.97]$	1000	–	–	P
AOTC: dependent single, upper kink	$[-2.25, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
Tuition deduction: joint, lower threshold	$[-2.25, 5.52]$	$[-4.20, +\infty]$	1000	–	–	J

Continued on next page

Table 4 (continued)

Policy	L1: reported 95%	L2: endpoints	n_2	L3: endpoints	n_3	Scope
HOPE: dependent single, lower kink	$[-6.45, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
AOTC: joint, upper kink	$[-17.06, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
HOPE and Lifetime Learning credits	$[-\infty, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
Cannabis legalization: education channel	–	–	–	$[-\infty, +\infty]$	1000	R

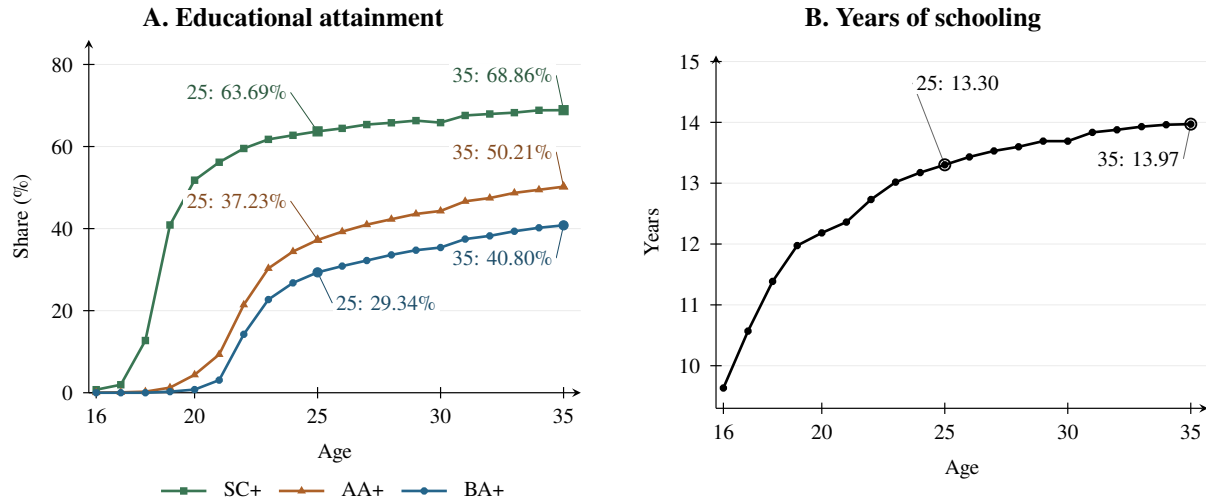
Notes: L1 shows recovered source 95 percent intervals. L2/L3 span the 2.5th percentile of candidate minima to the 97.5th percentile of maxima over the specified input repetitions. Counts n_2, n_3 are usable defined-MVPF repetitions out of 1,000; an asterisk marks incomplete counts. These summaries have no demonstrated confidence coverage for the continuous path class. Infinite endpoints include self-financing and audited positive-cost ratio limits. A dash means no recovered L1 interval or no admitted L3; NR means joint draws are unavailable for the full L3 inputs. Scope codes: J, registered joint source/accounting draws; P, constructed education perturbations; C, fixed-return conditional inputs; X, economic inputs with education fixed; R, synthetic correlated shocks. All condition on ACS calibration. Source intervals and reconstructed summaries need not share inputs or coverage targets. Cannabis reports education-channel L3; its last-horizon L2 is a diagnostic. Appendix F gives methods and exclusions. Order and superscripts follow Figure 8.

Table 5. Reference and single-parameter reconstructed MVPFs

Policy	Reference	Permanent (K)		Temporary (b)		Temporary (d)	
	MVPF	ΔK	MVPF	Δb (%)	MVPF	Δd	MVPF
Florida grant	7.42	0.371	18.40 [†]	—	—	—	—
DC tuition aid	22.98	—	—	13.7	9.15	—	—
Adams	0.72	-0.0244	0.77	-1.18	1.29	0.425	1.31
STBF	1.75	0.459	2.07	—	—	—	—
Wisconsin grant	1.43	0.146	2.20 [†]	6.47	0.94	—	—
Kalamazoo	1.93	0.416	2.23 [†]	—	—	—	—
Michigan CC	29.46	0.0343	19.52	3.55	0.86	—	—
CUNY Pell	1.39	0.0197	2.02	0.686	0.96	-0.215	0.92
Ohio Pell	2.49	—	—	—	—	—	—
Tennessee HOPE	1.86	0.0575	4.59	2.28	0.83	—	—
Tennessee Pell	0.85	-0.0188	0.47	-0.808	1.11	0.246	1.14
Georgia HOPE	4.00	0.321	9.25	—	—	—	—
Social Security	4.86	—	—	—	—	—	—
Appropriations: spending	4.00	0.0822	5.56	3.92	1.01	—	—
Early Head Start	0.35	—	—	—	—	—	—
Texas CC	349.51	0.248	SF [†]	—	—	—	—
Adult Pell	2.18	0.0815	SF	—	—	—	—
Appropriations: tuition	1.02	8.23×10^{-4}	1.03 [†]	0.0386	1.00 [†]	-0.0154	1.00 [†]
HOPE credit	12.58	0.128	SF	6.05	1.28	—	—
HOPE ind. single: lower	18.86	1.96×10^{-4}	SF	0.00946	0.07	-0.00414	-0.05
HOPE dep. single: lower	-1.64	-2.72×10^{-4}	-3.14	-0.0131	0.77	0.00575	1.73
HOPE dep. single: upper	11.83	3.15×10^{-4}	SF	0.0152	1.60	-0.00665	0.94
HOPE joint: lower	SF	0.00124	SF	0.0599	SF	-0.0263	14.13
HOPE joint: upper	2.37	4.74×10^{-4}	SF	0.0229	0.55	-0.0100	0.30
Tuition ded. single: lower	SF	0.0262	SF	2.02	SF	—	—
Tuition ded. joint: lower	-0.02	-0.0169	-1.26	-0.611	0.99	0.183	1.14
HOPE / Lifetime Learning	-8.81	-0.195	-4.85	—	—	—	—
AOTC single: upper	-0.02	-6.64×10^{-5}	-1.62	-0.00364	0.95	0.00174	1.12
AOTC single: lower	SF	0.00153	SF	0.0840	1.39	-0.0405	0.08
AOTC joint: upper	-1.77	-4.39×10^{-4}	-3.21	-0.0241	-0.10	0.0115	0.33
AOTC joint: lower	SF	4.68×10^{-4}	SF	0.0257	8.46	-0.0123	2.53
AOTC ind. single: lower	6.75	3.17×10^{-5}	SF	0.00174	0.96	-8.31×10^{-4}	0.86
AOTC instrument	10.05	9.39×10^{-5}	SF	0.00390	1.00	-0.00136	0.78
Tennessee Promise	SF	Joint	SF	—	—	—	—
College placement	SF	0.0545	SF	5.11	SF	—	—
Tuition ded. single: upper	SF	0.000	SF	0.00	SF	0.000	SF
Tuition ded. joint: upper	0.77	0.000	0.77	0.00	0.77	0.000	0.77

Notes: Each pair varies the named parameter family; the other two remain at H^0 values. K changes eventual attainment; b and d effects eventually vanish. ΔK and Δd are in years; Δb is the proportional speed change in percent. Tennessee Promise uses joint plateau shifts of (3.502, 4.976, 3.349) percentage points for SC+, AA+, and BA+, respectively. [†] marks the same L2 solution also satisfying L3; — denotes no admissible match. SF denotes self-financing ($W > 0$, $N \leq 0$), represented by $+\infty$ in Figure 10. The table includes all 37 reference calculations; cannabis has no whole-policy reference. Appendix B.5 defines the matching procedure. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

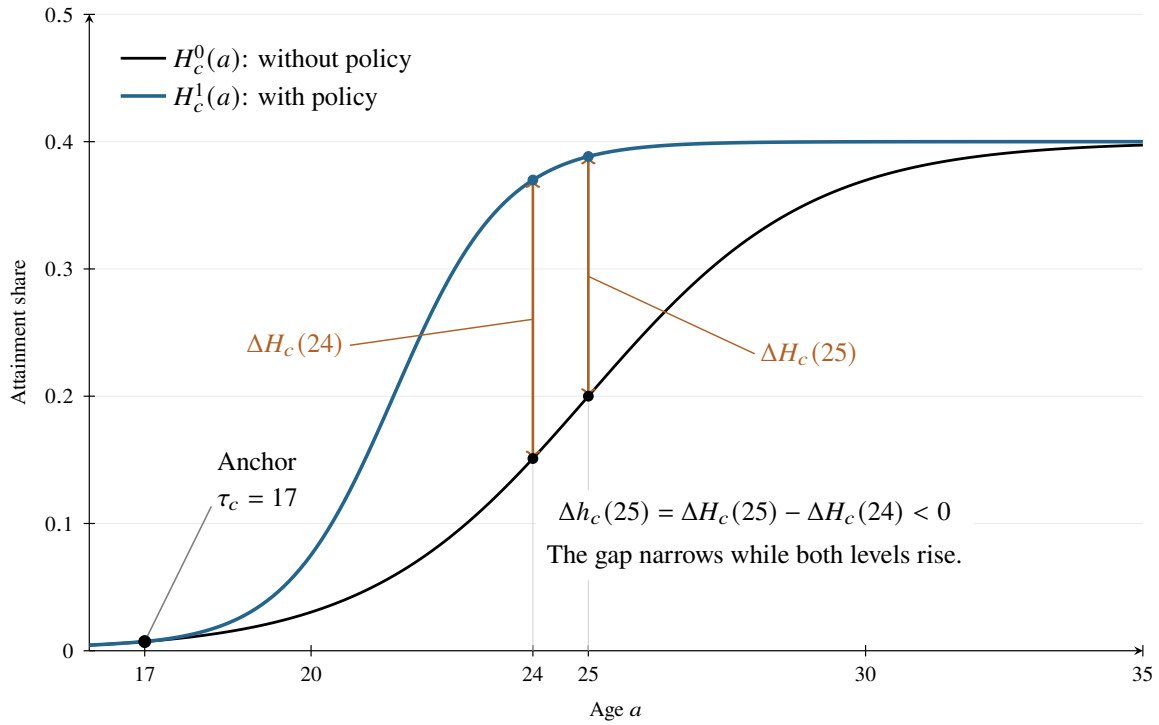
Figure 1. Education profiles for a fixed set of birth cohorts



Notes: ACS 2000–2022, birth cohorts 1985–1987, ages 16–35; all sexes, birthplaces, and racial and ethnic groups are included. Person weights apply within cohort–age cells; cohorts receive fixed one-third weights. Repeated cross sections do not fix composition within birth cohorts. SC+ denotes some college or higher, an attendance proxy; AA+ and BA+ denote cumulative degree thresholds. Schooling years use the EDUCD crosswalk. Labels report ages 25 and 35. Appendix A.5 documents these descriptive profiles.

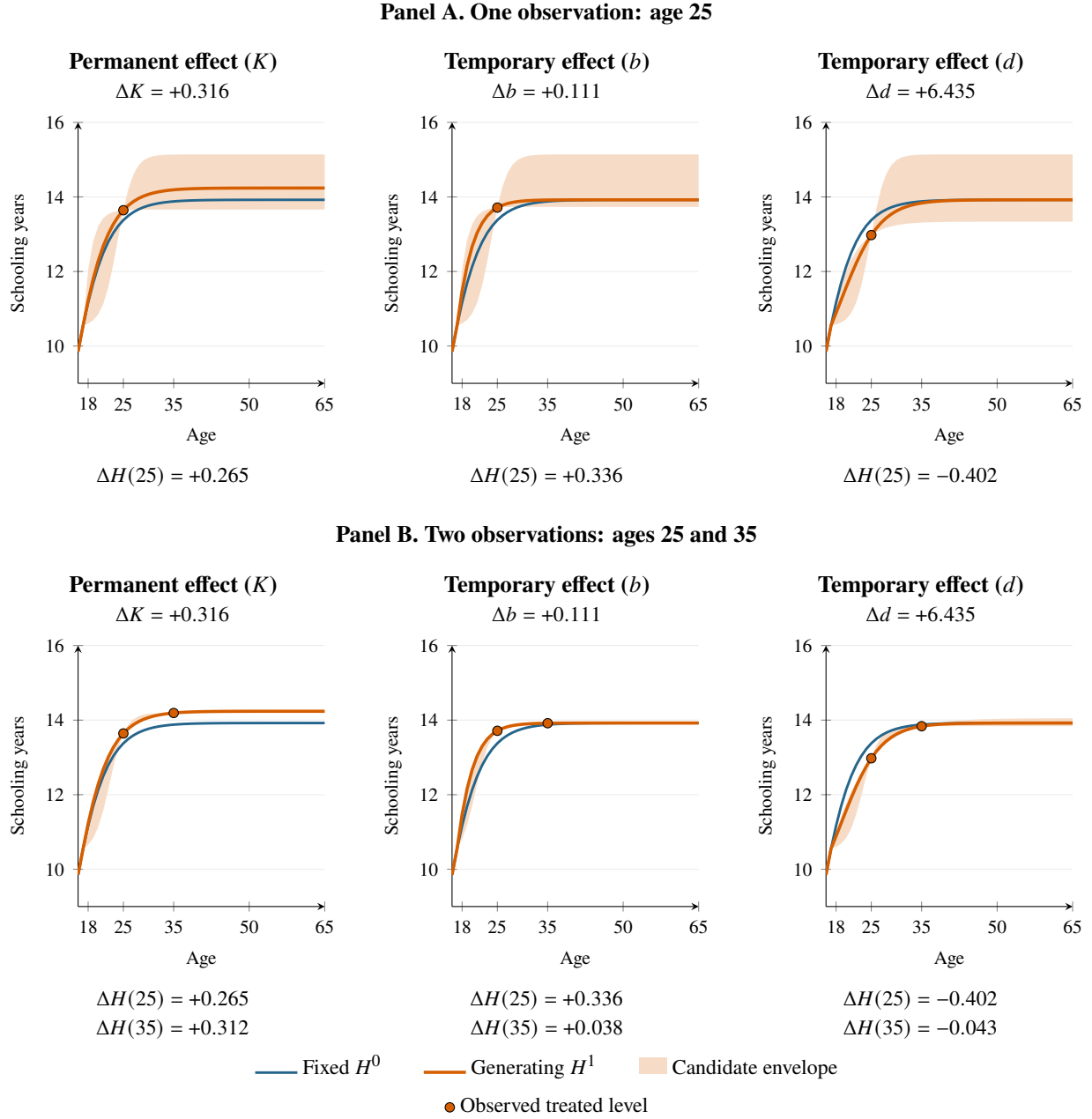
Source: Author’s calculations from ACS microdata.

Figure 2. Educational attainment, treatment effects, and the pre-policy anchor



Notes: The hypothetical reference is $H_c^0(a) = 0.4\Lambda[0.5(a - 25)]$. The treated path uses Equation (2), $(K, b, d) = (0.4, 1, 21.5)$, and the same age-17 level. Arrows mark attainment gaps at 24 and 25. Both levels rise while the gap narrows, giving $\Delta h_c(25) < 0$; both plateaus equal 0.4. These are illustrative attainment shares, not estimated effects.

Figure 3. Compatible schooling paths with one and two observed ages

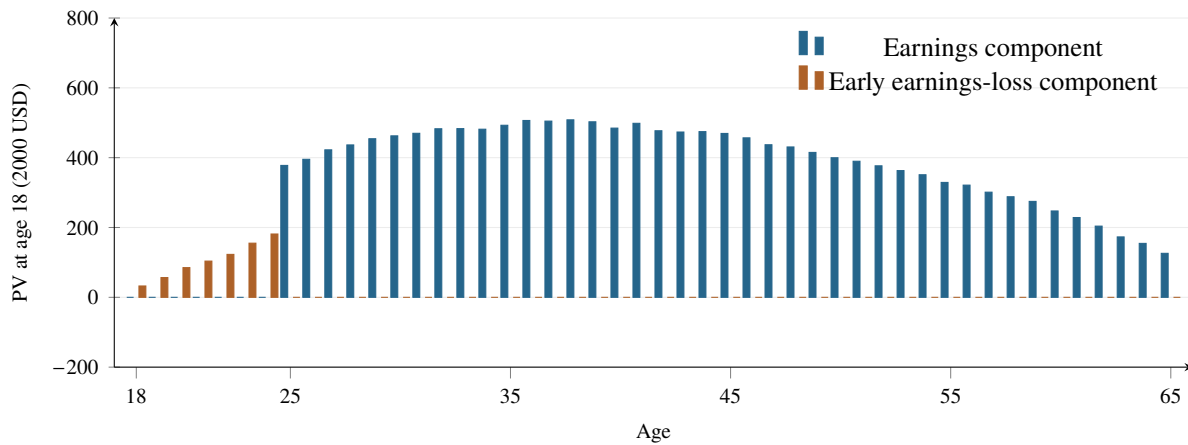


Notes: Panel A matches age 25; Panel B adds age 35 for the same generating scenario. Each column varies one generating parameter to its conditional-interval midpoint. Permanent and temporary describe that generating path. Blue is H^0 ; orange-red is the generating H^1 . Shading gives pointwise candidate envelopes with all three parameters free and a common age-17 anchor. Edges can combine different paths; shading is neither a confidence band nor a probability distribution. Baseline ages above 35 are extrapolated. Scales are common.

Figure 4. Single-effect and full-path valuation with a permanent schooling gain

A. Single age-25 effect

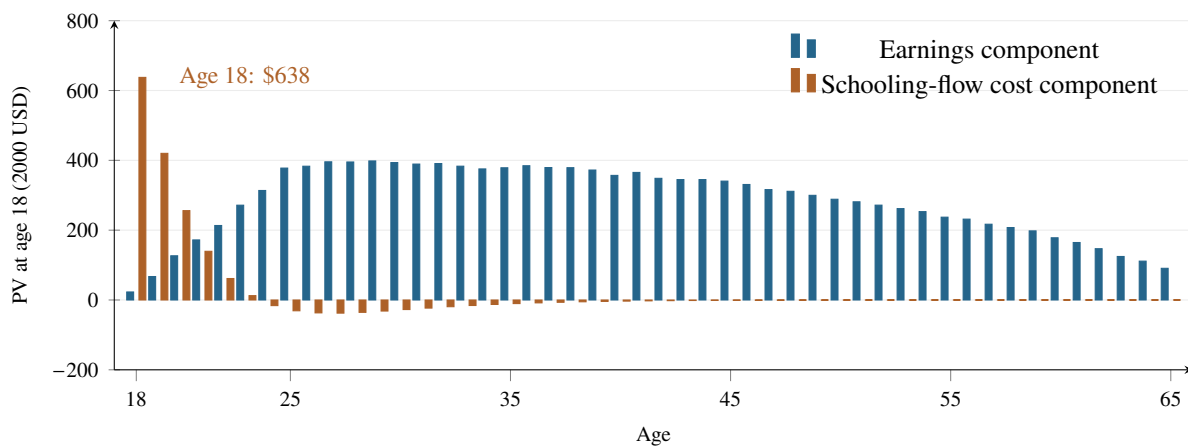
Education NPV: \$15,390



Earnings PV: \$16,127; early earnings-loss PV: \$737. NPV subtracts orange from blue.

B. Full schooling path

Education NPV: \$12,397

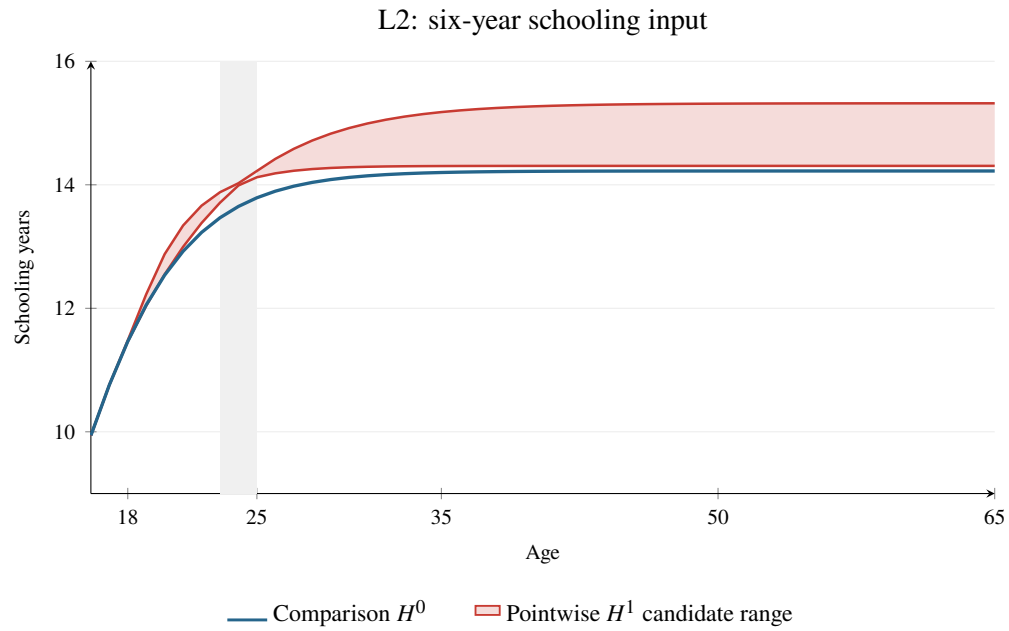


Earnings PV: \$13,588; net schooling-flow cost PV: \$1,191.

Positive costs: \$1,527; relative cost savings: \$336.

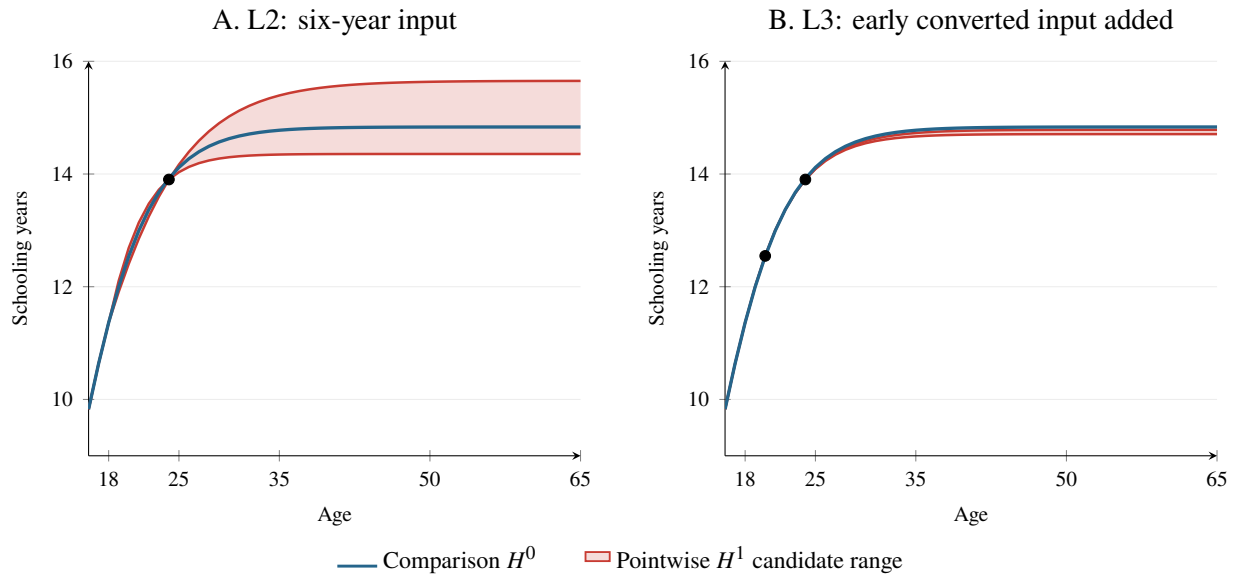
Notes: Both panels use $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$; Figure B2 shows the level paths. Panel A applies the signed earnings premium to the age-25 effect throughout life. Panel B uses age-specific schooling stocks and acquisition flows. Blue bars denote the earnings component. Orange bars denote the signed early-earnings adjustment in Panel A and schooling-flow opportunity cost in Panel B; negative orange values in Panel B are relative cost savings. NPV subtracts orange from blue. Values are discounted to age 18 at 3 percent, in 2000 dollars. The shared age-17 anchor excludes a pre-policy schooling difference.

Figure 5. STBF: schooling paths compatible with the six-year input



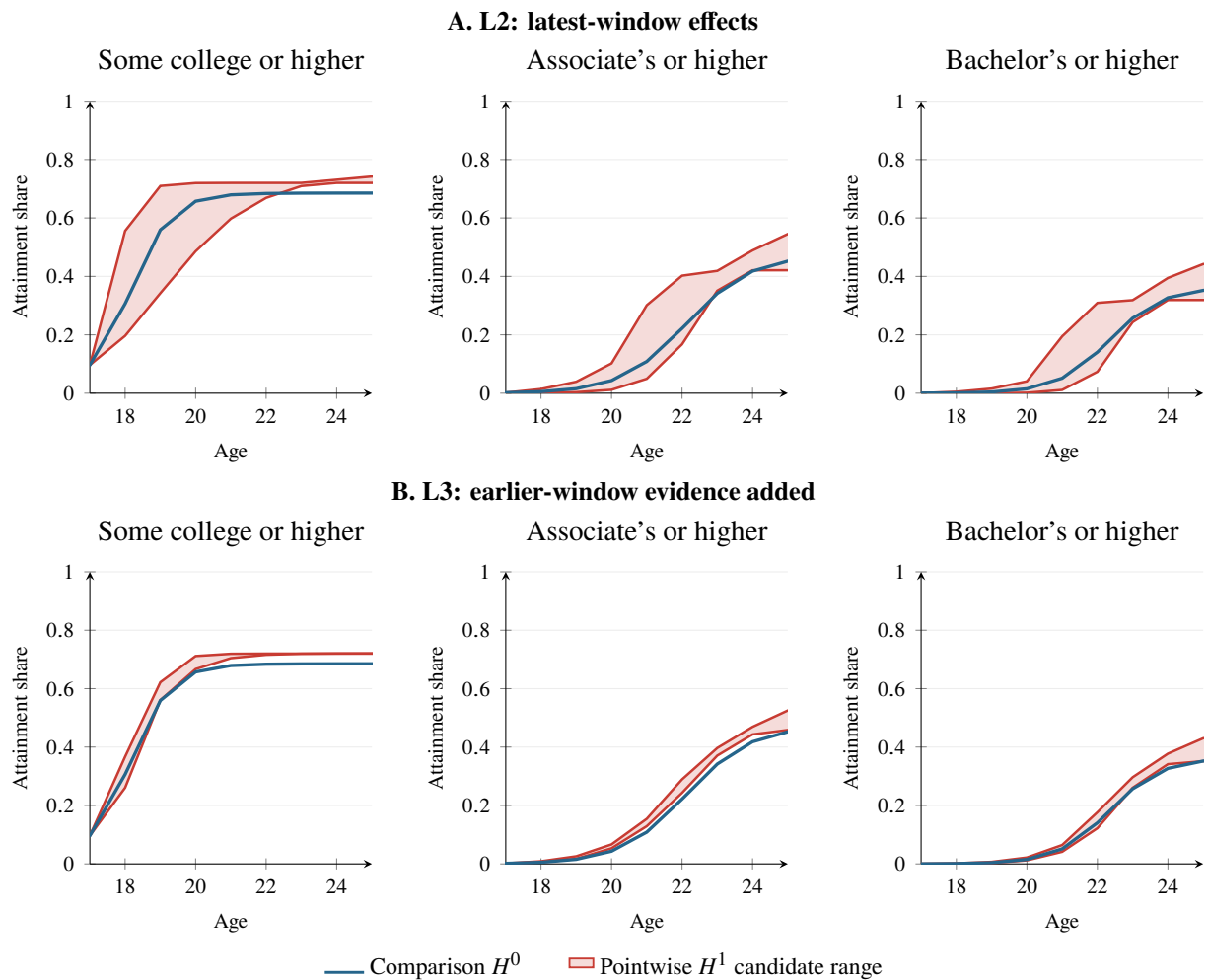
Notes: L2 matches the weighted HS+6 schooling-equivalent input of 0.3648 years. The gray strip marks branch endpoints at ages 23–25. Blue is H^0 ; red boundaries and shading give the pointwise minimum and maximum H^1 among the candidates. Boundaries may combine different paths and are not confidence limits. No comparable additional schooling-time moment supports L3 under the retained conversion. Appendix D.6 documents the Nebraska baseline and conversion.

Figure 6. Adams: adding the early converted education input



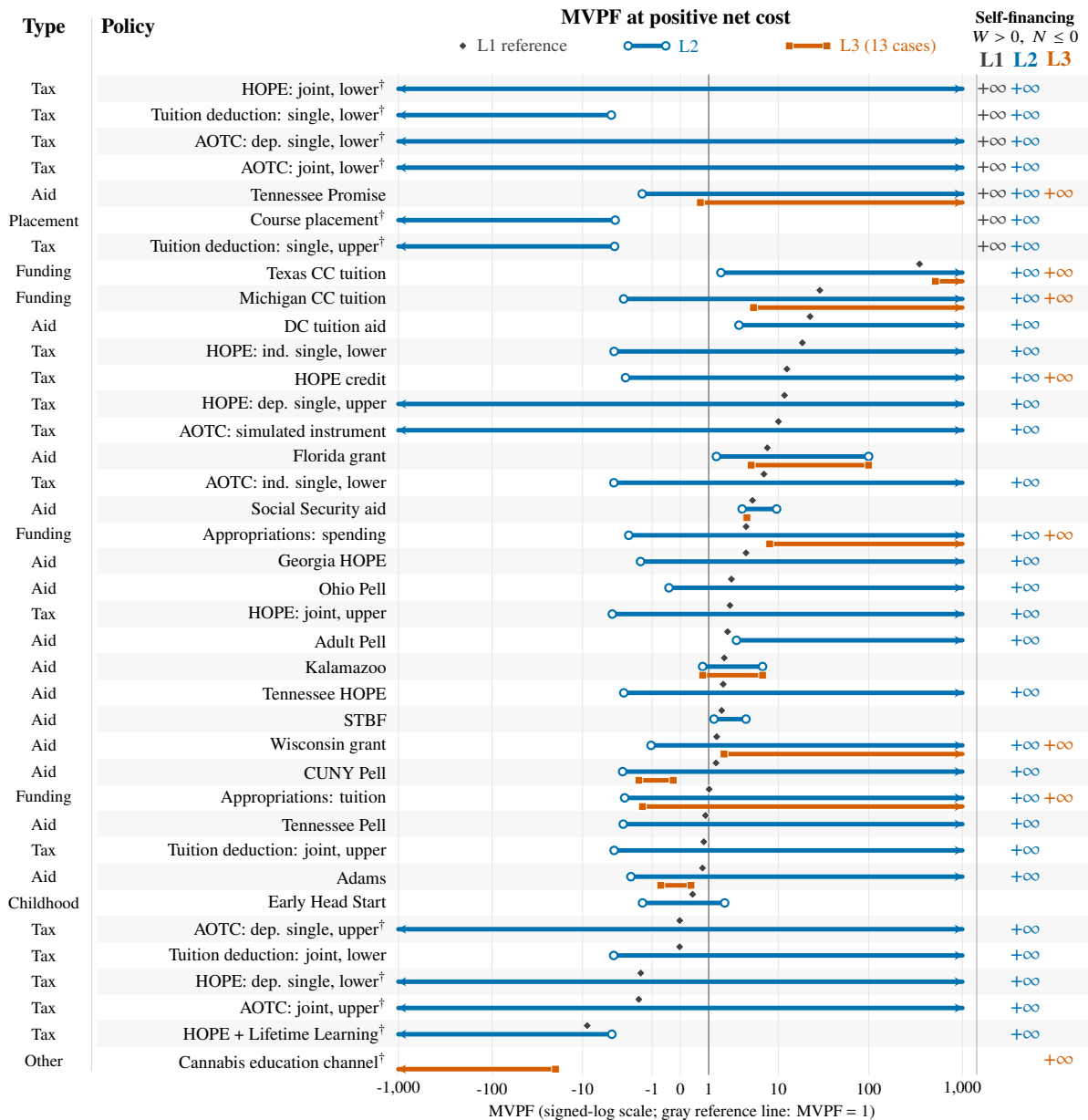
Notes: L2 matches -0.018 schooling-equivalent years at HS+6 (model age 24). Panel B additionally assigns 0.007 completed-schooling-equivalent years to HS+2 (age 20); this L3 restriction retains the early source window under an explicit temporal allocation. Dots mark converted restrictions. Blue is H^0 ; red boundaries and shading give the pointwise H^1 candidate range. Its edges may combine different paths and are not confidence limits. Scales are common. Appendix D.7 documents the calibration and allocation.

Figure 7. Tennessee Promise: treated-attainment ranges around the observed age windows



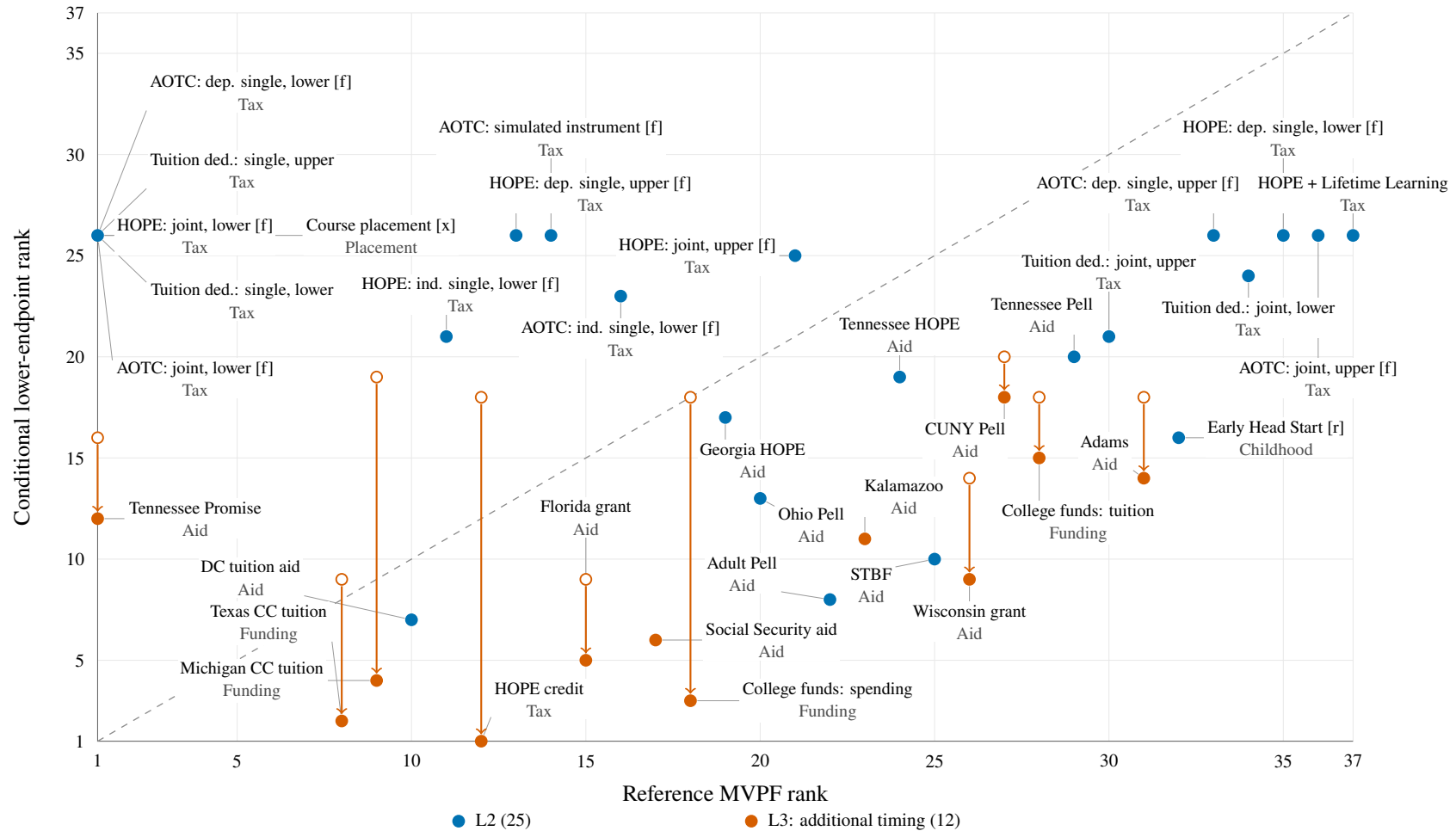
Notes: Blue is H^0 ; red boundaries and shading give the pointwise minimum and maximum H^1 among joint credential candidates. The boundaries may combine different paths across ages and credentials. L2 matches the three effects averaged over ages 23–24. L3 adds fifteen earlier-window marginal permutation intervals from Schoonover et al. (2026), with a one-pixel inward extraction rule. Both panels use the same baseline, age-17 anchor, support, completion caps, and scales. These ranges are finite-search envelopes, not confidence bands or probability distributions. Appendix D.8 documents the restrictions; the appendix atlas shows ages through 65.

Figure 8. Education continuation ranges, ordered by the reference MVPF



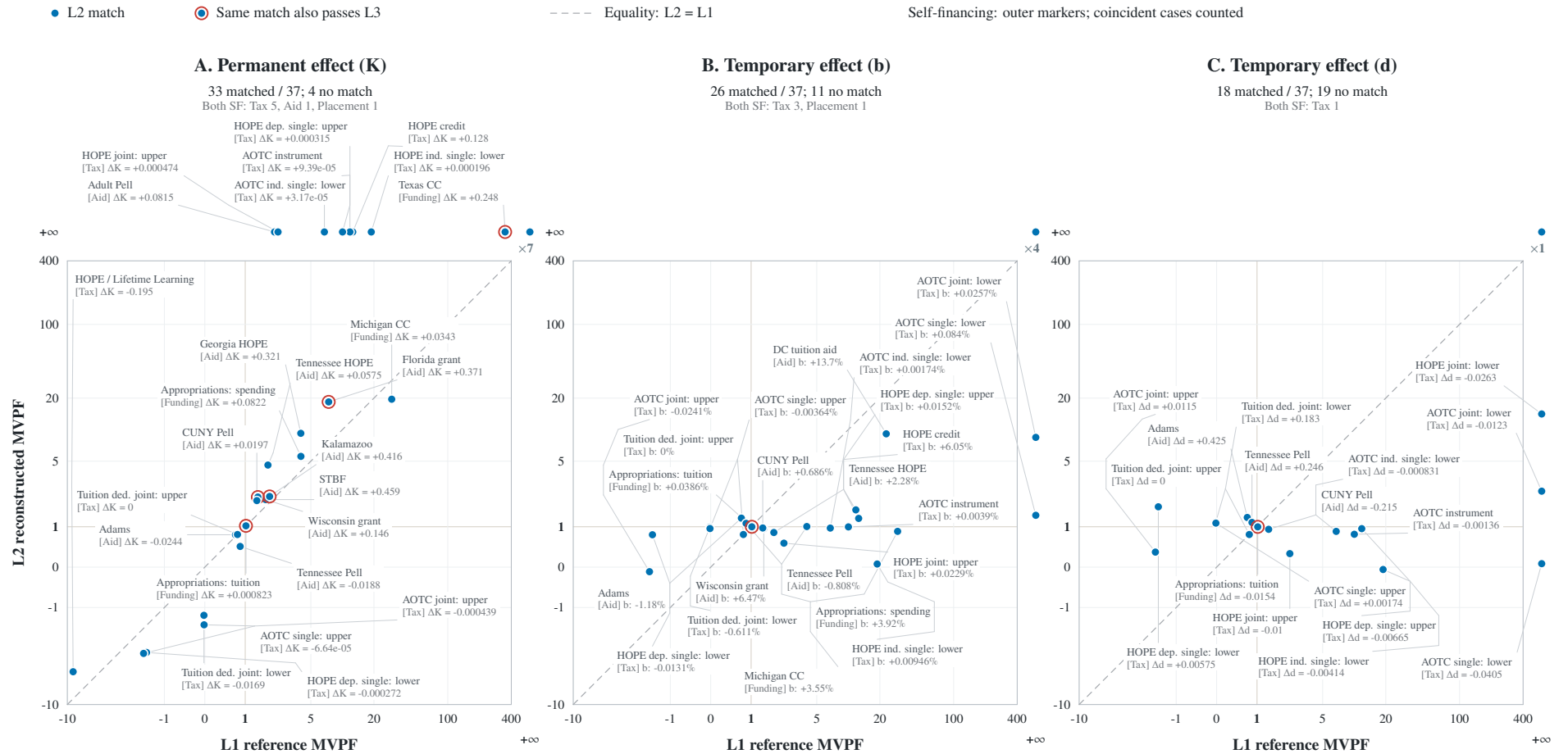
Notes: Rows follow descending L1. Types: Aid, direct student aid; Tax, education tax benefits; Funding, tuition and public-college funding; Placement, assessment and course placement; Childhood, early-childhood services; Other, another policy's education channel. Gray diamonds denote L1; blue circles and red squares bound positive-cost L2 and L3 spans. Separate +∞ marks denote self-financing ($W > 0, N \leq 0$); arrows include audited positive-cost ratio limits. Gaps do not establish infeasible intermediate ratios. Daggers mark undefined-ratio paths. Missing L3 means no added restriction is admitted under the retained mapping, not no source dynamics. Superscripts f , r , and x identify tax-unit conversion, account alignment, and an earnings extension. Cannabis has no whole-policy L1 and shows primary L3. These candidate spans are not confidence intervals or certified bounds. Table 3 gives values. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

Figure 9. L1 and candidate-based lower-endpoint ranks for 37 policy calculations



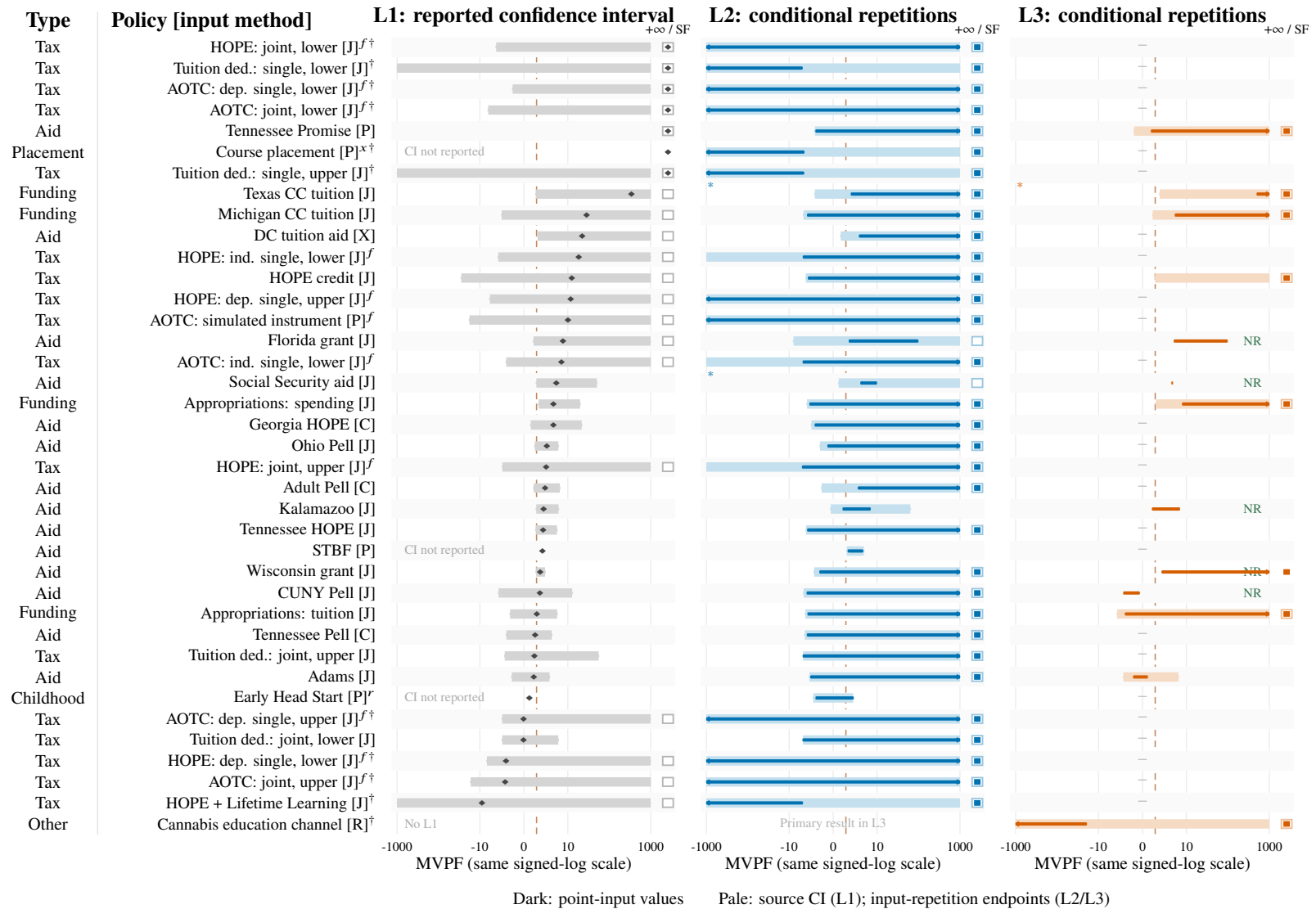
Notes: Higher lower endpoints rank first, followed by higher upper endpoints; admitted L3 replaces L2. Finite endpoints are rounded to two decimals, and equal pairs share a competition rank. Labels give policy names and Figure 8 types; [f], [r], and [x] mark tax conversion, account alignment, and an earnings extension. Eight tax calculations share unbounded endpoint pairs with four other calculations; coincident cases share a marker. Arrows replace only the indicated policy's L2 by L3. Ten calculations contain undefined-ratio paths; those paths, not the calculations, are omitted. Cannabis lacks L1. Table E2 gives all ranks. This is a candidate-based display rule, not a social-welfare or certified worst-case ordering.

Figure 10. Reference and reconstructed MVPFs under permanent and temporary effects



Notes: Blue points match L2; red rings identify the same path also satisfying L3. Labels give policy type, ΔK in schooling years, proportional b changes in percent, and Δd in years; the other two parameter families stay fixed. Finite axes have identical asinh scales and square dimensions. Self-financing ($W > 0, N \leq 0$) appears outside them: top markers retain L1, right markers retain L2, and corner counts identify coincident self-financing pairs. Permanent effects change eventual attainment; temporary effects eventually vanish. Appendix B.5 gives the method; Table 5 gives values and missing matches. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

Figure 11. Point-input ranges, source intervals, and conditional input repetitions



Notes: Policy types follow Figure 8. Dark marks show fixed-input results; pale bands show reported L1 intervals or conditional L2/L3 endpoint summaries. Filled $+\infty$ marks denote self-financing; open squares denote infinite summary endpoints. Arrows extend beyond the axis; the guide marks one. Daggers mark undefined ratios; asterisks mark incomplete draws; NR means no joint L3 draws. Method codes: J, joint source/accounting draws; P, constructed education perturbations; C, fixed-return conditional inputs; X, economic inputs with education fixed; R, synthetic correlated shocks. Tennessee's normal perturbations are not source permutation draws. Superscripts follow Figure 8. Table 4 gives endpoints and counts. Reconstructed summaries have no demonstrated coverage for all continuations.

A Literature Review and Descriptive Education Profiles

This appendix documents the two sources of descriptive evidence in Section 2: a review of when higher-education effects are measured and used in longer-run valuation, and ACS profiles of educational attainment across ages and birth cohorts. The literature review counts distinct papers; the policy applications elsewhere in the manuscript count policy–estimand calculations. For each reviewed paper, the accompanying records identify the source version, education outcomes, observation windows, relevant passages and exhibits, and inclusion decision. The review takes the original estimates as inputs and does not reassess each study’s causal identification.

A.1 Search sample and inclusion

The journal search covers 2000–2025 in the *American Economic Review*, *Quarterly Journal of Economics*, *Journal of Political Economy*, *Review of Economic Studies*, *Review of Economics and Statistics*, *AEJ: Applied Economics*, *AEJ: Economic Policy*, *Journal of Human Resources*, *Journal of Labor Economics*, *Journal of Public Economics*, *Economics of Education Review*, and *Education Finance and Policy*. The 674 journal-search records are the supplied, preliminarily screened higher-education records in the full-text manifest, not all articles published in those journals. Full-text inclusion and duplicate checks retain 447 for analysis. The register preserves each supplied record and its inclusion decision; the exercise is a documented review of this selected search sample, not a census of education research. Eleven additional studies underlying welfare calculations and one unpublished author manuscript on marijuana legalization bring the retained sample to 459 papers. After deduplication of the additions against the journal search, the combined register contains 686 records. The remaining 227 records include exclusions and duplicates.

Inclusion requires an original estimate relating a policy, intervention, or external exposure to a higher-education outcome observed within a finite window. The education estimate can be secondary to the paper’s main analysis. Studies reporting only test scores, aspirations, earnings, or high-school outcomes are excluded, as are purely descriptive comparisons, reviews, and editorials. The review imposes no fixed maximum follow-up length: an education outcome observed many years after exposure still provides finite-window evidence. Working and published versions of the same study count once, while distinct studies of one program can count separately. The resulting sample is selected. In particular, the additional welfare-source papers were included for their relevance to valuation, so their frequency cannot be interpreted as the prevalence of valuation in the wider literature.

A.2 Education Outcomes, Estimation Times, and Endpoints

The review distinguishes college entry (CE), associate-degree outcomes (AA), and bachelor’s-degree or higher outcomes (BA+). CE requires first entry or ever entering college; current enrollment or persistence alone does not qualify. AA preserves the source study’s distinction between an associate’s degree and at least an associate’s degree. The overall category also includes credits, completed schooling, graduate entry, and other qualifying education measures. A paper can contribute to several outcome groups, so the group counts cannot be added to obtain a total number of papers.

The timing count records distinct reported effect-estimation times or pooled windows within a comparable series. Effects reported at years 3 and 6 count as two times; effects reported separately at years 3, 4, 5, and 6 count as four. A single coefficient pooled over years 3–6 counts as one window. Alternative specifications,

treatment arms, and subgroups at the same horizon do not add times. The coding retains the time origin, reported horizons, selected series, and supporting table or figure. Independent samples or unmatched time origins are not combined into a single longitudinal sequence. The overall and outcome-specific counts can differ because they summarize different education evidence within the same paper.

Figure A2 and Table A1, Panel A, summarize these counts. All 459 papers have an overall count: 249 report one time or window and 210 report more than one. CE includes 237 eligible papers, of which 236 have a coded count; the remaining paper is excluded only from that panel's percentage denominator. AA has 40 coded papers and BA+ has 190. Among the BA+ papers, 132 report one time or window, or 69.5 percent. A single window can span several years, so these statistics describe the number of reported estimates over time rather than the length of follow-up or the duration of a policy effect.

An observation endpoint is recorded separately from the timing count and assigned to a defined clock. Attained age dates the person's education outcome. Individual follow-up measures elapsed time since an origin such as random assignment, high-school exit, or college entry. Policy or cohort event time dates an observation relative to a reform or exposed cohort and need not track the same people. These clocks are kept separate throughout. A six-year completion outcome is not assigned age 24 without evidence on the relevant entry age; academic terms are converted to years only when the source calendar supports that conversion.

Every paper receives one overall endpoint classification, reported in Table A1, Panel B. An *exact numerical endpoint* supplies a value on at least one defined clock (366 papers). An *interval or bound* restricts the endpoint without supplying a unique value (38). A *stage with unaligned timing* identifies an education stage or observation window without a common numerical endpoint (46). An *unresolved endpoint* has insufficient or conflicting information about the cutoff (9). All four categories remain in the literature sample. For example, an admission-stage effect can have one identifiable estimation window even when the paper does not report elapsed time between assignment and admission.

Figure A3 relates the number of reported times or windows to numerical endpoints for all education outcomes; Figure A4 presents the same comparison separately for CE, AA, and BA+. The overall panels contain 104 age pairs, 258 follow-up pairs, and 29 event-time pairs, representing 366 distinct papers. These counts overlap because a paper can supply more than one clock. The other 93 papers remain in the sample and in the endpoint classification, without an imputed coordinate. Bubble area is proportional to the number of papers at each coordinate. The displayed Spearman correlations are computed from the individual paper records, including ties, rather than treating each bubble as one observation. Correlations are shown only when a panel contains at least ten papers and both coordinates vary. The figures describe reporting patterns, not causal relationships or changes in treatment effects. In the overall panels, counts and endpoints can summarize different education outcomes within a paper; the outcome-specific panels narrow that comparison to the recorded education category.

A.3 Interpretation and Education-Based Lifetime Valuation

The review separates statements about finality from arguments about the adequacy of the observation window. A *finality or permanence statement* discusses whether an education difference is final or persistent, preserving the source authors' qualifications (18 papers). A *completion or window-adequacy argument only* explains why education is expected to be complete or follow-up sufficiently long, without explicitly asserting a permanent treatment effect (22 papers). The second category is assigned only when no qualifying finality

statement is recorded. The combined count of 40 therefore measures the presence of these arguments, not the number of studies establishing permanent effects.

Education-based lifetime or career valuation requires a quantitative projection from education evidence over a lifetime or working career. It includes projected earnings, fiscal effects, benefits relative to costs, and MVPF calculations; a complete welfare calculation is not required. Qualitative references to education returns, lifetime calculations unrelated to education estimates, and education costs totaled over a few years do not qualify. Thirty-one papers meet this definition. One also belongs to an interpretation category, so 70 papers satisfy at least one of these criteria and 389 have neither detected in the reviewed material (Table A2, Panel A). The latter classification does not imply that a paper contains no discussion of long-run outcomes.

Panel B of Table A2 links the 31 valuation papers to the education outcomes entering their calculations. This is a narrower question than the paper-level reporting counts above. It asks whether the valued education outcome is observed at one or several ages or individual follow-up times; it does not count earnings observations, different educational outcomes at different deadlines, or successive exposed cohorts as additional observations of one attainment path. When several outcomes enter a valuation, the classification uses the richest comparable series among those outcomes. A paper can therefore supply several inputs to a calculation while reporting only one window for each. For example, credits measured in one academic year and transfer measured in the next supply different education margins, whereas bachelor's completion reported at years 4, 5, and 6 supplies repeated observations of a degree-attainment margin. The table records available education information, not a claim that every reported time enters the authors' valuation formula.

Eighteen papers report one age, follow-up, or stage-specific window per valued education series; 12 report multiple outcome times. The remaining paper values annual institutional degree awards indexed by spending-shock time, which does not supply an individual completion deadline. Papers in the first group may still report policy-event dynamics across successive cohorts. Five of the six age-based papers supply an exact upper age, and 17 of the 20 follow-up papers supply an exact last deadline. Their endpoint spans are shown within each clock, without pooling ages and elapsed years or imputing missing cutoffs. The other follow-up records use a pooled 6–8-year window or upper bounds; the age record without an exact endpoint and the four stage-specific records remain in the table. A pooled adult-age window is not a lifetime education trajectory, and a multiple-time classification does not imply that final attainment is observed.

The accompanying 31-paper ledger records the valued outcome, reported times, time origin, endpoint qualifications, source version, and supporting pages or exhibits. It also preserves event-time information separately. These linked classifications supplement the 459-paper records rather than overwriting their broader definitions. They describe the selected valuation papers and should not be read as evidence that the original authors ignored dynamics or that their valuations are necessarily incorrect.

A.4 Source Versions and Replication Records

The review counts use recorded source versions, while the main text cites published articles where available. STBF record 628 uses the September 2020 working paper; the application uses Angrist et al. (2022), whose six-year degree results and graduation dynamics appear in Table III and Figure III. STBF applies the framework of Hendren and Sprung-Keyser (2020) but is not one of their earlier policy reconstructions. Version differences remain explicit in the evidence ledger.

For Adams, record 10 uses the May 2013 working paper and its roughly 30-percentage-point, \$300,000 illustration for marginal college switchers. The published Cohodes and Goodman (2014) use about 27 per-

centage points and \$270,000. Neither illustration is the 2.5-percentage-point eligibility effect or the separate policy-average MVPF in Hendren and Sprung-Keyser (2020). Tennessee record 684 uses the May 2026 version of Schoonover et al. (2026); the main BA-or-higher outcome at ages 23–24 is distinct from exact BA and the alternative ages 22–24 specification. Appendix D documents these welfare accounts, their inputs, and their authors’ discussion of dynamics and extrapolation.

Restricting the review to the journal sample leaves 130 one-window papers among 186 coded BA+ papers, or 69.9 percent, compared with 132 of 190 papers, or 69.5 percent, in the combined sample. Education-based lifetime or career valuation appears in 28 of 447 journal papers and 31 of 459 papers overall. These comparisons show how the reported counts change when the additional sources are excluded. The replication files contain the paper register, timing and interpretation decisions, endpoint classifications, evidence locations, source versions, and coordinates underlying the figures. The exhibit code aggregates those recorded judgments without recoding the papers. Bubble plots use the September 21 endpoint review, which revisited the 120 previously unplotted papers and retained the earlier reviewed coordinates for the others.

A.5 ACS sample and construction of the age profiles

Figure 1 uses ACS microdata selected within survey years 2000–2022, birth cohorts 1985–1987, and ages 16–35. Cohort is indexed by survey year minus reported integer age. The selected observations span 2001–2022, and every cohort contributes all 20 ages: 1985 over 2001–2020, 1986 over 2002–2021, and 1987 over 2003–2022. The resulting sample contains 1,978,860 respondent records from all 50 states and the District of Columbia. There are no restrictions on birthplace, race, Hispanic origin, sex, or marijuana policy status. Positive person weights and valid detailed educational attainment (EDUCD) are required; neither condition excludes a record within this selection. All 60 cohort–age cells are observed.

Person weights are applied within cohort–age cells, and the three cohorts receive equal weights at every age. Let S_{ca} be the respondents in cohort c at age a , w_i the IPUMS person weight, and H_{ik} respondent i ’s value of education measure k . For a group g containing three birth cohorts C_g , the profile is

$$\bar{H}_{gk}(a) = \frac{1}{3} \sum_{c \in C_g} \frac{\sum_{i \in S_{ca}} w_i H_{ik}}{\sum_{i \in S_{ca}} w_i}. \quad (12)$$

For Figure 1, $C_g = \{1985, 1986, 1987\}$. Fixed cohort weights prevent changes in the mixture of birth cohorts across ages. The underlying ACS observations are repeated cross sections, so the profiles need not hold immigration, survival, survey coverage, or other within-cohort composition constant. They describe the education of residents observed at each age, including people born outside the United States.

In the ACS profiles, SC+ is some-college-or-higher attainment, used as a proxy for college attendance; it is not current enrollment or a directly observed date of first entry. AA+ and BA+ denote cumulative degree thresholds. The EDUCD crosswalk assigns 12 schooling years to high-school completion, 12.5 to less than one year of college, 13 to at least one year of college without a degree, 14 to an associate’s degree, 16 to a bachelor’s degree, 18 to a master’s degree, 19 to a professional degree, and 20 to a doctorate. Lower-grade categories use the reported grade or the documented midpoint. YS is therefore an approximation based on attainment categories. The replication files provide the crosswalk, cohort–age means, respondent counts, weight sums, survey years, and source information. Using a fitted ACS profile as an untreated potential-outcome path in a policy application requires a separate calibration assumption.

A.6 Education growth after age 25 across cohorts

Figure A1 compares four nonoverlapping groups of birth cohorts: 1976–1978, 1979–1981, 1982–1984, and 1985–1987. The comparison uses ages 25–35, which are observed for all twelve cohorts within the 2000–2022 survey window. An earlier cohort cannot generally be observed at every age from 16 to 35 in these data: complete coverage of that interval would require a birth year between 1984 and 1987. The common 25–35 interval therefore permits an observed comparison of education growth after age 25 without filling in missing earlier ages. The sample contains 4,516,181 respondent records in 132 cohort–age cells and uses the same population definition, education crosswalk, person weights, and fixed one-third cohort weights as Figure 1.

Each curve shows the increase relative to its cohort group’s age-25 level. Degree and attendance changes are displayed in percentage points; schooling changes are displayed in years. This descriptive age difference is distinct from the policy effect ΔH used elsewhere in the paper. The accompanying cohort-growth data retain all levels at ages 25 and 35 and their differences.

Across the four groups, the age-25-to-35 increase in BA+ rises from 6.22 to 11.46 percentage points. The corresponding AA+ increase rises from 7.77 to 12.99 points, and the increase in approximate schooling rises from 0.447 to 0.669 years. All three increments increase successively across these groups. The college-attendance proxy follows a different pattern: its increments are 5.72, 7.28, 6.09, and 5.17 percentage points. Thus the pattern concerns later degree attainment and schooling accumulation, rather than a uniform increase across every education measure. These level differences do not measure the fraction of individual degree holders who completed after age 25, and the comparison does not isolate a causal effect of birth year or economic conditions.

B Education-Path Framework and Simulation

The Section 3 exercise asks which treated years-of-schooling paths remain compatible with stipulated age-specific effects and how their education-channel values differ under a common rule. It is a deterministic compatibility calculation. This appendix defines the observations and path restrictions, derives the matching and valuation formulas, and reports the experimental design, numerical checks, and complete results. The comparison path is the equal-cohort mean of the three fitted ACS profiles documented in Appendix C. Averaging the three cohort parameters does not make that comparison path itself a single logistic curve.

B.1 Schooling Stocks, Flows, and Observations

For a specified cumulative education outcome, $H_c^j(a)$ is its population mean at age a under policy state j . It is a share for a binary attainment outcome and is measured in schooling years for the years-of-schooling simulation. Its first difference records acquisition between adjacent ages. The effect $\Delta H_c(a)$ is the difference between the two stocks, whereas $\Delta h_c(a)$ is the difference between their acquisition flows, as defined in Equation (1). Even if both levels are nondecreasing, either effect can have either sign. For example, a treated group can acquire more schooling early and less later while retaining a positive stock difference throughout. Figure 2 illustrates these definitions with hypothetical degree-attainment shares. Both level paths increase, but the gap falls between ages 24 and 25, so $\Delta h_c(25) < 0$. These curves explain the notation and are not estimated policy effects or the ACS calibration. In the numerical schooling-year simulation, the first possible

effect age is $a_0 = 18$: the model sets $\Delta H_c(a) = 0$ before age 18 but allows $\Delta H_c(18)$ to differ from zero. The first flow is consequently $\Delta h_c(18) = \Delta H_c(18)$, relative to the zero gap at age 17.

The observation operator must retain what the source actually measures. An effect on attainment at a specified age enters as

$$\mathcal{M}_{a_*}(\Delta H_c) = \Delta H_c(a_*). \quad (13)$$

A population-weighted window average instead enters as

$$\mathcal{M}_{\mathcal{A}}(\Delta H_c) = \sum_{a \in \mathcal{A}} w_a \Delta H_c(a), \quad (14)$$

where the source's weights sum to one over $\mathcal{A}$. An effect on acquisition between two ages restricts the change in the stock gap:

$$\mathcal{M}_{a_1, a_2}^{\text{flow}}(\Delta H_c) = \Delta H_c(a_2) - \Delta H_c(a_1). \quad (15)$$

Observing two stock effects separately imposes two constraints, unlike observing only their difference. A completion rate within six years of entry is not automatically a stock measured at age 24: an attained-age mapping requires information on entry ages and the outcome definition. The experiments below use exact stock effects at one or two attained ages and require no such age bridge. They hold the target population fixed across those observations. For each compatible candidate I retain the terminal effect $\Delta H_c(65)$, the annual sum $\sum_{a=18}^{65} \Delta H_c(a)$, and the education NPV in Equation (25). The annual sum has schooling-year–age-year units and is distinct from additional completed schooling at the terminal age.

B.2 Logistic Parameters and Feasible Scenario Intervals

Write $p = (K, b, d) = (K_c + \Delta K_c, b_c + \Delta b_c, d_c + \Delta d_c)$ for the total treated-path parameters, and let $S = H_c^0(\tau_c)$. Define $L(a) = \Lambda[b(a - d)]$ and $F(a) = [L(a) - L(\tau_c)]/[1 - L(\tau_c)]$. The continuation in Equation (2) is $H_c^1(a) = S + (K - S)F(a)$ after τ_c . Holding the anchor fixed, its plateau derivative is $\partial H_c^1(a)/\partial K = F(a)$. For either shape parameter $v \in \{b, d\}$, the derivative is

$$\frac{\partial H_c^1(a)}{\partial v} = (K - S) \frac{[1 - L(\tau_c)] \partial_v L(a) - [1 - L(a)] \partial_v L(\tau_c)}{[1 - L(\tau_c)]^2}. \quad (16)$$

Here $\partial_b L(a) = (a - d)L(a)[1 - L(a)]$ and $\partial_d L(a) = -bL(a)[1 - L(a)]$. In particular, $\partial_d H_c^1(a) = -(K - S)b[1 - L(a)]F(a) < 0$ after the anchor when $K > S$ and $b > 0$. Figure B1 varies each total parameter with a common starting level. K changes the eventual plateau and scales the remaining acquisition $K - S$; b changes its speed; and d changes its timing subject to the common start. A nonzero K change thus produces a permanent education effect, whereas changes in b or d alone produce temporary effects with the same eventual attainment. These labels describe the generating path, not every candidate compatible with its observed effect. The negative derivative with respect to d means that a higher d postpones attainment while preserving the anchor and eventual plateau. Because normalization is affine in $L(a)$, the post-anchor continuation has its inflection and maximum instantaneous acquisition rate at $a = d$ when $d > \tau_c$. If $d \leq \tau_c$, that peak lies outside the continuation domain, and its acquisition rate declines after the anchor. The reference value $d = 12.564854$ is below the simulation's age-17 anchor, so it is not a post-policy age of peak acquisition. None of these parameters is an individual time to degree.

An equivalent interpretation uses the age at which the path reaches a specified attainment level. Let a_q satisfy $H_c^1(a_q) = S + q(K - S)$, where $q \in (0, 1)$ is the fraction of remaining attainment accumulated after the anchor. The normalized continuation implies

$$a_q = \tau_c + \frac{1}{b} \log \left(\frac{1 + q \exp[b(d - \tau_c)]}{1 - q} \right). \quad (17)$$

Its derivative with respect to d is $\partial a_q / \partial d = q \exp[b(d - \tau_c)] / \{1 + q \exp[b(d - \tau_c)]\}$, which lies strictly between zero and one. Holding K , b , and the anchor fixed, a higher d therefore increases the age required to reach each such level. The delay depends on q and is smaller than the change in d ; it is not a uniform horizontal shift. Here q refers to progress along the population attainment path, not the fraction of individuals who graduate. The age at which half of the remaining attainment has accumulated is $a_{1/2} = \tau_c + b^{-1} \log\{2 + \exp[b(d - \tau_c)]\}$, which generally differs from d .

The anchor implements an assumption about exposure in a fixed population. In the age-based experiment, individuals are unaffected through age 17, and a policy may first alter their measured attainment at age 18. The post-policy continuation therefore passes through the comparison level at 17, rather than attaching a curve whose value already differs there. This eliminates a mechanical level discontinuity at the transition; it does not require a zero effect at the first affected annual observation or equal derivatives at the anchor. In applications, eligibility, announcement, and exposure timing determine which age can serve as the last unaffected observation. An earlier anticipatory response would invalidate a later anchor. Equality refers to two potential outcomes for the same target population, not to equality of raw means in two selected samples. A common anchor for people first exposed at different ages requires a genuinely unaffected common age or an explicit cohort-specific construction before aggregation.

Each candidate is constrained to $0 \leq H_c^1(a) \leq 25$ schooling years at every annual age from 18 through 65. Over the same ages, its annual acquisition satisfies

$$0 \leq H_c^1(a) - H_c^1(a - 1) \leq 1.5. \quad (18)$$

The common-anchor construction imposes $H_c^1(a) = H_c^0(a)$ through age 17, so the first increment is the increase from that shared level along the normalized continuation. The 1.5-year cap is a maintained bound on changes in the harmonized schooling measure, not a threshold estimated from causal policy responses. Nondecreasing attainment restricts the levels and does not require a nondecreasing treatment effect. The parameter domain ensures $K > S$ and $b > 0$. Membership in this class establishes the stated mathematical restrictions, while the plausibility of its exposure and schooling assumptions remains a separate economic question.

For each coordinate $j \in \{K, b, d\}$, I fix the other two at their reference values and intersect its domain with all path restrictions. The reference triple p^0 is the mean of the fitted cohort parameters documented in Appendix C. The conditional feasible intervals are $I_K = [13.337015, 15.139521]$ for the plateau, $I_b = [0.120000, 0.617632]$ for speed, and $I_d = [8.000000, 30.000000]$ for timing. A scenario at position u in $I_j = [L_j, U_j]$ sets $p_j = L_j + u(U_j - L_j)$, leaving the other two parameters at p^0 . The midpoint scenarios use $u = 1/2$ and increments 0.3160 schooling years, 0.1113 inverse age-years, and 6.4351 age-years. The timing midpoint consequently represents later accumulation. This rule selects locations within the conditional feasible intervals without equalizing observed effects or selecting on NPV. The grid is a design for examining the maintained model, not an estimate of a typical policy response.

B.3 Exact Matching and Numerical Search

I characterize compatibility using a constrained minimum-distance criterion. A candidate path implies the moments $\mathbf{g}_c(\theta_c; a_0) = \mathcal{M}\{H_c^1(\cdot; \theta_c, a_0) - H_c^0(\cdot)\}$. The criterion measures their discrepancy from the observed moments:

$$Q_c(\theta_c; \mathbf{m}, a_0) = [\mathbf{m} - \mathbf{g}_c(\theta_c; a_0)]' W [\mathbf{m} - \mathbf{g}_c(\theta_c; a_0)], \quad (19)$$

where W is positive definite and the search is restricted to paths in C . With three free path parameters and one or two effects observed after the anchor, minimizing Q_c need not select a unique path. For exact moments, compatible paths have $Q_c = 0$, and every positive-definite W gives the same zero-residual set. The simulation solves the age-specific equations directly and retains verified zero-residual candidates. Sampling error in reported effects requires separate inference. The anchor is built into every candidate and is not counted as another observed policy effect.

Let $S = H_c^0(\tau_c)$, let (K, b, d) denote total treated-path parameters, and define $F(a; b, d) = [\Lambda\{b(a-d)\} - \Lambda\{b(\tau_c - d)\}] / [1 - \Lambda\{b(\tau_c - d)\}]$. For one effect at $a_* > \tau_c$, the implied treated level is $y = H_c^0(a_*) + m_{a_*}$. Conditional on (b, d) , exact matching gives

$$K = S + \frac{y - S}{F(a_*, b, d)}. \quad (20)$$

I evaluate a 401×401 grid in (b, d) , recover K , and retain candidates satisfying the level, increment, and parameter restrictions. For each output metric, local constrained optimization then searches for smaller and larger values from up to two separated feasible starting candidates, using scaled coordinates, at most 100 iterations, and an objective tolerance of 10^{-11} . The observed moment remains imposed algebraically.

For two effects at $\tau_c < a_1 < a_2$, let $y_j = H_c^0(a_j) + m_{a_j}$, $z_j = y_j - S$, $R = z_2/z_1$, and $t_j = \exp[-b(a_j - \tau_c)]$. Since $F(a_j; b, d) = (1 - t_j)/(1 + qt_j)$ for $q = \exp[b(d - \tau_c)]$, define $r = R(1 - t_1)/(1 - t_2)$. Solving the two matching equations gives $q = (r - 1)/(t_1 - rt_2)$. The implied timing parameter is

$$d = \tau_c + \frac{\log q}{b}, \quad (21)$$

and the plateau is

$$K = S + z_1 \frac{1 + qt_1}{1 - t_1}. \quad (22)$$

The search is one-dimensional in b . I use 1,601 grid values, refine detected feasible-interval boundaries by root finding, and refine candidate metric extrema locally within feasible intervals. A candidate must have $q > 0$, finite parameters, and satisfy all maintained restrictions. The registered generating paths have $0 < z_1 < z_2$, so the formulas are not used for degenerate observations at the anchor or with equal treated levels. I include the stipulated generating path in each library to verify nonemptiness. This is a deterministic compatibility exercise, not an estimator-performance experiment.

A candidate is accepted only if its largest moment residual is at most 10^{-8} schooling years and every path or parameter restriction holds to tolerance 10^{-9} . If a local optimizer stops just outside a boundary, I move its proposal a small fraction toward its verified starting point, recover K from the exact matching equation, and check the conditions again; the feasibility tolerance is not relaxed. Independently feasible outputs are retained even when the optimizer does not report convergence, because these outputs supply candidate paths rather than certified extrema. Libraries are deduplicated using parameter values rounded to 12 decimal places. Every verified two-age candidate is also supplied to both component single-age libraries, after which their

metric endpoints and pointwise path envelopes are recomputed. This preserves numerical nesting without treating the finite search as a proof of completeness.

B.4 Education-Channel Earnings and Opportunity-Cost Valuation

The valuation accounts retain the notation, discount factor, and age support in Section 3.3. A source education effect is converted into completed schooling years per member of the target population, $m_{YS} = T(m^{\text{source}})$, preserving the source policy unit and its outcome-specific bridge for credits, enrollment, or credentials. A schooling stock observed at age a^* gives $m_{YS} = \Delta H_c(a^*)$.⁴¹ With the signed dollar premium $\pi_{YS}^{\text{HSK}}(a) = Y_0(a)r_{a-a_0+1}^{\text{HSK}}$, the benchmark is

$$\text{NPV}_c^{\text{point}}(\Delta H_c; a_0, A, \rho) = \sum_{a=a_0}^A D_\rho(a; a_0) \pi_{YS}^{\text{HSK}}(a) \Delta H_c(a^*). \quad (23)$$

For the retained calibration, $r_t^{\text{HSK}} = -0.0667$ for $t = 1, \dots, 7$ and $r_t^{\text{HSK}} = 0.1126$ thereafter, following the early earnings decline and later gains estimated by Zimmerman (2014) and used by Hendren and Sprung-Keyser (2020). These coefficients describe the harmonized benchmark; Hendren and Sprung-Keyser's policy calculations also use policy-specific education inputs, observed earnings where available, and other accounting terms. The signed premium already incorporates early forgone earnings. Directly subtracting an additional schooling opportunity cost from Equation (23) would count that loss twice.

The path account values the stock $\Delta H_c(a)$ and its increment,

$$\Delta h_c(a) = \Delta H_c(a) - \Delta H_c(a-1). \quad (24)$$

At the first affected age, the common anchor gives $\Delta h_c(a_0) = \Delta H_c(a_0)$. Define the positive dollar premium $\pi_{YS}^+(a) = Y_0(a)r^+$, where $r^+ = 815/7241 \approx 0.112554$. Both $\pi_{YS}^{\text{HSK}}(a)$ and $\pi_{YS}^+(a)$ are annual dollar premia per completed schooling year and use the same earnings profile. Applying the positive premium at ages 18–24 is a mechanical backcast for the decomposition, not an observed positive earnings return at those ages. With the same discount factor, valuation ages, and target population as the benchmark, Equation (6) is

$$\text{NPV}_c^{\text{path}}(\Delta H_c; a_0, A, \rho) = \sum_{a=a_0}^A D_\rho(a; a_0) [\pi_{YS}^+(a) \Delta H_c(a) - c_{\text{HSK}} \Delta h_c(a)]. \quad (25)$$

The flow term records the treated-minus-comparison cost of acquisition. It can be negative when the comparison group acquires more schooling at that age.

The common c_{HSK} is a calibration for this decomposition, not a universal policy-specific cost estimated by Hendren and Sprung-Keyser. In the reference calculation, one additional year is acquired evenly from ages 18 through 24: $\Delta h^{\text{ref}}(a) = 1/7$ for these seven ages and zero otherwise, with $\Delta H^{\text{ref}}(a) = \sum_{u=18}^a \Delta h^{\text{ref}}(u)$ and $\Delta H^{\text{ref}}(17) = 0$. I apply the two premium schedules to this same reference stock and choose c_{HSK} to equate their present values after the explicit flow cost:

$$\sum_{a=18}^A D_\rho(a; 18) \pi_{YS}^{\text{HSK}}(a) \Delta H^{\text{ref}}(a) = \sum_{a=18}^A D_\rho(a; 18) [\pi_{YS}^+(a) \Delta H^{\text{ref}}(a) - c_{\text{HSK}} \Delta h^{\text{ref}}(a)]. \quad (26)$$

⁴¹For a window average, replace the fixed input by $\sum_{u \in \mathcal{A}} w_u \Delta H_c(u)$ using source weights. Other outcomes retain their conversion T and observation operator.

Solving for the cost gives

$$c_{\text{HSK}} = \frac{\sum_{a=18}^A D_{\rho}(a; 18) [\pi_{\text{YS}}^+(a) - \pi_{\text{YS}}^{\text{HSK}}(a)] \Delta H^{\text{ref}}(a)}{\sum_{a=18}^A D_{\rho}(a; 18) \Delta h^{\text{ref}}(a)}. \quad (27)$$

With the frozen earnings schedules and $\rho = 0.03$, the calibration gives $c_{\text{HSK}} = \$4,172.26$ in 2000 U.S. dollars per additional schooling-year flow. In Equation (26), the signed premium multiplies the age-varying reference stock $\Delta H^{\text{ref}}(a)$; in the single-effect benchmark, it multiplies the fixed observed gap $H_c^1(a^*) - H_c^0(a^*)$. The calibration therefore equates two valuations of one reference acquisition path, rather than imposing an identity between the point and path NPVs. For other timing paths, Equation (25) remains a maintained valuation rule.

The calculation treats an additional completed year recorded at age a as a proxy for schooling that displaces earnings at that age. Completion and enrollment need not coincide, and a person who leaves school need not work. The opportunity-cost assumption should therefore be reconsidered in each policy application. The simulation holds the same $Y_0(a)$, r^+ , c_{HSK} , ρ , age support, and 2000-dollar units fixed across candidate paths. This education-channel value excludes tuition, taxes, transfers, public education resources, and other components of willingness to pay and net government cost; those enter the policy accounting separately.

Figure 4 compares the two valuation accounts for a joint-change path with $\Delta K_c = 0.2625$, $\Delta b_c = 0.0300$, and $\Delta d_c = -0.4500$. Figure B2 plots the exact level paths entering those accounts. This separate illustration uses the same pre-policy anchor and admissibility restrictions as the experiments. Its maximum annual schooling increment is 0.766 years, below the 1.5-year cap. The schooling gap is 0.153 years at age 18, 0.364 at age 25, 0.280 at age 35, and 0.263 at age 65. The difference initially grows, and the comparison group later partly catches up, while a positive attainment difference remains at the terminal age. This is a lasting-gain example with changes in accumulation timing as well as the plateau.

In the single-effect panel, $a^* = 25$, so the fixed gap $H_c^1(25) - H_c^0(25)$ enters the income projection at every valuation age. Blue bars display the positive part of $D_{\rho}(a; 18) \pi_{\text{YS}}^{\text{HSK}}(a) [H_c^1(25) - H_c^0(25)]$ and orange bars the absolute magnitude of its negative part. The full-path panel instead uses $D_{\rho}(a; 18) \pi_{\text{YS}}^+(a) [H_c^1(a) - H_c^0(a)]$ for blue bars and $D_{\rho}(a; 18) c_{\text{HSK}} \Delta h_c(a)$ for orange bars. Both panels have identical axes, discounting, and dollar units, and NPV equals the sum of blue bars minus the sum of orange bars. Orange bars below zero in the full-path panel represent relative cost savings at ages when the comparison group acquires more schooling. Under this accounting, the single-effect value is approximately \$15,390 and the full-path value is \$12,397: \$13,588 in earnings value less \$1,191 in net opportunity cost. These values illustrate the maintained valuation rules; they do not establish that either predicts actual earnings more accurately. Their difference reflects both the schooling-path input and its valuation mapping; the reference calibration in Equation (26) does not remove either distinction.

The first cost follows from additional acquisition after the common anchor. Between ages 17 and 18, completed schooling rises by 0.613 years along H_c^0 and by 0.766 years along H_c^1 . Their difference is $\Delta h_c(18) = \Delta H_c(18) = 0.153$. At the discount base, $D_{\rho}(18; 18) = 1$, so multiplying this flow by $c_{\text{HSK}} = \$4,172.26$ produces the approximately \$638 orange bar. The coefficient is common across ages, not an unusually high age-18 wage. Later positive flows contribute \$889 in discounted costs and negative flows contribute \$336 in savings. Total positive cost charges are therefore \$1,527, while net opportunity cost is \$1,191.⁴² The initial charge follows from the assumed acquisition-to-work mapping; it is not an observed employment loss.

⁴²Net costs deduct later savings, so the initial share differs from its share of gross charges. Without discounting, the signed sum is $c_{\text{HSK}} [\Delta H_c(65) - \Delta H_c(17)]$. Dropping the initial charge while retaining later savings would break this accounting identity.

B.5 Single-Parameter Explanations of Policy Education Inputs

The policy exercise asks whether an observed education input can be matched by a permanent effect through K alone or a temporary effect through b or d alone. It differs from the deterministic experiments in Section 3: the policy input is held fixed, and the active parameter is solved for rather than selected to generate a new input. Each calculation retains its geographic and cohort calibration, observation operator, policy unit, and valuation account. A weighted stock input and a difference between two schooling stocks are both admissible inputs to this exercise. Neither is replaced by an effect at a convenient single age. The eleven finite tax calculations retain their policy-dose conversion and acquisition-window mapping.

I preserve every fitted component of the comparison path. Let j index those components within policy i , with fixed weights ω_{ij} and baseline parameters $z_{ij}^0 = (K_{ij}^0, \log b_{ij}^0, d_{ij}^0)$. For parameter family $q \in \{K, b, d\}$, the experiment uses

$$z_{ij}^{1,q}(t) = z_{ij}^0 + te_q, \quad (28)$$

where e_q selects the corresponding coordinate. A plateau or timing experiment adds the same t to every component's K or d . A speed experiment multiplies each component's b by $\exp(t)$, so its labels report $100[\exp(t) - 1]$ percent. The other two parameter families, component weights, and pre-policy anchors remain fixed. Applying the anchored continuation component by component and then averaging reproduces H_i^0 exactly at $t = 0$. This construction preserves baseline heterogeneity. A nonzero K shift changes the limiting attainment level and is therefore permanent. Changing b or d with each component's K fixed leaves the limiting aggregate gap at zero and is therefore temporary, although the gap can persist beyond the valuation horizon. It is a separate diagnostic and need not be a coordinate slice of the main search's common-parameter specification.

Let $\Delta H_i^q(\cdot; t)$ denote the resulting aggregate education effect. The matching calculation retains values in

$$\mathcal{T}_{iq} = \left\{ t : H_i^{1,q}(\cdot; t) \in C_i, \quad \mathcal{M}_i\{\Delta H_i^q(\cdot; t)\} = \mathbf{m}_i^{\text{L2}} \right\}. \quad (29)$$

Here C_i includes the level and annual-acquisition restrictions, parameter bounds, timing support, and calibrated covariance ellipsoid. The ellipsoid is applied to the weighted parameter center in $(K, \log b, d)$ coordinates. Holding two coordinates fixed can substantially shorten the remaining interval relative to that coordinate's marginal bounds. The numerical routine scans this conditional interval, brackets sign changes, and checks local minima of the absolute matching residual for tangencies. Reported solutions have input residuals below 10^{-8} in the input's units. No input standard errors are added to obtain a match. Missing solutions remain missing rather than being assigned zero or an unrestricted parameter value.

Tennessee Promise has three education inputs. Its family experiment permits a separate change in the active parameter for SC+, AA+, and BA+, holding the other two parameters of each credential path fixed. The three latest-window effects must be matched jointly, and the paths must satisfy credential ordering and the late-completion restrictions. Thus, its plateau point represents three credential-specific plateau changes, reported in percentage points, rather than one schooling-year shift. Only this family has a jointly admissible solution. Red rings in the policy figures indicate that an already matched L2 solution also satisfies the admitted additional L3 restrictions. They do not represent a second optimization, an additional time-allocation assumption, or a confidence region.

Figure B3 compares the reference earnings NPV with the earnings NPV of each matching path. The saved path value is first converted into the monetary and discount-base units used for L1 using its existing accounting bridge. Both coordinates are then divided by the same positive reference program expenditure.

This normalization permits a comparison within each policy without mixing its currency years or policy doses. The reference earnings component is recovered from the original account; it is not inferred from the MVPF ratio. Algorithmic placement has no source lifetime-earnings projection and is excluded from this comparison, rather than assigned a reference NPV of zero. Cannabis has no whole-policy L1. The paired samples contain 32, 25, and 18 calculations for K , b , and d , respectively. Table B3 reports the normalized values and parameter changes for all 37 reference calculations, including unpaired cases. A self-financing welfare account can still have a finite earnings NPV and remains in these plots.

The income comparisons retain each application’s valuation rule, including its opportunity-cost treatment and any disclosed earnings scaling. They therefore differ from the common synthetic-cohort calibration used in the Section 3 simulations. Differences from a source earnings projection may reflect both the education continuation and the valuation mapping. For example, the Adams reference earnings loss is about \$1,204 in the account’s 2004-dollar units. Matching its converted -0.018 schooling-year input with plateau, speed, or timing changes gives losses of about \$1,109, \$188, and \$150. The latter two temporary-effect paths converge to the comparison plateau, reducing the duration of the earnings loss; the permanent-effect path retains a lower plateau. This comparison does not establish which path occurred or overturn the source’s evidence of dropout during follow-up.

The matched samples also differ across parameter families. In the separate subset of 14 calculations with one YS level input, the plateau, speed, and timing exercises yield 11, 7, and 3 matches. This subset allows cohort-age weights but excludes acquisition contrasts and joint credentials. For Florida, the required input is 0.1819 schooling years, whereas the saved speed and timing scans reach at most 0.0914 and 0.0089 years. The absent points therefore reflect the input and the conditional support. They do not show that timing responses are impossible outside this specification. The full parameter sweeps, matching results, exclusions, earnings-account bridges, and policy classifications are retained in the replication files.

B.6 Two Experiments and Summary Measures

The first experiment changes the generating process while fixing the observed age at 25. For each parameter j , I use $u = 0, 0.01, \dots, 1$ in the scenario rule from Appendix B.2 and additionally retain its reference value, yielding 102 registered scenarios per parameter and 306 rows. The two parameters not varied in a generating scenario remain at their reference values. I then discard that knowledge of the generating mechanism and search the common three-parameter class for paths matching the scenario’s age-25 effect. The three reference rows describe the same generating path and are retained within each mechanism’s reporting grid; they are not three independent pieces of evidence. The complete finite grids remain in the replication results; they do not exhaust the continuum of real-valued parameters.

The second experiment holds a generating process fixed and changes the observed ages. It uses five interval positions, $u \in \{0, 0.25, 0.5, 0.75, 1\}$, for each parameter, giving the 15 scenarios in Table B1. For each scenario the design set contains all 18 single ages from 18 through 35 and all $\binom{18}{2} = 153$ distinct age pairs, producing $15 \times 171 = 2,565$ scenario–design comparisons. The age-25 results for these 15 scenarios are shared with the first experiment, leaving 2,856 distinct scenario–design identifiers across the two output files. The three midpoint scenarios in Figure 3 are unchanged between Panels A and B. A fixed generating process implies age-specific effects that generally differ across ages; it does not mean that the same numerical treatment effect is imposed at every age. Every verified pair candidate is also inserted into both

component single-age libraries. This ensures that their numerical summaries respect the nesting implied by the observation restrictions.

For a scenario s and observation design D , let $\widehat{\mathcal{P}}_s(D)$ denote its finite library of verified candidates. For a candidate treated path h , write $\text{NPV}_c^{\text{path}}(h - H_c^0)$ for the value of its associated education-effect path under Equation (25), holding a_0 , A , and ρ fixed. Its smallest NPV is $\widehat{\text{NPV}}_s^-(D) = \min_{h \in \widehat{\mathcal{P}}_s(D)} \text{NPV}_c^{\text{path}}(h - H_c^0)$, and its largest is $\widehat{\text{NPV}}_s^+(D) = \max_{h \in \widehat{\mathcal{P}}_s(D)} \text{NPV}_c^{\text{path}}(h - H_c^0)$. The reported width is

$$W_s(D) = \widehat{\text{NPV}}_s^+(D) - \widehat{\text{NPV}}_s^-(D). \quad (30)$$

For comparisons with observing age 25 alone, the proportional width reduction is $1 - W_s(D)/W_s(\{25\})$, calculated within each scenario before summarizing. A zero benchmark width would make that proportional measure undefined rather than justify recording a zero reduction. Median widths and median proportional reductions use the three interior positions, $u = 0.25, 0.5, 0.75$. Largest widths and sign counts use all five positions, keeping the two boundary cases visible. A pair with a positive candidate above \$0.01 and a negative candidate below $-\$0.01$ is recorded as admitting both NPV signs. The cents-level threshold avoids treating numerical zero as a sign reversal. These are equal-weight summaries of the experimental scenarios, not estimates of the frequency of mechanisms among real policies.

B.7 Numerical Checks and Interpretation

All 306 first-experiment rows and 2,565 second-experiment rows were completed. The largest observed-moment residual among retained candidates was 3.55×10^{-15} schooling years, and the most negative recorded constraint margin was -9.92×10^{-10} , within the declared numerical tolerance. The audit checked 2,295 pair designs for nesting in their component single-age libraries; the largest recorded containment discrepancy was 1.50×10^{-10} in the numerical metric comparisons. An independent calculation from the stored endpoint parameters recomputed 17,136 metric-endpoint records using the normalized logistic difference directly. Its largest NPV difference was 4.87×10^{-10} dollars, and its largest terminal-schooling and cumulative-gap differences were 9.05×10^{-14} schooling years and 6.63×10^{-13} schooling-year-age units, respectively. Some endpoint records correspond to the same path because multiple metrics or designs can select it.

The local optimizer made 6,708 recorded constrained-refinement calls: 4,526 converged and passed feasibility checks, 1,690 did not report convergence but supplied independently verified feasible candidates, and 492 outputs were rejected. Boundary repair toward a verified seed, as described in Appendix B.3, was used for 1,102 accepted outputs without relaxing the tolerances. Feasible outputs contribute witnesses and candidate extrema regardless of their stopping flag; they are not certified optima. A pilot comparison of the 201×201 and 401×401 single-age grids, together with 801 and 1,601 pair-grid values, changed the checked NPV endpoints by at most about \$0.021. That pilot supports local numerical stability in the designs checked, not completeness of the continuous search. No sampling bootstrap or Monte Carlo coverage calculation is part of these two experiments.

The numerical checks establish that saved candidates meet the maintained restrictions, not that those restrictions hold for real policies. ACS fit is informative within observed ages, while extrapolation above age 35, population transport, the logistic family, empirical support, and the flow-to-cost mapping remain assumptions. Finding opposite-sign values establishes conditional ambiguity; finding only one sign does not certify sign identification. Finite-bank envelopes can switch candidate identity across ages, so their edges

need not be feasible paths. No probabilities or confidence levels are attached to them. Alternative families, calibrations, and noisy observations require separate analyses.

B.8 Results across the Complete Experimental Grids

Table B2 summarizes all 102 age-25 scenarios per parameter. The complete 306-row file retains the generating parameter, observed effect, terminal schooling and cumulative-gap ranges, NPV endpoints, and generating-path NPV. Endpoints for different metrics can belong to different paths. Median NPV widths are \$53,017, \$50,163, and \$67,244 for the plateau, speed, and timing scans; 61, 62, and 102 scenarios, respectively, admit both signs. These counts summarize the chosen grid and assign no probabilities to policy mechanisms.

Table 2 reports selected observation designs, and Figure B4 covers every single age and age pair. Median widths use the three interior generating scenarios; maximum widths use all five, preserving the most permissive boundary or interior result. Diagonal cells represent one age and upper-triangle cells two distinct ages. All panels share one dollar-width scale. The five scenarios' individual heatmaps and all 2,565 scenario-design rows remain in the replication material, allowing the aggregate patterns to be checked against every generating path.

The two panels of Figure 3 use the same midpoint scenarios and common axes, so their visual difference comes from the additional age-35 constraint and the accompanying candidate search. For these three scenarios the NPV-width reductions are 93.5–97.8 percent. The candidates found for the plateau and speed examples have positive NPVs, while those for the timing example have negative NPVs. These midpoint results do not describe every registered scenario: the additional age-35 observation still admits both signs in three of the five timing scenarios. Other age pairs need not produce the same reduction. Within the registered interior scenarios, adding age 24 to age 25 gives larger median proportional reductions than adding age 20 in each parameter group, despite the smaller age separation. The result concerns where observations constrain this restricted curve family, and should not be generalized to an optimal sampling design under noisy estimates without a separate analysis.

C Comparison Schooling Paths and Empirical Support

C.1 Construction of the Comparison Curve

For the schooling-year exercise in Section 3, I hold fixed a comparison path formed from three ACS birth-cohort fits. Let $\hat{H}_{0q}(a)$ be the fitted mean completed schooling profile for birth cohort q . The reference path is

$$H_c^0(a) = \frac{1}{3} \sum_{q=1985}^{1987} \hat{H}_{0q}(a). \quad (31)$$

The subscript c in the simulation therefore refers to this synthetic target group, not to one of the three birth cohorts. The ACS figure in Figure 1 shows descriptive attainment at observed ages; Equation (31) instead averages fitted cohort curves and is used as a fixed input to the path search. Neither object is an estimated policy counterfactual for a specific case in Section 4.

The 60 person-weighted cohort–age means and their population weights come from the sample and education crosswalk documented in Appendix A.5. All three cohorts contribute every age from 16 through 35. The following fit uses these cell means, with no additional demographic restriction.

Let $\overline{YS}_{qa}$ denote the person-weighted mean in cell (q, a) and ω_{qa} its population weight normalized to sum to one over the 20 ages within cohort q . I estimate each cohort’s parameters separately by nonlinear weighted least squares:

$$(\widehat{K}_q, \widehat{b}_q, \widehat{d}_q) = \arg \min_{K, b, d} \sum_{a=16}^{35} \omega_{qa} \left[\overline{YS}_{qa} - K \Lambda\{b(a - d)\} \right]^2. \quad (32)$$

The fitting bounds are $K \in [0.01, 25]$, $b \in [0.001, 5]$, and $d \in [6, 75]$. Five starting values lead to the same retained objective in each cohort; each fit allows at most 3,000 function evaluations with numerical tolerances of 10^{-13} . These fitting bounds differ from the narrower treated-path search domain in Section 3. Table C1 reports the cohort sizes, parameter estimates, and weighted root mean squared errors. I then average fitted curves with fixed one-third weights, rather than allowing cohort shares to change with age.

C.2 Observed Ages and Extrapolation

Each selected cohort contributes all ages 16–35, so age 25 and age 35 are both within every cohort’s observed support. Figure C1 compares the mean fitted curve with the mean of the observed cohort–age schooling measures over that common range. Cohort-specific weighted root mean squared errors are 0.112–0.115 schooling years, and the largest absolute cell residual is 0.327 years. This is an in-sample descriptive fit diagnostic. Every comparison-path value above age 35 is extrapolated. The age-17 fitted value is the shared level at which candidate continuations are anchored, under the assumption that the hypothetical policy first affects measured attainment at $a_0 = 18$. Fixing cohort weights removes changes in the included birth-cohort shares; it does not turn repeated cross sections into a panel or establish the profile as the untreated path for a policy population. The simulation conditions on the fitted curve rather than propagating its sampling or extrapolation uncertainty.

C.3 Parameter Reference and Restrictions

The reference triple used to express $\theta_c = (\Delta K_c, \Delta b_c, \Delta d_c)$ is the arithmetic mean of the three fitted cohort parameter triples. Its values are approximately (13.922226, 0.257500, 12.564854). Its logistic curve is not generally equal to the mean of three logistic curves in Equation (31); their largest difference over ages 16–65 is 0.000412 schooling years. After anchoring the reference-parameter continuation at the common age-17 level, its largest discrepancy from H_c^0 over ages 18–65 is 0.000226 schooling years. Thus, the zero-increment reference exactly shares the anchor but does not impose an exactly zero effect at later ages.

The plateau restriction $K \in [13.3370, 15.1395]$ is the weighted first-to-99th percentile interval of a historical bank of 716 state-by-birth-cohort logistic fits. The bank covers the 50 states and the District of Columbia and birth cohorts 1980–1994, using ACS observations from 2000–2023. Its source construction retains U.S.-born non-Hispanic White, non-Hispanic Black, and Hispanic respondents, valid education records, positive person weights, and recorded male or female sex. State–cohort profiles are fitted by population-weighted nonlinear least squares. Retained fits have at least 14 age cells spanning at least 13 years, begin by age 20,

reach at least age 29, converge, and lie inside the fitting bounds.⁴³ The interval restricts candidate eventual schooling to a broad central range of historical fitted profiles. Its transfer to the current full-population, 1985–1987 baseline is a maintained calibration assumption: the populations and survey windows differ, and the interval is neither a confidence interval for that baseline nor validation of policy-induced endpoints.

The speed and timing domains, $b \in [0.12, 0.65]$ and $d \in [8, 30]$, and the 1.5-year annual acquisition cap are experimental design restrictions rather than estimated quantiles from the current cohort sample. The simulation does not impose the historical bank’s timing-index bounds or joint parameter ellipsoid. Appendix B.6 describes the parameter sweeps within these restrictions. The reported widths are conditional on this domain, the baseline, and the valuation calibration; sensitivity to a population-matched support bank, alternative bounds, or other path families has not been evaluated.

D Policy Welfare Accounting and Education-Path Replacement

The welfare reconstruction replaces specified education-related components of a source account while retaining its policy contrast and the remaining accounting assumptions. This appendix defines the units and component boundaries, maps an education path into private income and fiscal returns, and states the extra information needed to let other education costs vary. The formulas describe the accounting procedure; they do not establish that a candidate path is the true policy response or that a finite search has recovered all compatible paths.

D.1 Policy Units and the Reference Ledger

Index the policy contrast by i . L1 reports WTP_i^{ref} , N_i^{ref} , and their associated MVPF, using the components in Equations (8) and (9). Each entry uses the same target population, treatment contrast, currency year, and discount base. Values may be stated per eligible person, per recipient, or per specified change in aid, but all components within an account must use the same unit. Multiplying every component by the same positive scale leaves the MVPF unchanged; mixing components from different populations or treatment margins does not. An intention-to-treat effect and an effect for students induced to switch institutions therefore require a documented conversion before they can enter one account. Monetary values from different cases also require conversion to common dollar units before their levels, rather than their dimensionless ratios, are compared.

The terms B_i^{ref} and G_i^{ref} collect the private and public components outside the replaced earnings projection and the separately recorded education payments. They are not necessarily pure cash transfers and statutory program costs. Depending on the source, they may include other measured private benefits, effects on public spending, or fiscal receipts. The private payment $C_i^{\text{priv,ref}}$ and public cost $C_i^{\text{pub,ref}}$ must follow the source’s tuition and subsidy convention. For example, if the benefit of a grant is recorded separately in B_i^{ref} , the same grant cannot also be deducted from the private tuition charge to create a second benefit. Public education costs must likewise exclude payments already included in G_i^{ref} . Where the source nets private tuition receipts from institutional expenditure, that convention is retained. The original component sum, not a generic cost-per-degree formula, determines the reference values.

⁴³Plateau weights sum person weights over each profile; quantiles interpolate cumulative normalized weights. The resulting limits are 13.337015059499 and 15.139521154274. Observations span ages 16–43 collectively, so plateaus remain fitted asymptotes rather than observed lifetime attainment.

Interpreting income changes as willingness to pay retains the welfare assumptions of the source calculation. In their college-admission example, Hendren and Sprung-Keyser (2020) treat the earnings gain as a return to human capital rather than compensation for greater effort, and value the income change net of taxes and education expenses. A benefit requiring additional work effort, an unmeasured consumption value of college, or an effect on other people’s earnings would require a corresponding welfare adjustment. Scholarship transfers also depend on whether the recipient would have attended without the award. These assumptions are not consequences of the logistic path specification. For the three main applications, the reference ledger must follow the Adams calculation in Hendren and Sprung-Keyser (2020) and the STBF and Tennessee Promise calculations in Angrist et al. (2022) and Schoonover et al. (2026), respectively.

The accounting check is to recover the source’s willingness to pay and net cost by summing its recorded components before substituting any path. Feeding the original replaced components back into Equations (10) and (11) must then return those same totals. This is an arithmetic identity check, not evidence that the original continuation belongs to the candidate logistic family. If a source does not separately report a component needed for the subtraction, recovering it requires a documented calculation or an explicit additional calibration. A missing component cannot be silently set to zero. The case records report original education inputs and timing in Table D1, followed by the reference and replacement components in Table D2.

D.2 From Attainment to Disposable Income and Fiscal Returns

The observation operator in Equation (3) determines what a candidate education path must reproduce. A stock effect at one age, an average over an age window, and an effect on schooling acquired during an interval constrain different features of the path. Converting credits or credentials into schooling years is a separate step and must preserve the source’s treatment margin. In particular, a change in the probability of earning a bachelor’s degree is not automatically four additional years per person: the appropriate conversion depends on schooling already counted through enrollment, partial completion, and other credentials. The anchor must precede policy exposure or anticipation for the target population. Neither age 17 nor the simulated 1985–1987 comparison group is imposed on every application merely because it is used in Section 3.

Let $e_i^s(a)$ be the annual pretax earnings effect in account $s \in \{\text{ref}, \text{path}\}$, and let $\mathcal{A}_i$ be the set of ages included in the replaced projection. The discount factor is $D_i(a) = D_{\rho_i}(a; a_{d,i})$, with source-specific rate ρ_i and discount-base age $a_{d,i}$. That base need not equal the first age at which attainment can change. All annual earnings effects are expressed in the same currency year, and their gross present value is

$$\text{NPV}_i^{\text{earn},s} = \sum_{a \in \mathcal{A}_i} D_i(a) e_i^s(a). \quad (33)$$

For a schooling-year path under the stock–flow rule in Equation (6), the annual input is

$$e_i^{\text{path}}(a; \Delta H_i) = \pi_{i,\text{YS}}^+(a) \Delta H_i(a) - c_i(a) \Delta h_i(a). \quad (34)$$

Here $\pi_{i,\text{YS}}^+(a)$ is the annual dollar premium per additional completed schooling year, and $c_i(a)$ is the earnings loss assigned to an additional schooling-year acquisition flow. The difference $\Delta h_i(a) = \Delta H_i(a) - \Delta H_i(a-1)$ retains its meaning from Equation (1). This formula requires an acquisition-to-work mapping: it does not follow from the attainment data alone. A credential path instead needs its own earnings mapping or a stated conversion to schooling years. The simulation’s opportunity-cost coefficient is not a universal coefficient for the policy applications.

Early earnings losses enter only once. If the retained earnings schedule already contains the losses associated with attending school, adding a second forgone-earnings term would duplicate them. A stock–flow decomposition therefore replaces that signed schedule with the positive premium and explicit flow term in Equation (34); an account retaining the signed schedule does not also subtract the flow cost. If early earnings effects are directly observed and retained, the candidate projection must identify which later component is being replaced. The observed component cannot remain in the fixed ledger while also being included in the replacement NPV. Timing-dependent forgone earnings and returns to completed education are distinct from tuition and public resource costs, which enter the accounts separately.

With a retained linear tax-and-transfer schedule, let $\vartheta_i(a)$ denote the fraction of an earnings change that accrues to the government through taxes and reduced transfers. The disposable-income and fiscal components are

$$\text{NPV}_i^{\text{after},s} = \sum_{a \in \mathcal{A}_i} D_i(a) [1 - \vartheta_i(a)] e_i^s(a) \quad (35)$$

and

$$\text{NPV}_i^{\text{tax},s} = \sum_{a \in \mathcal{A}_i} D_i(a) \vartheta_i(a) e_i^s(a). \quad (36)$$

The two components sum to Equation (33). Negative earnings effects reduce private income and, under a positive rate, reduce fiscal revenue. The same schedule is used in the reference and path calculations when the comparison is intended to hold taxation fixed. If the source uses different short- and long-run rates, the age-specific rates remain in the sum; a single average rate generally cannot reproduce both components for every candidate path. Nonlinear tax rules or separate transfer responses require their source-specific mapping rather than these linear formulas. In particular, applying a nonlinear tax schedule to mean earnings need not recover mean taxes without distributional information.

When $\vartheta_i(a) = \bar{\vartheta}_i$ is constant, the bookkeeping is especially transparent. Writing $\Delta E_i = \text{NPV}_i^{\text{earn,path}} - \text{NPV}_i^{\text{earn,ref}}$, the replacement changes willingness to pay by $(1 - \bar{\vartheta}_i)\Delta E_i$ and net fiscal cost by $-\bar{\vartheta}_i\Delta E_i$. The private gain and public revenue gain partition the same pretax earnings change. This special case explains the signs in Equations (10) and (11); it does not impose one tax rate across policies. If the original income forecast and the candidate forecast use different return or opportunity-cost rules, their difference incorporates that accounting change as well as the education continuation. Separating the two effects requires evaluating a specified reference path under the same candidate valuation rule, when such a reference path can be recovered from the source.

D.3 Education Payments, Resource Costs, and the Scope of Replacement

Holding the non-earnings components fixed isolates the earnings-channel comparison; it does not establish that these components would remain unchanged under every education path. Longer enrollment can affect tuition, instruction costs, and aid duration, while changes in institution type can affect costs even at the same schooling level. Revaluing these components requires source-specific links between attainment, enrollment, aid eligibility, and unit costs. Private payments would then be subtracted from WTP and public resource costs added to net cost. The main replacement holds them fixed unless a case supplies the required link. In particular, annual degree attainment does not measure time enrolled, and using $\Delta h_i(a)$ to represent resource use requires an additional assumption beyond its role as a proxy for forgone work.

With fixed conversion rules, prices, and valuation coefficients, the account can also be collected into a ratio of affine functions of path summaries:

$$\text{MVPF}_i(\Delta H_i) = \frac{\alpha_i + \mathbf{w}_i^\top \mathbf{z}_i(\Delta H_i)}{\beta_i + \mathbf{n}_i^\top \mathbf{z}_i(\Delta H_i)}, \quad N_i(\Delta H_i) > 0. \quad (37)$$

The constants α_i and β_i collect retained components; $\mathbf{z}_i$ contains the specified discounted earnings and, where applicable, education-cost summaries. The coefficients $\mathbf{w}_i$ and $\mathbf{n}_i$ map those summaries into willingness to pay and net cost. For the main replacement alone, take $\mathbf{z}_i = (\text{NPV}_i^{\text{after,path}}, \text{NPV}_i^{\text{tax,path}})^\top$, $\mathbf{w}_i = (1, 0)^\top$, and $\mathbf{n}_i = (0, -1)^\top$, with $\alpha_i = \text{WTP}_i^{\text{ref}} - \text{NPV}_i^{\text{after,ref}}$ and $\beta_i = N_i^{\text{ref}} + \text{NPV}_i^{\text{tax,ref}}$. This is a representation of the retained accounting, not a claim that welfare is linear in the logistic parameters or that every education-policy valuation uses fixed coefficients. Returns that depend on education expansion, endogenous prices, or additional behavioral channels would require further modeling.

D.4 Comparing Paths Under a Common Welfare Account

L1 is the original welfare calculation. L2 imposes the education moments selected from that calculation and searches the maintained continuation class. L3 adds comparable source information while retaining the L2 moments and all other conditions. Write $\widehat{\mathcal{D}}_{i\ell}$ for the finite set of verified education-effect paths found for policy i at information layer $\ell \in \{2, 3\}$. The joint welfare values are

$$\widehat{\mathcal{V}}_{i\ell} = \left\{ \left(\text{WTP}_i^{\text{path}}(\Delta H_i), N_i^{\text{path}}(\Delta H_i) \right) : \Delta H_i \in \widehat{\mathcal{D}}_{i\ell} \right\}. \quad (38)$$

The numerator and denominator of each MVPF come from the same member of this set. Pairing the largest willingness to pay with the smallest net cost from different paths can create a ratio that no candidate attains. Likewise, the reference MVPF falling inside a candidate range does not establish that the reference continuation is a member of the path family. A later observation refines the admissible set only when it concerns the same population and outcome under a compatible time mapping. Changing the sample, functional form, or cost calibration defines an alternative specification rather than a nested information addition.

For positive net costs, $\text{MVPF}_i > 1$ if and only if $\text{WTP}_i - N_i > 0$. Finding two compatible, positive-cost paths on opposite sides of this threshold establishes ambiguity conditional on the maintained restrictions and account. Finding candidates on only one side does not establish that all compatible paths lie there. The self-financing region requires positive willingness to pay and nonpositive net cost and is reported separately. Ratios can also become arbitrarily large in magnitude along a sequence of positive-cost paths whose costs approach zero; an education effect crossing zero is not sufficient to establish such a sequence. A large finite ratio or a rounded zero denominator is therefore not a proof of an infinite bound. The analysis assigns no probabilities to candidate paths, and any sampling analysis of the source estimates remains distinct from this continuation exercise.

The threshold at one compares a dollar of measured private willingness to pay with a dollar of net public cost. It is not a universal policy-adoption rule. As Hendren and Sprung-Keyser (2020) show, a budget-neutral comparison of policies affecting different people also depends on the social weights attached to their beneficiaries and on the financing alternative. The cross-policy comparisons in Section 5 retain that distinction. Within a policy, the accounting exercise asks the narrower question of whether the same education evidence supports different threshold or fiscal classifications under a fixed valuation rule.

D.5 Geographic References, Source Control Levels, and Empirical Support

The geographic library uses the unrestricted 2000–2022 IPUMS extract: no race, sex, or birthplace restriction is imposed. It contains the 50 states, the District of Columbia, and a national aggregate. Person weights form outcome means within state–birth-cohort–age cells; birth cohort is survey year minus integer age. The 2000 component is Census sample 200001, followed by annual ACS samples in 2001–2022. The sample inventory is retained with the replication files. The full library supplies years of schooling and some-college-or-higher (SC+), associate’s-or-higher, bachelor’s-or-higher, and postgraduate outcomes. Case-specific geographic and cohort weights are fixed across ages, so changing the observed age does not change those chosen weights. This does not turn repeated cross sections into individual histories or eliminate migration and other within-cell composition changes. The library is an external description of education profiles, not an additional source of policy identification.

For each geographic cell and outcome, I fit a logistic profile to person-weighted age means, using observed ages 16–35 when the coverage supports estimation. State shapes are regularized toward their national counterparts. A cohort with insufficient young-adult support instead borrows a national shape and uses its available observations to calibrate the level. The data therefore involve both forward extrapolation for young cohorts and backward reconstruction for historical cohorts. The 1947 cohort used for Adult Pell is calibrated at ages 53–65, and the 1963 cohort used for Social Security student benefits at ages 37–59. Both borrow the 1978 national shape, itself fitted at ages 22–35. Their early-life paths are not observations of those cohorts at ages 16–35. Table E1 reports actual support and fit modes; the accompanying registry lists every state–cohort–outcome cell, donor, and level-calibration range. These records also distinguish Tennessee’s subsequent source-population and control-level adjustments. The library contains 10,920 fits across 52 geographic units, 35 birth cohorts, and six outcome definitions.

A published control mean is used only when its population, outcome, and clock match the effect under study. In Tennessee, the effects concern high school graduates and residence one year before interview. I therefore construct the corresponding Tennessee profiles using migration information and rescale each outcome to its reported age-23–24 control mean. The common cohort weights remain fixed. This operation supplies an external level and shape; it does not recover the regression’s conditional untreated prediction, and conditioning on high school graduation is itself part of the source population definition. In STBF and Adams, observed degree-completion means cannot identify a total-completed-schooling reference curve. In particular, STBF’s control mean of 3.93 years enrolled is not converted into an observed total schooling level by adding twelve. Those two main schooling exercises retain the corresponding state’s full-population schooling profile. The distinction between source-matched levels and external population proxies is explicit in the case registry.

The empirical support is constructed before searching policy paths. Let $x = (K, \log b, d)^\top$. For each outcome I retain fitted profiles for the 1983–1987 donor cohorts in the 51 state-level regions. A raw logistic profile with its own level at the anchor can be written exactly in the anchored normalized form, using the same parameter triple; a separate post-anchor refit is therefore unnecessary. Such refits are saved as diagnostics because many schooling fits have weakly determined timing parameters at the numerical boundary. The main covariance averages the relevant state’s within-donor covariance and the pooled-state covariance with equal weights. National targets use the pooled covariance; multi-state targets average the state covariances with the fixed target weights. The support center is the weighted parameter location of the target cohort’s fitted

reference, and the restriction is

$$(x - \bar{x}_i)^\top \Sigma_i^{-1} (x - \bar{x}_i) \leq 11.34486673. \quad (39)$$

The cutoff retains the earlier 99-percent chi-square scale only as an empirical calibration convention. It is not a confidence region for a treatment response and assigns no probability to candidate paths. Centering historical dispersion on a different population or cohort is a maintained transport restriction. The wider sensitivity doubles Σ_i and retains every previously verified main candidate.

Timing restrictions refer to fractions of the increase after the anchor, rather than to the unanchored curve's half-plateau age. For anchor τ , the age at fraction q of the remaining logistic increase is

$$a_q(b, d; \tau) = \tau + \frac{1}{b} \log \left\{ \frac{1 + q \exp[b(d - \tau)]}{1 - q} \right\}. \quad (40)$$

For $q = 0.1$ and 0.9 , pooled donor deviations between the 0.5th and 99.5th percentiles are transported around the corresponding target reference. The wider sensitivity multiplies these deviations by $\sqrt{2}$. Schooling paths remain nondecreasing and acquire at most one year per annual step after the anchor; probability paths stay between zero and one. These restrictions describe the admitted economic continuations. They are not validated causal restrictions merely because they are informed by ACS profiles.

The numerical search uses a deterministic Sobol shape grid and a 97×97 regular grid in $(\log b, d)$. For a fixed pair, the retained schooling moment is linear in K , which is solved algebraically. The joint-support inequality also gives an analytic admissible interval for K . The main schooling search uses 2^{14} Sobol pairs; the Tennessee pools use 2^{13} per credential. An exact second schooling moment is imposed by bracketing and refining roots in d at 97 values of b . Valid roots enter both L3 and L2. Appendix E.9 documents the larger grids used in the collection-wide numerical sensitivity. This design avoids a separate high-dimensional likelihood optimization for each draw. The saved candidates satisfy the retained moment tolerance of 2×10^{-7} ; all point-input paths are independently rebuilt and checked. These checks establish feasibility of saved witnesses, not exhaustive coverage of the continuous set.

D.6 STBF: Schooling-Equivalent Inputs and Auxiliary Degrees

The main input retains the randomized-offer contrast for the 2012–2014 four-year-college strata. The archived replication workbook reports a schooling-time effect of 0.3647777438 years in cell HOPE tax credit of sheet T3, whereas printed Table III reports 0.360. I retain the workbook estimate; the difference is not exactly publication rounding. This variable measures postsecondary time enrolled. Using it as a completed-schooling-equivalent increment is an additional bridge, rather than a direct observation of total completed years of schooling. The reference earnings calculation uses credentials and time enrolled, whereas the replacement applies the common schooling stock–flow rule. The difference between the reference and path calculations consequently reflects both continuation and valuation assumptions. The workbook's six-year bachelor's effect is 0.0813350 and its any-degree effect is 0.0518224. The first-year college-entry coefficient from a broader set of entry cohorts is not relabeled as a six-year observation for this population.

The L1 calculation in Angrist et al. (2022) assumes that the scholarships are publicly funded. The earnings model uses 2008–2019 ACS observations for Nebraska-born respondents aged 18–65, with at least a high-school degree excluding GEDs and at most a bachelor's degree. It relates earnings to educational attainment and potential experience, then applies the experimental control group's demographic

weights. The estimated credential and time-enrolled effects imply a discounted earnings gain of \$21,150 per treated applicant. With the \$32,250 scholarship transfer and 20 percent tax rate, willingness to pay is $32,250 + 0.8(21,150) = 49,170$, and net public cost is $32,250 - 0.2(21,150) = 28,020$. Their ratio is 1.75. The separately reported \$2,390 increase in educational spending is not added again to this stated account.

Let $r \in \{2012, 2013, 2014\}$ denote entry year and $s \in \{17, 18, 19\}$ high-school age. The branch birth cohort is $r - s$, its last unaffected age is s , and its endpoint is $s + 6$. The branch weight is $\omega_{rs} = v_s/3$, where $v = (0.0652070, 0.5758797, 0.3589133)$ is the retained age calibration, not an experimentally recovered set of participant shares. Nebraska supplies the branch-specific reference curves. The common treated parameter triple must satisfy

$$\sum_{r,s} \omega_{rs} [H_{rs}^1(s+6) - H_{rs}^0(s+6)] = 0.3647777438. \quad (41)$$

The three entry years and age weights are the same at every age. The observed window is therefore a weighted collection of ages 23–25, not an effect at a selected midpoint. The main bank contains 4,648 candidates.

The auxiliary degree check uses separate Nebraska credential shapes, scaled to the source’s control means at HS+6: 0.6361746 for bachelor’s completion and approximately 0.693 for any degree, the latter obtained from the rounded no-degree share. The any-degree measure is represented by an AA-or-higher proxy, not by adding the separately reported AA and BA shares, which may overlap. A saved pair matches both credential effects and satisfies bachelor’s attainment no greater than any-degree attainment at every modeled age. Neither this pair nor the source’s degree mechanisms supplies a joint mapping to the YS continuation. The same auxiliary pair can therefore accompany every admitted YS path. I consequently report STBF as an L2-only schooling reconstruction: Figure 5 shows the L2 paths. There is no duplicated L3 envelope. Adding a schooling L3 would require another comparable schooling-time moment or an explicit cross-outcome linkage; an unchanged projection obtained without that linkage is not additional schooling information.

The replacement values schooling over ages 18–65, discounted to 18 at 3 percent, using the positive schooling premium and the calibrated forgone-earnings cost of 4,172.2626 in 2000 dollars per schooling-year increment. The price conversion to 2019 dollars is $255.7/172.2$. The retained transfer and tax rate yield $WTP = 32,250 + 0.8E$ and $N = 32,250 - 0.2E$. Putting the reference $E = 21,150$ into that account recovers willingness to pay of \$49,170 and net cost of \$28,020. Appendix E.9 separates the geographic reference, empirical support, and explicit opportunity-cost assumptions. Removing the latter charge is a valuation sensitivity, not evidence about the true amount of forgone work.

The retained L2 candidates have willingness to pay between \$39.02 thousand and \$80.46 thousand, and net cost between \$20.20 thousand and \$30.56 thousand. Figure 5 shows the pointwise schooling range alongside H^0 . Table D3 separately reports paired welfare values for the complete low- and high-earnings-NPV paths S1 and S2. Neither the envelope edges nor component extremes from different paths are treated as a single feasible welfare calculation.

D.7 Adams: Education Conversion and Additional Source Timing

The Adams reconstruction retains the eligibility RD contrast for the 2005–2006 graduating classes and the source program used by Hendren and Sprung-Keyser (2020). The source observes college entry immediately after high school and within two years, persistence in the fourth year, and graduation within four and six years. The retained conversion assigns one year to additional two-year enrollment, two years to additional four-year enrollment, half a year to two-year completion, and one further year to bachelor’s completion. The corre-

sponding multipliers are (1, 2, 0.5, 1), giving $0.007 + 2(0) + 0.5(0) - 0.025 = -0.018$ schooling-equivalent years per eligible student. The baseline is the full-population Massachusetts profile for birth cohort 1987, with age 18 as the last unaffected observation. L2 imposes

$$\Delta H_{\text{Adams}}(24) = -0.018. \quad (42)$$

L3 preserves this endpoint and places the separately reported early entry contribution at age 20:

$$\Delta H_{\text{Adams}}(20) = 0.007. \quad (43)$$

These are temporal allocations of the source schooling conversion, not directly estimated annual YS effects. The source's local population near the eligibility threshold is not interchangeable with all Massachusetts residents; that external-baseline approximation is retained explicitly. Its control bachelor's mean of 0.631 provides degree context, but does not identify a comparable total-schooling control curve.

The root search gives 18 L3 paths, all contained in the 5,667 L2 candidates. The source supplies two distinct windows, and the retained conversion assigns schooling-equivalent values to their inputs. L3 additionally maintains that the converted entry contribution is acquired by the early modeled endpoint. It does not claim that the source directly estimates annual completed schooling. Michigan's early-entry and later-credential reconstruction follows the same rule. By contrast, a second time point obtained solely by splitting one entry estimate using an assumed duration is excluded. The original authors' evidence favors dropout over delay within follow-up; later recovery among candidate paths remains an unobserved continuation, not a rejection of that evidence.

The monetary account starts from the reference earnings NPV $E^{\text{ref}} = -1,204.04$, willingness to pay of \$1,364.23, and net cost of \$1,896.35, all in 2004 dollars. The retained linear fiscal mapping uses the effective earnings tax-and-transfer share $t = 0.1979114476$. This is the ratio of the earnings-related fiscal change to the gross earnings change in L1, rather than a statutory income-tax rate. Its implied reference fiscal effect is $tE^{\text{ref}} \simeq -238.29$, and the private income effect is $(1-t)E^{\text{ref}} \simeq -965.74$. The negative fiscal effect represents revenue lost when projected earnings fall. Applying the same effective share to alternative earnings paths is a maintained restriction; reproducing the source's aggregate fiscal effect does not recover a full age-specific tax system.

To replace these income components, first remove them from the reported totals. Define w_0 as the retained private component outside earnings. Then

$$w_0 = \text{WTP}^{\text{ref}} - (1-t)E^{\text{ref}} \simeq 1,364.23 - (-965.74) = 2,329.97. \quad (44)$$

In the source program, this component is the scholarship value for students whose enrollment behavior is unchanged, net of changes in private education payments. It is not an additional earnings benefit. Define n_0 as fiscal cost before the earnings-related revenue effect. It is

$$n_0 = N^{\text{ref}} + tE^{\text{ref}} \simeq 1,896.35 - 238.29 = 1,658.06. \quad (45)$$

This amount combines direct scholarship expenditure with the source's changes in public education spending. Net cost exceeds this pretax amount because the reference earnings loss reduces revenue. The source program computes the private-payment and public-spending changes from enrollment duration and institution choice. Holding w_0 and n_0 fixed therefore preserves those source calculations rather than allowing education costs to adjust to every alternative path.

Inserting the candidate earnings NPV E gives $WTP = w_0 + (1 - t)E$ and $N = n_0 - tE$. Equivalently, willingness to pay changes from its reference total by $(1 - t)(E - E^{\text{ref}})$ and net cost by $-t(E - E^{\text{ref}})$. Substituting $E = E^{\text{ref}}$ recovers both reference totals. All displayed dollar values are rounded; the calculations retain the exported coefficients. The reference projection incorporates early earnings losses through its signed schooling-return schedule, while the path rule uses the positive schooling premium and explicit forgone earnings. The L1 comparison consequently changes this timing rule as well as the continuation; L2 and L3 use the same path valuation. The point earnings kernel is stored in 2000 dollars, so its 188.9/172.2 conversion is applied once when reporting E in 2004 dollars. The exported WTP and net-cost kernels already contain that conversion.

Under the L3 time mapping, earnings NPVs range from -4.93 to -2.14 thousand dollars. WTP crosses zero, while net cost remains positive between \$2.08 thousand and \$2.63 thousand. Since $n_0 - w_0 = -671.91$, a positive-cost MVPF above one would require a smaller earnings loss than these paths imply. Appendix E.9 examines baseline and valuation alternatives. This L3 enters the collection summaries, ordering, and propagation of the registered joint input repetitions.

D.8 Tennessee Promise: Matched Credential Paths and Earlier Windows

The Tennessee source uses 2000–2024 ACS data and overlapping age groups among high school graduates. Exposure is the share of a cohort eligible at age 17, with geographic assignment based on state of residence one year earlier. The regressions include state and survey-year fixed effects. The reconstruction uses fixed equal weights across the 1992–1998 birth cohorts, which reach age 17 during the county-to-statewide expansion in 2009–2015. These are the model cohorts, not a claim that the source regressions use only those birth years. Each outcome’s Tennessee age shape is fitted to the corresponding external population and rescaled so that its age-23–24 mean equals the source control mean: 0.685 for some college or higher, 0.380 for associate’s or higher, and 0.292 for bachelor’s or higher. The treated curves share their outcome-specific age-17 reference levels. This reproduces selected population levels, not a fitted untreated DID trajectory; the control mean may itself combine treatment states, covariates, and cohorts differently from the regression contrast.

For $j \in \{C, A, B\}$, the retained source effects are $m_j \in \{0.035, 0.040, 0.027\}$. I impose

$$\frac{\Delta H_j(23) + \Delta H_j(24)}{2} = m_j. \quad (46)$$

Equal age weights approximate the unavailable regression operator. At every annual age, both reference and treated levels satisfy $H_B^s(a) \leq H_A^s(a) \leq H_C^s(a)$. Each credential has its own anchored parameter triple and state-centered support. The constituent shape pools are combined with a deterministic Sobol design, supplemented with combinations near earnings extremes and earlier-window fits. The main bank contains 86,298 complete joint candidates. Each reported welfare pair uses one such joint candidate; separate credential extrema are never combined as if they were one feasible path.

The additional late-completion restriction is relative to those who have not yet attained the outcome. Let $\bar{H}_{jc}$ be a donor cohort’s mean level at ages 23–24. Define the historical completion fraction

$$R_j = \max_{c \in \{1983, \dots, 1987\}} \frac{K_{jc} - \bar{H}_{jc}}{1 - \bar{H}_{jc}}. \quad (47)$$

If necessary, the main allowance also includes the fitted target reference’s own fraction. The candidate restriction is

$$K_j^1 \leq \bar{H}_j^1 + (1 - \bar{H}_j^1)R_j. \quad (48)$$

The donor profiles use the same Tennessee high-school-graduate denominator. This rule avoids treating a statewide attainment share as a universal ceiling for a selected population. It still assumes that remaining completion among the treated can be bounded by the chosen donor experience, whose plateau is itself extrapolated. The main bachelor's ceiling is 0.4537. Table D4 gives the other quantities; Appendix E.9 separately varies this allowance and the parameter support, so the two restrictions are not changed together.

L3 compares predicted adjacent-age averages with all fifteen earlier coefficients, retaining the three latest-window effects exactly. The source's Table 2 reports clustered SEs, but its inference uses policy permutations because uncertainty can be understated with one treated state. I recover the corresponding main-specification marginal permutation intervals from Figure A5, panels B, D, and F (PDF page 62). The filled-circle series uses the same overlapping two-year windows as Table 2, despite plotting them at their first age. The numerical screen moves both extracted caps inward by one native-image pixel. This preserves a margin against reading error without expanding an interval until a path fits. Neither the source coefficient, the ACS reference, the Logistic family, the cohort weights, nor the empirical support is changed. Table D5 records the intervals and predictions of the complete L3 paths. The resulting 46,502 L3 candidates are also members of L2. Intersecting marginal intervals does not recover a simultaneous 95 percent region, particularly when adjacent windows overlap; the source's joint permutation draws remain unavailable.

The source welfare account applies Tennessee-specific education returns to project earnings and taxes. It assumes 2.3 years of Promise participation and five years to a bachelor's degree. Direct scholarship spending averages \$2,760 before discounting; the source reports total public cost before tax offsets of \$14,793. Tuition relief for students who would have enrolled without the program and projected after-tax earnings for induced students enter willingness to pay. The reported totals are \$72,837 for willingness to pay, -\$10,053 for net public cost, and \$24,845 for additional taxes. The authors qualify the use of average education returns for induced students and the public-expenditure assumptions. The reconstruction preserves the monetary units and policy unit of this account.

The valuation retains reconstructed Tennessee age-specific earnings differences between adjacent credential categories, $\pi_C(a) = y_{SC}(a) - y_{HS}(a)$, $\pi_A(a) = y_{AA}(a) - y_{SC}(a)$, and $\pi_B(a) = y_{BA}(a) - y_{AA}(a)$. The resulting earnings component is

$$E(\Delta \mathbf{H}) = \lambda \sum_{a=18}^{64} D_{0.03}(a; 18) \sum_{j \in \{C, A, B\}} \pi_j(a) \Delta H_j(a). \quad (49)$$

The schedule differences retain their signs and receive no additional schooling-flow opportunity-cost term. The reported tax gain and the retained 26 percent tax rate imply reference gross earnings of $E^{\text{ref}} = 24,845/0.26 = 95,557.69$. The scale $\lambda = 4.883134$ makes constant endpoint gaps reproduce this total. It calibrates the reconstructed earnings schedule to the gross earnings NPV in L1, rather than estimating participation or the causal return for induced students. Without the scale, none of the main retained candidates is self-financing (Appendix E.9).

The private remainder is obtained by removing the reference after-tax earnings contribution from total willingness to pay:

$$w_0 = 72,837 - 0.74 \left(\frac{24,845}{0.26} \right) \approx 2,124.31. \quad (50)$$

Thus, \$2,124.31 is the private transfer and tuition component left after the earnings contribution is removed from the reported total. It is recovered from the account, not an additional estimate of a policy effect. Adding

the projected tax receipts back to net public cost gives

$$n_0 = -10,053 + 24,845 = 14,792. \quad (51)$$

This is the reconstructed pretax fiscal cost, containing scholarships and education spending. It differs by one dollar from the source’s displayed \$14,793 total because the reported components do not sum exactly at their published precision. I retain the reported net-cost and tax totals for an internally consistent reconstruction.

The candidate account is therefore $WTP = 2,124.31 + 0.74E$ and $N = 14,792 - 0.26E$, using unrounded remainders in the calculations. Equivalently, it starts from the reported willingness to pay and net cost and replaces the private and fiscal shares of E^{ref} with those of E . Returning E^{ref} recovers the source totals. Fiscal break-even occurs at $E = n_0/0.26 = 56,892.31$; with positive willingness to pay, net cost at or below zero yields the self-financing classification. Grant duration, tuition, and public education resources remain in the fixed remainders. Their invariance across paths is a restriction on this comparison, not a prediction about how enrollment costs respond to each education continuation.

Among the L2 candidates, willingness to pay ranges from -52.78 to 196.40 thousand dollars and net cost from -53.47 to 34.08 thousand dollars. L3 narrows these spans to [8.04, 179.00] and [-47.35, 12.71] thousand dollars, respectively. Figure 7 displays the pointwise attainment ranges across the joint candidates alongside H^0 . The extraction-precision results are consolidated in Appendix E.9. Compatible L3 paths are found under the tested rules, but the position of the lower endpoint relative to one depends on graphical precision.

D.9 Source-Input Repetitions

The three applications use the 1,000 prespecified repetitions summarized in Table 4. STBF perturbs its schooling-time coefficient with a researcher-normal rule; Adams retains linked education and accounting draws. Tennessee uses synthetic normal perturbations of the displayed clustered-SE coefficients and shifts each earlier permutation interval with its perturbed coefficient, holding widths fixed. This does not recover the source’s joint permutation distribution, and its warning about clustered SEs also limits these perturbations. The calibration and path restrictions are fixed. All 1,000 draws have verified L2 paths; Adams and Tennessee also have 1,000 verified L3 paths. Adams’ repeated early and late inputs retain the same conversion and temporal allocation as its point-input L3. Appendix F defines endpoint quantiles, source-draw differences, and the limits of their statistical interpretation for the full collection.

E Full Policy Collection and Case Registry

The collection compares 38 policy calculations using their registered education inputs, population calibrations, and welfare accounts. This appendix records the case definitions, L1 values and admitted reconstruction inputs, numerical checks, and priority rule underlying Section 5. The 35 cases outside the main applications receive separate input profiles. Full-precision operators, complete path witnesses, and archived diagnostic comparisons remain in the replication files; source-input uncertainty is treated in Appendix F.

E.1 Cases, Source Families, and Calibration

Table E1 lists all 38 calculations. The collection retains 34 earlier calculations and adds algorithmic college placement, two further tuition-deduction contrasts, and the cannabis education account. Several calculations

can come from one paper. The source-account cases represent 22 research families, with one further family for the author cannabis study. The grouping combines related Bulman–Hoxby versions and tax-schedule calculations, the spending and tuition calculations associated with Deming–Walters, and the Kalamazoo Promise research versions. The source-authors string is retained separately. A family-level witness count records whether at least one member supplies the specified paths; it is not an estimate of a policy-population frequency.

The geographic registry selects the source’s treated population rather than combining it mechanically with control states. Birth cohorts and mixture weights remain fixed across candidates. Geographic matching improves comparability, but an ACS mean is not an identified untreated trajectory for a source study’s selected population. The source-control status records that distinction, including Tennessee’s high-school-graduate denominator and control-level calibration. Education conversion, age mapping, the anchored logistic continuation, empirical support, and transport of welfare prices remain maintained assumptions. Reproduction of a source coefficient or account is distinct from validation of those assumptions.

The file `source_moments_expanded.csv` records each encoded moment’s outcome, layer, target, standard error when available, and nonzero age–cohort operator weights. The case contracts additionally preserve comparison paths, anchoring ages, accounting kernels, and input provenance. A converted coefficient is not relabeled as observed completed schooling. In particular, current enrollment and accumulated education are different outcomes; the source-specific bridge remains part of each reconstruction. Full-precision inputs permit the matching equation to be reproduced without inferring an age from a figure or replacing a window by its midpoint.

E.2 Education Inputs and Observation Windows

These profiles document the 35 calculations not developed in Section 4. Each first identifies the source outcome, observation window, and numerical effect; then states the schooling conversion and restriction used in L2; and finally lists the additional inputs in L3 or explains why that layer is absent. The three main applications are documented in Appendices D.6, D.7, and D.8. Unless stated otherwise, the reference conversion is that of Hendren and Sprung-Keyser (2020) and the associated Policy Impacts calculation. Here $\Delta H(a)$ denotes the schooling-years effect for the case’s target population, except in Early Head Start, where it denotes the BA+ attainment effect. Probability coefficients are reported in probability units unless explicitly expressed in percentage points. Credits divided by 24, enrollment multiplied by an assumed duration, and credentials assigned additional schooling years are conversions of source estimates, not directly observed years-of-schooling effects. Table E1 records the matched comparison populations. I distinguish the source’s observation window from the ages assigned in the reconstruction throughout.

For the point-input results, L2 retains the stated converted input as an equality. An admitted L3 preserves that equality and adds comparable source observations. The added restrictions are exact for Adams, Michigan, CUNY, Social Security student benefits, and the first-year HOPE tax-credit component. Florida, Wisconsin, Kalamazoo, the two college-appropriation calculations, Texas tuition, and cannabis instead allow each added moment to lie within two marginal standard errors of its reported value; those screens are not joint 95 percent confidence regions. The profiles give the centers of these screens. The accompanying moment registry records the standard errors, full age–cohort weights, and observation operators. A case with no L3 can still contain several source coefficients: an extra restriction requires separately observed, comparable timing,

rather than splitting a single observation by an assumed duration. Converted restrictions with distinct source windows retain the stated temporal and population mappings.

Florida Student Access Grant

Castleman and Long (2016) estimate the effect of eligibility for the Florida Student Access Grant near the financial-aid cutoff. Their credit outcomes cover the first semester and the first four years after high school. Table 4 reports effects of 0.355 credits in the first semester and cumulative effects of 1.098, 2.666, 3.847, and 4.366 credits after years one through four. L2 uses only the fourth-year total, imposing $\Delta H(22) = 4.366/24 = 0.181917$ schooling years under a representative entry age of 18. L3 retains this endpoint and adds the first-semester and first-, second-, and third-year inputs. Their converted values are 0.014792, 0.045750, 0.111083, and 0.160292 schooling years at assigned ages 18.5, 19, 20, and 21, respectively, subject to the marginal-error screens above. Thus, L3 restricts when the credit gain accumulates before the same fourth-year total. The source also reports subsequent credential outcomes; these are not imposed as additional credit observations without a further credential-to-schooling conversion.

District of Columbia Tuition Assistance Grant

Abraham and Clark (2006) study the District of Columbia Tuition Assistance Grant's effect on college enrollment. The retained outcome is first-time freshman enrollment among students who completed high school in the preceding 12 months. The reference calculation uses an enrollment increase of 0.089, or 8.9 percentage points, and assumes two additional schooling years per induced entrant. L2 therefore imposes $\Delta H(20) = 2(0.089) = 0.178$ schooling years. Age 20 is the assigned endpoint of that duration conversion, not a directly observed schooling outcome for all source students. There is no L3: placing 0.089 schooling years at age 19 would divide the same assumed two-year total between years, rather than add a separately estimated follow-up.⁴⁴

Wisconsin Scholars Grant

Goldrick-Rab et al. (2016) estimate the effects of the randomized Wisconsin Scholars Grant on semester credits, cumulative credits, and completion for three entering cohorts. The retained cumulative-credit estimates are 0.9 credits after three years for cohort 1 and 2.1 credits in the pooled observation of cohort 2 after three years and cohort 3 after two years. L2 combines these inputs with weights 0.314570 and 0.685430 and divides by 24, yielding 0.071772 schooling years. It imposes one weighted age-cohort restriction, rather than treating that pooled result as an observation at a single age.⁴⁵ L3 preserves this restriction and separately screens the 0.9- and 2.1-credit cumulative inputs. It also uses Table 5's semester estimates to form annual credit effects: 0.5, 0.2, and 0.1 credits in years one, two, and three for cohort 1; 0.4 and 1.0 credits in years one and two for pooled cohorts 2–3; and 0.5 credits in year three for cohort 2 alone. Each is divided by 24 and restricts the change in ΔH across the corresponding year, with entry assigned to age 18. The third year of

⁴⁴The reference uses published Table 7A's 0.089 coefficient. The recovered 2003 working paper is an earlier version and does not independently verify that coefficient or its SE.

⁴⁵L2 uses $0.314570\Delta H_{1990}(21) + 0.342715\Delta H_{1991}(21) + 0.342715\Delta H_{1992}(20)$, with unrounded weights in computation. The 475 and 1,035 group counts motivate the retained group weights; the pooled cohorts are split equally as a calibration assumption, not as recovered regression weights. Entry at 18 and common response parameters are also assumptions. The third cohort's unobserved third year has zero observation weight.

cohort 3 is unobserved and is not imputed. These separate regression estimates enter marginal-error screens, rather than being forced to add up exactly.

Kalamazoo Promise

Bartik et al. (2021) estimate Kalamazoo Promise eligibility effects on enrollment, attempted credits, and credentials. The credit estimates used here are 3.236, 4.309, and 6.562 additional attempted credits after two, three, and four years, respectively. The reference calculation retains the fourth-year estimate rounded to 6.56 and converts it at 24 credits per schooling year. L2 imposes $\Delta H(22) = 6.56/24 = 0.273333$. L3 keeps this exact fourth-year input and adds targets of $3.236/24 = 0.134833$ at age 20 and $4.309/24 = 0.179542$ at age 21, each subject to its marginal-error screen. These ages assume entry at 18. The source also observes enrollment within six and 12 months and later credentials, including four- and six-year completion. Those outcomes are not interchangeable with attempted-credit totals under the retained conversion. Because the cohorts with sufficient follow-up differ across the credit horizons, interpreting them as restrictions on a common response also requires population comparability.

Michigan community-college tuition

Acton (2021) estimates the effect of a \$1,000 reduction in local community-college tuition for Michigan high-school graduates. For the 2009–2011 graduation cohorts, enrollment within one year changes by 0.036 at local community colleges, -0.021 at other community colleges, -0.004 at vocational colleges, and -0.003 at four-year colleges. Completion outcomes observed in 2017 change by -0.002 for certificates, 0.003 for associate's degrees, and 0.011 for bachelor's degrees. The reference conversion weights these seven probability effects by (1, 1, 1, 2, 0.5, 1, 2) schooling years, respectively, giving 0.029 years. L2 imposes $\Delta H(25) = 0.029$; age 25 represents the pooled 2017 completion observation, which occurs six to eight years after high school across the source cohorts. L3 retains the early entry window separately, imposing $\Delta H(19) - \Delta H(18) = 0.005$ in addition to the L2 equality. The 0.005 is $0.036 - 0.021 - 0.004 + 2(-0.003)$ under the same reference conversion. Assigning this duration-weighted contribution to the first modeled year is a temporal allocation, not an observed completed-schooling increment. Unlike a split of one isolated entry observation, it distinguishes the source's entry and later credential windows. The alternative semester and credit outcomes are not added to a composite that already uses enrollment and completion.

CUNY Pell grants

Marx and Turner (2018) estimate the effects of an additional \$1,000 of Pell Grant aid on first-year CUNY students entering in 2007–2011. Table 8 reports 0.212 additional academic credits earned in the current year and 0.223 cumulative credits three years after entry. L2 uses the latter, imposing $\Delta H(21) = 0.223/24 = 0.009292$ schooling years under an entry-age mapping of 18. L3 retains that equality and adds $\Delta H(19) = 0.212/24 = 0.008833$ exactly in the point-input exercise. The difference between the two inputs therefore concerns the timing of earned credits under the same conversion. The first- and third-year estimates use 38,100 and 32,271 observations, so their interpretation as points on one response path requires comparability of the available samples. The table's separate returning-student estimate, 0.492 current-year credits, concerns another risk set and is not a third observation of the first-year entrants.

Ohio Pell grants

Bettinger (2004) studies withdrawal between the 1999–2000 and 2000–2001 academic years in Ohio. The reference input is the Table 5.3, column 4 estimate for students initially enrolled at four-year colleges: a \$1,000 increase in imputed Pell aid reduces the withdrawal probability by 0.064, or 6.4 percentage points. The reference conversion reverses this sign and assigns one additional schooling year to persistence, yielding 0.064 expected schooling years. L2 imposes $\Delta H(20) = 0.064$, with age 20 assigned to the persistence endpoint. There is no L3. Other columns change the estimation sample or specification for the same transition; they do not observe later attainment for this target group. In particular, the retained estimate is the four-year-student contrast in Table 5.3, not an additional follow-up from Table 5.5.

Tennessee HOPE scholarship

Bruce and Carruthers (2014) study Tennessee HOPE eligibility and post-high-school college choice for 2006–2009 graduates. In the retained low-GPA eligibility comparison, Table 3 reports a -0.022 effect on two-year college enrollment, a reduction of 2.2 percentage points. The reference calculation uses a one-year schooling-duration contrast for the sector response, reversing the sign to obtain 0.022 schooling years. L2 imposes $\Delta H(20) = 0.022$ at the assigned endpoint. This is the reference conversion of the eligibility estimate; it is not rescaled into an effect per scholarship recipient. There is no L3 because the sector choices and robustness estimates do not supply later completed-schooling observations for the same contrast. The records end in 2009, so follow-up differs across graduation cohorts. The program’s requirement to enroll within 16 months of high-school graduation is an eligibility rule, not a common observed follow-up duration.

Tennessee Pell eligibility

Carruthers and Welch (2019) estimate initial-fall college-entry responses at the Pell eligibility cutoff for 2006–2009 Tennessee high-school graduates. The reference calculation retains probability coefficients of -0.009200 , 0.002450 , and 0.000514 for Tennessee public, out-of-state public, and private four-year institutions, respectively. The public two-year coefficients are 0.002260 in Tennessee and 0.001570 out of state. The private for-profit enrollment coefficient is 0.001360 ; the reference conversion groups this margin with two-year entry. The resulting four-year and two-year-equivalent sums are $P_4 = -0.006236$ and $P_2 = 0.005190$. Applying the reference durations of two schooling years for four-year entry and one for two-year entry gives $2P_4 + P_2 = -0.007282$. L2 matches this total at the assigned second post-high-school year, averaging over graduation ages 17, 18, and 19 rather than placing everyone at one attained age.⁴⁶ There is no L3: $P_4 + P_2 = -0.001046$ is a sum of the same initial-entry coefficients, not a separately observed first-year schooling total. Imposing it at an earlier age would assume how the converted duration is allocated over time.

Georgia HOPE scholarship

Cornwell et al. (2006) estimate Georgia HOPE effects on first-time freshman enrollment using a state-year panel from 1988–1997. The reference inputs include log-enrollment effects of 0.086 for public four-year

⁴⁶Graduation-age weights are 0.065207 , 0.575880 , and 0.358913 ; each graduation cohort receives one-quarter weight. The second-year restriction therefore uses ages 19–21. These are calibration weights, not additional source effects.

colleges, 0.132 for private four-year colleges, and -0.045 for public two-year colleges, together with a reduction of 560 in recent high-school graduates leaving the state for college.⁴⁷ These are enrollment flows across successive cohorts, not follow-up outcomes for one entering class. The reference conversion transforms each log coefficient using $\exp(\beta) - 1$, applies pre-policy sector counts and its recipient denominator, and accounts for reduced out-of-state attendance. This gives net enrollment responses of 0.055988 in four-year colleges and -0.014674 in two-year colleges. With durations of two and one years, the schooling input is $2(0.055988) - 0.014674 = 0.097302$. L2 imposes that total at age 20, equally averaging the calibrated 1975–1979 birth cohorts. There is no L3: assigning the sector sum of 0.041314 to age 19 would allocate the assumed duration, rather than incorporate a separately reported attainment window.

Social Security student benefits

Dynarski (2003) estimates how Social Security student-benefit eligibility affects college attendance and completed schooling. Table 3 reports 0.564 additional completed schooling years by age 23. By age 28, the lower-bound treatment of attrition gives 0.679 years; excluding attriters gives 0.754, and assigning their last observed schooling gives 0.727. L2 retains the lower-bound estimate used for L1, imposing $\Delta H(28) = 0.679$. L3 adds $\Delta H(23) = 0.564$ exactly while retaining the same age-28 convention. These are two actual observation ages, so no enrollment-duration conversion or assumed division across years is required. The 0.754 and 0.727 estimates are alternative treatments of missing outcomes at age 28, not additional follow-up periods, and are not imposed alongside the selected lower-bound input.

College appropriations: spending

Deming and Walters (2017) estimate how public-college resources affect institutional enrollment using institution-year data from 1990–2013. The reference input comprises enrollment elasticities of 0.304, 0.796, 0.845, and 0.830 in the spending-change year and the next three years. Following the reference calculation, I sum these coefficients, divide by the 2.02 persistence adjustment, and scale by \$1,000 relative to average spending of \$29,856.23. The resulting schooling-equivalent input is 0.046013 years. L2 imposes $\Delta H(22) - \Delta H(18) = 0.046013$. Ages 18–22 place the aggregate enrollment response on the modeled schooling path; the source does not follow an individual birth cohort through those ages. L3 retains the four reported lag responses separately. Applying the same conversion to each gives annual schooling-equivalent targets of 0.005041, 0.013199, 0.014011, and 0.013762 at modeled ages 19–22, each within two marginal standard errors, while the L2 total stays exact. This transports the institutional policy-time sequence to the representative population’s acquisition years; it does not recover an individually observed cohort path. The reference calculation already uses all four coefficients in its sum, so L3 adds their separate dynamic restrictions rather than previously omitted coefficients.

Early Head Start

Hess (2025) estimates the effects of county access to Early Head Start on subsequent educational attainment. Table 1.2 distinguishes partial from full exposure and population intent-to-treat from participant effects. For full exposure, the retained bachelor’s-completion estimate is 0.168 for participants, or 16.8 percentage points;

⁴⁷These values follow the Hendren and Sprung-Keyser replication. The accessible 2004 working paper differs from the published specification, including the private four-year coefficient; it does not independently verify every retained value.

the corresponding intent-to-treat estimate is 0.012. L2 models the BA+ outcome directly and imposes a weighted age-22 gap of 0.168 over the calibrated 1989–1999 birth cohorts. It does not multiply the coefficient by schooling years. Age 22 is the reconstruction’s representative completion endpoint: the source pools adult attainment from linked Census–ACS records and limits its exposure cohorts so that college graduation can be observed from that age. There is no L3 because no additional comparable age-specific BA+ estimate is retained for the same exposure group. Partial-exposure estimates, sex-specific effects, and the pooled some-college outcome change the estimand rather than adding another BA+ observation age.

Texas community-college tuition

Denning (2017) estimates the effects of community-college district access and its associated tuition reduction for Texas high-school graduation cohorts from 1994–2005. The enrollment inputs cover the high-school graduation year and each of the next six years. The community-college probability effects, in time order, are 0.032, 0.036, 0.027, 0.024, 0.0095, 0.0083, and 0.014. The university effects at the same times are –0.00045, 0.010, 0.0082, 0.0089, 0.017, 0.0068, and 0.0082. Online Appendix Table A4 reports bachelor’s-completion effects of 0.0033 after six years and 0.0078 after eight years. The reference schooling conversion sums the seven annual enrollment responses across sectors, obtaining 0.209450 years, and adds $2(0.0078 - 0.0033) = 0.009000$ years for the later completion increment. L2 therefore imposes $\Delta H(26) = 0.218450$, using a graduation-age mapping of 18. L3 preserves that equality and screens the earlier cumulative sums: 0.031550, 0.077550, 0.112750, 0.145650, 0.172150, 0.187250, and 0.209450 years at assigned ages 18–24, respectively. The eight-year completion input supplies the age-26 endpoint; no year-seven enrollment effect is imputed. L3 thus retains separately reported timing under the reference enrollment-to-schooling conversion.

Adult Pell Grant introduction

Seftor and Turner (2002) estimate the effect of introducing federal aid for adult students on current college enrollment among people ages 22–35. The retained probability effects are 0.015 for men and 0.014 for women. The reference calculation weights them by sample sizes of 47,407 and 62,405, giving a pooled enrollment response of 0.0144317, and assumes two additional schooling years per induced enrollment. L2 imposes $\Delta H(30) - \Delta H(28) = 2(0.0144317) = 0.0288634$. Ages 28 and 30 are representative placements of this assumed duration; the source does not observe the same adults’ cumulative schooling at these two ages. There is no L3. Estimates for different age groups or policy periods do not supply a later observation of the same target population, and dividing the two-year conversion into separate annual effects would add an assumption rather than an observed window.

College appropriations: tuition

Deming and Walters (2017) also estimate institution-level enrollment responses to tuition changes over 1990–2013. The four retained log-enrollment/log-tuition coefficients are –0.016, 0.066, 0.031, and –0.073 in the current year and the next three years. The reference transformation divides their sum, 0.008, by 2.02 and multiplies by 1,000/8,599.86, yielding 0.000460519 schooling years. L2 imposes $\Delta H(22) - \Delta H(18) = 0.000460519$, preserving the reference calculation’s policy-direction convention.⁴⁸ L3 adds the separate

⁴⁸The positive input preserves the released Hendren and Sprung-Keyser tuition-cut coding. Applying a literal negative tuition change would reverse its sign; that is an alternative specification, not additional timing information.

converted lag targets -0.000921 , 0.003799 , 0.001785 , and -0.004202 for modeled annual acquisition at ages 19–22, screened within two marginal standard errors, while retaining the exact L2 sum. As in the spending calculation, this is conditional on transporting institution-level policy time to the representative age path. The coefficients already enter the reference total; the added restriction preserves their sequence.

HOPE tax credit

Turner (2011) estimates enrollment responses to tax-based student aid among people ages 18–19. The reference inputs are a 0.235-percentage-point increase in first-year full-time enrollment per \$100 of aid and a 0.210-percentage-point increase in second-year persistence conditional on completing the first year.⁴⁹ Using the reference subsidy of \$1,104 and 1.5 additional years for the second margin gives $(1,104/100)[0.00235 + 1.5(0.00210)] = 0.060720$ schooling years. L2 imposes $\Delta H(22) - \Delta H(18) = 0.060720$. The four-year age interval is the retained placement of this converted total, not four observed years of individual attainment. L3 preserves the first-year input separately as $\Delta H(20) - \Delta H(18) = (1,104/100)(0.00235) = 0.025944$, while retaining the L2 total. The remaining 0.034776 follows the same 1.5-year conversion of the second margin. The first component's placement in the earlier age block and the reference combination of conditional risk sets are maintained mappings, not separately estimated unconditional cohort effects. The source's overall enrollment effect, 0.320 percentage points per \$100, and its both-years indicator, 0.372 percentage points, overlap the retained margins and are not extra follow-up observations.

The eleven HOPE/AOTC contrasts use the finite policy-unit conversion in Appendix E.4.1. For the ten kink estimates, a fixed schedule displacement calibrated to one point-estimate credit dollar gives an attendance response of β_i/κ_i ; the two-year conversion gives $2\beta_i/\kappa_i$ schooling years. The assumed acquisition years are 19–20 for the age-19 cell and 20–21 for the age-20 cell. Only current attendance is observed: neither the second year nor contemporaneous sector outcomes supplies L3. The case profiles below retain each coefficient, first stage, and converted restriction. All use the common finite anchored path support and corresponding valuation schedules. The simulated-instrument coefficient is already per credit dollar and is treated separately.

HOPE credit: independent single filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is 3×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1156229. The retained two-year duration converts it to 0.000051893 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1987–1988 ACS cohorts.

HOPE credit: dependent single filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is -5×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1386706. The retained two-year duration converts it to -0.000072113 schooling-equivalent years. L2

⁴⁹The conditional second-year input follows the Hendren and Sprung-Keyser replication. The accessible 2010 working paper reports a different, unconditional second-year specification; it is not an independent verification of this input.

imposes this converted total through the joint cohort–age operator in Equation (53), using national 1987–1988 ACS cohorts.

HOPE credit: dependent single filers, phase-out end

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is 6×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1441341. The retained two-year duration converts it to 0.000083256 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1987–1988 ACS cohorts.

HOPE credit: joint filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is 12×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.0731243. The retained two-year duration converts it to 0.000328208 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1987–1988 ACS cohorts.

HOPE credit: joint filers, phase-out end

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is 4×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.0638205. The retained two-year duration converts it to 0.000125352 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1987–1988 ACS cohorts.

Tuition deduction: single filers, phase-start

Hoxby and Bulman (2016) study college attendance around federal tuition-deduction thresholds. For single filers at the deduction’s start, the reference calculation retains a current-attendance effect of 0.005, or 0.5 percentage points, from its Table 5 input. The assumed two-year duration gives 0.010 schooling years, and L2 imposes $\Delta H(23) - \Delta H(21) = 0.010$. The source also reports Table 7 effects of (0.020, 0.014, 0.008, –0.011, 0.007) in years zero through four relative to expected on-time high-school graduation. These are actual additional reported windows, but they are not admitted as nested L3 in the current calculation. The Table 7 year-one coefficient differs from the retained Table 5 input, and the common sample/specification and age mapping have not been reconciled. The full five-value sequence is preserved as a separate timing exercise; it does not silently replace the L2 input or its observation operator.

Tuition deduction: joint filers, phase-start

Hoxby and Bulman (2016) report a current-attendance effect of –0.003, or –0.3 percentage points, for joint filers at the tuition-deduction start in the input retained by the reference calculation. Applying its two-year duration gives –0.006 schooling years. L2 imposes $\Delta H(20) - \Delta H(18) = -0.006$. Table 7 separately reports

attendance effects of $(0.016, -0.003, 0.010, 0, -0.001)$ in years zero through four relative to expected on-time high-school graduation. Although the year-one coefficient agrees with the retained input in this contrast, the dynamic sequence is an attendance measure, whereas L2 constrains a two-year schooling increment. A compatible mapping that adds these observations while preserving the original restriction is not established. There is therefore no admitted L3; all five reported values remain in the separate timing exercise. The source clock is expected graduation time, not years since a policy event.

HOPE and Lifetime Learning tax credits

Long (2004) studies the introduction of the HOPE and Lifetime Learning tax credits using repeated cross sections. The reference input comes from current college enrollment among people ages 18–24 in Table 3.11: an odds ratio of 0.9342 and a baseline enrollment probability of 0.463. The retained reference approximation is $(0.9342 - 1) \times 0.463 = -0.0304654$ for the enrollment change. Multiplication by the assumed two schooling years gives -0.0609308 , so L2 imposes $\Delta H(23) - \Delta H(21) = -0.0609308$. The odds-ratio conversion is the reference approximation, rather than an exact conversion of a logistic odds ratio into a probability difference. Ages 21–23 place the converted total on the model path; they do not describe a two-wave source observation. There is no L3 because comparisons across calendar years or age groups do not supply repeated attainment observations for the same cohort.

AOTC: dependent single filers, phase-out end

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is -2×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.2164388. The retained two-year duration converts it to -0.000018481 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1991–1992 ACS cohorts.

AOTC: dependent single filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is 43×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.2019312. The retained two-year duration converts it to 0.000425888 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1991–1992 ACS cohorts.

AOTC: joint filers, phase-out end

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is -7×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1146557. The retained two-year duration converts it to -0.000122105 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1991–1992 ACS cohorts.

AOTC: joint filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is 7×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1074026. The retained two-year duration converts it to 0.000130351 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1991–1992 ACS cohorts.

AOTC: independent single filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is 1×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.2268326. The retained two-year duration converts it to 0.000008817 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (53), using national 1991–1992 ACS cohorts.

AOTC: simulated instrument

Bulman and Hoxby (2015), Table 16, report attendance responses of 1.4×10^{-5} per actual credit dollar at age 19 and -1.6×10^{-5} at age 22 over 2006–2011. L2 retains the age-19 estimate. Under the two-year conversion, the fixed one-dollar input is 0.000028 schooling-equivalent years, constraining the average $\Delta H_c(20) - \Delta H_c(18)$ across the 1987–1992 cohorts. This maps two assumed acquisition years onto finite logistic levels; it does not treat the second year as observed. The direct-credit component is one dollar because the source coefficient is already second stage. Appendix E.4.1 documents the correction from approximately 0.38, retained in every repetition. The age-22 result concerns different cohorts and does not form a longitudinal L3 for the age-19 population. L1 remains the reported welfare calculation, so its comparison with L2 includes the unit correction.

Algorithmic college placement

Bergman et al. (2026) randomize algorithmic rather than test-only course placement at seven New York community colleges. In the June 2026 version, Table 3 reports a pooled effect of 0.562 additional college credits earned over the available follow-up. The entering cohorts have different follow-up lengths through 2018–2019; this is not a common two-year endpoint. The reconstruction divides the pooled estimate by 24 and imposes $\Delta H(23) - \Delta H(21) = 0.562/24 = 0.023417$ in L2. Ages 21–23 are a representative placement of the converted increment. The source also reports 1.717 additional earned credits for the more mature fall-2016 entering cohort, observed for roughly two and a half years. This is not admitted as L3 because it changes the cohort relative to the pooled L2 estimate, rather than adding a later observation of the same target. First-term remedial or college-level course outcomes likewise cannot be imposed as total earned-credit stocks without another conversion. Thus, the main calculation has L2 only.

Tuition deduction: single filers, phase-end

Hoxby and Bulman (2016) provide the tuition-deduction phase-out-end contrast for single filers. The current-attendance coefficient retained in the reference Table 5 input is zero. With the same two-year duration con-

version, L2 imposes $\Delta H(20) - \Delta H(18) = 0$; this does not impose a zero effect at every subsequent age. Table 7 reports attendance effects of (0.008, 0.015, 0.008, 0.009, -0.002) in years zero through four relative to expected on-time high-school graduation. These values are preserved in the separate dynamic exercise. There is no admitted L3 because the year-one coefficient of 0.015 differs from the retained zero input, and the correspondence of the source specifications and observation operators remains unresolved. The source's additional windows are therefore documented rather than described as absent, but they are not imposed as nested restrictions on the existing L2 path.

Tuition deduction: joint filers, phase-end

Hoxby and Bulman (2016) report the tuition-deduction phase-out-end contrast for joint filers. The retained current-attendance input is zero, which the reference two-year conversion also maps to zero schooling years. L2 imposes $\Delta H(20) - \Delta H(18) = 0$. The source's Table 7 sequence is (0.010, 0, 0.015, 0.005, 0.002) in years zero through four relative to expected on-time high-school graduation. Its year-one coefficient agrees with the retained zero, but an attendance sequence is not the same object as L2's converted two-year schooling increment. A compatible nested mapping has not been established. There is consequently no admitted L3; all five source values remain in the separate timing exercise. The zero L2 increment leaves later continuation unrestricted except by the maintained path and empirical-support conditions.

Cannabis legalization: education channel

Li (2026) estimates recreational-cannabis legalization effects on bachelor's completion among first-time, full-time college entrants, using entering cohorts from 2006–2017 and exposure after college entry. The reported effects are -0.016, -0.011, and -0.018 on completion within four, five, and six years, respectively. The education-channel reconstruction assumes two additional schooling years per bachelor's completion, giving targets of -0.032, -0.022, and -0.036 schooling years. With entry assigned to age 18, these correspond to ages 22, 23, and 24. The primary result is L3: it retains the six-year target exactly and adds the four- and five-year targets with the marginal-error screens specified above, equally averaging the calibrated 1988–1999 birth cohorts. The six-year-only restriction is stored as an L2 diagnostic and is not presented as a separate main result. The source's three-year outcome for two-year institutions concerns a different entrant population and is not added to the bachelor's-completion path. Thus, the primary reconstruction uses all three comparable BA windows without combining unlike credential samples.

E.3 Welfare Witnesses and Information Layers

For finite-policy calculations, let $\mathcal{B}_{i\ell}$ be the saved finite bank for calculation i and layer ℓ . Crossing MVPF one at positive cost requires a path h^- with $N_i(h^-) > 0$ and $W_i(h^-) < N_i(h^-)$ and another path h^+ with $N_i(h^+) > 0$ and $W_i(h^+) > N_i(h^+)$. Both satisfy the same retained information and accounting restrictions. A harmful-costly versus self-financing comparison instead pairs $W_i(h^-) < 0$, $N_i(h^-) > 0$ with $W_i(h^+) > 0$, $N_i(h^+) \leq 0$. The file `paired_welfare_witnesses.csv` supplies candidate identifiers and paired values. It never combines marginal extrema. Table 3 supplies the separate component extrema; they are descriptive summaries and cannot be combined into a ratio.

CUNY and Social Security student benefits add an exact second observation at the point inputs. Adams and Michigan add their separately reported early entry components under the retained schooling conversion

and an explicit temporal allocation. The HOPE tax-credit case similarly separates first- and second-college-year components under the reference conversion of their distinct risk sets. CUNY adds first-year credits to its three-year input, conditional on comparability across the two available samples. Social Security adds the age-23 schooling estimate to the retained age-28 input; the latter uses the source's lower-bound treatment of attrition, rather than a complete-case estimate without missing outcomes. Florida, Kalamazoo, Texas tuition, and cannabis retain L2 exactly and screen additional reported moments using two marginal standard errors. Florida includes the first semester as well as the earlier annual credit totals, using the retained entry-age mapping to locate the first semester at age 18.5. Kalamazoo adds credits in years two and three to the retained fourth-year input; the mature cohorts available at those horizons differ. These marginal screens are not joint 95 percent regions. Tennessee Promise instead uses the source's marginal permutation intervals, extracted from Figure A5 with a one-pixel inward guard. Their intersection is not a joint region; the lower endpoint is sensitive to graph precision. These differences preclude interpreting component-width ratios as a cross-case score for data quality.

A main L3 restriction must preserve a time window actually reported by the source while retaining the education-conversion rule used in L2. Wisconsin qualifies because Table 5 reports semester credit effects and cumulative totals. Its annual flows sum adjacent observed semesters under the retained credits-to-schooling conversion. The later-cohort third-year restriction uses cohort 2 alone because cohort 3 is not observed that long; no missing year is imputed. The entry-age mapping, cohort weights, and common response parameters are already maintained in L2. Where covariance is unavailable, the annual-flow SE upper bounds sum the marginal semester SEs. The circle marker therefore identifies added source timing, not a reconstruction free of calibration assumptions.

L3 may preserve converted source dynamics rather than directly measured annual schooling. The two college-appropriation calculations use four reported policy-lag responses, retaining the reference persistence and enrollment conversion and assigning the resulting sequence to modeled acquisition years. This requires transport from institutions to a representative population. Adams, Michigan, and HOPE tax credit likewise retain explicit timing or risk-set mappings. These assumptions are recorded alongside the source information; the L3 label does not establish their validity. Tennessee Pell and Georgia HOPE remain L2 only because their proposed extra years merely split a single sector-entry observation using assumed durations. They add no separately observed window or dynamic response.

Every verified main L3 row is included in L2, and extrema are recomputed from the saved rows. Component-width ratios therefore cannot exceed one apart from numerical tolerance. Nesting of recorded banks does not establish that either exhausts the continuous feasible set. Failure to verify a candidate remains a search outcome. A span between finite endpoints does not assert that every intermediate value has been verified. No distribution over candidates or over the continuous continuation class is specified.

E.4 Monetary Units and Reference Components

Table 3 uses a positive reference program-cost component as a display denominator wherever its provenance is established. The same denominator divides willingness to pay and net cost in L1 and both reconstructions of a calculation. It is not the path-dependent net cost, and this scaling leaves each MVPF unchanged. The component follows the native account, such as a scholarship's discounted transfer or a tax credit's direct revenue cost; it need not represent an identical policy dose across applications. In tax-schedule calculations, the original policy unit can be a small change in a credit or deduction. A large

normalized component can therefore combine a broad continuation range with a small reference expenditure. It should not be read as a large absolute dollar amount for an average program participant. The registry data/chapter5/reference_accounts_and_units.csv records the divisor, source file, source field, and monetary base for all 38 calculations. The underlying point-input ranges and their units are preserved separately from the rounded table.

The reference component pair is taken from the same source account used to identify L1, with unrounded ledger values retained when they match that record. Reported reference ratios are not silently replaced by ratios recomputed from rounded components. For Early Head Start, the library account gives $E^{\text{ref}} = 21,944$, $W^{\text{ref}} = 3,906$, and $N^{\text{ref}} = 11,216$. Its fiscal sum is $24,257 + 1,540 - 10,478 - 4,103 = 11,216$: program cost plus the community-college cost change, less university savings and additional taxes. The dissertation reports net cost of about 12,755. Its displayed calculation appears to add the 1,540 community-college term again after including it in the net education-cost change. I use the library ledger consistently and retain the dissertation amount as an alternative account, rather than mix their components. The source price year remains unresolved.

For the Early Head Start reconstruction, let $E_i(h)$ be the existing discounted BA+ earnings profile multiplied by 0.2788902. This fixed scale reproduces 21,944 when the retained 0.168 BA+ effect is applied throughout the valuation ages. I then use $W_i(h) = E_i(h) - 18,038$ and $N_i(h) = 15,319 - (4,103/21,944)E_i(h)$. Both reference components are reproduced exactly. The scale remains fixed across paths and all 1,000 education-input repetitions. It is a reference-anchoring assumption, not an independently identified age-specific return or a recovery of the missing price year. Monetary values remain in the source-account units. The saved point paths yield approximately $[-1.53, 1.96]$, with positive cost throughout; substituting the dissertation's reference cost shifts the range to approximately $[-1.40, 1.67]$. The path constraints are unchanged.

Algorithmic college placement retains the June 2026 source components, $W = 220$ and $N = -90$. The reconstruction adds education-related earnings and taxes that the original account deliberately omitted. It remains an explicitly marked extension in the full ordering, rather than evidence that continuation alone overturns an otherwise identical account. Cannabis is separate because it has no whole-policy reference and values only the education channel.

E.4.1 Marginal Tax Inputs in a Finite Policy Experiment

The eleven HOPE/AOTC reconstructions use the same finite, anchored logistic attainment levels as the other applications. Their source coefficients require an additional policy-unit conversion. Table 14 of Bulman and Hoxby (2015) reports changes in attendance-income slopes at tax-schedule kinks for ages 19–20, in 2007 for HOPE and 2011 for AOTC. Table 13 supplies the corresponding credit-income first stages. Let β_i be the attendance coefficient and $\kappa_i > 0$ the credit first stage, both signed in the reference policy direction. I fix a schedule displacement $q_i = 1/\kappa_i$, calibrated to raise the credit by one dollar at the point estimates. I assume that the local attendance and accounting responses transport linearly over this displacement. This specifies a finite counterfactual; the source derivative alone does not identify its exact effect. Keeping the reference two-year enrollment-to-schooling conversion gives the input

$$m_i^{\text{YS}} = 2\beta_i q_i = \frac{2\beta_i}{\kappa_i}. \quad (52)$$

For the Table 16 simulated-instrument estimate, the attendance coefficient is already per actual credit dollar, so $q_i = 1$ without another first-stage division. These conversions define schooling-equivalent inputs, not directly observed completed-schooling effects.

The source ages determine the cohort calibration; assumed duration is stated separately. Nationwide ACS profiles use the 1987–1988 birth cohorts for HOPE, 1991–1992 for the AOTC kinks, and 1987–1992 for the simulated-instrument input observed at age 19 during 2006–2011. Equal cohort-cell weights are maintained because the source regression weights have not been recovered. In the annual acquisition convention, current enrollment at age a contributes to schooling in years a and $a + 1$ under the two-year duration assumption. The converted total therefore restricts $\Delta H_c(a + 1) - \Delta H_c(a - 1)$. For HOPE, the joint restriction is

$$\frac{1}{2} [\Delta H_{i,1987}(21) - \Delta H_{i,1987}(19)] + \frac{1}{2} [\Delta H_{i,1988}(20) - \Delta H_{i,1988}(18)] = m_i^{\text{YS}}. \quad (53)$$

The AOTC kink operator replaces these cohorts by 1991 and 1992. The simulated-instrument operator averages $\Delta H_c(20) - \Delta H_c(18)$ across its six age-19 cohorts. The second acquisition year is a duration allocation, not another source observation, and does not create L3. All tax paths assume no attainment effect through age 17, allowing first effects at age 18. This exposure restriction is maintained rather than estimated. Earlier placements at ages 21–23 or 25–27 are not treated as source observation windows. Independent-filer earnings schedules beginning at 25 remain valuation assumptions. The age-22 simulated-instrument result refers to a different cohort composition and remains outside the main L3 set.

The restored paths retain the current full-population ACS calibration, the transported joint parameter support, completion-timing restrictions, nondecreasing levels, and the one-schooling-year annual acquisition cap. A common finite treated parameter triple is used across each application's cohort components, with their separate baseline anchor levels. Thus both H^0 and H^1 are finite attainment paths throughout the comparison. For example, the HOPE independent-single contrast has $\beta_i = 3 \times 10^{-6}$ and $\kappa_i = 0.1156229$. Its fixed schedule displacement is about 8.649 income-margin units, and its two-year input is approximately 0.000051893 schooling years. This is a local-linear calibration of a one-dollar credit change, not the effect of adopting the whole tax program. Matching an ACS support restriction does not establish the plausibility of every policy response inside that support.

The monetary conversion follows the same fixed displacement. For the ten kink contrasts, the non-earnings components become $w_{0i}^F = q_i w_{0i}^{\text{source}}$ and $n_{0i}^F = q_i n_{0i}^{\text{source}}$. The per-schooling-year earnings and fiscal schedules retain their units and values. The account evaluates the finite cohort-average gap, $\overline{\Delta H}_i(a)$, as

$$W_i^F(h) = w_{0i}^F + \sum_a w_i(a) \overline{\Delta H}_i(a). \quad (54)$$

The corresponding net cost is $N_i^F(h) = n_{0i}^F + \sum_a n_i(a) \overline{\Delta H}_i(a)$. The reported L1 ratio and interval are unchanged. Its WTP, net cost, and program-cost display divisor receive the same normalization, so their normalized values are unchanged as well. For the simulated instrument, I retain the unit correction that sets the direct credit payment to one dollar rather than the first-stage amount of approximately 0.38. At each input repetition, the correction adds $1 - \text{tax_value}$ to net cost and $(1 - \text{tax_value}) \min(1 - \beta_i, 1)$ to the non-earnings WTP component. Its L1–L2 comparison therefore includes this accounting correction as well as continuation.

All 1,000 archived joint input IDs are retained. The point normalization q_i stays fixed across repetitions; education, tax-credit, cost, and earnings inputs still vary together according to their registered scheme. In particular, five negative first-stage draws are retained, rather than discarded or divided into the education

coefficient. Their credit changes are not constrained to equal one dollar away from the point estimates. The simulated-instrument repetitions remain a constructed reference-input exercise, not draws from the source’s sampling distribution. Converting monetary units from dollars to cents leaves the same physical policy experiment, path constraints, and MVPF unchanged. Changing the size of the finite policy experiment is instead a substantive sensitivity, because the ACS support restricts levels rather than responses per dollar.

A limited dose sensitivity repeats the finite-path screening for point credit changes of 0.50, 1, and 2, scaling the education input and non-earnings account components together. At all three doses, each tax calculation has verified negative-WTP, positive-cost paths and positive-WTP, nonpositive-cost paths. This checks the existence of these conflicting classifications on the tested grid; it does not establish invariant endpoints or welfare bounds across policy sizes. Currency-unit checks separately preserve the physical policy experiment and recover the same schooling restriction and MVPF.

The files `finite_tax_input_ledger.csv`, `independent_point_checks.csv`, and the paired-witness records give the source coefficients, first stages, joint age-cohort operators, equation errors, and complete finite paths. WTP and net-cost extrema are finite. Some MVPF endpoints nevertheless remain infinite because verified finite paths approach zero net cost from above with nonzero WTP. Those ratio limits are distinguished from self-financing states. Equal endpoint pairs retain tied ranks; no additional responsiveness cap or path probability is introduced to separate them.

E.5 Candidate-Based Lower-Endpoint Ordering

Figure 9 ranks all 37 calculations with a reference MVPF. Twelve use L3: Florida, Adams, Wisconsin, Kalamazoo, Michigan, CUNY Pell, Social Security student benefits, both college-appropriation calculations, HOPE tax credit, Texas CC tuition, and Tennessee Promise. The other twenty-five use L2, including the eleven finite tax reconstructions just defined. Early Head Start uses the aligned account, and algorithmic placement retains its marked earnings-channel extension. Cannabis has no whole-policy reference. Table E2 reports every rank and source window; the restricted 24-case ordering remains in the data.

For each selected finite-path specification, let $\underline{M}_i$ and $\overline{M}_i$ be the lower and upper endpoints of defined MVPFs, including audited one-sided limits and the self-financing convention. Let r_2 round finite numbers to two decimals and leave infinities unchanged. The display orders policies lexicographically in ascending order of

$$(-r_2(\underline{M}_i), -r_2(\overline{M}_i)). \quad (55)$$

Equal endpoint pairs receive the same competition rank, one plus the number of strictly preferred calculations. Reference ranks compare the recorded L1 values, preserving ties. Repeating the full ordering with three- or four-decimal endpoint rounding changes the rank of one calculation. The displayed precision therefore matters for very close finite endpoints. This numerical stability check does not establish global extrema: a newly discovered adverse path could change an ordering based on candidate minima.

An open marker replaces only that policy’s selected L3 set by L2, holding other selected sets fixed. Ten of the twelve added-information cases improve under the rule and two remain unchanged. The arrows describe the rule’s response to restrictions, not an increase in the policy’s true welfare. Ten ranked calculations admit paths with nonpositive WTP and nonpositive net cost. Those unassigned-ratio paths, rather than the calculations containing them, are omitted from the ordering; their monetary components remain in the collection table. The rule assigns neither artificial finite values to infinity nor probabilities to paths, and it is not a social-welfare or certified maximin ranking.

E.6 Reported Timing and Its Use in Reconstruction

Table E3 distinguishes source timing coverage from imposed path restrictions. It counts distinct age or individual-follow-up windows for the relevant contrast and outcome family. An overlapping two-year age interval counts once; additional credentials, estimators, subgroups, and tax contrasts at that window do not create more horizons. Institutional policy lags and college-grade risk sets retain their own clocks. The count measures neither precision nor independent observations.

Among the 37 calculations with L1, 17 have one verified window, five have two, and 11 have at least three. Four use other clocks: Georgia’s state-year freshman flows, the two Deming–Walters institution-level calculations, and Turner’s college-grade and pooled enrollment outcomes. The published Tennessee HOPE paper now verifies one initial-college-choice window. Its 16-month eligibility deadline is not a common observed follow-up, and its alternative sectors and specifications do not constitute additional attained-education times. The unranked cannabis application reports completion at three horizons. All 38 calculations therefore receive a coverage record, and all 37 calculations with L1 enter the expanded ordering with their specification groups identified.

Several multiple-window studies still require a population, outcome, or specification bridge before their results constrain one path. Absence of L3 does not imply that a source lacks dynamic evidence. Nor does L3 imply that the reference valuation ignored an estimate: Texas can use separately reported timing while retaining coefficients already used in its aggregate conversion. Adams’ early entry contribution, by contrast, requires an added completion-time assignment and remains a sensitivity. The registry separates reference inputs, source windows, imposed restrictions, and their empirical or assumed origins, with source locations and hashes. It does not treat the number of windows as a causal determinant of ranking changes. Several policy contrasts also share a study family.

E.7 Source Comparability and Numerical Verification

Additional coefficients constrain a common path only when the population, policy contrast, outcome, and clock can be reconciled with L2. For the four tuition-deduction contrasts, contemporaneous coefficients differ between Tables 5 and 7 despite both being described as baseline; the source text does not resolve that specification difference. The AOTC simulated-instrument exercise changes the observation operator and population combination. Their timing exercises remain separate specifications, not nested L3. For algorithmic college placement, the verified pooled credit effect of 0.562 (SE 0.292) and earliest-cohort effect of 1.717 concern different populations. Its registered two-year mapping is a representative proxy for mixed follow-up. DC tuition assistance retains a coefficient traceable through the L1 calculation, but the recovered 2003 version does not independently verify the final 2006 coefficient or its missing SE. The case profiles preserve these limits instead of turning every additional source column into a new path restriction.

The main L3 banks retain Florida’s first-semester and annual credit totals and Wisconsin’s reported semester credit flows. Their full constraint sets lack registered joint input draws, so earlier, narrower-screen repetition results are not presented as intervals for the expanded L3. Appendix F.4 uses the available marginal uncertainty in a separate sensitivity calculation. Tennessee’s earlier-window screen uses the source’s permutation intervals, with graph-extraction sensitivity reported in Appendix D.8. There are thirteen admitted and verified point L3 banks, including the cannabis education channel.

For finite-policy calculations, all saved point candidates are independently evaluated at annual ages 15–65, checking their anchors, nondecreasing attainment, credential nesting where required, encoded moments, empirical support, and paired welfare values. Every verified L3 candidate also enters L2. Searches supplement shape grids with multiple-start component optimization and equations for zero WTP, WTP equal to cost, zero net cost, and selected ratios. At a zero-cost root with nonzero WTP, decreasing positive-cost targets check the one-sided limit. These checks prevent a sparse grid from obscuring narrow positive-cost branches, but do not certify global finite endpoints or connectedness of the feasible set. The same checks apply to registered input repetitions. An unsuccessful search remains distinct from evidence of infeasibility.

The replication records retain the per-case outcome review, moment operators, completion and pole checks, and independent path reconstructions. Earlier specifications and numerical-search comparisons are archived separately from the current tables. Historical endpoints obtained under different ACS calibrations are not imported into the present results. The October revision also recomputes Wisconsin point paths and all 1,000 L2 repetitions using the joint cohort–follow-up operator in Appendix E.2. The source-input sequence, external baseline, and welfare account are unchanged. A before/after ledger records the small endpoint changes; the L2 and L3 welfare classifications are unchanged.

E.8 The Complete Path Display

Figure E1 displays comparison and treated schooling levels for all 36 schooling-equivalent calculations. Figure E2 covers Early Head Start’s BA+ path and Tennessee Promise’s three joint credential paths. Together they cover all 38 finite-policy calculations, including the three main applications. Black lines show H^0 . Blue and red boundaries and shading show the pointwise minimum and maximum H^1 in the saved L2 and admitted L3 sets. These ranges describe attainment without selecting paths by their welfare values. Tennessee’s ranges are computed from joint candidates that satisfy credential nesting; the cannabis education account displays only its primary L3.

The envelope edges can switch candidate identity across ages and credentials and need not themselves form feasible paths. They summarize the current finite search, not a probability distribution, sampling interval, or certified boundary of all possible continuations. Complete candidate paths and their paired welfare values remain in the replication records. All schooling panels share one age and schooling scale, and all credential panels share one age and attainment-share scale. The source profiles in Appendix E.2 and the expanded operators distinguish observed follow-ups from their assigned ages. Neither a later crossing nor a widening modeled gap establishes an individual behavioral mechanism or shows that the source researchers overlooked dynamics.

E.9 Point-Input Sensitivity across All Applications

The sensitivity exercise asks whether the welfare differences remain when the calibration or valuation changes, holding the reported education inputs at their point values. It covers all 38 calculations and consolidates the checks for STBF, Adams, and Tennessee Promise. Figure E3 gives the overview, and Table E4 reports numerical spans. The common comparison uses L3 for the thirteen calculations with admitted additional source restrictions and L2 for the other twenty-five. Cannabis contributes its education-channel L3, without a whole-policy L1. Both information sets, where available, remain in the accompanying data. These counts therefore differ from the L2-only comparison for the 37 calculations with L1 in Section 5.

E.9.1 Changes to Calibration, Valuation, and Search

The narrower-support check multiplies the empirical covariance matrix by 0.5 and the transported completion-timing deviations by $\sqrt{0.5}$. The wider check uses factors of 2 and $\sqrt{2}$. Both retain the source moments, anchors, natural attainment limits, and annual acquisition caps. They contract or expand dispersion around the same comparison profile; they do not represent alternative confidence levels. A geographic check replaces state or multistate profiles and covariance calibration with the national counterparts for the same birth cohorts. It applies to 16 calculations; the other 22 already use national profiles. Source operators and population weights are unchanged. Tennessee retains its high-school-graduate denominator and source-control-level matching in this geographic comparison. Its late-completion allowance is held fixed in the common support checks and varied separately.⁵⁰

The earnings checks multiply the path-dependent earnings and associated fiscal channels by $q \in \{0.75, 1.25\}$. Fixed transfers and non-earnings remainders remain unchanged, as do the separately encoded education-resource kernels. In terms of the signed account components, the exercise uses $W_i^{(q)}(h) = w_{0i} + qE_{Wi}(h) + C_{Wi}(h)$ and $N_i^{(q)}(h) = n_{0i} + qE_{Ni}(h) + C_{Ni}(h)$. Here E_{Wi} and E_{Ni} are the private and fiscal earnings contributions, while C_{Wi} and C_{Ni} are the remaining path-dependent education-cost contributions. The original kernels are recovered exactly at $q = 1$. The earnings contribution includes the account's maintained forgone-work treatment; this is a valuation-scale sensitivity, not an estimated change in a causal return to schooling. Removing an explicit forgone-work charge is reported separately where that component is identified.

The numerical check raises the scalar Sobol shape grid from 2^{14} to 2^{16} pairs and supplements it with searches for monetary extrema and welfare thresholds. Tennessee increases its credential pools and joint combination search while retaining credential nesting and the source permutation screens. All alternatives use the same finite anchored logistic family. When the baseline or support changes, candidates are re-evaluated and the search is repeated; valuation changes reprice paths and supplement searches under the new account. Complete annual paths, source restrictions, and paired welfare values are independently checked. A failed search would be reported separately from infeasibility; every common specification here has verified paths. No new input repetitions are run, and these results are neither endpoint confidence intervals nor global bounds.

E.9.2 Which Welfare Comparisons Survive?

Under the main specification, 21 of the 38 selected information sets contain positive-cost MVPFs on both sides of one. Narrower support reduces that count to 18; wider support raises it to 23. Reducing the earnings-channel valuation to 75 percent leaves 20, and increasing it to 125 percent leaves 21. The national-profile comparison gives 22 when the 22 unchanged national cases are included. The denser search preserves all 38 cases' threshold and self-financing classifications, although it finds additional paths with unassigned ratios in two tax calculations. Thus, neither the remaining variation nor its economic importance is explained simply by the original grid density. Finite-search stability still does not certify an endpoint.

The restrictions matter for particular judgments. Narrower support moves all found Kalamazoo and Ohio Pell positive-cost candidates above one and all found Early Head Start candidates below one. Wisconsin loses its self-financing candidates under narrower support or the lower earnings scale. Replacing its Wis-

⁵⁰The library's donor and observed-age conventions still apply, including backward reconstruction for historical cohorts. A national comparison is a calibration alternative, not a validation of population transport.

consin profile with the national profile instead admits positive-cost ratios below one. These are changes in the evidence supplied by the specified searches, not proofs that other paths are impossible. Across all common specifications, 19 calculations continue to admit both negative-WTP paths with positive public cost and positive-WTP paths with nonpositive cost, compared with 23 under the main specification. Several tax calculations share source studies and restrictions, so these frequencies describe calculations rather than independent replications. The persistence of opposing path pairs supports the conditional ambiguity claim; the cases that change show why empirical restrictions and earnings calibration must remain visible.

E.9.3 Source-Specific Checks

The final panel of Table E4 collects the additional checks that have a source-specific interpretation. STBF substitutes the printed 0.360-year input for the unrounded replication input and removes the explicit forgone-work charge. Its found positive-cost MVPFs remain above one throughout the common and specific checks. Adams uses its separately reported early entry input in L3, with the explicit time allocation described in Appendix D.7. Its main ratio span is approximately $[-0.62, 0.30]$. The common checks retain this allocation while varying empirical support, calibration, or earnings values. Its opportunity-cost comparison changes valuation without adding an education observation.

Tennessee varies the digitized permutation boundaries, the matching of source control levels, the late-completion allowance, and the reference earnings multiplier. The zero-, one-, and two-pixel inward rules give lower L3 ratios of approximately 0.29, 0.63, and 1.22. Thus, the position relative to one depends on graphical precision even when the later effects are held fixed. Removing the reference earnings multiplier eliminates self-financing among the found paths. The checks separate restrictions on education continuation from the earnings schedule used to value it; they do not establish the source's joint sampling distribution. The eleven finite tax experiments additionally vary the point credit change to 0.50 and 2, changing schooling inputs and fixed monetary components together. Early Head Start substitutes the dissertation's net-cost reference. Cannabis compares education-cost sectors, forgone-work and tax-base conventions, and equal state weights. These checks retain each application's economic scope. The cannabis fiscal classification continues to concern education alone.

The machine-readable results report NPV, WTP, net cost, positive-cost ratio spans, and fiscal classifications for every tested specification and both information sets where defined. Spans of separate monetary components must not be combined to form a ratio: each classification uses one paired path. These exercises leave the logistic functional form, core outcome conversions, exposure anchors, and causal interpretation of the source estimates largely maintained. They do not validate those assumptions, and the one-at-a-time changes do not establish robustness to all simultaneous departures.

F Conditional Input Repetitions and Additional Information

The uncertainty exercise separates variation across education paths from variation under the specified source-input and accounting perturbations. Each of the 38 policy calculations uses 1,000 prespecified L2 input repetitions and the same observation operators and response restrictions as its point-input calculation. L3 repeats the added restrictions where joint input draws are available; five cases retain point-input L3 only for their current full constraint sets. The geographic comparison curves, empirical support, education conversions, and specified valuation relationships remain fixed. Table 4 places the resulting MVPF endpoint summaries along-

side available source estimates and reported 95 percent intervals. Table F1 reports willingness to pay and net cost separately, including cases in which the ratio is poorly behaved near zero net cost. These are within-case comparisons: monetary years, policy units, populations, and the uncertainty included in the source accounts can differ across cases.

F.1 Conditional Endpoint Summaries

For repetition r , let $\widehat{\mathcal{D}}_{i\ell}^{(r)}$ be the finite verified path bank for case i and information layer ℓ . The numerical search combines the bank with targeted zero-cost checks. Its lower and upper summaries, including numerically audited one-sided ratio limits, are denoted by $\underline{M}_{i\ell}^{(r)}$ and $\overline{M}_{i\ell}^{(r)}$. I report the order-statistic summary

$$\left[q.025 \left(\{ \underline{M}_{i\ell}^{(r)} \}_{r \in \mathcal{R}_{i\ell}} \right), q.975 \left(\{ \overline{M}_{i\ell}^{(r)} \}_{r \in \mathcal{R}_{i\ell}} \right) \right], \quad (56)$$

where $\mathcal{R}_{i\ell}$ contains repetitions with a verified finite-path set and a defined ratio. The valid count is always reported relative to the original 1,000 repetitions. The same operation is applied to WTP and net-cost endpoints. For the eleven tax calculations, the policy normalization in Appendix E.4.1 stays fixed across repetitions. Infinite entries from the positive-benefit, nonpositive-cost convention and from numerically audited one-sided positive-cost limits remain in the order statistics. When both components are nonpositive, the components remain reportable but that path is not assigned a self-financing MVPF. Component extrema are computed separately and are never combined to manufacture a ratio.⁵¹

Equation (56) describes the conditional input exercise; it is not a demonstrated 95 percent confidence region for the entire continuous continuation set. Finite search can miss extrema, selected input draws can lack a verified candidate, and external calibration uncertainty is not resampled. Moreover, the input construction differs according to what the source releases. Registered joint source and accounting draws are retained where available. STBF, Early Head Start, algorithmic college placement, the simulated-instrument education input, and Tennessee Promise instead use researcher-constructed perturbations of reported education effects. DC tuition assistance varies economic inputs while its education effect remains fixed because a corresponding source SE is unavailable. Georgia HOPE and adult Pell retain conditional exercises that hold earnings-return parameters fixed. The case registry distinguishes these methods rather than labeling every row an original-study bootstrap.⁵²

The source comparison is likewise qualified by what can be recovered. Reported MVPF intervals are available for 34 of the 38 calculations. The main comparison uses the frozen Policy Impacts Library snapshot of 20 September 2026; for Tennessee Promise, Schoonover et al. (2026), Section 5.2.2, explicitly report both endpoints as infinite. No comparable reported interval was recovered for STBF, Early Head Start, or algorithmic college placement, and the author cannabis study does not supply an original MVPF calculation. These entries are left blank. The comparison preserves the source’s reported method and account rather than relabel a conditional reconstruction as its published interval. Differences in width can reflect continuation, source inputs, dependence assumptions, valuation calibration, and coverage targets simultaneously.

⁵¹Quantiles use the inverse empirical distribution without interpolating finite and infinite values. With 1,000 complete repetitions, endpoints use the 25th minimum and 975th maximum order statistics. Each ratio still pairs components from one path and draw.

⁵²Registered draw dependence is preserved; failures are not replaced. Newly verified L3 paths also enter the corresponding L2 sets. Denser search finds no additional paths for the 48 unresolved draws, without proving infeasibility. No distribution over paths is imposed.

F.2 Additional Information and Unsuccessful Searches

Additional timing and repeated inputs require separate availability checks. Table E1 identifies thirteen admitted L3 specifications and 25 calculations without one. All thirteen have verified point-input candidates; Table 4 records which also have joint input repetitions. Appendix E.7 explains the population, outcome, and timing requirements. Thus an absent L3 interval can mean that no additional restriction is admitted or that repeated inputs are unavailable, rather than that the source reports no dynamics.

Missing additional information and failure to find a compatible path have different implications. All 38,000 prescribed L2 repetitions complete, and 37,952 return finite-path sets. The 48 exceptions are in the Social Security student-benefit calculation; they remain recorded as unsuccessful searches, not replaced draws or proofs of infeasibility. One Texas tuition draw has verified WTP and net-cost values but no defined MVPF under the maintained convention. Florida, Wisconsin, Kalamazoo, CUNY, and Social Security lack joint input mappings for their full additional-information sets, so no corresponding main L3 repetition interval is supplied. Earlier Florida and Wisconsin summaries remain archived under their narrower screens. Tennessee has genuine earlier age windows and now uses the source's Figure A5 main-result permutation intervals. All 1,000 registered synthetic normal-input repetitions have verified L3 paths after shifting each interval with its perturbed coefficient and holding interval widths fixed. These are not draws from the source permutation distribution. The source warning about underestimated clustered SEs also applies to the normal perturbations, whose scales use those displayed SEs, with a 0.0005 floor for the two bachelor's coefficients whose SEs print as zero. This floor is used in the current 1,000 repetitions and is an imposed reporting-precision convention, not a recovered sampling standard error. The interval intersection and its endpoint quantiles therefore have no demonstrated joint 95 percent interpretation. The former two- and three-SE diagnostics remain archived. Adams, Michigan, both appropriations calculations, and HOPE tax credit also retain their existing 1,000 linked-input L3 repetitions. These propagate the recorded inputs through the same conversion, time allocation, and population mappings as their point reconstructions. Those mappings remain fixed assumptions, not additional uncertainty recovered from the source draws. Tennessee Pell and Georgia HOPE remain L2 only because splitting a single entry observation by assumed duration supplies no additional source window.

F.3 Cannabis: An Education-Channel Welfare Account

The cannabis calculation values the education channel of the registered policy contrast without claiming to value legalization as a whole. Its retained bachelor's-completion effects at four, five, and six years after the source's reference event are -0.016 , -0.011 , and -0.018 . A fixed two-schooling-years-per-degree bridge converts these to -0.032 , -0.022 , and -0.036 schooling-equivalent years. The registered age mapping places them at model ages 22, 23, and 24, with age 18 as the last unaffected observation; it is a maintained mapping rather than individually observed exposure ages. L2 uses the last moment and L3 screens the earlier moments within two marginal SEs while retaining L2 exactly. The baseline uses the 14 adopting states and the 1988–1999 birth cohorts, with equal cohort weights and fixed state weights from the age-18 population. These state weights are external calibration weights, not recovered weights from the source's causal estimator. Appendix E.9 includes an equal-state point-calibration sensitivity.

Let G_i be the discounted earnings gain before the separate forgone-work adjustment, F_i the discounted schooling-acquisition flow, and $O_i = c_i F_i$ the corresponding forgone earnings, all computed from the same

candidate path. Under the fixed effective tax rate, $T_i = t_i G_i$. The education account is

$$W_i^{\text{edu}} = G_i - O_i - T_i - p_i F_i, \quad (57)$$

where p_i is private payment per additional full-time-equivalent education year. Its net public cost is

$$N_i^{\text{edu}} = g_i F_i - T_i, \quad (58)$$

where g_i is the public education-resource cost net of private payment. The main schedule uses $t_i = 0.189$, $c_i = 4,172.26$, $p_i = 5,997.02$, and $g_i = 8,557.30$ in 2000 dollars. Private payments and public resource costs come from the 2015 public four-year-college inputs in the replication of Hendren and Sprung-Keyser (2020), aggregated within states by full-time-equivalent enrollment, deflated, and combined with the same fixed geographic weights. The schedule is held fixed across the 1,000 repetitions.⁵³

The paired components, rather than a single ratio, are the central reported result for this case. Under the main cost schedule, L2 WTP ranges from approximately $-\$9.87$ thousand to $\$14.20$ thousand and net education cost from $-\$0.74$ thousand to $\$0.57$ thousand. For L3, the corresponding ranges are $[-\$4.02, \$0.51]$ thousand and $[-\$0.05, \$0.20]$ thousand. Both negative-WTP, positive-cost paths and positive-WTP, education-fiscal-surplus paths are found. Appendix E.9 shows that both types remain in the saved point banks when costs are based on all public colleges or community colleges, when the separate forgone-work charge is removed, when it enters the tax base, and when states receive equal weights. Ratios become very large near zero net cost; a finite numerical endpoint there is not a certified finite bound on the continuous ratio. The difference $W_i^{\text{edu}} - N_i^{\text{edu}} = G_i - O_i - (p_i + g_i)F_i$ supplies a complementary education-resource comparison because the tax transfer cancels.

Other legalization components remain outside this account. For example, adding a specified direct none-education cost C_i would give net cost $C_i + N_i^{\text{edu}}$, so, for a positive-benefit path, the largest such cost consistent with fiscal self-financing is $-N_i^{\text{edu}}$ and the net-benefit break-even cost is $W_i^{\text{edu}} - N_i^{\text{edu}}$. These are conditional accounting thresholds, not estimates of the actual policy budget. Cannabis excise revenue, health, crime, enforcement, and consumption value would require separate evidence before a total-policy MVPF could be reported. In particular, a fiscal surplus in the education account does not establish that legalization itself is self-financing.

The cannabis repetition results also have a narrower interpretation than a new microdata bootstrap. They extend the registered calibration using 15 equal-weight synthetic Rademacher clusters and an imposed cross-moment correlation parameter of 0.95, scaled by the retained marginal SEs. These are researcher input perturbations, not recovered cluster influence functions or a re-estimation of the earlier CBASE model. The resulting endpoint summaries in Tables 4 and F1 condition on that perturbation rule, the degree-to-schooling bridge, and the fixed education-cost schedule. Keeping this limitation explicit permits an education-related valuation while leaving unsupported noneducation components outside the calculation.

F.4 A Separate Sensitivity without Joint Input Draws

For Florida, Wisconsin, Kalamazoo, CUNY, and Social Security student benefits, Table F2 examines a simultaneous region formed from the reported marginal standard errors. For J encoded source moments, the

⁵³The Delta Cost inputs price one schooling-equivalent increment as one full-time-equivalent education year and treat average expenditure as marginal cost. Both are calibration assumptions: attainment effects do not identify enrollment sector, tuition, or forgone work. Public resource cost is total expenditure less student payment. The main tax rule uses gross earnings gains; a sensitivity deducts forgone-work costs first.

normal Bonferroni critical value is $\Phi^{-1}(1 - 0.05/(2J))$. Each moment may vary within its estimate plus or minus that value times its SE. Wisconsin’s annual credit sums and aggregate input use conservative SE upper bounds obtained by summing the relevant marginal SEs, since their covariance is unavailable. Florida includes the recovered first-semester coefficient. Under the marginal normal approximations, the union bound gives an approximate 95 percent simultaneous region for the encoded input means without requiring their covariance matrix. This approximation remains conditional on the source’s population and attrition conventions; it does not resolve missing outcomes or transport between different samples. All encoded inputs, including the L2 endpoint, vary inside the region, while ACS calibration and accounting prices remain fixed.

The table reports the extrema among paths found within this input region. A complete projection of a valid input region could support welfare-set inference under the maintained model, but a finite search need not recover that projection. The reported spans are therefore sensitivity results, not certified welfare confidence limits. They are kept outside the main input-repetition panels and the point-input ranking. This distinction permits use of the available marginal uncertainty without fabricating a joint draw distribution or assigning probabilities to continuation paths.

G Policy-Specific Male and Female Education Paths

This appendix uses published subgroup treatment effects to construct male and female continuations within seven policies. The purpose is to distinguish differences in the size of an education effect from differences in when education is acquired. No restriction presumes that women graduate later or face larger timing losses. These degree-channel exercises preserve the original subgroup outcomes; they do not replace the pooled-policy education inputs or welfare accounts used elsewhere in the paper.

G.1 Source Effects and the Reference Populations

Table G1 records the retained point inputs and the resulting standardized education valuations. The college-aid exercises use reported degree outcomes. For Adams, Table 7, Panel D, column 5 of Cohodes and Goodman (2014) reports effects on graduation from any four-year college within four years of high school. The sample contains the 2005–2008 high-school classes, rather than the 2005–2006 classes used for the pooled six-year result. The source test does not reject equality of the male and female coefficients. For STBF, Online Appendix Figure A4 of Angrist et al. (2022) reports six-year BA effects for the four-year applicant strata, with both estimates rounded to 0.08. The reported year-six gender contrast is 0.001, with a standard error of 0.005. Equal displayed coefficients therefore do not establish identical underlying estimates. The figure also reports earlier degree outcomes, but its samples differ by year. I retain the six-year inputs and do not combine the earlier values into a sex-specific L3 without an additional sample-transport assumption.

For Wisconsin, Table 7 of Goldrick-Rab et al. (2016) reports the first cohort’s administrative-sample four-year BA effect as 4.7 percentage points, with a treatment-by-male interaction of 0.0 points at the printed precision. Both subgroup inputs are therefore 0.047 in this conditional calculation. The gender level coefficient is not a treatment effect. This specification differs from the table’s smaller financial-aid sample and from the pooled credit-based account used in the collection. The table does not supply the covariance needed to recover a male subgroup standard error. For Michigan community-college tuition, Table 7, Panel B, column 7 of Acton (2021) reports the effects of in-district residence on BA completion. These are not per-dollar tuition slopes. The degree sample contains the 2009–2011 high-school classes and excludes degrees

obtained in high school; its outcomes are observed through 2017. The source test does not reject equality of the male and female BA effects. The enrollment columns use additional cohorts and do not supply an extra BA observation for the same population.

Early Head Start retains the full-exposure college-graduation ATT in Table 1.2 of Hess (2025): 0.130 for men and 0.212 for women, with source standard errors of 0.040 and 0.050. The corresponding ITT estimates are 0.009 and 0.015. The ATT inputs cannot be described as effects on the entire population exposed to county program availability. The source pools adult attainment; the reconstruction assigns age 22 as the completion endpoint and uses the main case's BA-channel calibration. I retain its national 1989–1999 cohort mixture and full-population denominator, rather than imposing the high-school-graduate restriction used for the college-aid references. Transporting a national age profile to the program's participants remains an external calibration assumption.

For Tennessee, Schoonover et al. (2026) report BA-or-higher effects of 0.012 for men and 0.041 for women at ages 23–24. The female estimate is statistically significant at the 10 percent level; the male estimate is not statistically significant. This contrast does not itself test equality between the two effects. Figure 4F supplies estimates and marginal permutation intervals for the earlier overlapping two-year age groups, 18–19 through 22–23. The reference population follows the main case's high-school-graduate denominator and prior-year Tennessee residence. The fixed 1992–1998 cohort mixture is retained, separately by sex. No pooled control mean is relabeled as a sex-specific control mean.

For Kalamazoo, Table 8 of Bartik et al. (2021) reports within-district IV effects on obtaining a BA/BS within six years after high-school graduation: 0.041 for men and 0.132 for women, with heteroskedasticity-robust standard errors of 0.070 and 0.077. The source test does not reject equality of the sex-specific BA/BS effects. The corresponding pre-Promise eligible-group means are 0.295 and 0.304. This is not the alternative between-district estimand in the source's Table 9. It is also not the attempted-credit input used in the pooled Kalamazoo Promise account. The subgroup exercise retains the reported degree outcome directly.

All calibrations use the read-only ACS 2000–2022 extract, valid person weights and education codes, and all race and birthplace categories. Means are calculated within sex-cohort-age cells, retaining fixed cohort weights that are identical across sexes within a policy. The college-aid references use high-school graduates; Early Head Start uses the full population. Geography is current residence except for Tennessee's known prior-year residence. Massachusetts birth cohorts 1987–1990 approximate the Adams high-school classes under graduation at 18, mapping four-year completion to age 22. Wisconsin's 2008 entering cohort is approximated by birth cohort 1990 and entry at 18, also mapping its four-year result to age 22. These state profiles are proxies for the respective scholarship populations, not measured control paths at the RD cutoff or among grant-eligible college entrants.

STBF retains the main case's fixed entry-age mixture for the 2012–2014 cohorts and applies the six-year degree moment at the corresponding ages 23, 24, and 25. The same age and cohort weights apply to both sexes; subgroup-specific entry-age distributions were not recovered. Michigan's 2009–2011 high-school classes are approximated by birth cohorts 1991–1993. Completion through 2017 then enters a fixed equal-weight mixture at ages 26, 25, and 24, rather than a single midpoint. Kalamazoo's 2006–2008 post-policy graduating cohorts are approximated by birth cohorts 1988–1990 under graduation at 18. Six-year follow-up is mapped to age 24, and the state's sex-specific age shape is rescaled to the source subgroup mean at that age. This transports a pre-policy eligible-group level and is not an identified untreated path for the marginal IV population. Source age distributions would permit less restrictive mappings.

Logistic reference curves are fitted separately by cohort and sex using available cohort–age means through age 35. State fits retain the national-shape shrinkage rule. Recent cohorts without sufficient later-age support borrow a fitted shape of the same sex and fit their own level. The five added exercises reuse the saved aggregate ACS cells and sex-reference fits after checking geography, denominator, cohort coverage, fit status, and monotonicity. They use 96 successful reference and late-attainment donor fits; their largest weighted root-mean-square fitting error is approximately 7.5 percentage points. Tennessee and Kalamazoo retain their existing fitted references and saved results. Figures G1, G2, and G3 display all seven policies. These external means need not equal a source study’s untreated potential outcomes. The common-reference sensitivity below removes the difference between the two transported age profiles.

G.2 Continuation Restrictions and Earlier Source Information

For subgroup $s \in \{m, f\}$, the treated continuation follows the anchored Logistic specification in Section 3. Each component equals its reference curve through the retained degree-channel anchor, is nondecreasing, and lies in $[0, 1]$. The college-aid anchors use the calibrated entry ages. Early Head Start retains the main case’s approximation of zero BA attainment through age 19; that is a degree-outcome approximation, not a claim that childhood treatment has no earlier effects. The subgroup moment restriction is

$$\mathcal{M}_i\{H_s^1 - H_s^0\} = \widehat{\delta}_{is}, \quad (59)$$

where the operator retains each policy’s component ages and weights. Tennessee averages ages 23 and 24, while the other operators use the mappings described above. The subgroup coefficients are treated as fixed point inputs, including their published rounding. Sampling uncertainty is not eliminated by this conditioning. The pooled-policy repetitions do not become subgroup draws or confidence intervals.

The geographic joint-support covariance is shared across sexes and transported to each reference center, using the same state/pooled construction as the main reconstruction. A common residual-attainment cap limits the fraction of those without a BA at the observed window who can acquire one later. The cap takes the largest residual share among the two subgroup reference curves and the corresponding 1983–1987 sex-specific donor curves. Donor attainment is evaluated using the source operator’s age weights. The cap is fixed before the candidate search and is identical for men and women within a policy. This avoids imposing a systematically more permissive continuation on either sex; it remains an empirical transport restriction, not a proven policy-response bound.

Tennessee’s earlier subgroup intervals are recovered from the embedded image of Figure 4F. I retain the image coordinates and scale, round the recovered limits outward to 0.001, and add a 0.002 outward allowance for graphical resolution. The extraction includes marker pixels if they extend beyond a very short interval, so this screen is deliberately conservative relative to the digitized plot. Earlier model moments must fall inside all five such intervals. They are not imposed as exact point estimates, converted to normal standard errors, or interpreted jointly as a 95 percent confidence region. The author’s textual last-window estimates remain the exact inputs. Both endpoint-only and screened results are saved. No other policy in this subgroup exercise receives a second degree moment by distributing one coefficient across assumed years or combining different outcomes or samples. The pooled application instead reports zero-, one-, and two-pixel inward sensitivities. The subgroup allowance deliberately avoids excluding paths because of coarse figure resolution; it does not share the pooled extraction rule or establish joint coverage. These degree-channel exercises do not alter the pooled L3 classifications.

G.3 Separating Timing from the Terminal Effect

A larger early effect for women does not by itself show a greater delay loss. To separate timing from the eventual gain, I additionally impose the same terminal BA effect and valuation prices on the two sexes. Earlier attainment can bring earnings gains forward, while the date of acquisition also affects forgone-work costs. Both enter the standardized education NPV. I use the Section 3 earnings and forgone-work schedule and fix a two-schooling-year equivalent for an incremental BA relative to some college. This conversion is an explicit standardization, not an estimate of observed schooling or the full credential mix for either sex. The resulting BA-channel account is

$$V_{is}^{BA} = 2 \sum_{a=18}^{65} D_{.03}(a; 18) [\pi_{YS}^+(a) \Delta H_{is}^{BA}(a) - c_{HSK} \Delta h_{is}^{BA}(a)] . \quad (60)$$

Prices are common across sexes and cases; values are in 2000 dollars discounted to age 18. The account excludes sex-specific wage returns, labor supply, changes in other qualifications, transfers, and fiscal costs. It therefore values a standardized education channel and does not supply a complete sex-specific MVPF. Direct summation of acquisition flows is checked against the equivalent stock-kernel calculation.

The unrestricted point-input spans in Table G1 mix timing and eventual attainment. For the timing comparison, each subgroup must additionally satisfy

$$\Delta H_{im}^{BA}(65) = \Delta H_{if}^{BA}(65) = \kappa, \quad (61)$$

for a fixed terminal-effect grid. Adams uses $\kappa \in \{-0.15, -0.10, -0.05, 0\}$, allowing the same terminal reduction or zero effect after its negative four-year inputs. The other policies use $\kappa \in \{0.02, 0.05, 0.10, 0.15\}$. These grid points are comparisons, not quantiles of a path distribution. Table G2 reports every value and the number of verified paths for both sexes. The common-reference sensitivity uses the equally weighted mean of the two subgroup reference curves within each component, retaining the source effects, age weights, support construction, residual cap and terminal grid.

Female-minus-male NPV endpoints compare independent complete paths at the same κ , common prices, and common exposure-age weights within the policy. Each endpoint retains a pair of parameterized path witnesses. A negative difference supplies a witness of lower female education NPV at that terminal effect. It would not establish that such a path occurred, that it is more likely, or that pregnancy or marriage caused it. Figure G4 summarizes the smallest and largest paired differences across each policy's tested grid. Its endpoints can come from different terminal effects, but each male–female pair shares one terminal effect. The figure does not rank policies or assign probabilities to the continuations.

Six policies have a verified equal-terminal comparison under separate reference curves at one or more grid values. Adams, STBF, Wisconsin, and Michigan community-college tuition each permit both signs of the NPV difference at an individual fixed terminal value. Early Head Start and Tennessee have positive differences in the saved comparisons. For Tennessee, when both terminal effects are five percentage points, the saved paths give women a standardized education NPV approximately \$37–\$137 higher than men's, in 2000 dollars. The common-reference sensitivity has the same direction. The positive comparisons describe the paths found and do not establish that every compatible continuation favors women. Wisconsin admits both sexes at a two-point terminal gain, while no female path is found at five points. The separate-reference Kalamazoo banks do not yield a common terminal value on the four-point grid. Under the common reference, a ten-point terminal gain is supported for both sexes, with higher female NPV in the saved comparisons.

STBF and Wisconsin, which have identical rounded point inputs, permit both NPV orderings under the common reference. Source precision, population transport, finite search, age mapping, and common-price standardization limit conclusions about subgroup welfare.

These findings motivate examining subgroup dynamics without imposing a female-delay interpretation. Marriage, childbearing, work, and college access may influence education timing, but the reconstructed paths do not identify those mechanisms. Tennessee’s authors themselves discuss delayed transitions in schooling, work, and family formation. Their evidence does not establish that marriage or childbirth caused women to delay graduation. Distinguishing that mechanism requires additional subgroup or longitudinal evidence.

H Replication Files and Numerical Reproduction

The replication package separates numerical results from manuscript presentation. Compiling the paper reads frozen tables and native TikZ/PGFPlots figures; it does not rerun path estimation or input repetitions. Full-precision inputs and complete output grids remain available when their printed summaries are condensed. Raw ACS person records are not redistributed in the Overleaf project; source manifests and aggregate fitting inputs are supplied.

H.1 Literature Coding and Simulation

Literature registers in `audit/chapter2/` retain inclusion decisions, outcome-specific timing, interpretation codes, endpoints, and evidence locations. The 31-paper valuation ledger is in `reproducibility/literature_valuation_timing/`. Source versions and PDF fingerprints distinguish working and published papers. Descriptive figure data are in `data/`; the exhibit programs aggregate the frozen coding.

The two Section 3 experiments are in `reproducibility/chapter3_two_experiments/`. Inputs include the 60 full-population ACS cohort–age means, crosswalk, fitted curves, parameter support, and valuation schedule. Its `results/` directory contains all 306 age-25 scan rows and 2,565 observation-design rows, their stipulated effects, and parameterized extrema. `experiment_engine.py --run` rebuilds searches; `summarize_and_audit.py` independently recomputes endpoints. Appendix B.7 defines the numerical checks. Figure inputs are in `data/chapter3/`; individual-scenario heatmap code remains in `figures/chapter3/`. The compact appendix tables are produced by `scripts/build_compact_appendix_tables.py` without changing numerical inputs.

H.2 Policy Paths, Input Repetitions, and Subgroups

Within `reproducibility/`, policy contracts and numerical checks are in `policy_bootstrap1000/` and `chapter5_synthesis/`. Displayed results are in the `data/` subdirectories `chapter4/bootstrap1000/` and `chapter5/`. They retain all 38,000 case–draw identifiers, including unsuccessful searches and undefined ratios, and distinguish admitted L3 from alternative specifications. Expanded moment operators, account units, geographic registries, paired witnesses, complete-path parameters, and source-interval provenance permit the printed results to be traced to their inputs. The local archive additionally retains the full frozen draw contracts and candidate banks.

The seven-policy subgroup exercise is in `reproducibility/chapter5_gender_seven/`, with the Tennessee and Kalamazoo source extraction in `reproducibility/chapter5_focus_and_gender/`. These

directories retain subgroup effects, cached ACS cells, fitted references, common valuation prices, every terminal-effect grid point, complete candidate parameters, and witnesses. Independent checks evaluate the anchored paths directly and value acquisition flows separately from the search program's stock kernel. They are point-input exercises and do not modify the pooled policy accounts or their 1,000 repetitions. The exhibit manifest records which archived displays were replaced by concise summaries and where their complete numerical content remains available.

H.3 Current Revision and Archived Results

The October 2026 revision is indexed in `data/revision_20261002/`. It supplies Wisconsin's corrected joint observation operator and before/after endpoints, all 1,000 revised L2 repetition summaries, the historical marginal-tax unit ledger, and the cell-level calibration-support register. `data/chapter5/case_registry.csv` identifies the 24 comparisons without the additional unit or account adjustments, eleven finite tax reconstructions, two aligned or extended accounts, and the education-channel calculation. The current admission file lists thirteen L3 specifications. The October 6 source-window review, five restored contracts, and validation results are in `reproducibility/l3_revision_20261006/`; Adams' converted early-window restriction is included. Rebuilding the displays uses `scripts/build_candidate_ordering.py` for the full 37-case and restricted 24-case orders and the current publication builders, not an archived ranking script. The complete local candidate arrays and fixed draw contracts remain available through the authoritative manifest and hashes. Older files under historical replication directories describe their original specifications and are not substitute inputs for the revised displays.

The portable Wisconsin revision is in `reproducibility/revision_20261002/`. It includes the corrected case contract, the compressed fixed-draw contract, the calibration library, numerical engines, and frozen point and repetition results. `python run_wisconsin.py` verify checks every released point path and paired monetary account; its README separates fresh point search from repetition refinement using released grid endpoints. Generated files are written to its ignored `computed/` directory. The revision status ledger distinguishes verified source inputs from unresolved publication-version and monetary-unit qualifications.

The current finite tax restoration is indexed in `data/tax_finite_20261002/`. It supplies the coefficient and monetary unit ledger, source-age joint operators, finite path checks, and 11,000 recomputed input-repetition endpoints. All active case contracts again use finite attainment levels. The earlier local-derivative specification is historical and is not consumed by current tables or figures. Early Head Start retains the correction aligning its monetary components to L1. `reproducibility/tax_finite_20261002/` records the workflow and input hashes; full local candidate arrays are indexed in the authoritative result manifest. The previous manuscript is recoverable at `backup/pre-finite-tax-restoration-20261002`. Figures and tables use current point and repetition results from the same finite specification.

The single-parameter policy diagnostics are indexed in `data/one_parameter/`. They supply matched parameter changes, reference and path earnings values, monetary bridges, fiscal states, exclusions, and full-precision plotting coordinates. Policy types are recorded in `data/chapter5/policy_categories.csv`. The source algorithm and frozen native figure scenes are in `reproducibility/one_parameter/`; its README identifies the local dependencies needed for fresh estimation. `python scripts/build_one_parameter_figures.py` rebuilds the six manuscript figure inputs from the frozen scenes without estimat-

ing paths. These diagnostics preserve baseline component heterogeneity and do not replace the main candidate ranges or input repetitions.

The collection-wide point-input sensitivities are in `data/all38_sensitivity/`. They contain 404 case-specification-information-set rows, the 266 cells in the common overview, and independent path and accounting checks. `reproducibility/all38_point_sensitivity/` supplies the specification plan, frozen contracts, and numerical source code; its README identifies the local calibration libraries and candidate archives required for fresh searches. `python scripts/build_all38_sensitivity.py` regenerates the native overview and numerical table from the frozen CSV files without those estimation dependencies. The local archive retains complete candidate parameters and annual witness paths for all 302 current case-specification combinations. These exercises hold source inputs fixed and do not rerun the 1,000 input repetitions.

The active ranking display reads `data/chapter5/information_rank_scatter.csv`; the older `priority_scatter.csv` supplies display names only. Current repetition summaries use `data/chapter4/bootstrap1000/bootstrap_endpoint_quantiles.csv`, including the revised tax and Early Head Start accounts. The similarly named file under `data/chapter5/` is an earlier snapshot, not the input to the current repetition exhibits.

Table A1. Literature coverage, estimation times, and observation endpoints

<i>A. Reported effect-estimation times or windows</i>						
Outcome	Full sample				Journal sample	
	Eligible	Coded	One	One (%)	One/coded	One (%)
All education outcomes	459	459	249	54.2	244/447	54.6
College entry (CE)	237	236	193	81.8	189/231	81.8
Associate's degree (AA)	40	40	32	80.0	32/39	82.1
Bachelor's degree or higher (BA+)	190	190	132	69.5	130/186	69.9

<i>B. Availability of a numerical observation endpoint</i>			
Endpoint classification	Papers	Percent	
Exact numerical endpoint	366	79.7	
Interval or one-sided bound	38	8.3	
Stage identified; clocks not aligned	46	10.0	
Endpoint missing or conflicting evidence	9	2.0	
Total	459	100.0	

Notes: Outcome groups overlap. A time or pooled window need not represent one year; one CE paper lacks a category-specific count. The journal sample excludes 11 added welfare-source papers and one author manuscript. Panel B assigns all 459 papers to one category. Attained age, individual follow-up, and policy event time remain separate clocks. Appendix A.2 gives coding rules.

Table A2. Interpretation of finite-window evidence and lifetime valuation

<i>A. Interpretation and valuation</i>				
Category	Full sample		Journal sample	
Finality or permanence statements	18		17	
Completion or window-adequacy arguments only	22		22	
Either interpretation category	40		39	
Education-based lifetime or career valuation	31		28	
Both interpretation and valuation	1		1	
At least one category	70		66	
Neither detected	389		381	
<i>B. Timing of the valued education outcome in 31 papers</i>				
Outcome clock	One window	Multiple windows	Papers	Last endpoint Range (<i>n</i>)
Attained age	5	1	6	22–54 (5)
Individual follow-up	9	11	20	1.5–11 (17)
Education stage only	4	0	4	—
Institutional event time	—	—	1	5 (1)
Total	18	12	31	

Notes: The samples contain 459 papers overall and 447 journal papers. Finality statements and completion arguments are mutually exclusive; valuation can overlap either. Neither detected refers to the reviewed version. Panel B counts comparable times for the valued education outcome. Endpoint ranges use exact values only, with counts in parentheses; institutional flows remain separate from individual follow-up. Non-exact endpoints remain in the sample.

Table B1. Generating scenarios for the observation-design experiment

Scenario	K	b	d	$\Delta H_c(25)$	$\Delta H_c(35)$	NPV
K000	13.3370	0.2575	12.5649	-0.4904	-0.5777	-24.47
K025	13.7876	0.2575	12.5649	-0.1126	-0.1328	-5.62
K050	14.2383	0.2575	12.5649	+0.2652	+0.3121	+13.22
K075	14.6889	0.2575	12.5649	+0.6430	+0.7570	+32.06
K100	15.1395	0.2575	12.5649	+1.0209	+1.2018	+50.91
b000	13.9222	0.1200	12.5649	-1.1271	-0.5345	-18.46
b025	13.9222	0.2444	12.5649	-0.0644	-0.0121	-0.67
b050	13.9222	0.3688	12.5649	+0.3361	+0.0378	+3.17
b075	13.9222	0.4932	12.5649	+0.4722	+0.0426	+4.43
b100	13.9222	0.6176	12.5649	+0.5189	+0.0430	+4.96
d000	13.9222	0.2575	8.0000	+0.0787	+0.0072	+0.74
d025	13.9222	0.2575	13.5000	-0.0295	-0.0027	-0.27
d050	13.9222	0.2575	19.0000	-0.4017	-0.0429	-3.67
d075	13.9222	0.2575	24.5000	-1.2601	-0.1990	-12.28
d100	13.9222	0.2575	30.0000	-2.1886	-0.7111	-27.49

Notes: Scenarios use the 0, 25, 50, 75, and 100 percent positions of each conditional feasible interval. Only the named generating parameter changes; all candidate parameters may vary. Paths share the age-17 anchor. NPV uses the common earnings and opportunity-cost rule, in thousands of 2000 dollars per person. Averaging fitted curves differs from evaluating the continuation at average parameters.

Table B2. Complete age-25 scans: summary across generating parameters

Generating parameter	Parameter interval	Age-25 effect interval (YS)	Scenarios	Candidate NPV width			Both signs found
				Minimum	Median	Maximum	
K	[13.337, 15.140]	[-0.490, 1.021]	102	24.40	53.02	67.40	61/102
b	[0.120, 0.618]	[-1.127, 0.519]	102	42.98	50.16	73.07	62/102
d	[8.000, 30.000]	[-2.189, 0.079]	102	59.93	67.24	89.68	102/102

Notes: Each row uses 101 equally spaced interval positions plus the reference value. Only the named generating parameter changes; all candidate parameters may vary when matching age 25. Widths are thousands of 2000 dollars per person. A “both signs” count requires NPVs below $-\$0.01$ and above $\$0.01$. Counts describe a deterministic design, not probabilities. The replication file retains all 306 scenarios and their extrema.

Table B3. Reference and single-parameter reconstructed earnings NPVs

Policy	Reference	Permanent (K)		Temporary (b)		Temporary (d)	
	NPV	ΔK	NPV	Δb (%)	NPV	Δd	NPV
Florida grant	8.63	0.371	16.77 [†]	—	—	—	—
DC tuition aid	8.95	—	—	13.7	5.71	—	—
Adams	-1.07	-0.0244	-0.98	-1.18	-0.17	0.425	-0.13
STBF	0.66	0.459	0.88	—	—	—	—
Wisconsin grant	0.59	0.146	1.20 [†]	6.47	0.10	—	—
Kalamazoo	1.10	0.416	1.32 [†]	—	—	—	—
Michigan CC	9.45	0.0343	8.91	3.55	1.66	—	—
CUNY Pell	0.56	0.0197	1.08	0.686	0.12	-0.215	0.07
Ohio Pell	2.53	—	—	—	—	—	—
Tennessee HOPE	1.31	0.0575	2.90	2.28	0.23	—	—
Tennessee Pell	-0.30	-0.0188	-0.69	-0.808	-0.06	0.246	-0.03
Georgia HOPE	2.81	0.321	6.68	—	—	—	—
Social Security	2.79	—	—	—	—	—	—
Appropriations: spending	2.70	0.0822	3.32	3.92	0.44	—	—
Early Head Start	n.a.	—	—	—	—	—	—
Texas CC	11.16	0.248	13.83 [†]	—	—	—	—
Adult Pell	2.47	0.0815	11.16	—	—	—	—
Appropriations: tuition	0.03	8.23×10^{-4}	0.03 [†]	0.0386	4.72×10^{-3} [†]	-0.0154	3.14×10^{-3} [†]
HOPE credit	5.95	0.128	12.20	6.05	1.50	—	—
HOPE ind. single: lower	6.44	1.96×10^{-4}	24.61	0.00946	-0.75	-0.00414	-1.01
HOPE dep. single: lower	-4.14	-2.72×10^{-4}	-15.57	-0.0131	-1.21	0.00575	-0.65
HOPE dep. single: upper	5.06	3.15×10^{-4}	19.06	0.0152	1.48	-0.00665	0.80
HOPE joint: lower	23.75	0.00124	89.45	0.0599	6.96	-0.0263	3.75
HOPE joint: upper	9.72	4.74×10^{-4}	36.60	0.0229	2.85	-0.0100	1.53
Tuition ded. single: lower	6.88	0.0262	17.77	2.02	1.80	—	—
Tuition ded. joint: lower	-1.66	-0.0169	-4.63	-0.611	-0.26	0.183	-0.09
HOPE / Lifetime Learning	-57.97	-0.195	-178.01	—	—	—	—
AOTC single: upper	-1.40	-6.64×10^{-5}	-5.05	-0.00364	-0.32	0.00174	-0.16
AOTC single: lower	31.04	0.00153	111.58	0.0840	6.99	-0.0405	3.52
AOTC joint: upper	-11.54	-4.39×10^{-4}	-41.50	-0.0241	-2.61	0.0115	-1.31
AOTC joint: lower	12.01	4.68×10^{-4}	43.19	0.0257	2.71	-0.0123	1.36
AOTC ind. single: lower	1.86	3.17×10^{-5}	6.76	0.00174	-0.25	-8.31×10^{-4}	-0.33
AOTC instrument	5.97	9.39×10^{-5}	20.01	0.00390	1.13	-0.00136	0.45
Tennessee Promise	6.46	Joint	7.34	—	—	—	—
College placement	n.a.	0.0545	n.c.	5.11	n.c.	—	—
Tuition ded. single: upper	0.00	0.000	0.00	0.00	0.00	0.000	0.00
Tuition ded. joint: upper	0.00	0.000	0.00	0.00	0.00	0.000	0.00

Notes: Each pair varies the named parameter family; the other two remain at H^0 values. K changes eventual attainment; b and d effects eventually vanish. ΔK and Δd are in years; Δb is the proportional speed change in percent. Tennessee Promise uses joint plateau shifts of (3.502, 4.976, 3.349) percentage points for SC+, AA+, and BA+, respectively. [†] marks the same L2 solution also satisfying L3; — denotes no admissible match. NPVs use aligned account units per dollar of reference expenditure. n.a.: missing expenditure normalizer (Early Head Start) or reference earnings projection (college placement); n.c.: matched path without a comparable pair. Figure B3 plots the comparable pairs. Appendix B.5 defines the matching procedure. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

Table C1. ACS cohort support and comparison-path fits

Birth cohort	Records	Ages	K	b	d	RMSE
1985	630,409	16–35	13.8926	0.2533	12.5107	0.112
1986	659,595	16–35	13.9037	0.2614	12.6170	0.113
1987	688,856	16–35	13.9704	0.2578	12.5669	0.115

Notes: All three cohorts contribute 20 ages in the full-population ACS extract. Fits use person-weighted cell means and normalized cell population weights. The comparison curve averages the three fits with fixed one-third weights. RMSE is in schooling years. These descriptive fits do not validate extrapolation beyond age 35.

Table D1. Source education inputs, clocks, and reconstruction roles

Education input	Effect	SE	Clock / role
<i>STBF: randomized offer, 2012–2014 four-year-college strata</i>			
Schooling-equivalent input	0.3648	0.0413	HS+6; L2 schooling
Any degree (AA+ proxy)	0.0518	0.0150	HS+6; auxiliary endpoint
Bachelor's degree	0.0813	0.0155	HS+6; auxiliary endpoint
<i>Adams: eligibility RD, 2005–2006 graduating classes</i>			
Two-year enrollment	0.0070	0.0050	HS+2; conversion
Four-year enrollment	0.0000	0.0070	HS+2; conversion
Two-year completion	0.0000	0.0040	HS+6; conversion
Four-year completion	-0.0250	0.0090	HS+6; conversion
Combined schooling (derived)	-0.0180	0.0240	HS+6; L2
Early schooling (derived)	0.0070	0.0050	HS+2; allocation sensi- tivity
<i>Tennessee Promise: age-window effects of program exposure</i>			
Some college+, ages 18–19	0.03500	0.0040	earlier-window screen
Some college+, ages 19–20	0.03200	0.0040	earlier-window screen
Some college+, ages 20–21	0.02500	0.0040	earlier-window screen
Some college+, ages 21–22	0.02700	0.0030	earlier-window screen
Some college+, ages 22–23	0.02800	0.0040	earlier-window screen
Some college+, ages 23–24	0.03500	0.0040	L2
AA+, ages 18–19	0.00200	0.0010	earlier-window screen
AA+, ages 19–20	0.01400	0.0020	earlier-window screen
AA+, ages 20–21	0.03100	0.0030	earlier-window screen
AA+, ages 21–22	0.02600	0.0030	earlier-window screen
AA+, ages 22–23	0.02900	0.0040	earlier-window screen
AA+, ages 23–24	0.04000	0.0040	L2
BA+, ages 18–19	-0.00100	0.0000 ^a	earlier-window screen
BA+, ages 19–20	0.00100	0.0000 ^a	earlier-window screen
BA+, ages 20–21	0.00001	0.0010	earlier-window screen
BA+, ages 21–22	-0.00200	0.0020	earlier-window screen
BA+, ages 22–23	0.01200	0.0030	earlier-window screen
BA+, ages 23–24	0.02700	0.0040	L2

Notes: Degree effects are in probability units; 0.01 is one percentage point. STBF schooling is converted time enrolled; Adams schooling uses retained enrollment/completion multipliers. STBF's deposited input is 0.3648, versus 0.360 printed. Tennessee coefficients and clustered SEs reproduce rounded Table 2 values; its L3 instead uses source permutation intervals (Appendix D.8). ^a A displayed-zero SE receives a 0.0005 floor in the current 1,000 normal-input repetitions as an imposed reporting-precision scale. No joint confidence region is implied.

Table D2. L1 welfare components and the earnings replacement

Component	STBF	Adams	Tennessee Promise
Gross earnings NPV	21,150.00	-1,204.04	95,557.69
Retained private remainder	32,250.00	2,329.97	2,124.31
Retained pretax fiscal cost	32,250.00	1,658.06	14,792.00
L1 WTP	49,170.00	1,364.23	72,837.00
L1 net cost	28,020.00	1,896.35	-10,053.00
Tax / transfer rate	0.2000	0.1979	0.2600

For each case: $WTP^{\text{path}} = w_0 + (1 - t)E(\Delta H)$ and $N^{\text{path}} = n_0 - tE(\Delta H)$. Transfers, tuition, and nonincome fiscal components in w_0, n_0 remain fixed.

Notes: Remainders include retained transfers and education costs. Inserting L1 earnings recovers its WTP and net cost to rounding. STBF and Adams use explicit forgone earnings in 2019 and 2004 dollars. Tennessee uses joint credential increments and a 4.883134 earnings scale, without an additional flow cost. Its component-based fiscal remainder is \$14,792 versus \$14,793 rounded in the source. Appendices D.6–D.8 detail the case-specific mappings.

Table D3. Verified paths and their paired welfare components

Path	Outcome	K	b	d	$\Delta H(65)$	WTP	Net cost	MVPF
S1	YS	14.3064	0.4287	14.9446	0.0825	39.02	30.56	1.28
S2	YS	15.3203	0.2106	12.8684	1.0961	80.46	20.20	3.98
A1	YS	14.3556	0.3491	14.2621	-0.4790	-12.67	5.36	-2.36
A2	YS	15.6534	0.1829	12.6669	0.8177	25.54	-4.07	$+\infty$
A3	YS	14.7082	0.2762	14.1597	-0.1264	-1.62	2.63	-0.62
A4	YS	14.7843	0.2554	12.7237	-0.0503	0.62	2.08	0.30
T1	SC+	0.7200	2.1047	18.5529	0.0348	-52.78	34.08	-1.55
	AA+	0.4235	2.2753	21.1605	-0.0485			
	BA+	0.3194	3.0000	20.9475	-0.0429			
T2	SC+	0.7527	0.7115	18.9492	0.0675	196.40	-53.47	$+\infty$
	AA+	0.5728	0.7691	22.1110	0.1008			
	BA+	0.4537	1.7560	22.9180	0.0914			
T3	SC+	0.7200	2.6286	18.3524	0.0348	8.04	12.71	0.63
	AA+	0.4647	1.2654	21.6065	-0.0073			
	BA+	0.3542	1.6451	22.0035	-0.0081			
T4	SC+	0.7209	1.2306	17.8067	0.0357	179.00	-47.35	$+\infty$
	AA+	0.5728	0.8891	22.3018	0.1008			
	BA+	0.4537	1.2989	22.7693	0.0914			

Notes: WTP and net cost are thousands of case-specific dollars. S1–S2, A1–A2, and T1–T2 are L2 low/high-NPV paths; A3–A4 are Adams L3 extremes under the stated temporal allocation; T3–T4 are Tennessee L3 extremes. Each Tennessee triple is one joint candidate. Paths satisfy retained moments within 2×10^{-7} and pass level, flow, and support checks. Parameters are rounded; full-precision residuals are archived. Age 65 is extrapolated; Tennessee valuation ends at 64.

Table D4. Tennessee Promise: population matching and continuation limits

Quantity	Some college+	AA+	BA+
Source control mean, ages 23–24	0.6850	0.3800	0.2920
Matched treated mean, ages 23–24	0.7200	0.4200	0.3190
Remaining-completion fraction cap	0.1402	0.2635	0.1978
Implied treated plateau ceiling	0.7593	0.5728	0.4537
Largest saved treated plateau	0.7527	0.5728	0.4537

Notes: The denominator is high-school graduates resident in Tennessee one year earlier. Source control means calibrate levels; cohort fits supply shapes. Completion caps use the largest remaining-completion fraction among five 1983–1987 donor cohorts, bounded below by the reference fraction. All candidates retain joint support, monotone levels, and credential nesting. Transporting donor behavior is an assumption; later plateaus are extrapolated.

Table D5. Tennessee Promise: source interval restrictions and complete-path predictions

Outcome	Age window	Effect	Lower	Upper	T3	T4
Some college+	18–19	3.500	0.807	6.020	0.807	1.941
	19–20	3.200	0.555	5.840	5.808	0.555
	20–21	2.500	-0.128	4.977	4.733	1.751
	21–22	2.700	-0.380	5.301	3.821	2.878
	22–23	2.800	-0.380	5.732	3.565	3.344
	23–24	3.500	0.124	6.775	3.500	3.500
AA+	18–19	0.200	-0.207	0.688	0.063	0.688
	19–20	1.400	0.613	2.366	0.613	1.470
	20–21	3.100	1.732	4.641	2.488	2.250
	21–22	2.600	0.017	5.312	5.312	2.512
	22–23	2.900	0.054	6.207	6.111	2.702
	23–24	4.000	0.539	7.513	4.000	4.000
BA+	18–19	-0.100	-0.344	0.185	-0.093	-0.044
	19–20	0.100	-0.185	0.503	-0.185	-0.185
	20–21	0.001	-0.714	0.873	0.178	-0.643
	21–22	-0.200	-2.354	2.090	2.090	-1.424
	22–23	1.200	-1.349	3.836	3.777	-0.751
	23–24	2.700	-0.397	5.847	2.700	2.700

Notes: Entries are percentage points. Effects reproduce source Table 2; intervals are digitized marginal permutation intervals from Figure A5, with a one-pixel inward guard. Ages 23–24 are matched exactly; fifteen earlier restrictions apply jointly to each candidate. T3–T4 are complete low/high-NPV paths. The lower MVPF is 0.29, 0.63, or 1.22 with zero-, one-, or two-pixel guards. Thus the comparison with one depends on extraction precision. Intersecting marginal intervals does not create a joint 95 percent region.

Table E1. Calibration populations, observed age support, and available L3 paths

Policy	Geography; birth cohorts	Outcome	Data ages	Fit	Shape cohort	L3 paths
Florida grant	Florida; 1982	YS	18–35	S	1982	4654
DC tuition aid	District of Columbia; 1981	YS	19–35	S	1981	–
Adams	Massachusetts; 1987	YS	16–35	S	1987	18
STBF	Nebraska; 1993, 1994, 1995, 1996, 1997	YS	16–29	B, S	1993, 1994	–
Wisconsin grant	Wisconsin; 1990, 1991, 1992	YS	16–32	S	1990–1992	3717
Kalamazoo	Michigan; 1988	YS	16–34	S	1988	4528
Michigan CC tuition	Michigan; 1992	YS	16–30	S	1992	97
CUNY Pell	New York; 1991	YS	16–31	S	1991	25
Ohio Pell	Ohio; 1981	YS	19–35	S	1981	–
Tennessee HOPE	Tennessee; 1989	YS	16–33	S	1989	–
Tennessee Pell	Tennessee; 1987, 1988, 1989, 1990, 1991, 1992	YS	16–35	S	1987–1992	–
Georgia HOPE	Georgia; 1975, 1976, 1977, 1978, 1979	YS	21–35	B, S	1978, 1979	–
Social Security aid	National; 1963	YS	37–59	B	1978	60
Appropriations: spending	National; 1983	YS	17–35	D	1983	453
Early Head Start	National; 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999	BA+	16–33	B, D	1989–1994	–
Texas CC tuition	Texas; 1987	YS	16–35	S	1987	688
Adult Pell	National; 1947	YS	53–65	B	1978	–
Appropriations: tuition	National; 1983	YS	17–35	D	1983	839
HOPE credit	National; 1981	YS	19–35	D	1981	72

Continued on next page

Table E1 (continued)

Policy	Geography; birth cohorts	Outcome	Data ages	Fit	Shape cohort	L3 paths
HOPE: ind. single, lower	National; 1987, 1988	YS	16–35	D	1987, 1988	–
HOPE: dep. single, lower	National; 1987, 1988	YS	16–35	D	1987, 1988	–
HOPE: dep. single, upper	National; 1987, 1988	YS	16–35	D	1987, 1988	–
HOPE: joint, lower	National; 1987, 1988	YS	16–35	D	1987, 1988	–
HOPE: joint, upper	National; 1987, 1988	YS	16–35	D	1987, 1988	–
Tuition deduction: single, lower	National; 1988	YS	16–34	D	1988	–
Tuition deduction: joint, lower	National; 1988	YS	16–34	D	1988	–
HOPE + Lifetime Learning	National; 1980	YS	20–35	D	1980	–
AOTC: dep. single, upper	National; 1991, 1992	YS	16–31	D	1991, 1992	–
AOTC: dep. single, lower	National; 1991, 1992	YS	16–31	D	1991, 1992	–
AOTC: joint, upper	National; 1991, 1992	YS	16–31	D	1991, 1992	–
AOTC: joint, lower	National; 1991, 1992	YS	16–31	D	1991, 1992	–
AOTC: ind. single, lower	National; 1991, 1992	YS	16–31	D	1991, 1992	–
AOTC: simulated instrument	National; 1987, 1988, 1989, 1990, 1991, 1992	YS	16–35	D	1987–1992	–
Tennessee Promise	Tennessee; 1992, 1993, 1994, 1995, 1996, 1997, 1998	SC+, AA+, BA+	16–30	B, S	1992–1994	46502
Course placement	New York; 1995	YS	16–27	B	1994	–
Tuition deduction: single, upper	National; 1988	YS	16–34	D	1988	–
Tuition deduction: joint, upper	National; 1988	YS	16–34	D	1988	–
Cannabis education channel	RML 14 states; 1988, 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999	YS	16–34	B, S	1988–1994	1137

Notes: Data ages span observed calibration cells, not complete coverage for every cohort. Fit modes: S, state shape shrunk toward a national fit; B, borrowed national shape with a fitted cohort level; D, direct national fit. The 1963 and 1947 cohorts are observed only at 37–59 and 53–65; their young-age profiles use the 1978 shape donor. Tennessee additionally uses source-control and target-population adjustments (Appendix D.8). YS denotes schooling-equivalent years; SC+, AA+, and BA+ are cumulative thresholds. A dash means no admitted L3. The cell-level register records each component’s support and donor.

Table E2. Source education inputs, observation windows, and priority ranks

Policy	Education input	Source observation window	L1	Own L2	Selected	Layer	Lower	Upper
HOPE: joint, lower kink [f] [†]	Current college attendance	Ages 19–20	1	26	26	L2	−∞	+∞
Tuition deduction: single, lower threshold [†]	Current college attendance	Years 0–4 from expected high-school graduation	1	26	26	L2	−∞	+∞
AOTC: dependent single, lower kink [f] [†]	Current college attendance	Ages 19–20 in 2011	1	26	26	L2	−∞	+∞
AOTC: joint, lower kink [f] [†]	Current college attendance	Ages 19–20 in 2011	1	26	26	L2	−∞	+∞
Tennessee Promise	Some college+, AA+, and BA+	Six overlapping age pairs, 18–19 through 23–24	1	16	12	L3	0.63	+∞
Algorithmic college placement [x] [†]	Earned credits	Pooled follow-up through 2018–2019; duration varies	1	26	26	L2	−∞	+∞
Tuition deduction: single, upper threshold [†]	Current college attendance	Years 0–4 from expected high-school graduation	1	26	26	L2	−∞	+∞
Texas CC tuition	Enrollment and BA completion	Enrollment through HS+6; BA at HS+6 and HS+8	8	9	2	L3	515.98	+∞
Michigan CC tuition	College entry and credentials	Entry within one year; credentials through 2017	9	19	4	L3	5.00	+∞
DC tuition assistance	College entry	Freshman entry within 12 months of high school	10	7	7	L2	3.20	+∞
HOPE: independent single, lower kink [f]	Current college attendance	Ages 19–20	11	21	21	L2	−4.07	+∞
HOPE tax credit	Enrollment and persistence	Ages 18–19; first and second college years	12	18	1	L3	+∞	+∞
HOPE: dependent single, upper kink [f]	Current college attendance	Ages 19–20	13	26	26	L2	−∞	+∞
AOTC: simulated instrument [f]	Current college attendance	Age 19 retained; age 22 reported separately	14	26	26	L2	−∞	+∞
Florida access grant	Earned credits	First semester and years 1–4 after high school	15	9	5	L3	4.67	99.62
AOTC: independent single, lower kink [f]	Current college attendance	Ages 19–20 in 2011	16	23	23	L2	−4.10	+∞
Social Security student benefits	Completed schooling years	Ages 23 and 28	17	6	6	L3	4.01	4.10
College appropriations: spending	Institution enrollment	Policy year and three lags, 1990–2013	18	18	3	L3	7.91	+∞
Georgia HOPE	Freshman entry by sector	State-year freshman flows, 1988–1997	19	17	17	L2	−1.66	+∞
Ohio Pell grants	College persistence	1999–2000 to 2000–2001	20	13	13	L2	−0.32	+∞

Table E2 (continued)

Policy	Education input	Source observation window	L1	Own L2	Selected	Layer	Lower	Upper
HOPE: joint, upper kink [f]	Current college attendance	Ages 19–20	21	25	25	L2	−4.30	+∞
Adult Pell introduction	Current college enrollment	Pooled ages 22–35	22	8	8	L2	2.95	+∞
Kalamazoo Promise	Attempted credits	Years 2, 3, and 4 after high school	23	11	11	L3	0.72	6.49
Tennessee HOPE	Two-year college entry	Initial enrollment, 2006–2009 high-school cohorts	24	19	19	L2	−3.00	+∞
STBF scholarship	Time enrolled	Six years after high school	25	10	10	L2	1.28	3.98
Wisconsin Scholars Grant	Earned credits	Semester inputs; cumulative years 2–3 after entry	26	14	9	L3	1.91	+∞
CUNY Pell grants	Earned credits	First and third years after college entry	27	20	18	L3	−1.77	−0.19
College appropriations: tuition	Institution enrollment	Policy year and three lags, 1990–2013	28	18	15	L3	−1.53	+∞
Tennessee Pell eligibility	College entry by sector	First fall after high school	29	20	20	L2	−3.06	+∞
Tuition deduction: joint, upper threshold	Current college attendance	Years 0–4 from expected high-school graduation	30	21	21	L2	−4.07	+∞
Adams Scholarship	Enrollment and AA/BA completion	Enrollment at HS+2; completion at HS+6	31	18	14	L3	−0.62	0.30
Early Head Start [r]	BA or higher attainment	Pooled adult attainment, eligible from age 22	32	16	16	L2	−1.53	1.96
AOTC: dependent single, upper kink [f] [†]	Current college attendance	Ages 19–20 in 2011	33	26	26	L2	−∞	+∞
Tuition deduction: joint, lower threshold	Current college attendance	Years 0–4 from expected high-school graduation	34	24	24	L2	−4.12	+∞
HOPE: dependent single, lower kink [f] [†]	Current college attendance	Ages 19–20	35	26	26	L2	−∞	+∞
AOTC: joint, upper kink [f] [†]	Current college attendance	Ages 19–20 in 2011	36	26	26	L2	−∞	+∞
HOPE and Lifetime Learning credits [†]	Current college enrollment	Pooled ages 18–24	37	26	26	L2	−∞	+∞

Notes: All 37 calculations with L1 are ranked. [f], [r], and [x] denote finite tax conversion, account alignment, and an earnings extension. Finite endpoints are rounded to two decimals before lower-endpoint-first ordering; equal endpoint pairs share a competition rank. Source windows need not equal assigned model ages. Undefined-ratio paths are omitted, not their calculations. Cannabis lacks L1 but remains in Table 3. The restricted 24-case order and rounding sensitivities are in the data. Ranks summarize candidates and are not social-welfare priorities.

Table E3. Reported education timing and its use in reconstruction

Cases	Windows	Source timing evidence	Current reconstruction
Florida access grant	3+	First-semester and years 1–4 credits	Reported timing retained as L3
DC tuition assistance	1	First college entry within 12 months of HS	L2 only; no added timing constraint
Adams Scholarship	3+	Entry, persistence, and BA through year 6	L3: separate early window under stated allocation
STBF scholarship	3+	Semester enrollment; BA in years 4–6	L2 only; no added timing constraint
Wisconsin Scholars Grant	3+	Semester credits; cumulative years 2/3	Reported timing retained as L3
Kalamazoo Promise	3+	Credits in years 2–4; degrees in years 4/6	Reported timing retained as L3
Michigan CC tuition	2	Entry within a year; later credentials	L3: early converted entry component
CUNY Pell grants	2	First-year and three-year credits	Reported timing retained as L3
Ohio Pell grants	1	Entry to next-year persistence	L2 only; no added timing constraint
Tennessee HOPE	1	First post-HS college enrollment	L2 only; no added timing constraint
Tennessee Pell eligibility	1	Initial-fall college-sector choice	L2 only; allocated timing excluded
Georgia HOPE	Other	State-year freshman flows	L2 only; allocated timing excluded
Social Security student benefits	2	Attainment at ages 23 and 28	Reported timing retained as L3
College appropriations	Other	Institution enrollment by policy lag	L3: lag responses under explicit time/population mapping
Early Head Start	1	Pooled adult attainment	L2 only; no added timing constraint
Texas CC tuition	3+	Annual enrollment at HS+0 through HS+6	Reported timing retained as L3
Adult Pell introduction	1	Pooled current adult enrollment	L2 only; no added timing constraint
HOPE tax credit	Other	College-grade and pooled enrollment	L3: separate first-year component under stated mapping
HOPE/AOTC income kinks	1	Contemporaneous credit-kink response	L2 only; no added timing constraint
Tuition-deduction thresholds	3+	Years 0–4 from expected HS graduation	Different operator; L2 in main plot
HOPE and Lifetime Learning credits	1	Pooled current college enrollment	L2 only; no added timing constraint
AOTC: simulated instrument	2	Attendance at ages 19 and 22	Different operator; L2 in main plot
Tennessee Promise	3+	Six age windows, 18–19 to 23–24	Reported timing retained as L3
Algorithmic college placement	2	Initial term; pooled and oldest-cohort follow-up	L2 only; no added timing constraint
Cannabis legalization: education channel	3+	BA completion within 4, 5, 6 years	Reported timing retained as L3; not ranked

Notes: Windows count source ages or follow-up periods, not coefficients. Institutional lags and college-grade contrasts are listed as Other. L3 retains L2 and adds source windows or dynamics under stated outcome, timing, and population mappings. A duration-only split of one input is not L3. Cannabis lacks L1. The timing audit records source locations.

Table E4. Point-input sensitivity: numerical comparisons

Policy	Set	Main	Narrow	Wide	National	Earnings 75%	Earnings 125%	Denser
<i>Direct student financial aid</i>								
Florida access grant	L3	[4.67, 99.62]	[5.45, 54.51]	[4.27, 216.14]	[3.46, 153.29]	[2.96, 10.16]	[7.49, > 10 ⁶] ^S	[4.67, 99.62]
DC tuition assistance	L2	[3.20, > 10 ⁶] ^S	— ^S	[−4.43, > 10 ⁶] ^S	[9.97, > 10 ⁶] ^S	[2.24, > 10 ⁶] ^S	[4.58, > 10 ⁶] ^S	[3.20, > 10 ⁶] ^S
Adams Scholarship	L3	[−0.62, 0.30]	[−0.49, 0.15]	[−0.79, 0.50]	[−0.47, 0.22]	[−0.27, 0.53]	[−0.91, 0.09]	[−0.62, 0.30]
STBF scholarship	L2	[1.28, 3.98]	[1.48, 3.05]	[1.16, 5.88]	[1.30, 4.57]	[1.20, 2.95]	[1.35, 5.38]	[1.28, 3.98]
Wisconsin Scholars Grant	L3	[1.91, > 10 ⁶] ^S	[2.09, 94.13]	[1.65, > 10 ⁶] ^S	[0.61, > 10 ⁶] ^S	[1.61, 291.26]	[2.25, > 10 ⁶] ^S	[1.91, > 10 ⁶] ^S
Kalamazoo Promise	L3	[0.72, 6.49]	[1.07, 4.23]	[0.61, 13.41]	[0.62, 15.23]	[0.74, 4.07]	[0.71, 10.96]	[0.72, 6.49]
CUNY Pell grants	L3	[−1.77, −0.19]	[−1.60, −0.52]	[−1.98, 0.36]	[−1.58, 0.45]	[−1.44, 0.02]	[−2.02, −0.39]	[−1.77, −0.19]
Ohio Pell grants	L2	[−0.32, > 10 ⁶] ^S	[3.91, > 10 ⁶] ^S	[−1.83, > 10 ⁶] ^S	[0.12, > 10 ⁶] ^S	[−0.15, > 10 ⁶] ^S	[−0.48, > 10 ⁶] ^S	[−0.32, > 10 ⁶] ^S
Tennessee HOPE	L2	[−3.00, > 10 ⁶] ^S	[−2.72, > 10 ⁶] ^S	[−3.20, > 10 ⁶] ^S	[−3.00, > 10 ⁶] ^S	[−2.79, > 10 ⁶] ^S	[−3.13, > 10 ⁶] ^S	[−3.00, > 10 ⁶] ^S
Tennessee Pell eligibility	L2	[−3.06, > 10 ⁶] ^S	[−2.81, > 10 ⁶] ^S	[−3.25, > 10 ⁶] ^S	[−3.09, > 10 ⁶] ^S	[−2.84, > 10 ⁶] ^S	[−3.20, > 10 ⁶] ^S	[−3.06, > 10 ⁶] ^S
Georgia HOPE	L2	[−1.66, > 10 ⁶] ^S	[0.30, 122.03]	[−3.58, > 10 ⁶] ^S	[−1.98, > 10 ⁶] ^S	[−1.20, 14.44]	[−1.99, > 10 ⁶] ^S	[−1.66, > 10 ⁶] ^S
Social Security student benefits	L3	[4.01, 4.10]	[4.04, 4.08]	[3.98, 4.13]	=	[2.79, 2.84]	[5.79, 5.97]	[4.01, 4.10]
Adult Pell introduction	L2	[2.95, > 10 ⁶] ^S	[9.46, > 10 ⁶] ^S	[0.28, > 10 ⁶] ^S	=	[2.06, > 10 ⁶] ^S	[4.20, > 10 ⁶] ^S	[2.95, > 10 ⁶] ^S
Tennessee Promise	L3	[0.63, > 10 ⁶] ^S	[0.64, > 10 ⁶] ^S	[0.63, > 10 ⁶] ^S	[−0.01, > 10 ⁶] ^S	[0.50, > 10 ⁶] ^S	[0.78, > 10 ⁶] ^S	[0.63, > 10 ⁶] ^S
<i>Education tax benefits</i>								
HOPE tax credit	L3	— ^S	— ^S	— ^S	=	— ^S	— ^S	— ^S
HOPE: independent single, start	L2	[−4.07, > 10 ⁶] ^S	[−4.07, > 10 ⁶] ^S	[−4.07, > 10 ⁶] ^S	=	[−4.07, > 10 ⁶] ^S	[−4.07, > 10 ⁶] ^S	[−4.07, > 10 ⁶] ^S
HOPE: dependent single, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	=	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: dependent single, end	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	=	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: joint, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[−95.92, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	=	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: joint, end	L2	[−4.30, > 10 ⁶] ^S	[−4.30, > 10 ⁶] ^S	[−4.34, > 10 ⁶] ^S	=	[−4.30, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−4.30, > 10 ⁶] ^S
Tuition deduction: single, start	L2	[< −10 ⁶ , −4.43] ^S	[< −10 ⁶ , −4.67] ^S	[< −10 ⁶ , −4.29] ^S	=	[< −10 ⁶ , −4.60] ^S	[< −10 ⁶ , −4.34] ^S	[< −10 ⁶ , −4.43] ^S
Tuition deduction: joint, start	L2	[−4.12, > 10 ⁶] ^S	[−4.06, > 10 ⁶] ^S	[−4.17, > 10 ⁶] ^S	=	[−4.07, > 10 ⁶] ^S	[−4.16, > 10 ⁶] ^S	[−4.12, > 10 ⁶] ^S
HOPE and Lifetime Learning	L2	[< −10 ⁶ , −4.36] ^S	[< −10 ⁶ , −4.43] ^S	[< −10 ⁶ , −4.29] ^S	=	[< −10 ⁶ , −4.51] ^S	[< −10 ⁶ , −4.28] ^S	[< −10 ⁶ , −4.36] ^S
AOTC: single, end	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	=	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: single, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	=	[< −10 ⁶ , > 10 ⁶] ^S	[−24.76, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: joint, end	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	=	[< −10 ⁶ , > 10 ⁶] ^S	[−1,000, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: joint, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	=	[< −10 ⁶ , > 10 ⁶] ^S	[−42.18, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: independent single, start	L2	[−4.10, > 10 ⁶] ^S	[−4.10, > 10 ⁶] ^S	[−4.10, > 10 ⁶] ^S	=	[−4.10, 1,000] ^S	[−4.10, > 10 ⁶] ^S	[−4.10, > 10 ⁶] ^S
AOTC: simulated instrument	L2	[< −10 ⁶ , > 10 ⁶] ^S	[−4.57, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	=	[< −10 ⁶ , > 10 ⁶] ^S	[−13.97, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
Tuition deduction: single, end	L2	[< −10 ⁶ , −4.02] ^S	[< −10 ⁶ , −4.05] ^S	[< −10 ⁶ , −3.99] ^S	=	[< −10 ⁶ , −4.04] ^S	[< −10 ⁶ , −4.00] ^S	[< −10 ⁶ , −4.02] ^S
Tuition deduction: joint, end	L2	[−4.07, > 10 ⁶] ^S	[−3.99, > 10 ⁶] ^S	[−4.13, > 10 ⁶] ^S	=	[−4.00, > 10 ⁶] ^S	[−4.11, > 10 ⁶] ^S	[−4.07, > 10 ⁶] ^S
<i>Tuition and public-college funding</i>								
Michigan CC tuition	L3	[5.00, > 10 ⁶] ^S	[8.34, > 10 ⁶] ^S	[2.41, > 10 ⁶] ^S	[2.42, > 10 ⁶] ^S	[2.92, 95.74]	[8.93, > 10 ⁶] ^S	[5.00, > 10 ⁶] ^S
College appropriations: spending	L3	[7.91, > 10 ⁶] ^S	[11.23, > 10 ⁶] ^S	[4.94, > 10 ⁶] ^S	=	[4.60, > 10 ⁶] ^S	[15.35, > 10 ⁶] ^S	[7.91, > 10 ⁶] ^S
Texas CC tuition	L3	[515.98, > 10 ⁶] ^S	— ^S	[57.82, > 10 ⁶] ^S	[49.80, > 10 ⁶] ^S	[11.91, > 10 ⁶] ^S	— ^S	[515.98, > 10 ⁶] ^S

Table E4 (continued): common sensitivity checks

Policy	Set	Main	Narrow	Wide	National	Earnings 75%	Earnings 125%	Denser
College appropriations: tuition <i>College assessment and course placement</i>	L3	$[-1.53, > 10^6]^S$	$[-1.40, > 10^6]^S$	$[-1.71, > 10^6]^S$	=	$[-1.17, 351.01]$	$[-1.81, > 10^6]^S$	$[-1.53, > 10^6]^S$
Algorithmic college placement <i>Early childhood services</i>	L2	$[< -10^6, -3.94]^S$	$[< -10^6, -3.96]^S$	$[< -10^6, -3.92]^S$	$[< -10^6, -3.94]^S$	$[< -10^6, -3.95]^S$	$[< -10^6, -3.93]^S$	$[< -10^6, -3.94]^S$
Early Head Start <i>Education channel of other policies</i>	L2	$[-1.53, 1.96]$	$[-0.94, 0.63]$	$[-1.57, 5.10]$	=	$[-1.45, 0.80]$	$[-1.61, 3.65]$	$[-1.53, 1.96]$
Cannabis: education channel	L3	$[< -10^6, -20.31]^S$	$[< -10^6, -20.35]^S$	$[< -10^6, -20.24]^S$	$[< -10^6, -19.82]^S$	$-^S$	$[< -10^6, -10.35]^S$	$[< -10^6, -20.31]^S$

Notes: Brackets give the span of positive-cost ratios among verified candidates. Superscript S marks self-financing separately. $> 10^6$ and $< -10^6$ indicate attained finite values beyond those display thresholds, not certified infinite bounds. = means that the main baseline is already national. Narrow and wide multiply the support covariance by 0.5 and 2 and timing deviations by their square roots. Income changes scale the earnings and associated fiscal channels; fixed transfers and education-resource components are held fixed. Each row uses the same point-input specification across columns.

Table E4. Point-input sensitivity: numerical comparisons (continued)

Policy	Specific change	Set	WTP (1,000)	Net cost (1,000)	Positive-cost MVPF
Adams Scholarship	No separate forgone-work charge	L3	[−1.95, 0.48]	[2.12, 2.71]	[−0.72, 0.22]
STBF scholarship	No separate forgone-work charge	L2	[39.76, 84.54]	[19.18, 30.37]	[1.31, 4.41]
STBF scholarship	Printed schooling input (0.360)	L2	[38.75, 80.25]	[20.25, 30.62]	[1.27, 3.96]
Tennessee Promise	Pooled late-completion allowance	L3	[8.04, 303.99]	[−91.27, 12.71]	[0.63, > 10 ⁶] ^S
Tennessee Promise	Zero inward pixels	L3	[4.03, 179.07]	[−47.38, 14.12]	[0.29, > 10 ⁶] ^S
Tennessee Promise	Two inward pixels	L3	[13.26, 178.92]	[−47.32, 10.88]	[1.22, > 10 ⁶] ^S
Tennessee Promise	No source-control level scaling	L3	[7.64, 174.44]	[−45.75, 12.85]	[0.59, > 10 ⁶] ^S
Tennessee Promise	No reference earnings multiplier	L3	[3.34, 38.35]	[2.07, 14.37]	[0.23, 18.56]
HOPE: independent single, start	Credit dose: 0.50 dollars	L2	[−49.51, 100.83]	[−24.72, 12.16]	[−1,000, > 10 ⁶] ^S
HOPE: independent single, start	Credit dose: 2 dollars	L2	[−49.49, 100.87]	[−24.72, 12.16]	[−4.07, > 10 ⁶] ^S
HOPE: dependent single, start	Credit dose: 0.50 dollars	L2	[−20.56, 39.43]	[−9.86, 5.13]	[< −10 ⁶ , 16.35] ^S
HOPE: dependent single, start	Credit dose: 2 dollars	L2	[−20.57, 39.41]	[−9.85, 5.13]	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: dependent single, end	Credit dose: 0.50 dollars	L2	[−21.79, 41.82]	[−10.46, 5.43]	[< −10 ⁶ , 1,000] ^S
HOPE: dependent single, end	Credit dose: 2 dollars	L2	[−21.77, 41.85]	[−10.46, 5.43]	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: joint, start	Credit dose: 0.50 dollars	L2	[−26.27, 50.51]	[−11.77, 6.11]	[< −10 ⁶ , 8.53] ^S
HOPE: joint, start	Credit dose: 2 dollars	L2	[−26.18, 50.64]	[−11.80, 6.09]	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: joint, end	Credit dose: 0.50 dollars	L2	[−28.16, 54.09]	[−12.60, 6.56]	[< −10 ⁶ , 1,000] ^S
HOPE: joint, end	Credit dose: 2 dollars	L2	[−28.12, 54.14]	[−12.61, 6.56]	[−4.30, > 10 ⁶] ^S
AOTC: single, end	Credit dose: 0.50 dollars	L2	[−27.11, 54.98]	[−12.81, 6.29]	[−19.02, > 10 ⁶] ^S
AOTC: single, end	Credit dose: 2 dollars	L2	[−27.12, 54.98]	[−12.81, 6.29]	[< −10 ⁶ , 17.97] ^S
AOTC: single, start	Credit dose: 0.50 dollars	L2	[−25.97, 52.79]	[−12.30, 6.02]	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: single, start	Credit dose: 2 dollars	L2	[−25.86, 52.95]	[−12.33, 6.00]	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: joint, end	Credit dose: 0.50 dollars	L2	[−33.85, 68.64]	[−15.68, 7.72]	[−26.71, > 10 ⁶] ^S
AOTC: joint, end	Credit dose: 2 dollars	L2	[−33.89, 68.58]	[−15.66, 7.73]	[−5.31, > 10 ⁶] ^S
AOTC: joint, start	Credit dose: 0.50 dollars	L2	[−32.98, 66.96]	[−15.30, 7.52]	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: joint, start	Credit dose: 2 dollars	L2	[−32.93, 67.02]	[−15.31, 7.51]	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: independent single, start	Credit dose: 0.50 dollars	L2	[−81.29, 176.11]	[−42.93, 19.82]	[−4.10, 1,001] ^S
AOTC: independent single, start	Credit dose: 2 dollars	L2	[−81.28, 176.12]	[−42.93, 19.82]	[−4.10, > 10 ⁶] ^S
AOTC: simulated instrument	Credit dose: 0.50 dollars	L2	[−37.08, 67.55]	[−15.81, 8.70]	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: simulated instrument	Credit dose: 2 dollars	L2	[−37.07, 67.57]	[−15.81, 8.70]	[−4.27, > 10 ⁶] ^S
Early Head Start	Dissertation net-cost reference	L2	[−25.56, 17.13]	[10.28, 18.26]	[−1.40, 1.67]
Cannabis: education channel	All-public-college costs	L3	[−4.32, 0.54]	[−0.06, 0.27]	[< −10 ⁶ , −16.11] ^S
Cannabis: education channel	Community-college costs	L3	[−4.68, 0.57]	[−0.07, 0.33]	[< −10 ⁶ , −13.91] ^S
Cannabis: education channel	Equal state weights	L3	[−4.46, 0.21]	[−0.04, 0.35]	[< −10 ⁶ , −12.76] ^S
Cannabis: education channel	No separate forgone-work charge	L3	[−4.52, 0.55]	[−0.05, 0.20]	[< −10 ⁶ , −22.82] ^S
Cannabis: education channel	Tax on net earnings	L3	[−4.12, 0.52]	[−0.04, 0.10]	[< −10 ⁶ , −39.64] ^S

Notes: Monetary components retain each case's source-account currency year and policy unit. Each component span is computed across paired paths; separate extrema must not be combined to form an MVPF. Tax-dose rows describe finite credit changes and can be small in absolute dollars. Adams L3 uses an explicit time allocation of the separately reported early entry input. The data file reports unrounded NPV, WTP, cost, ratios, and all L2/L3 results.

Table F1. Willingness to pay and net fiscal cost: fixed inputs and input repetitions

Policy	Layer	WTP: point	WTP: repetitions	Net cost: point	Net cost: repetitions
Florida grant	L2	[1.37, 29.32]	[-8.32, 51.26]	[0.29, 0.97]	[-4.89, 5.97]
Florida grant	L3	[4.13, 29.32]	NR	[0.29, 0.88]	NR
DC tuition aid	L2	[3.14, 51.92]	[0.77, 88.34]	[-4.68, 0.98]	[-13.26, 5.06]
Adams	L2	[-12.67, 25.54]	[-15.11, 28.15]	[-4.07, 5.36]	[-4.41, 5.69]
Adams	L3	[-1.62, 0.62]	[-6.47, 6.09]	[2.08, 2.63]	[1.02, 3.54]
STBF	L2	[39.02, 80.46]	[34.85, 83.88]	[20.20, 30.56]	[19.33, 31.58]
Wisconsin grant	L2	[-12.18, 42.94]	[-27.38, 69.23]	[-2.19, 11.59]	[-8.82, 15.18]
Wisconsin grant	L3	[11.05, 42.94]	NR	[-2.19, 5.78]	NR
Kalamazoo	L2	[14.60, 58.90]	[-3.38, 93.45]	[9.08, 20.15]	[1.36, 23.56]
Kalamazoo	L3	[14.60, 58.90]	NR	[9.08, 20.15]	NR
Michigan CC tuition	L2	[-7.73, 19.17]	[-15.45, 34.51]	[-4.54, 2.56]	[-8.35, 4.29]
Michigan CC tuition	L3	[1.13, 2.52]	[0.32, 4.59]	[-0.14, 0.23]	[-0.79, 0.66]
CUNY Pell	L2	[-24.67, 38.47]	[-46.87, 65.99]	[-8.80, 7.93]	[-15.45, 12.83]
CUNY Pell	L3	[-4.62, -0.28]	NR	[1.47, 2.62]	NR
Ohio Pell	L2	[-0.68, 32.70]	[-2.68, 56.04]	[-6.74, 2.10]	[-12.76, 2.54]
Tennessee HOPE	L2	[-21.39, 37.10]	[-37.61, 63.75]	[-8.36, 7.14]	[-15.07, 11.28]
Tennessee Pell	L2	[-22.69, 36.69]	[-33.75, 44.52]	[-7.74, 7.43]	[-8.77, 9.14]
Georgia HOPE	L2	[-3.89, 31.61]	[-5.52, 32.98]	[-0.36, 2.35]	[-0.53, 2.44]
Social Security aid	L2	[29.96, 44.45]	[2.73, 80.54]*	[4.65, 8.46]	[-1.35, 13.41]*
Social Security aid	L3	[31.90, 32.26]	NR	[7.86, 7.96]	NR
Appropriations: spending	L2	[-10.77, 28.35]	[-19.78, 48.39]	[-5.42, 4.23]	[-10.08, 6.27]
Appropriations: spending	L3	[4.22, 11.06]	[1.22, 19.28]	[-1.15, 0.53]	[-3.08, 1.14]
Early Head Start	L2	[-25.56, 17.13]	[-36.25, 17.19]	[8.74, 16.73]	[8.73, 18.71]
Texas CC tuition	L2	[2.90, 29.04]	[-4.54, 36.23]	[-5.13, 1.71]	[-6.55, 2.52]
Texas CC tuition	L3	[9.36, 13.94]	[1.45, 21.42]	[-1.18, 0.02]	[-2.17, 0.93]
Adult Pell	L2	[0.28, 1.79]	[-0.21, 2.43]	[-0.35, 0.10]	[-0.55, 0.24]
Appropriations: tuition	L2	[-13.01, 24.99]	[-24.14, 43.17]	[-4.92, 4.46]	[-9.03, 6.95]
Appropriations: tuition	L3	[-3.08, 6.35]	[-10.42, 14.93]	[-0.32, 2.01]	[-2.18, 3.47]
HOPE credit	L2	[-7.35, 23.91]	[-13.68, 42.30]	[-5.23, 2.59]	[-9.83, 4.06]
HOPE credit	L3	[8.22, 9.93]	[0.54, 22.27]	[-1.74, -1.31]	[-4.69, 0.49]
HOPE: ind. single, lower	L2	[-49.51, 100.85]	[-87.59, 175.33]	[-24.72, 12.16]	[-43.33, 21.40]
HOPE: dep. single, lower	L2	[-20.56, 39.43]	[-34.27, 65.55]	[-9.86, 5.13]	[-16.42, 8.53]
HOPE: dep. single, upper	L2	[-21.78, 41.83]	[-36.28, 69.61]	[-10.46, 5.43]	[-17.42, 9.02]
HOPE: joint, lower	L2	[-26.24, 50.55]	[-43.71, 84.27]	[-11.78, 6.10]	[-19.65, 10.13]
HOPE: joint, upper	L2	[-28.15, 54.11]	[-46.88, 90.07]	[-12.61, 6.56]	[-21.01, 10.88]
Tuition deduction: single, lower	L2	[-13.61, 27.59]	[-23.01, 46.13]	[-7.24, 3.07]	[-11.93, 5.36]
Tuition deduction: joint, lower	L2	[-41.45, 67.32]	[-70.29, 111.99]	[-15.29, 10.05]	[-25.97, 16.64]
HOPE + Lifetime Learning	L2	[-14.40, 6.31]	[-27.72, 21.62]	[-1.88, 3.30]	[-5.40, 6.30]
AOTC: dep. single, upper	L2	[-27.11, 54.98]	[-45.23, 91.45]	[-12.81, 6.29]	[-21.32, 10.45]
AOTC: dep. single, lower	L2	[-25.93, 52.84]	[-43.22, 87.63]	[-12.31, 6.02]	[-20.51, 10.01]

Continued on next page

Table F1 (continued)

Policy	Layer	WTP: point	WTP: repetitions	Net cost: point	Net cost: repetitions
AOTC: joint, upper	L2	[−33.87, 68.62]	[−56.47, 114.05]	[−15.68, 7.72]	[−26.11, 12.85]
AOTC: joint, lower	L2	[−32.96, 66.98]	[−54.96, 111.31]	[−15.30, 7.51]	[−25.48, 12.51]
AOTC: ind. single, lower	L2	[−81.29, 176.12]	[−144.51, 307.06]	[−42.93, 19.82]	[−75.44, 35.07]
AOTC: simulated instrument	L2	[−37.07, 67.56]	[−61.74, 112.35]	[−15.81, 8.70]	[−26.38, 14.45]
Tennessee Promise	L2	[−52.78, 196.40]	[−66.32, 206.02]	[−53.47, 34.08]	[−56.86, 38.80]
Tennessee Promise	L3	[8.04, 179.00]	[−11.41, 189.29]	[−47.35, 12.71]	[−51.02, 19.52]
Course placement	L2	[−10.36, 21.89]	[−12.22, 24.93]	[−5.68, 2.63]	[−6.46, 3.11]
Tuition deduction: single, upper	L2	[−31.56, 53.12]	[−53.45, 88.50]	[−13.51, 7.86]	[−22.60, 13.20]
Tuition deduction: joint, upper	L2	[−40.67, 68.77]	[−68.95, 114.55]	[−15.51, 9.99]	[−26.29, 16.51]
Cannabis education channel	L2	[−9.87, 14.20]	[−10.47, 14.91]	[−0.74, 0.57]	[−0.77, 0.59]
Cannabis education channel	L3	[−4.02, 0.51]	[−4.73, 1.88]	[−0.05, 0.20]	[−0.12, 0.23]

Notes: All components are in thousands of the retained case-specific dollars. Monetary years, populations, and policy units differ across cases. Point entries are finite-bank extrema; repetition entries summarize input-draw endpoint quantiles. WTP and cost extrema can use different paths and must not be divided to form an MVPE. Stars mark verified subsets, including 952 L2 draws in Social Security student benefits. Texas CC tuition has 1,000 verified component repetitions but only 999 with a defined ratio. Florida access grant, Wisconsin Scholars Grant, Kalamazoo Promise, CUNY Pell grants, and Social Security student benefits L3 lack joint draws; Tennessee Promise L3 uses all 1,000 registered normal-input perturbations with fixed source-permutation interval widths; these are not source permutation draws. All results condition on ACS calibration and the retained accounting. No probability is assigned to a path.

Table F2. Sensitivity to simultaneous marginal input regions without joint draws

Policy	Moments	Critical value	WTP	Net cost	Defined MVPF span
Florida grant	5	2.576	[−9.48, 35.26]	[0.16, 1.17]	[−8.11, 217.86]
Wisconsin grant	9	2.773	[−10.97, 48.15]	[−3.49, 11.29]	[−0.97, +∞]
Kalamazoo	3	2.394	[−5.63, 67.86]	[6.84, 25.21]	[−0.22, 9.93]
CUNY Pell	2	2.241	[−33.08, 44.90]	[−10.50, 10.15]	[−3.26, +∞]
Social Security aid	2	2.241	[−1.64, 61.99]	[0.03, 16.79]	[−0.10, 2.00×10^3]

Notes: For J encoded source moments, the Bonferroni critical value is $\Phi^{-1}(1 - 0.05/(2J))$. Marginal normal approximations give an approximate simultaneous 95 percent region for input means without their covariance matrix. Wisconsin's summed-credit and aggregate SE bounds sum the relevant marginal SEs; Florida includes its first-semester coefficient. All encoded inputs, including L2, vary inside the rectangle; ACS and accounting prices remain fixed. Entries are extrema among paths found, not certified welfare confidence limits. WTP and net cost are thousands of case-specific dollars. This sensitivity is separate from input repetitions and ranking.

Table G1. Published male and female degree effects and standardized education NPVs

Policy	Sex	Effect (pp)	SE (pp)	Paths	NPV span (\$1,000)
Adams Scholarship	Men	-0.8	0.9	2,325	[-14.73, 0.97]
	Women	-1.4	1.0	1,913	[-15.44, -0.60]
STBF scholarship	Men	8.0	–	5,173	[3.81, 12.14]
	Women	8.0	–	5,232	[4.06, 11.42]
Wisconsin Scholars Grant	Men	4.7	–	2,145	[-13.71, 7.71]
	Women	4.7	–	2,108	[-11.93, 2.96]
Kalamazoo Promise	Men	4.1	7.0	5,642	[-3.79, 6.38]
	Women	13.2	7.7	5,436	[7.74, 15.73]
Michigan CC tuition	Men	1.7	1.0	7,737	[-1.56, 6.26]
	Women	1.9	1.1	7,040	[-2.35, 4.71]
Early Head Start	Men	13.0	4.0	1,063	[-5.55, 11.34]
	Women	21.2	5.0	946	[-1.58, 12.81]
Tennessee Promise	Men	1.2	–	2,888	[-2.13, 10.19]
	Women	4.1	–	902	[-1.80, 6.77]

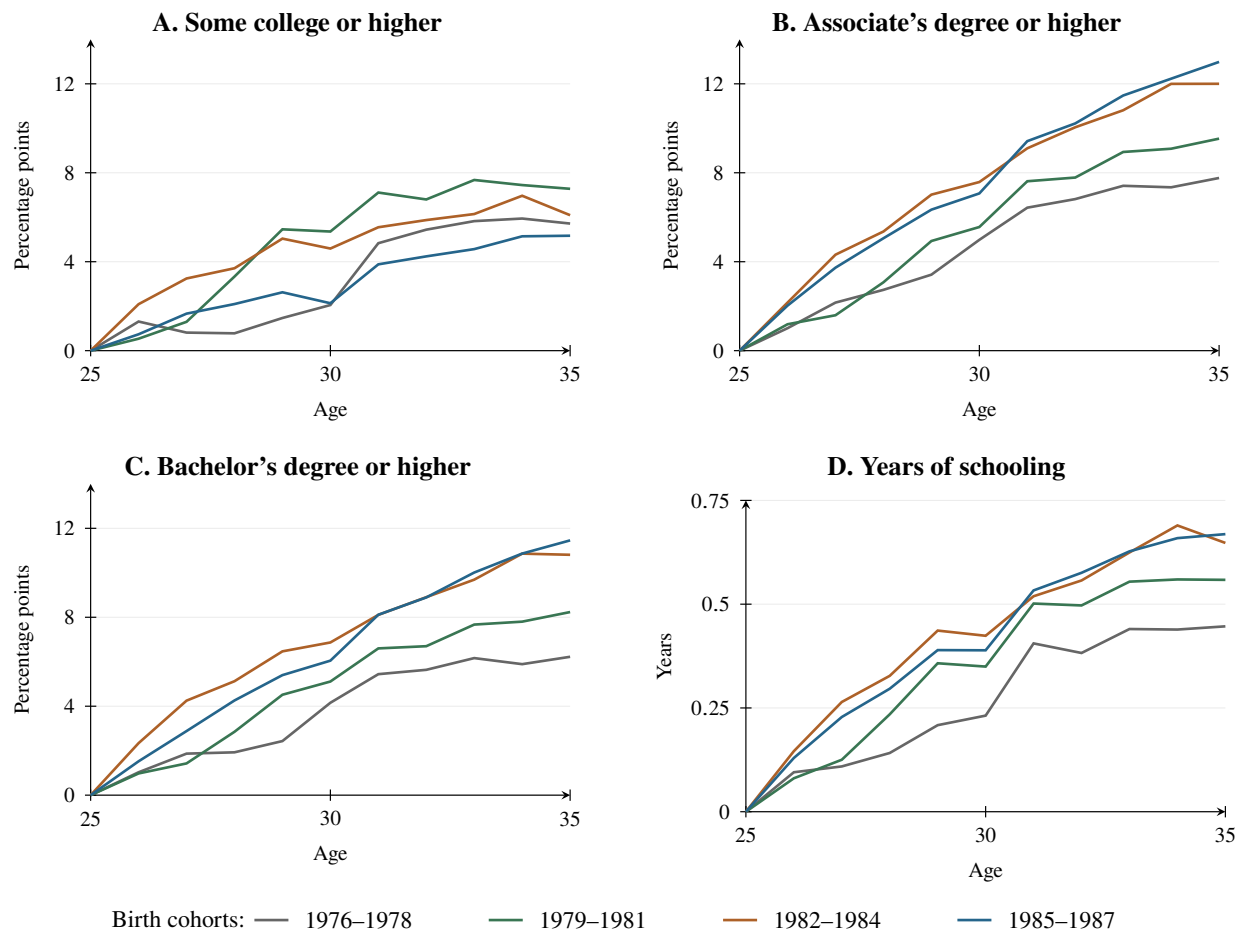
Notes: Effects and available source SEs are percentage points; dashes denote unrecovered SEs. Tennessee retains its earlier interval screen. Rounded STBF and Wisconsin effects need not be exactly equal. NPV spans combine timing and terminal-attainment variation. Valuation uses two schooling years per BA increment, common earnings/forgone-work prices, and 3 percent discounting to age 18 in 2000 dollars. These conditional education-channel values exclude transfers and fiscal costs. Appendix G.1 documents inputs.

Table G2. Male and female education NPVs at the same terminal effect

Policy	κ (pp)	Sex-specific reference curves		Common reference curve	
		Paths (M/F)	$V_f - V_m$ (\$)	Paths (M/F)	$V_f - V_m$ (\$)
Adams Scholarship	-15	32/64	[-267, -139]	43/42	[-69, 34]
	-10	62/64	[-328, -3]	44/46	[-158, 118]
	-5	93/54	[-407, 212]	69/75	[-284, 238]
	0	122/0	–	95/0	–
STBF scholarship	2	0/0	–	0/0	–
	5	71/72	[-290, 366]	72/72	[-331, 331]
	10	66/70	[-307, 273]	67/67	[-287, 287]
	15	0/0	–	0/0	–
Wisconsin Scholars Grant	2	74/83	[-64, 509]	84/84	[-249, 249]
	5	90/0	–	98/98	[-336, 336]
	10	0/0	–	0/0	–
	15	0/0	–	0/0	–
Kalamazoo Promise	2	25/0	–	24/0	–
	5	26/0	–	26/0	–
	10	0/19	–	25/20	[311, 676]
	15	0/18	–	0/19	–
Michigan CC tuition	2	63/65	[-298, 181]	63/64	[-223, 254]
	5	58/60	[-241, 124]	59/60	[-168, 199]
	10	0/0	–	0/0	–
	15	0/0	–	0/0	–
Early Head Start	2	45/43	[11, 88]	49/33	[112, 176]
	5	45/51	[10, 81]	50/44	[112, 170]
	10	41/59	[3, 110]	48/52	[110, 202]
	15	0/0	–	0/53	–
Tennessee Promise	2	21/11	[32, 163]	13/14	[44, 164]
	5	22/11	[37, 137]	13/16	[63, 162]
	10	21/0	–	0/17	–
	15	0/0	–	0/0	–

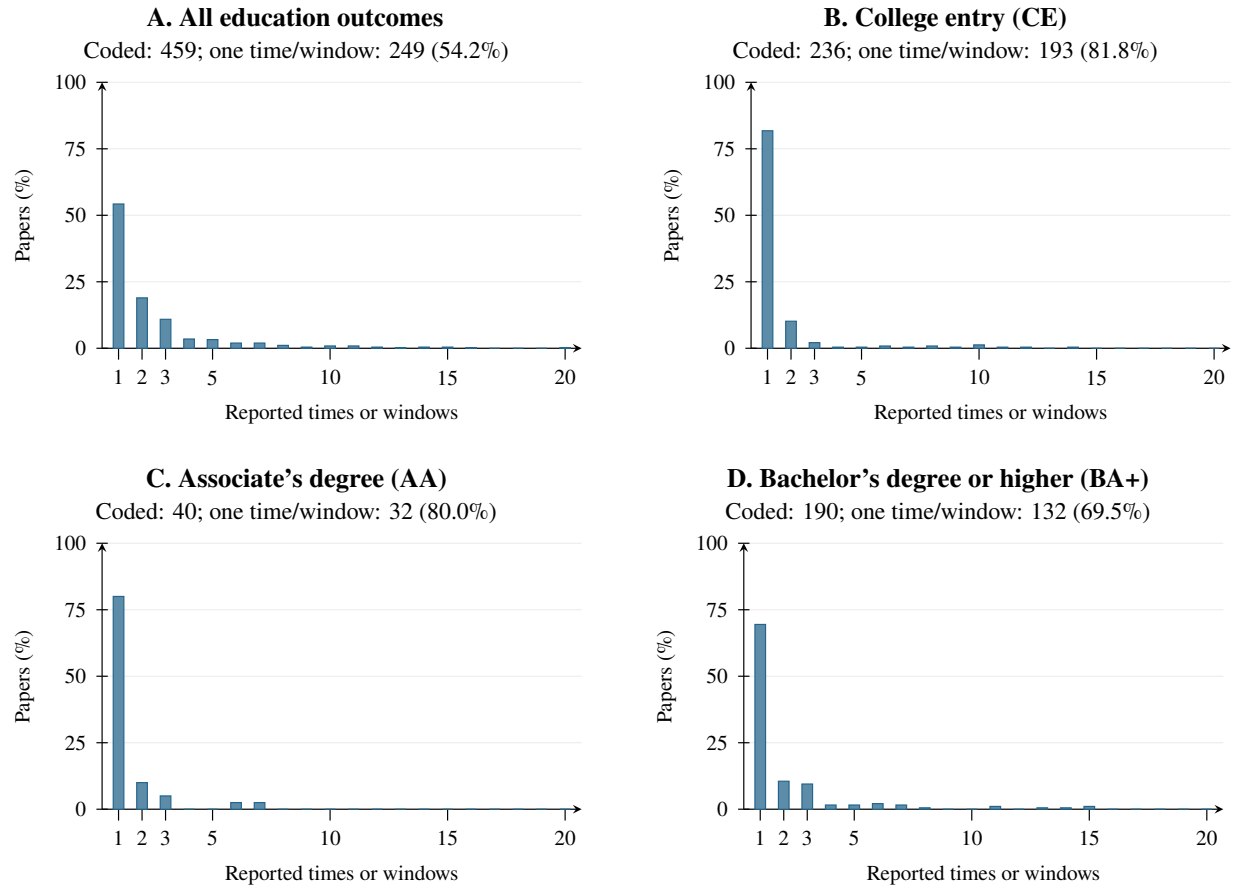
Notes: Pairs match sex-specific source effects and the same age-65 BA effect κ , at common prices. The common-reference check uses the mean of male and female baselines. Counts are verified paths; spans are female-minus-male NPV extrema, not confidence intervals. A dash means no verified pair, not proven incompatibility. Tennessee retains its interval screen. Comparisons are within policies; populations and estimands differ across policies.

Figure A1. Education growth after age 25 across birth cohorts



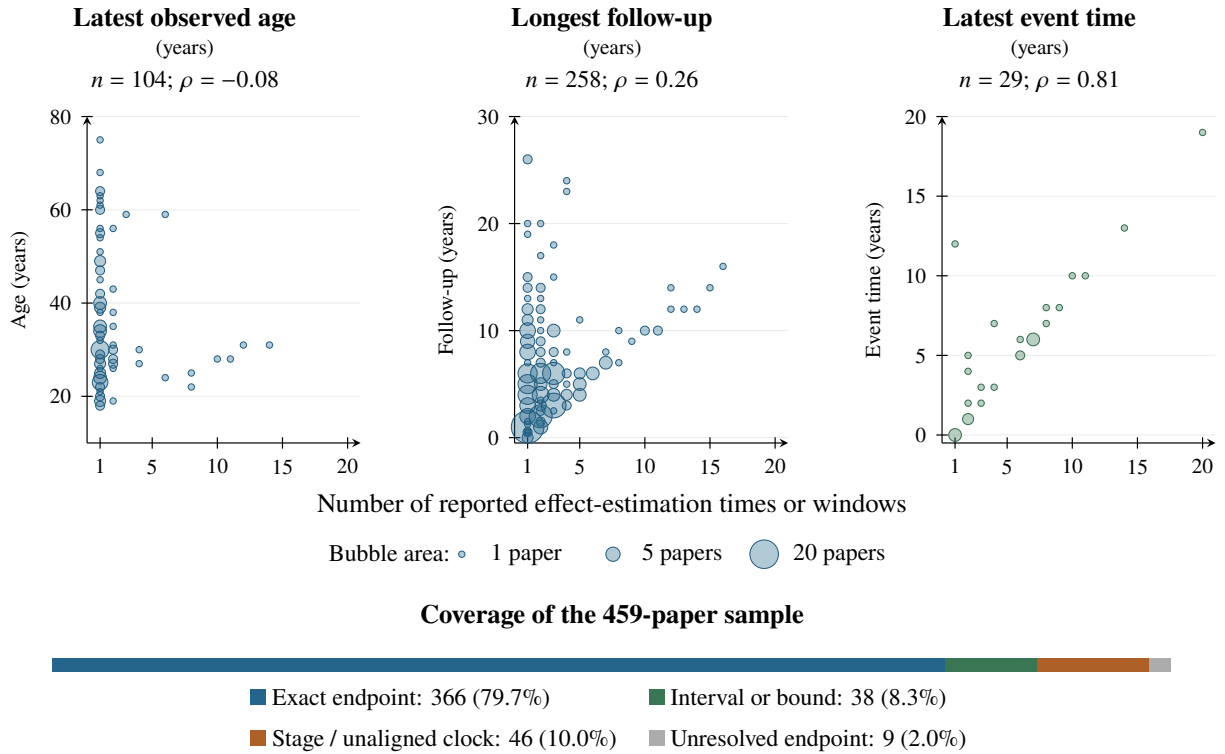
Notes: Each curve measures attainment relative to age 25 for its three-cohort group. ACS 2000–2022 includes all birthplaces and racial and ethnic groups. Person weights apply within cohort–age cells; cohorts have fixed one-third weights. Attainment panels share percentage-point scales; schooling uses years. SC+ is some college or higher; AA+ and BA+ are cumulative thresholds. Appendix A.6 documents the descriptive comparison.

Figure A2. Number of reported effect-estimation times or windows



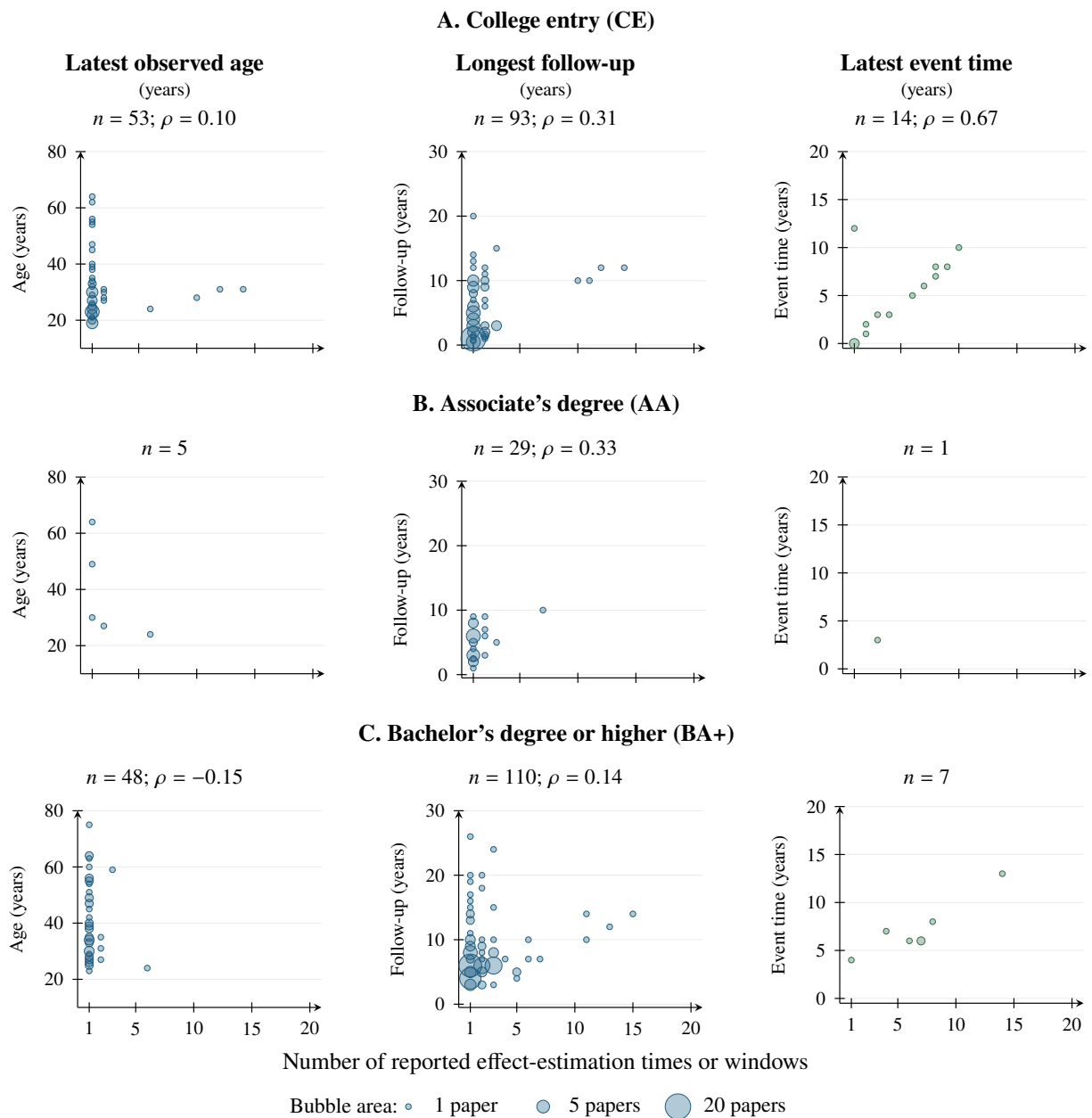
Notes: Percentages use papers with a usable count in each overlapping outcome group; one CE paper is excluded. A point or pooled window counts once; separately reported horizons count separately. Counts do not measure follow-up length or effect persistence. Axes are common. Table A1 gives counts; Appendix A.2 gives coding rules.

Figure A3. Estimation times and observation endpoints: all education outcomes



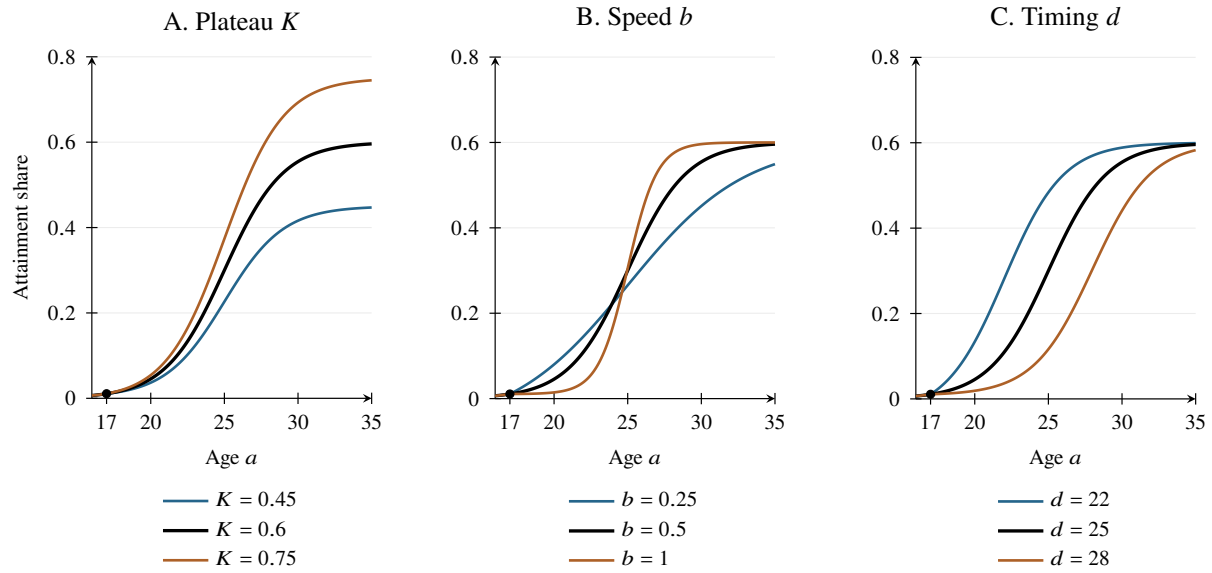
Notes: Bubble area counts papers at each coordinate. Panels separate attained age, individual follow-up, and policy event time, covering 366 distinct papers with overlap across clocks. The coverage bar includes all 459 papers; 93 lack an exact coordinate and receive no imputed endpoint. Correlations are paper-level Spearman correlations. Overall counts and endpoints may summarize different outcomes. Appendix A.2 gives coding rules.

Figure A4. Estimation times and observation endpoints by education outcome



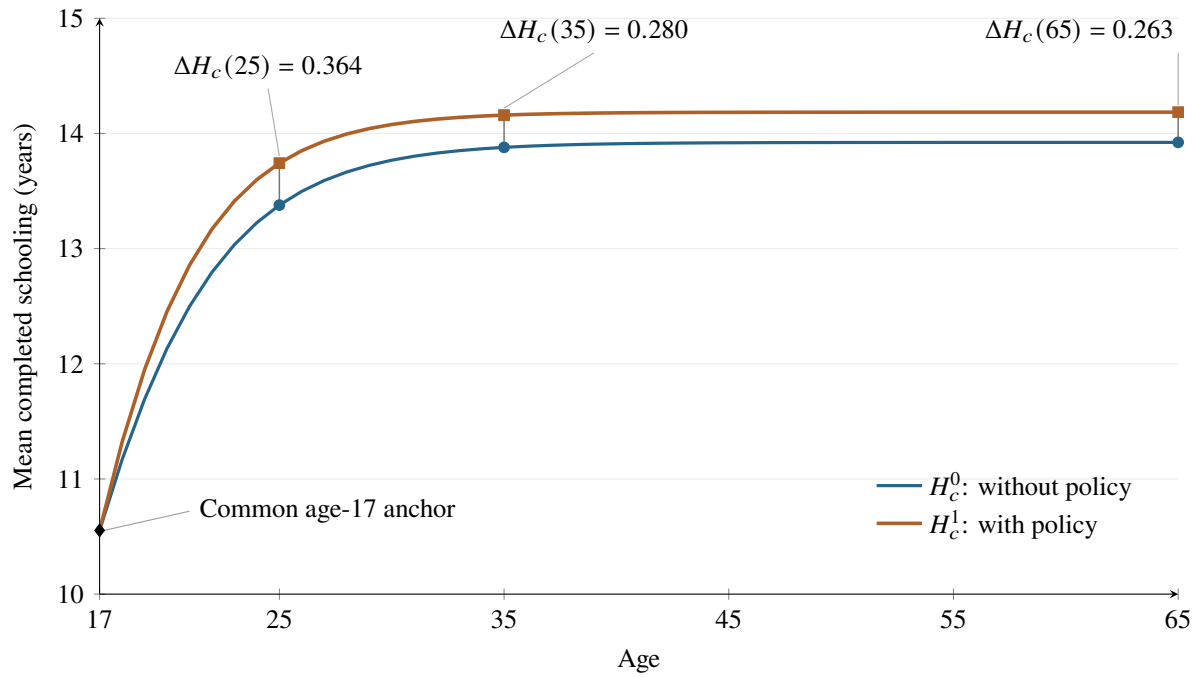
Notes: Rows show college entry, associate-degree outcomes, and BA+ outcomes; columns retain separate clocks. Scales are common within clocks; bubble areas share one count scale. Coverage is 146 CE, 34 AA, and 158 BA+ papers, with overlap. Spearman correlations require at least ten papers and variation in both coordinates. AA preserves each source's exact- or at-least-degree definition.

Figure B1. Parameter changes under a common pre-policy anchor



Notes: One parameter varies per panel; the others remain at $(K, b, d) = (0.60, 0.50, 25)$. All paths share the age-17 reference level. Black, blue, and orange denote reference, smaller, and larger parameter values. K governs the plateau; b and d govern speed and timing conditional on the anchor. In this normalized family, d need not mark half the plateau. Axes show hypothetical attainment shares.

Figure B2. Schooling paths underlying Figure 4



Notes: Blue is H_c^0 and orange is H_c^1 for Figure 4, with $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$. The baseline averages 1985–1987 cohort fits with one-third weights; paths share age 17. Labels show gaps at 25, 35, and 65. The early advantage narrows but remains positive at 65. Baseline ages above 35 are extrapolated.

128

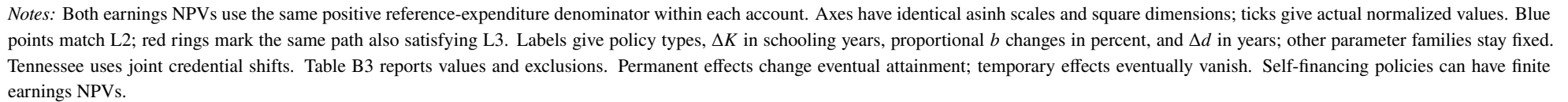
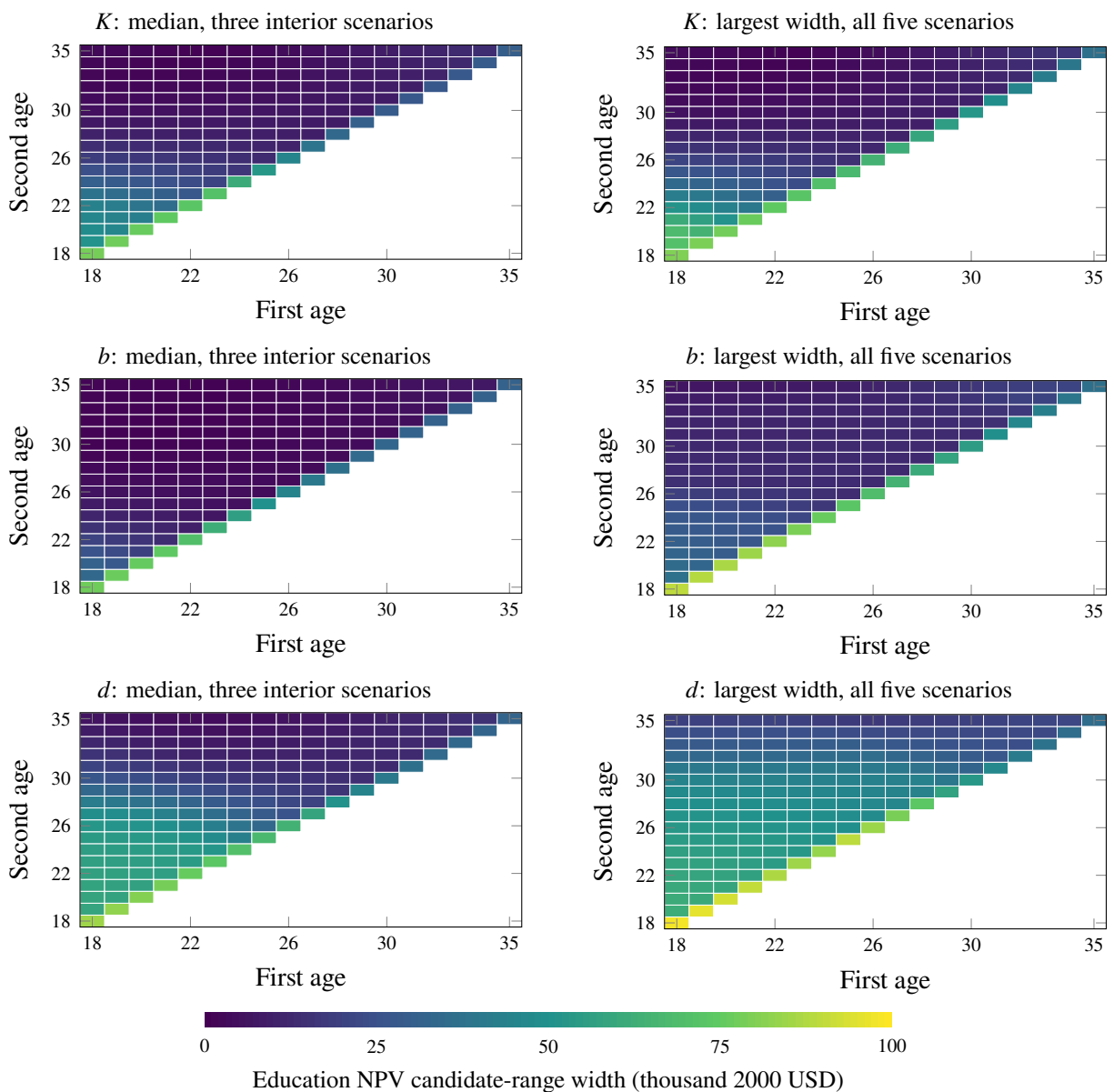
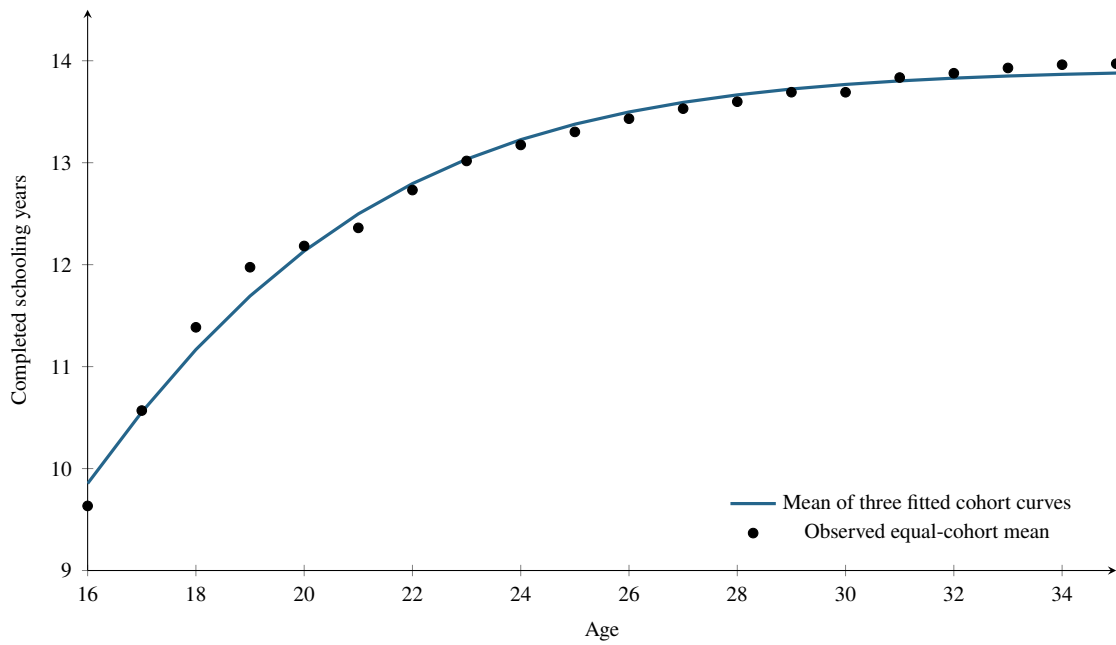


Figure B4. Observation timing and remaining education-NPV ranges



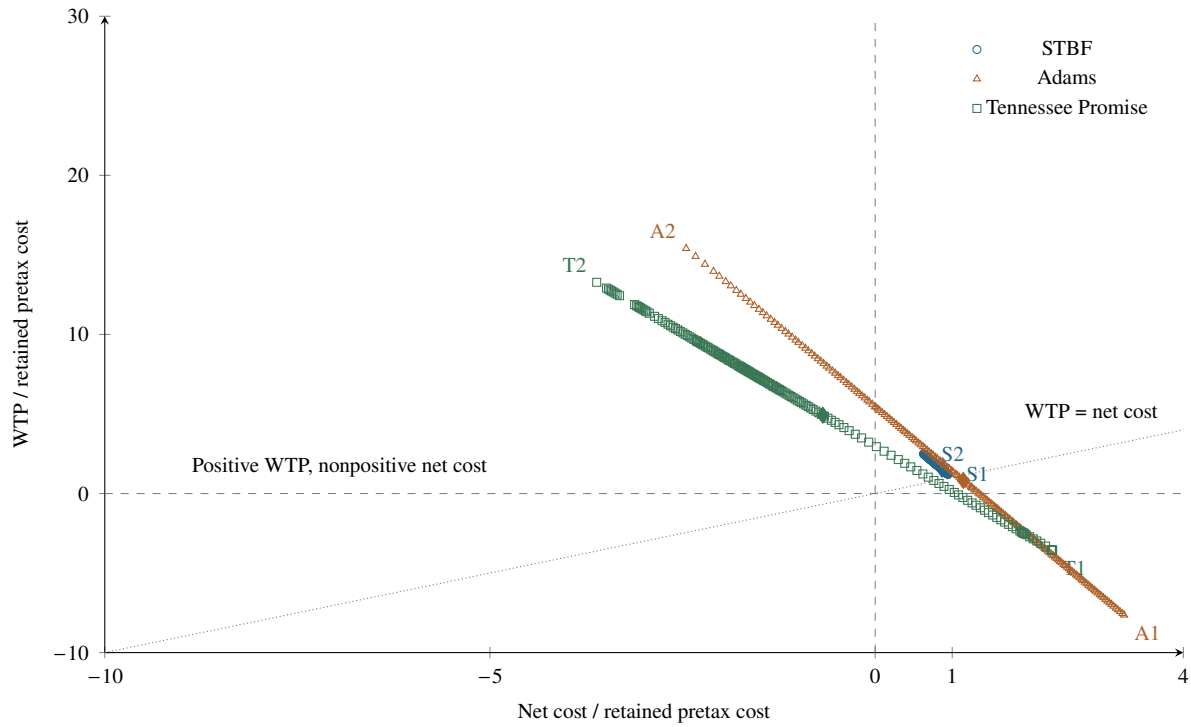
Notes: Diagonal cells use one observed age; upper-triangle cells use two. Widths are thousands of 2000 dollars per person on a common scale. Left panels take medians over three interior scenarios; right panels take maxima over all five. Values summarize verified candidates under common calibration, not confidence intervals or certified global bounds.

Figure C1. Comparison-path fit: ACS birth cohorts 1985–1987



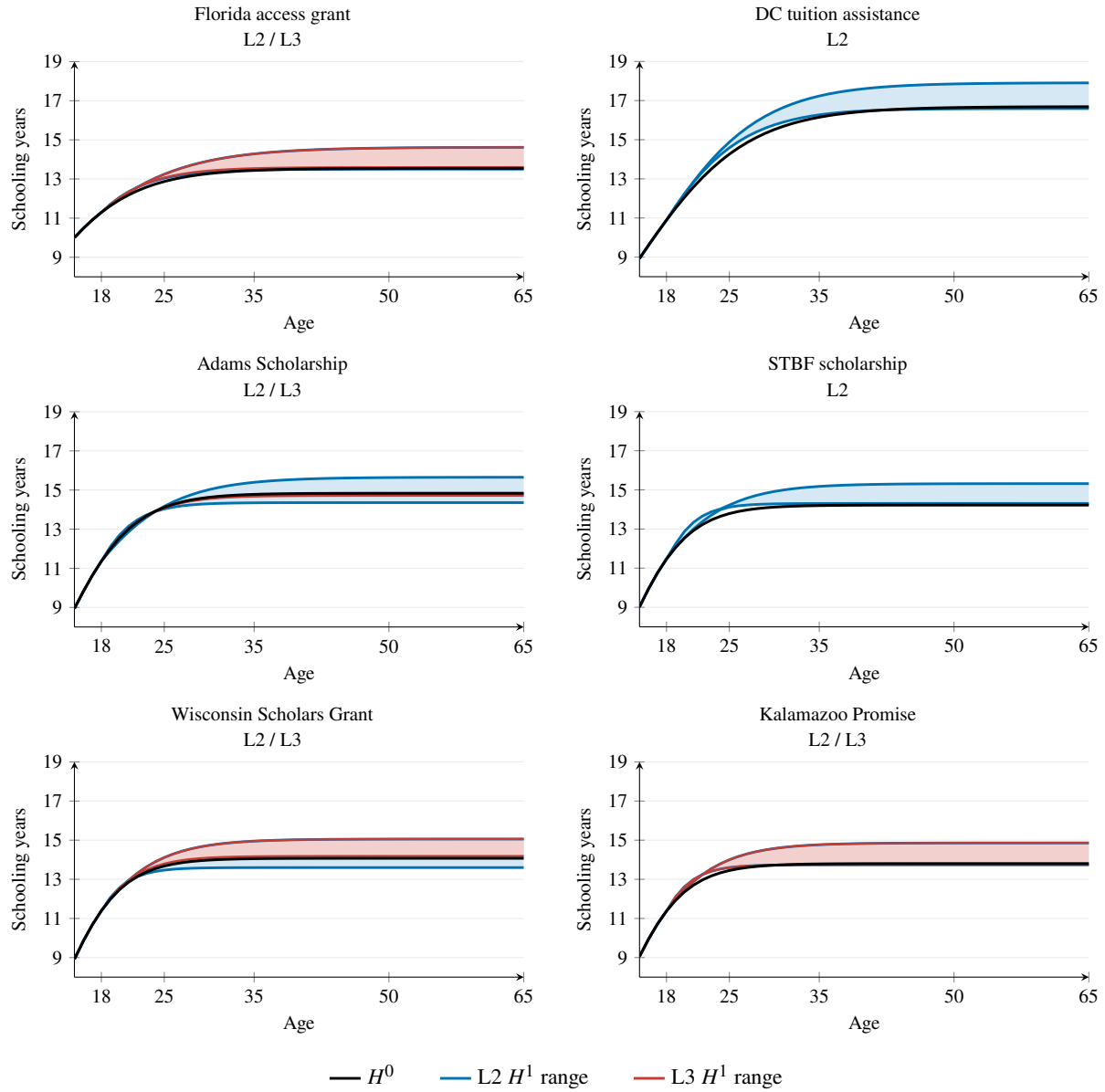
Notes: Observed support is ages 16–35. Cohort fits use person-weighted age means and normalized cell population weights. Fitted curves and observed means receive fixed one-third cohort weights. In-sample fit does not validate later extrapolation.

Figure D1. Education paths and joint welfare judgments



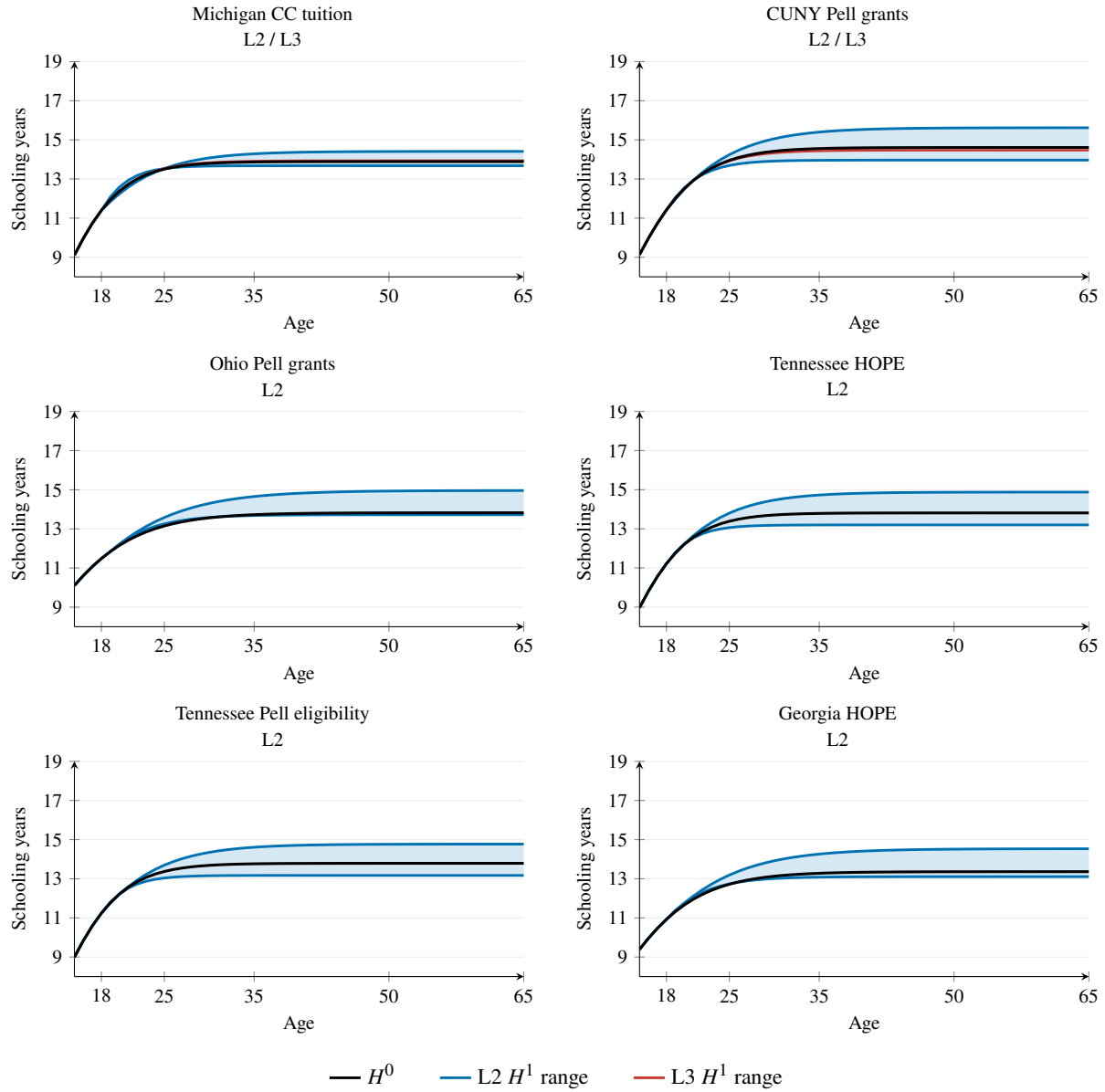
Notes: Each hollow point pairs WTP and net cost from one L2 path; displayed density has no probability interpretation. Diamonds denote L1 WTP and net cost. Both coordinates divide by the same retained positive cost: \$32,250 for STBF, \$1,658.06 for Adams, and \$14,792 for Tennessee. The diagonal denotes MVPF one only at positive net cost; $W > 0, N \leq 0$ denotes self-financing. Labels identify Table D3 paths.

Figure E1. Comparison and treated schooling paths



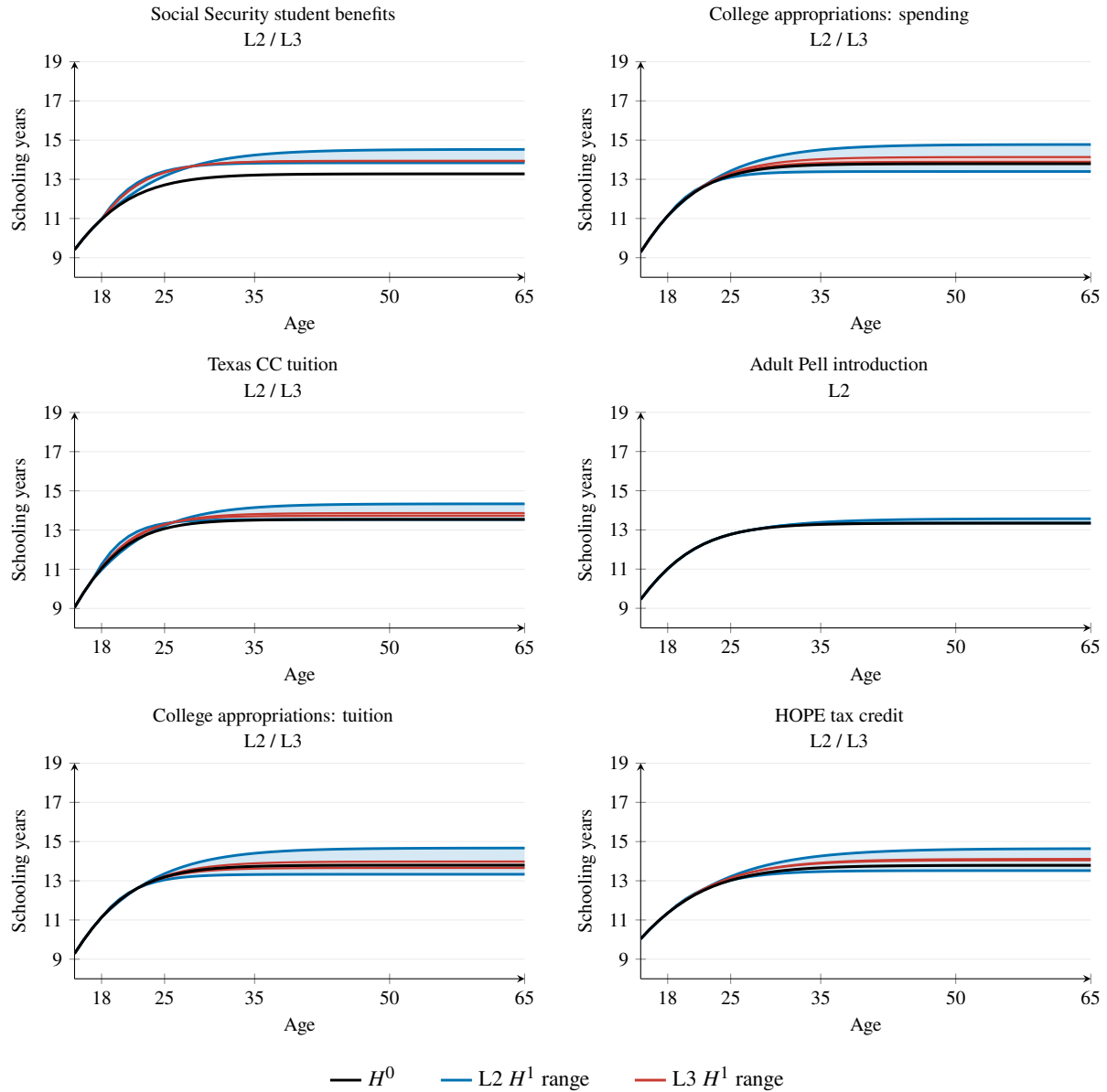
Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E1. Comparison and treated schooling paths (continued)



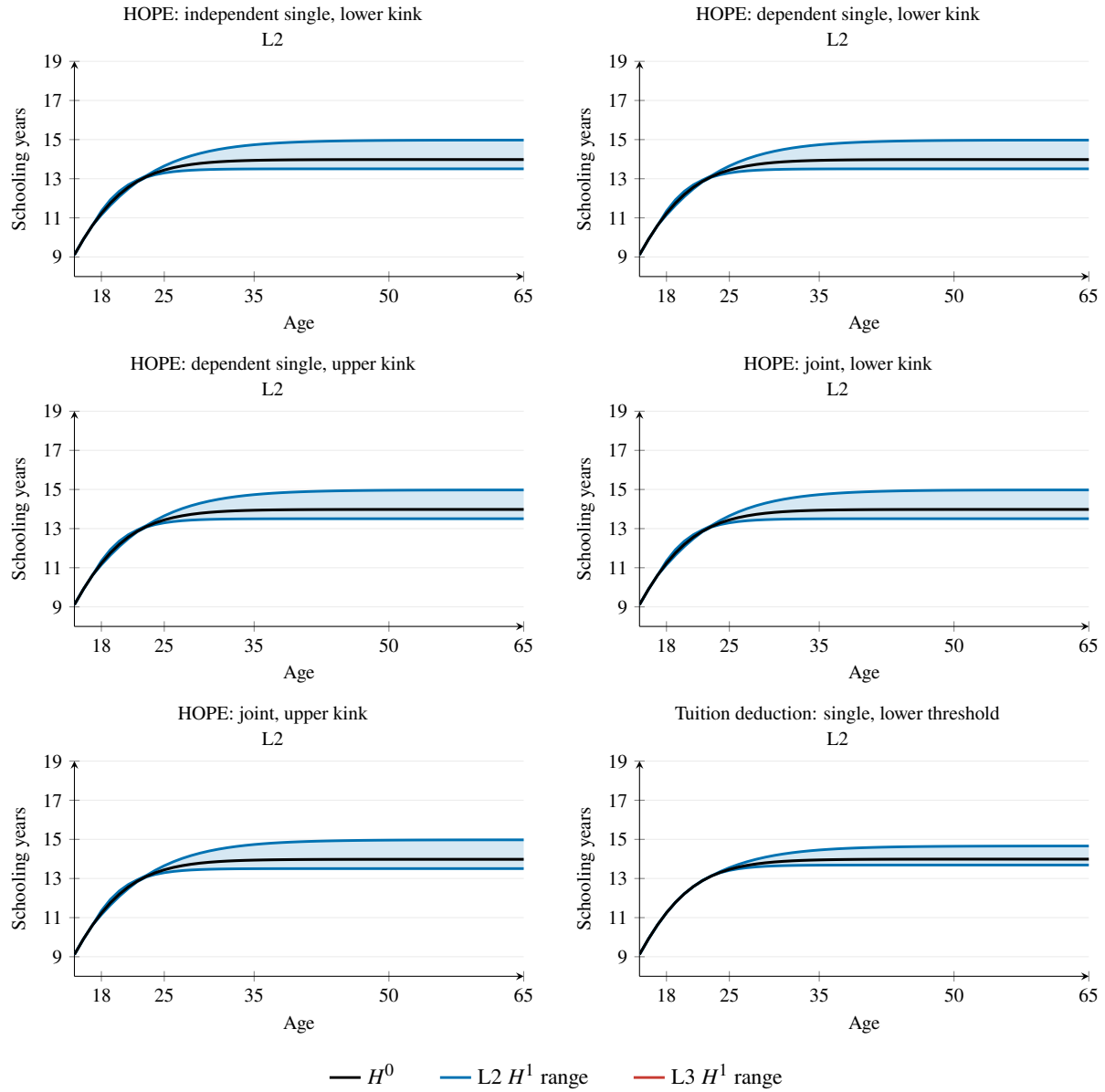
Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E1. Comparison and treated schooling paths (continued)



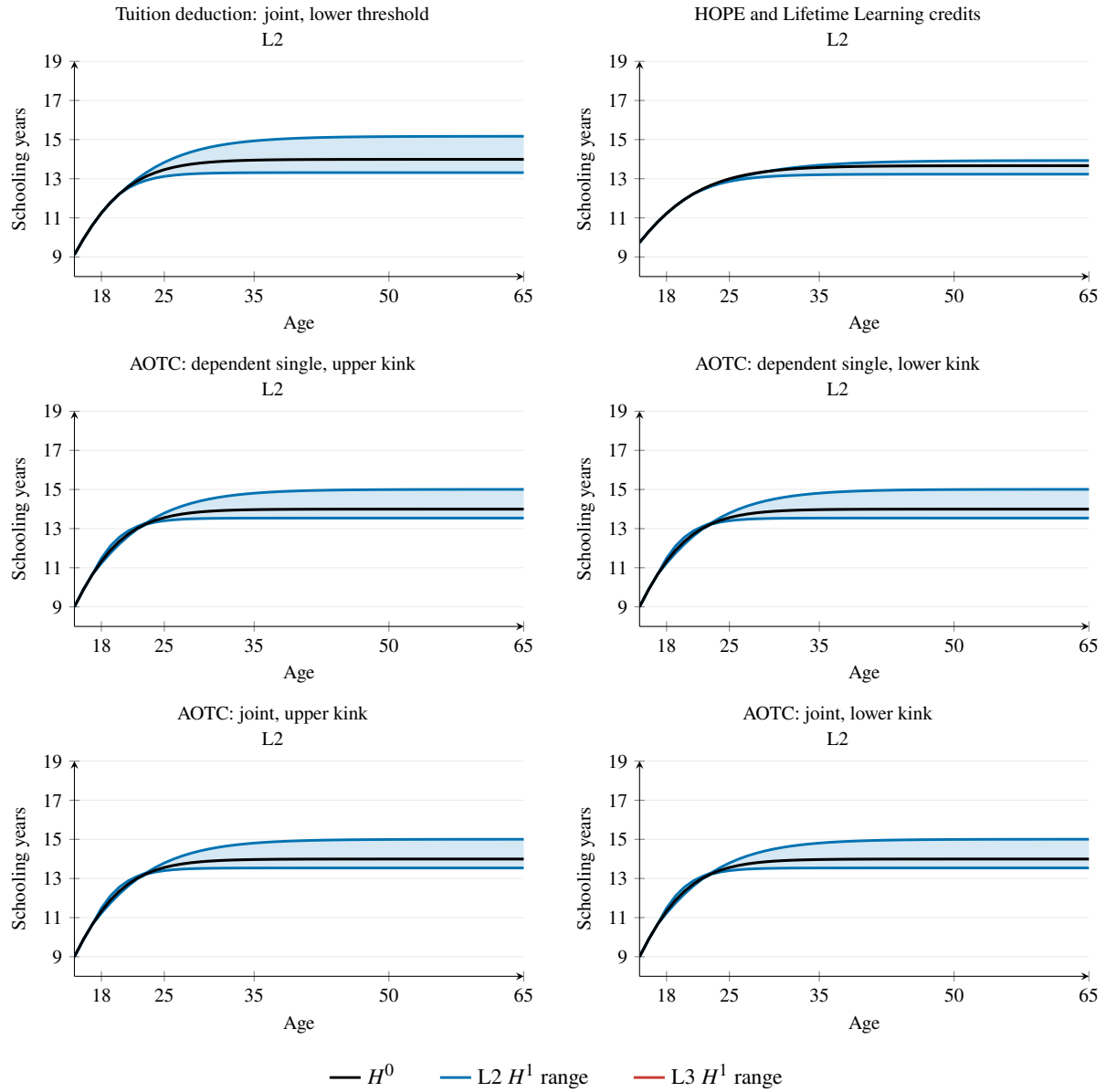
Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E1. Comparison and treated schooling paths (continued)



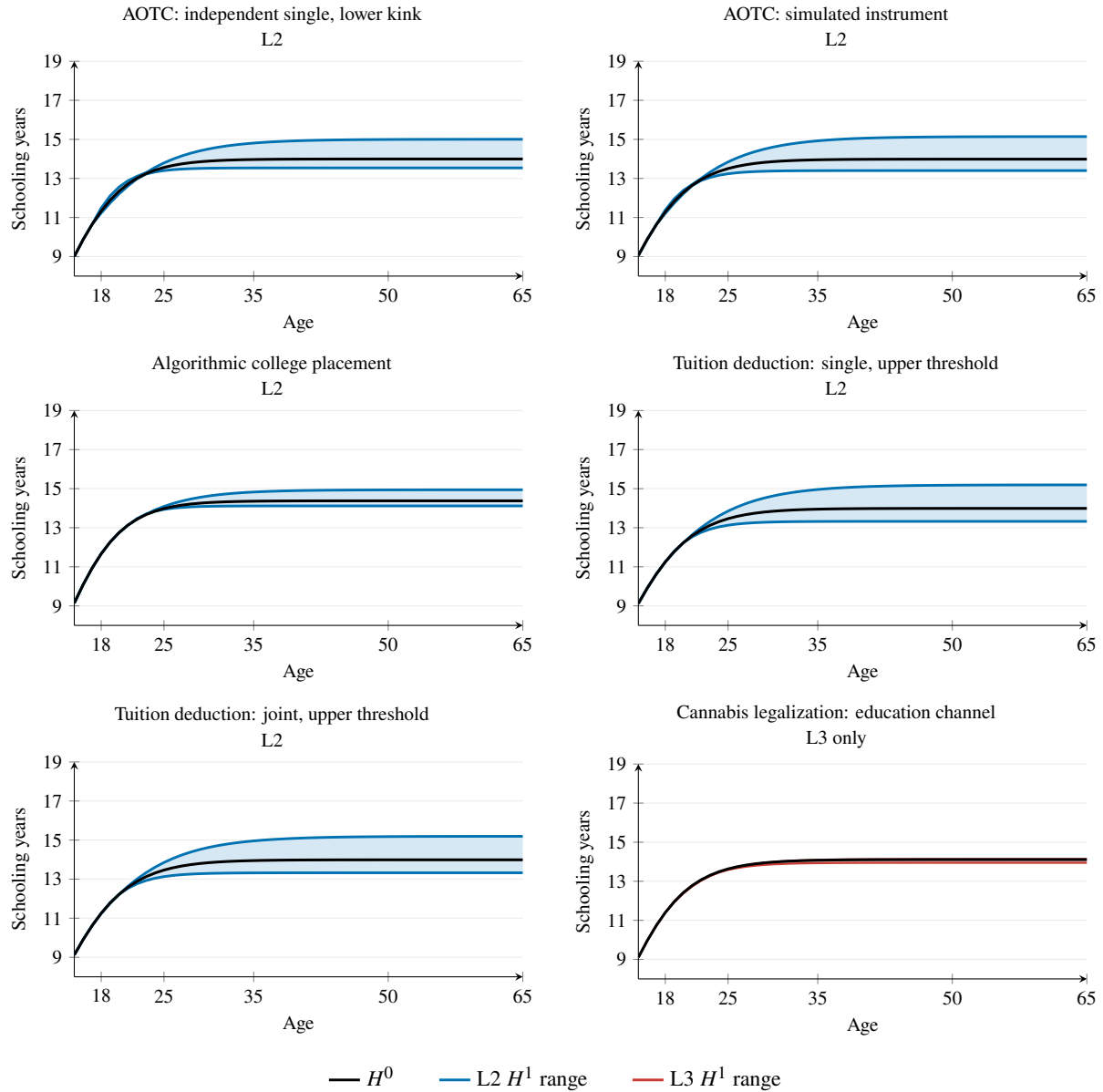
Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E1. Comparison and treated schooling paths (continued)



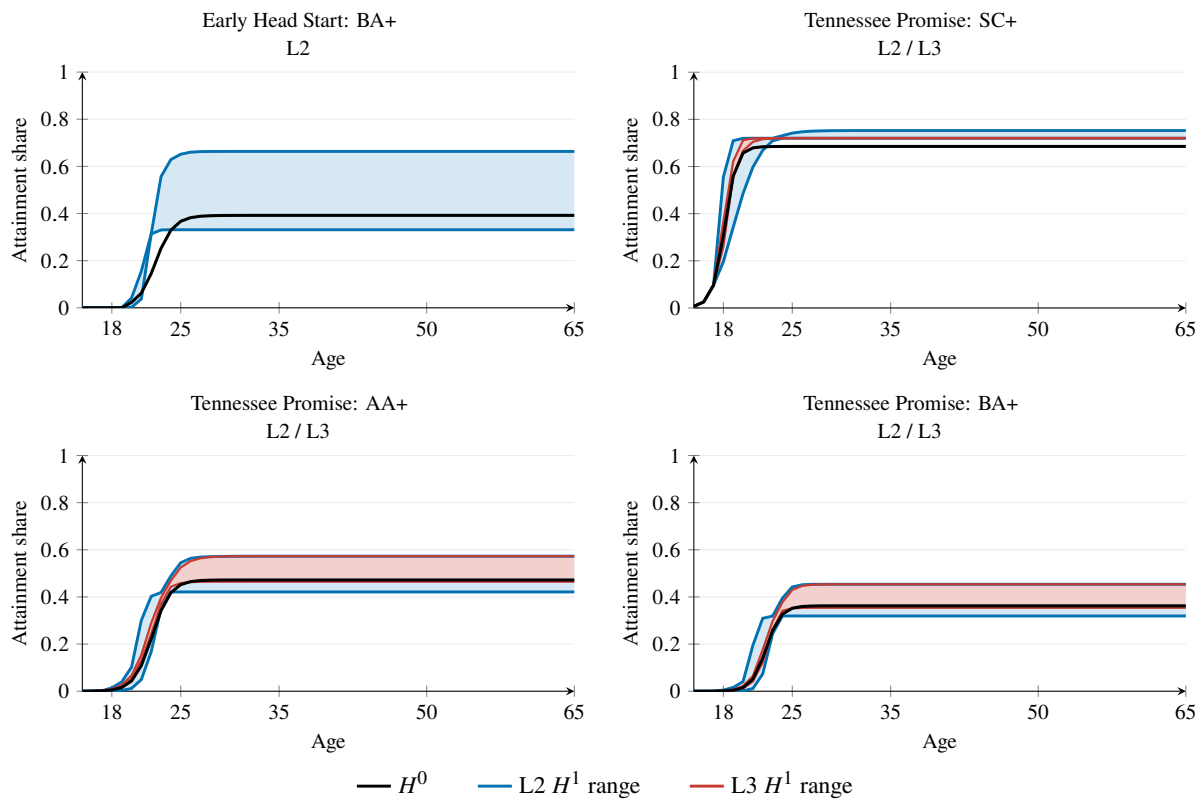
Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E1. Comparison and treated schooling paths (continued)



Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E2. Comparison and treated credential paths



Notes: Black is H^0 ; blue and red boundaries and shading show the pointwise H^1 ranges for L2 and L3. All panels share age and attainment-share scales. Tennessee ranges come from joint candidates satisfying credential nesting and the restrictions in Appendix D.8. Envelope edges may combine different candidates across ages and credentials; they need not form a feasible joint path. These finite-search ranges are not confidence bands or probability distributions. Targets and calibrations differ across the two applications.

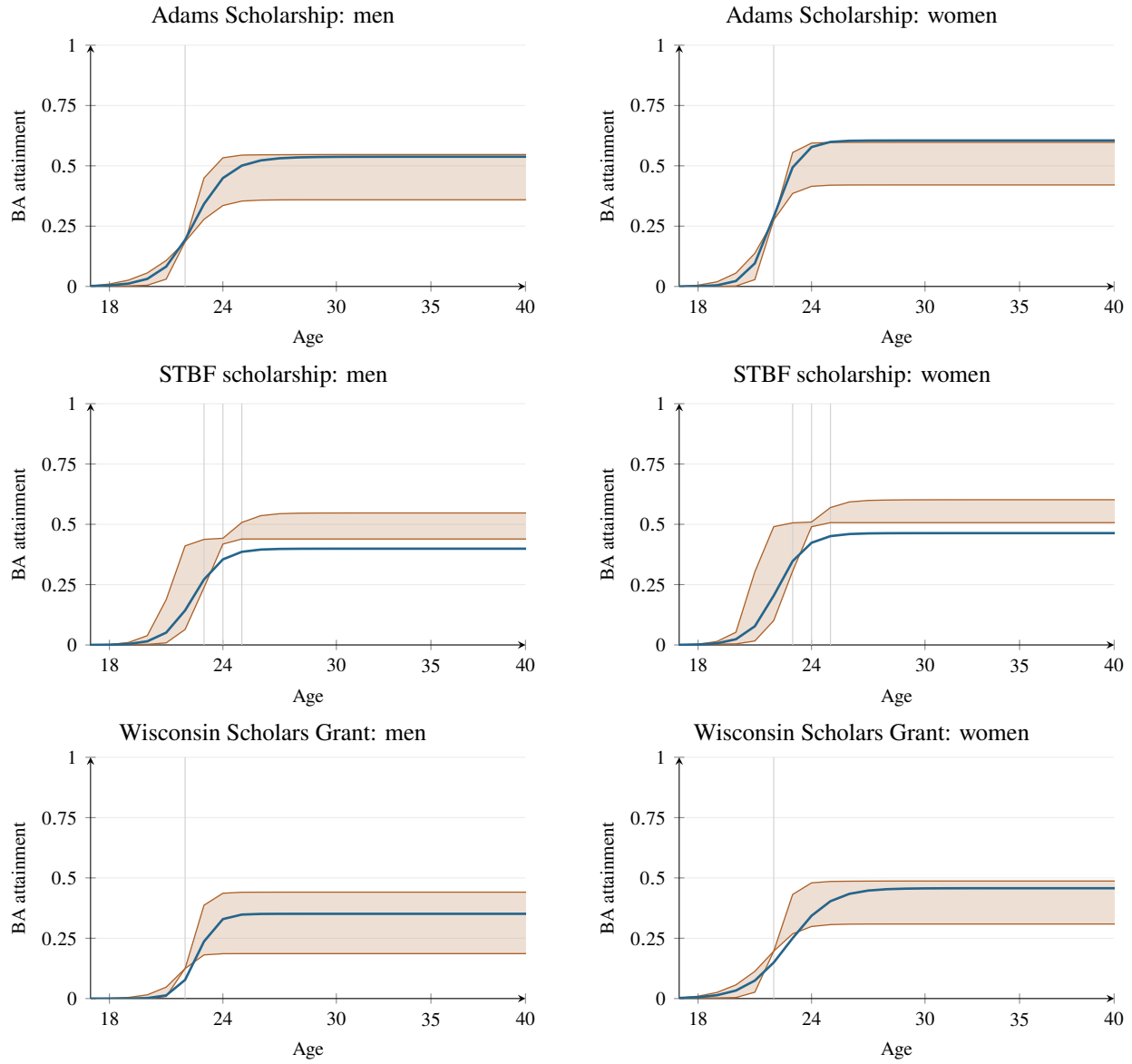
Figure E3. Point-input sensitivity across the policy collection

Policy calculation and information set	Sensitivity specification						
	Main	Narrow	Wide	National	Earnings 75%	Earnings 125%	Denser
Direct student financial aid							
Florida access grant (L3)						S	
DC tuition assistance (L2)	S	S	S	S	S	S	S
Adams Scholarship (L3)							
STBF scholarship (L2)							
Wisconsin Scholars Grant (L3)	S		S	S		S	S
Kalamazoo Promise (L3)							
CUNY Pell grants (L3)							
Ohio Pell grants (L2)	S	S	S	S	S	S	S
Tennessee HOPE (L2)	S	S	S	S	S	S	S
Tennessee Pell eligibility (L2)	S	S	S	S	S	S	S
Georgia HOPE (L2)	S		S	S		S	S
Social Security student benefits (L3)				=			
Adult Pell introduction (L2)	S	S	S	=	S	S	S
Tennessee Promise (L3)	S	S	S	S	S	S	S
Education tax benefits							
HOPE tax credit (L3)	S	S	S	=	S	S	S
HOPE: independent single, start (L2)	S	S	S	=	S	S	S
HOPE: dependent single, start (L2)	S	S	S	=	S	S	S
HOPE: dependent single, end (L2)	S	S	S	=	S	S	S
HOPE: joint, start (L2)	S	S	S	=	S	S	S
HOPE: joint, end (L2)	S	S	S	=	S	S	S
Tuition deduction: single, start (L2)	S	S	S	=	S	S	S
Tuition deduction: joint, start (L2)	S	S	S	=	S	S	S
HOPE and Lifetime Learning (L2)	S	S	S	=	S	S	S
AOTC: single, end (L2)	S	S	S	=	S	S	S
AOTC: single, start (L2)	S	S	S	=	S	S	S
AOTC: joint, end (L2)	S	S	S	=	S	S	S
AOTC: joint, start (L2)	S	S	S	=	S	S	S
AOTC: independent single, start (L2)	S	S	S	=	S	S	S
AOTC: simulated instrument (L2)	S	S	S	=	S	S	S
Tuition deduction: single, end (L2)	S	S	S	=	S	S	S
Tuition deduction: joint, end (L2)	S	S	S	=	S	S	S
Tuition and public-college funding							
Michigan CC tuition (L3)	S	S	S	S		S	S
College appropriations: spending (L3)	S	S	S	=	S	S	S
Texas CC tuition (L3)	S	S	S	S	S	S	S
College appropriations: tuition (L3)	S	S	S	=		S	S
College assessment and course placement							
Algorithmic college placement (L2)	S	S	S	S	S	S	S
Early childhood services							
Early Head Start (L2)				=			
Education channel of other policies							
Cannabis: education channel (L3)	S	S	S	S	S	S	S

■ Both sides of one ■ Only above one found ■ Only below one found

Notes: Colors refer to positive-cost MVPFs; gray means none was found. S marks a separate self-financing candidate. Equal signs indicate an already national baseline. Each alternative changes one specified assumption, with point inputs fixed. L3 is used where admitted; otherwise L2. Finite searches cannot establish absence of other paths. Table E4 reports values; Appendix E.9 defines the changes.

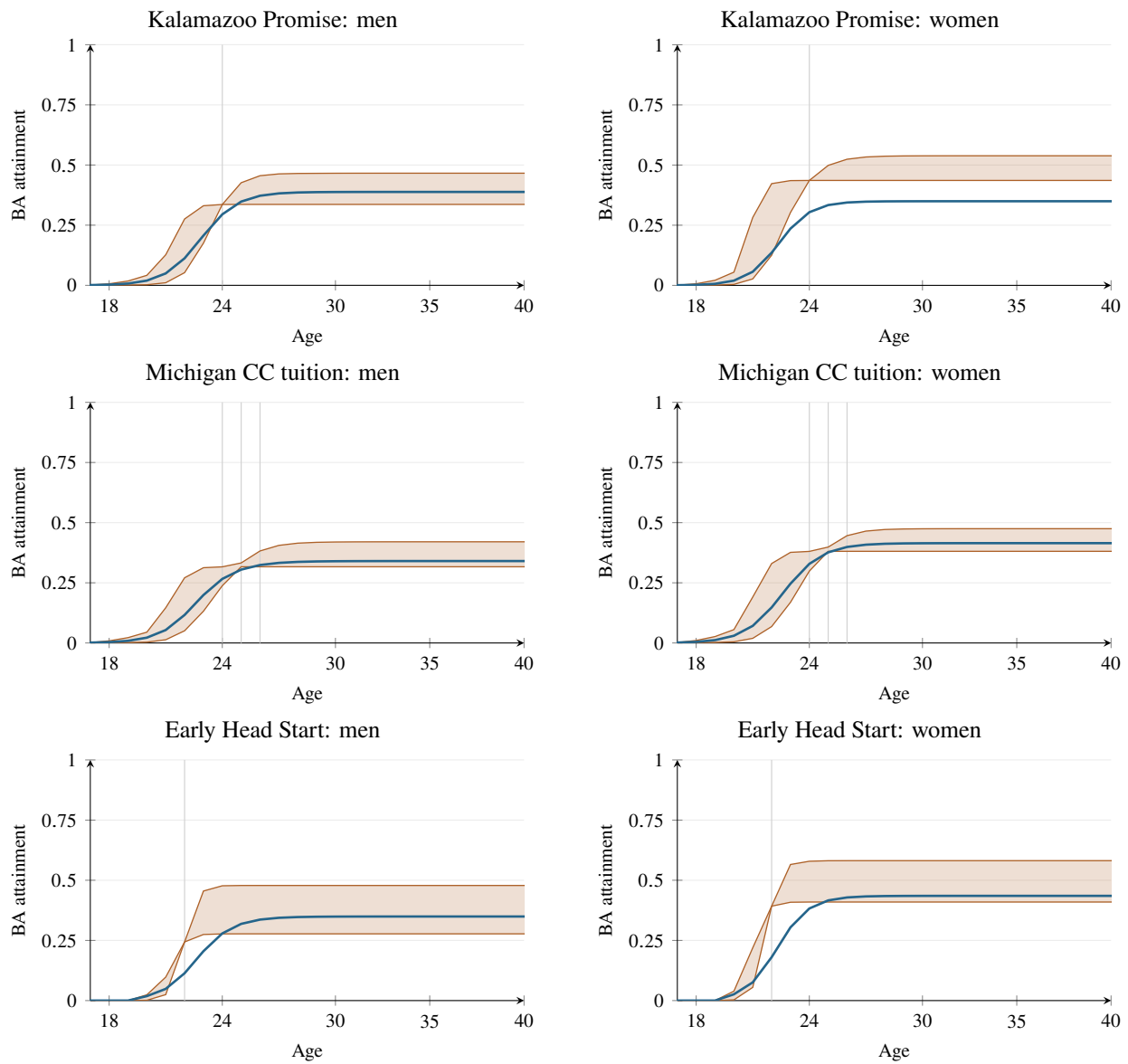
Figure G1. Male and female degree continuations: Adams Scholarship, STBF scholarship, Wisconsin Scholars Grant



Blue: H^0 Orange: candidate H^1 envelope

Notes: Blue is the sex-specific ACS comparison profile; orange boundaries and shading give pointwise H^1 candidate ranges. Edges need not form complete paths and are not confidence bands. Gray guides mark ages in the weighted source restrictions. Scales are common; ages above 35 are extrapolated. Appendix G.1 documents the degree inputs and populations.

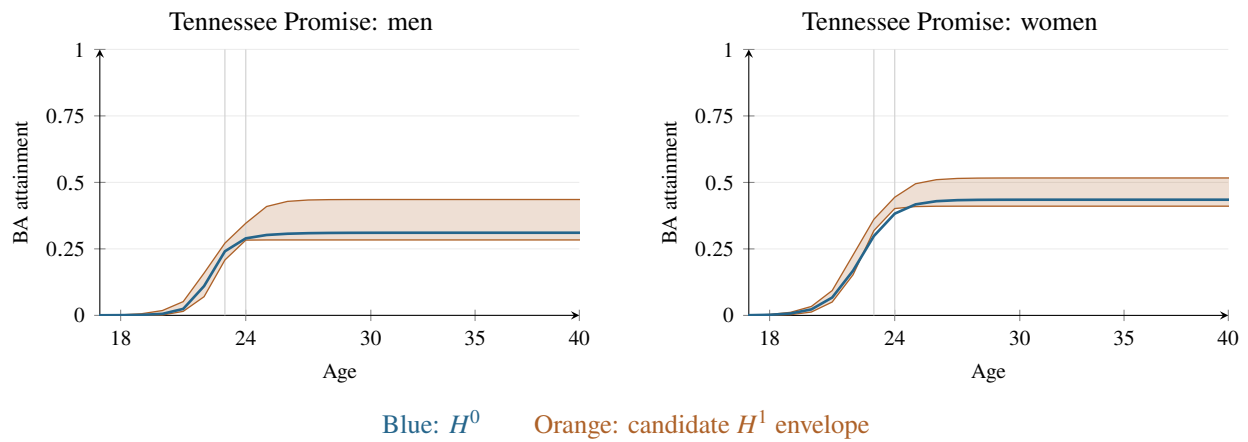
Figure G2. Male and female degree continuations: Kalamazoo Promise, Michigan CC tuition, Early Head Start



Blue: H^0 Orange: candidate H^1 envelope

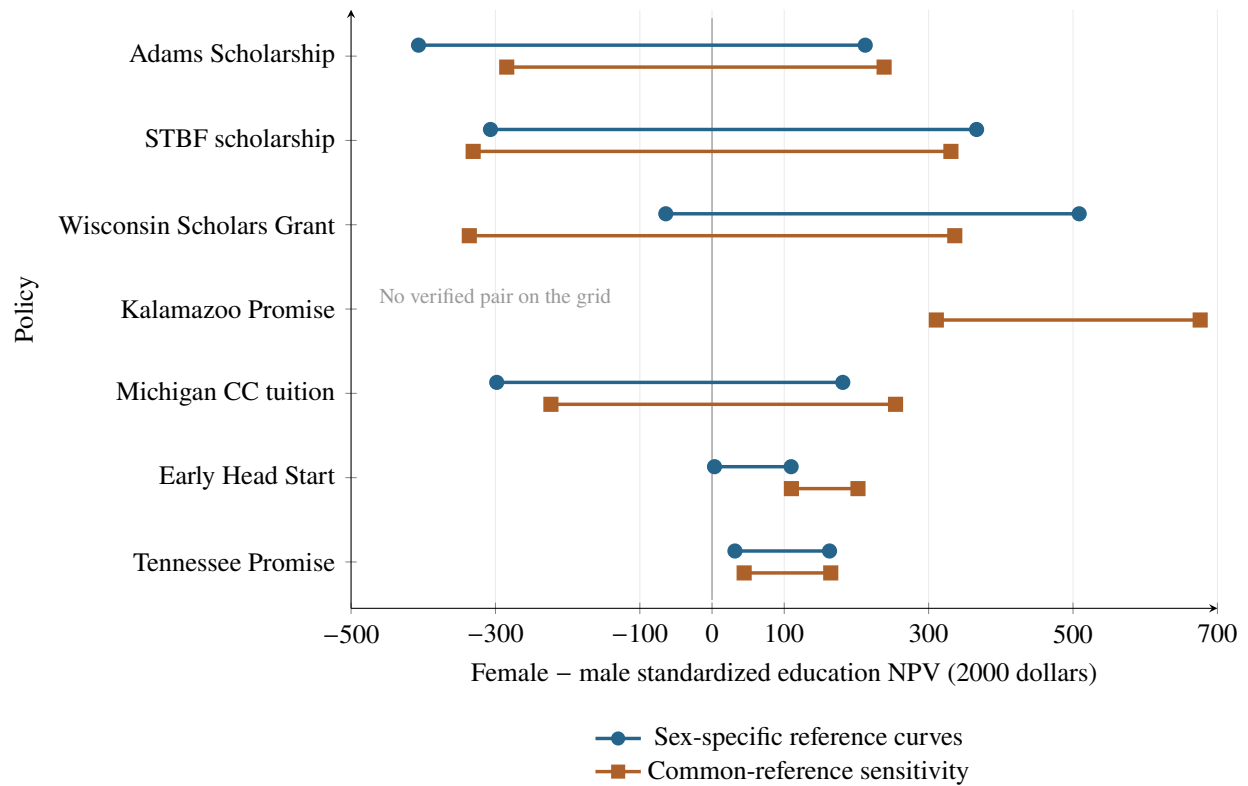
Notes: Blue is the sex-specific ACS comparison profile; orange boundaries and shading give pointwise H^1 candidate ranges. Edges need not form complete paths and are not confidence bands. Gray guides mark ages in the weighted restrictions. Scales are common; ages above 35 are extrapolated. Appendix G.1 explains Kalamazoo's control-level scaling and Early Head Start's degree-stock approximation.

Figure G3. Male and female degree continuations: Tennessee Promise



Notes: Blue is the sex-specific ACS comparison profile; orange boundaries and shading give pointwise H^1 candidate ranges. Edges need not form complete paths and are not confidence bands. Gray guides mark ages in the weighted restrictions. Ages above 35 are extrapolated. Appendix G.1 explains Tennessee's source intervals and outward graphical allowance, distinct from the pooled inward-extraction sensitivity.

Figure G4. Sex comparisons after fixing terminal education effects and valuation prices



Notes: Pairs match sex-specific source effects and the same age-65 BA effect. Blue spans use sex-specific baselines; orange spans use their mean. Positive differences favor women. Endpoints pool the tested terminal grid, holding terminal effects fixed within each pair. Missing spans mean no verified pair. Valuation uses common prices, two schooling years per BA increment, and 3 percent discounting to age 18 in 2000 dollars. Spans are numerical extrema, not probabilities or confidence intervals. Table G2 reports the grid.